THE EYE

VOLUME 7

Data Platform

The governed data, knowledge, memory, strategic state, product, trust, quality, lifecycle, and operating platform for The Eye

OBSERVE / UNDERSTAND / PREDICT / SIMULATE / DECIDE / LEARN

CONTROLLED DATA PLATFORM BASELINE / V1.0

Inherited from Volume 0 - Product Constitution, Volume 3 - Technical Architecture, Volume 4 - Engineering Specification, Volume 5 - AI Architecture, and Volume 6 - Infrastructure Architecture

2 August 2026

Document Control

Volume 7 is the controlled Data Platform baseline of The Eye. It governs how the frozen Strategic Intelligence Operating System acquires, contracts, identifies, versions, persists, processes, connects, serves, protects, corrects, recovers, exports, and retires data.

CONTROL	VALUE
Product	The Eye
Tagline	The Operating System for Strategic Intelligence
Controlled document	Volume 7 - Data Platform
Edition	1.0
Issue date	2 August 2026
Status	Controlled Data Platform Baseline
Binding sources	Volume 0 - Product Constitution v1.0; Volume 3 - Technical Architecture v1.0; Volume 4 - Engineering Specification v1.0; Volume 5 - AI Architecture v1.0; Volume 6 - Infrastructure Architecture v1.0
Primary audience	CTO, CIO, CDO, CISO, chief data and analytics officers, principal data and platform architects, data engineering, knowledge engineering, AI platform engineering, privacy, security, operations, governance, and customer technical authorities
Normative scope	Data authority; sources; ingestion; contracts; schemas; canonical objects; identity; temporal truth; provenance; correction; storage; processing; Knowledge Graph; Enterprise Memory; Digital Twins; data products; serving; privacy; quality; lifecycle; recovery; delivery; economics; acceptance; governance
Downstream inheritance	Volume 8 - Product Requirements Document; data domain specifications; source and product contracts; schemas; ontologies; asset and pipeline manifests; data quality plans; migration plans; operational runbooks; implementation repositories

Normative language

SHALL, MUST, SHALL NOT, and MUST NOT are binding. MAY permits an implementation choice inside a frozen authority, object, temporal, truth-state, provenance, policy, correction, failure, and evidence boundary. Requirements carry stable DP-CC-NNN identities. Conformance is scoped to exact contracts, schemas, code, data versions, policies, infrastructure, customer domain, deployment profile, consequence class, and period.

CONTROLLING RULE

Volume 0 controls constitutional meaning. Volume 3 controls product architecture. Volume 4 controls engineering realization. Volume 5 controls AI architecture. Volume 6 controls infrastructure. Volume 7 controls the Data Platform beneath and across those authorities. Data technology may transport, process, persist, and serve the system; it may not redefine truth, human authority, canonical ownership, or strategic meaning.

CONSTITUTIONAL INHERITANCE / C-003 · C-004 · C-008 · C-009 · C-011 · C-012 · C-035 · C-052

Data Platform Charter

The Eye operates one governed data foundation across the closed strategic loop. The Data Platform exists to preserve evidence, temporal truth, connected meaning, institutional memory, customer control, correction, replay, portability, and explainable strategic use.

DATA COMMITMENT	BINDING PLATFORM CONSEQUENCE
Evidence before interpretation	Immutable source evidence, rights, integrity, custody, time, and coverage are preserved before transformation or strategic use.
Contracts before structures	Datasets, streams, objects, graphs, indexes, features, APIs, and exports implement versioned semantic contracts.
Canonical ownership	Every authoritative lifecycle has one accountable component; projections remain rebuildable and cross-owner writes are prohibited.
Time and truth remain explicit	Event, observation, validity, record time, truth state, confidence, uncertainty, contradiction, and correction survive every path.
Knowledge stays connected	The Knowledge Graph anchors identity, relationships, provenance, strategy, memory, twins, foresight, and decisions without absorbing unlike authority.
Data use is purpose-bound	Identity, classification, rights, privacy, residency, retention, export, model use, and audit controls apply at every boundary.
Failure is visible state	Partial, stale, missing, denied, quarantined, disputed, recovering, and unreconciled data states are observable and testable.
One contract across deployments	SaaS, Private Cloud, On-Premise, disconnected, and air-gapped profiles preserve object, policy, correction, export, replay, and human-authority semantics.

Definition of Data Platform complete

- Every source, asset, object, contract, schema, ontology, pipeline, product, projection, exchange, policy, quality rule, lifecycle action, release, and owner has stable identity.
- Every material data object preserves tenant, time, truth state, provenance, policy, quality, correction, and accountable ownership through its full lifecycle.
- Every consumer can resolve exact inputs, versions, omissions, freshness, uncertainty, permissions, and evidence at the point of use.
- Every durable data class demonstrates availability, integrity, backup, restore, regional recovery, migration, export, deletion, and historical reconstruction evidence.
- No successful pipeline, query, index, model, or visualization is represented as strategic truth when authority, evidence, policy, quality, or temporal state is incomplete.

CONSTITUTIONAL INHERITANCE / C-002 · C-009 · C-010 · C-011 · C-012 · C-015 · C-035 · C-037 · C-052

Contents

Native Word heading styles provide navigation. Chapter, requirement, component, asset, contract, schema, object, source, zone, product, control, failure, metric, lifecycle, and decision identifiers are stable references for architecture, engineering, assurance, operations, customers, and downstream volumes.

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Executive Data Platform Statement

Volume 7 defines the governed data system that allows The Eye to observe the world, construct connected knowledge, remember institutional context, represent strategic state, reason about futures, support human decisions, and learn without rewriting history.

The Data Platform is organized around source contracts and immutable evidence; canonical data contracts and intelligence objects; four-dimensional temporal truth; explicit truth state and provenance; accountable state tiers; the Knowledge Graph and Enterprise Memory; governed twins and strategic packages; reusable data products; purpose-bound access; measurable quality; and evidence-driven lifecycle operations. Technologies remain replaceable behind stable component, asset, pipeline, product, and deployment contracts. Human authority, customer control, uncertainty, correction, replay, portability, and strategic meaning do not move with a storage or processing engine. Data quality is not successful processing; it is the demonstrated ability to preserve authority, temporal truth, provenance, policy, connected meaning, uncertainty, customer control, correction, replay, portability, and accountable recovery through collection, transformation, serving, failure, attack, disconnection, change, and retirement.

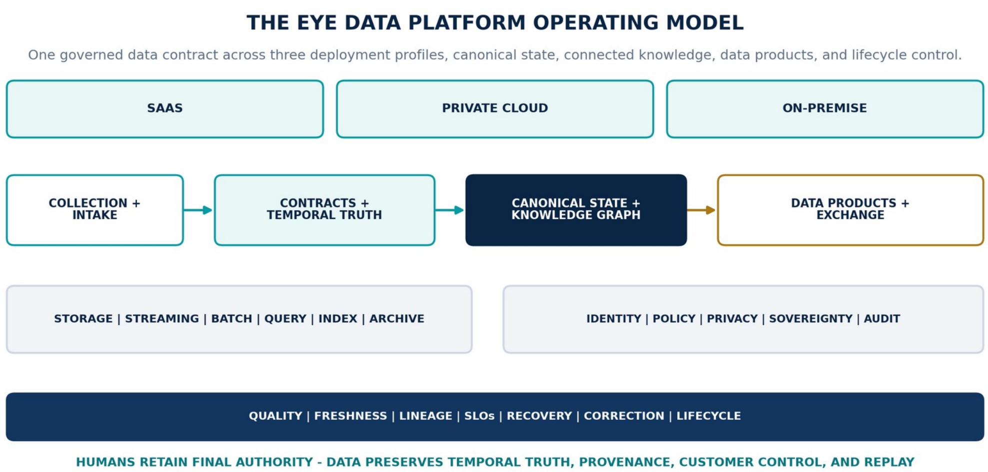

Figure 1. The Eye Data Platform operating model. SaaS, Private Cloud, and On-Premise implement the same governed data semantics through collection and intake, contracts and temporal truth, canonical state and the Knowledge Graph, data products and exchange, governed storage and processing, trust, quality, recovery, lifecycle control, and human authority.

DATA PLANE	ARCHITECTURE OBLIGATION
Acquisition and evidence plane	Register authorized sources; collect external and enterprise data; inspect, quarantine, preserve originals, measure coverage, and ingest corrections.
Contract and semantic plane	Govern schemas, canonical objects, identity, temporal meaning, truth state, provenance, policy metadata, compatibility, and non-destructive correction.
State and processing plane	Implement authoritative, evidence, analytical, operational, graph, memory, twin, stream, queue, index, feature, archive, and recovery classes.
Knowledge and strategic-state plane	Connect entities, relationships, events, objectives, risks, opportunities, twins, forecasts, scenarios, simulations, decisions, commitments, outcomes, and lessons.
Product and interoperability plane	Publish APIs, events, analytics, retrieval, context, features, datasets, exports, and marketplace packages through governed data products.
Trust, quality, and lifecycle plane	Enforce access, purpose, privacy, sovereignty, confidentiality, quality, freshness, audit, retention, deletion, resilience, release, economics, and acceptance.

CONSTITUTIONAL INHERITANCE / C-002 · C-005 · C-009 · C-010 · C-011 · C-012 · C-015 · C-035 · C-037 · C-049

PART I

Data Authority and Operating Model

Normative authority, ownership, data products, classification, traceability, and system context.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

1 Data Platform Authority and Scope

Data Platform Authority and Scope establishes the governed data capability necessary to translate the frozen product, engineering, AI, and infrastructure baselines into governed data capabilities without changing canonical ownership or strategic meaning. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

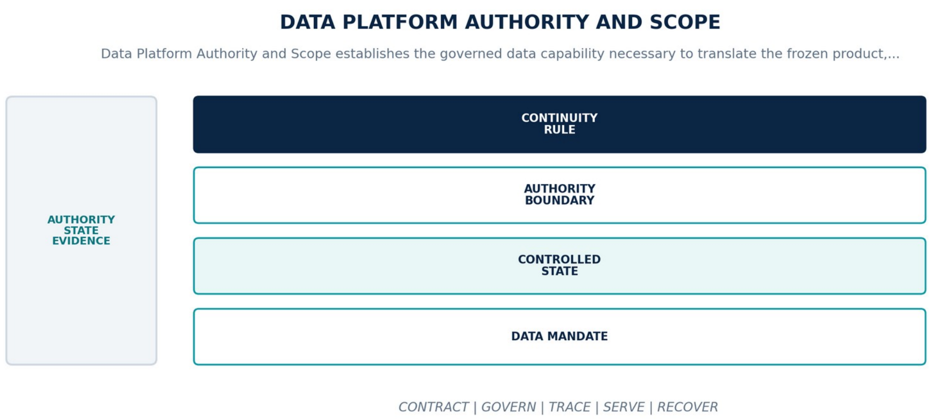

Figure 2. Data Platform Authority and Scope. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Translate the frozen product, engineering, AI, and infrastructure baselines into governed data capabilities without changing canonical ownership or strategic meaning.
Controlled state	Data capability identity, accountable owner, canonical layer, authority class, lifecycle state, deployment scope, control evidence, and conformance disposition.
Authority boundary	The Data Platform may transport, persist, process, index, and serve governed data but may not create an eleventh product layer, grant itself decision rights, or replace a canonical system of record.
Continuity rule	When a data capability, asset, pipeline, store, or product cannot resolve to an approved authority and ownership contract, the platform shall deny registration or production admission, preserve the disputed state, and route the conflict through architecture and data governance.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-01-001	The data-platform realization of Data Platform Authority and Scope SHALL translate the frozen product, engineering, AI, and infrastructure baselines into governed data capabilities without changing canonical ownership or strategic meaning.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-01-002	Every production instance SHALL declare and continuously reconcile data capability identity, accountable owner, canonical layer, authority class, lifecycle state, deployment scope, control evidence, and conformance disposition through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-01-003	The implementation SHALL enforce this authority boundary: the Data Platform may transport, persist, process, index, and serve governed data but may not create an eleventh product layer, grant itself decision rights, or replace a canonical system of record.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-01-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Platform Authority and Scope.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-01-005	When a data capability, asset, pipeline, store, or product cannot resolve to an approved authority and ownership contract, the platform SHALL deny registration or production admission, preserve the disputed state, and route the conflict through architecture and data governance; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-01-006	Production admission and continued operation SHALL require complete capability inventory, authority mapping, ownership acceptance, cross-deployment conformance, and signed operational evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A data capability, asset, pipeline, store, or product cannot resolve to an approved authority and ownership contract	Deny registration or production admission, preserve the disputed state, and route the conflict through architecture and data governance.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared data capability identity, accountable owner, canonical layer, authority class, lifecycle state, deployment scope, control evidence, and conformance disposition.
- Positive and negative boundary tests prove that the Data Platform may transport, persist, process, index, and serve governed data but may not create an eleventh product layer, grant itself decision rights, or replace a canonical system of record.
- Fault exercises inject a data capability, asset, pipeline, store, or product cannot resolve to an approved authority and ownership contract and verify that the platform can deny registration or production admission, preserve the disputed state, and route the conflict through architecture and data governance.
- Independent review accepts complete capability inventory, authority mapping, ownership acceptance, cross-deployment conformance, and signed operational evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Platform Authority and Scope is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 1, Ch. 4, Ch. 6, Ch. 11, Ch. 72. Primary Volume 5 AI anchors: Ch. 1, Ch. 6, Ch. 8, Ch. 13, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 1, Ch. 3, Ch. 6, Ch. 31, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-008 · C-009 · C-052

2 Specification Hierarchy and Data Conformance

Specification Hierarchy and Data Conformance establishes the governed data capability necessary to enforce the control order from constitutional meaning through data contracts, schemas, pipelines, products, releases, and observed operation. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

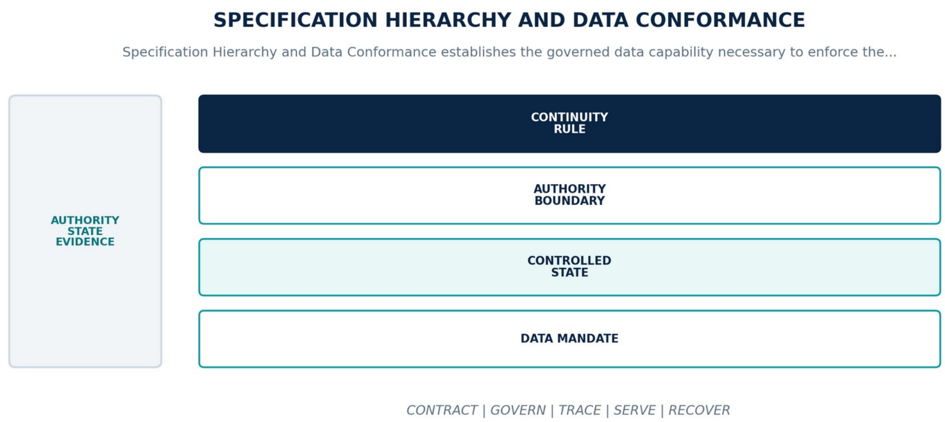

Figure 3. Specification Hierarchy and Data Conformance. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Enforce the control order from constitutional meaning through data contracts, schemas, pipelines, products, releases, and observed operation.
Controlled state	Normative data artifact identity, authority, version, effective time, supersession, exception, dependency closure, and affected release set.
Authority boundary	Lower-level data specifications may add precision but may not weaken temporal truth, provenance, customer control, access, retention, human authority, or canonical ownership.
Continuity rule	When two controlling artifacts conflict, a dependency becomes unresolved, or a data exception expires, the platform shall hold affected data release and downstream promotion, select the highest authority, expose impacts, and require explicit amendment or closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-02-001	The data-platform realization of Specification Hierarchy and Data Conformance SHALL enforce the control order from constitutional meaning through data contracts, schemas, pipelines, products, releases, and observed operation.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-02-002	Every production instance SHALL declare and continuously reconcile normative data artifact identity, authority, version, effective time, supersession, exception, dependency closure, and affected release set through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-02-003	The implementation SHALL enforce this authority boundary: lower-level data specifications may add precision but may not weaken temporal truth, provenance, customer control, access, retention, human authority, or canonical ownership.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-02-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Specification Hierarchy and Data Conformance.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-02-005	When two controlling artifacts conflict, a dependency becomes unresolved, or a data exception expires, the platform SHALL hold affected data release and downstream promotion, select the highest authority, expose impacts, and require explicit amendment or closure; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-02-006	Production admission and continued operation SHALL require machine-readable conformance manifests, conflict tests, dependency impact queries, exception expiry exercises, and independent review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Two controlling artifacts conflict, a dependency becomes unresolved, or a data exception expires	Hold affected data release and downstream promotion, select the highest authority, expose impacts, and require explicit amendment or closure.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared normative data artifact identity, authority, version, effective time, supersession, exception, dependency closure, and affected release set.
- Positive and negative boundary tests prove that lower-level data specifications may add precision but may not weaken temporal truth, provenance, customer control, access, retention, human authority, or canonical ownership.
- Fault exercises inject two controlling artifacts conflict, a dependency becomes unresolved, or a data exception expires and verify that the platform can hold affected data release and downstream promotion, select the highest authority, expose impacts, and require explicit amendment or closure.
- Independent review accepts machine-readable conformance manifests, conflict tests, dependency impact queries, exception expiry exercises, and independent review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Specification Hierarchy and Data Conformance is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 3, Ch. 6, Ch. 23, Ch. 72. Primary Volume 5 AI anchors: Ch. 2, Ch. 6, Ch. 14, Ch. 70, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 2, Ch. 6, Ch. 48, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-035 · C-051 · C-052

3 Data Principles and Semantic Parity

Data Principles and Semantic Parity establishes the governed data capability necessary to preserve one data contract and one strategic meaning across SaaS, Private Cloud, and On-Premise while permitting controlled physical implementation variance. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

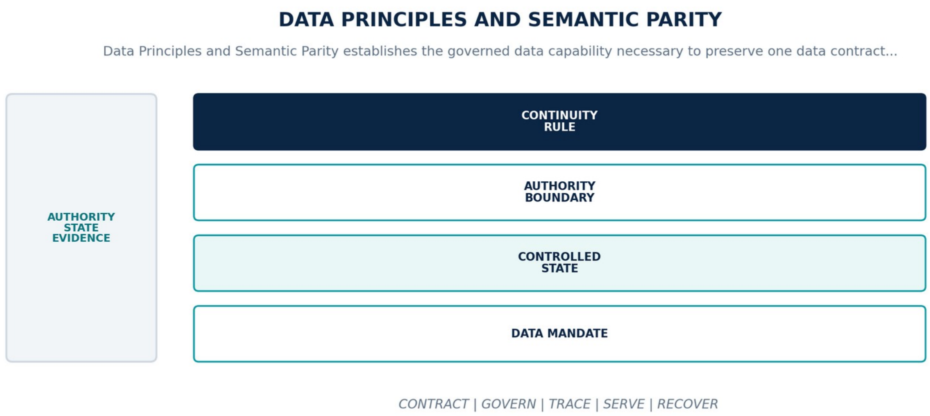

Figure 4. Data Principles and Semantic Parity. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Preserve one data contract and one strategic meaning across SaaS, Private Cloud, and On-Premise while permitting controlled physical implementation variance.
Controlled state	Deployment capability profile, semantic parity claim, permitted variance, contract suite, fixture set, evidence validity, and exception state.
Authority boundary	Providers, storage engines, processing frameworks, and operating models may vary; object, time, truth, provenance, policy, correction, export, failure, and replay semantics may not.
Continuity rule	When a deployment profile cannot pass the common data-contract and lifecycle conformance suite, the platform shall mark the affected capability unavailable or explicitly degraded and prohibit any claim of semantic parity until evidence closes.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-03-001	The data-platform realization of Data Principles and Semantic Parity SHALL preserve one data contract and one strategic meaning across SaaS, Private Cloud, and On-Premise while permitting controlled physical implementation variance.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-03-002	Every production instance SHALL declare and continuously reconcile deployment capability profile, semantic parity claim, permitted variance, contract suite, fixture set, evidence validity, and exception state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-03-003	The implementation SHALL enforce this authority boundary: providers, storage engines, processing frameworks, and operating models may vary; object, time, truth, provenance, policy, correction, export, failure, and replay semantics may not.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-03-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Principles and Semantic Parity.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-03-005	When a deployment profile cannot pass the common data-contract and lifecycle conformance suite, the platform SHALL mark the affected capability unavailable or explicitly degraded and prohibit any claim of semantic parity until evidence closes; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-03-006	Production admission and continued operation SHALL require identical golden data fixtures, contract diffs, export and re-import tests, lifecycle replay, and accountable acceptance for every deployment profile, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A deployment profile cannot pass the common data-contract and lifecycle conformance suite	Mark the affected capability unavailable or explicitly degraded and prohibit any claim of semantic parity until evidence closes.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared deployment capability profile, semantic parity claim, permitted variance, contract suite, fixture set, evidence validity, and exception state.
- Positive and negative boundary tests prove that providers, storage engines, processing frameworks, and operating models may vary; object, time, truth, provenance, policy, correction, export, failure, and replay semantics may not.
- Fault exercises inject a deployment profile cannot pass the common data-contract and lifecycle conformance suite and verify that the platform can mark the affected capability unavailable or explicitly degraded and prohibit any claim of semantic parity until evidence closes.
- Independent review accepts identical golden data fixtures, contract diffs, export and re-import tests, lifecycle replay, and accountable acceptance for every deployment profile for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Principles and Semantic Parity is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 5, Ch. 14, Ch. 23, Ch. 24, Ch. 30. Primary Volume 5 AI anchors: Ch. 7, Ch. 13, Ch. 20, Ch. 26, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 3, Ch. 8, Ch. 9, Ch. 10, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-007 · C-011 · C-012 · C-038 · C-042 · C-051

4 Data Ownership, Stewardship, and Accountability

Data Ownership, Stewardship, and Accountability establishes the governed data capability necessary to assign singular accountability for data authority, stewardship, technical custody, quality, access, lifecycle, and customer obligations. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

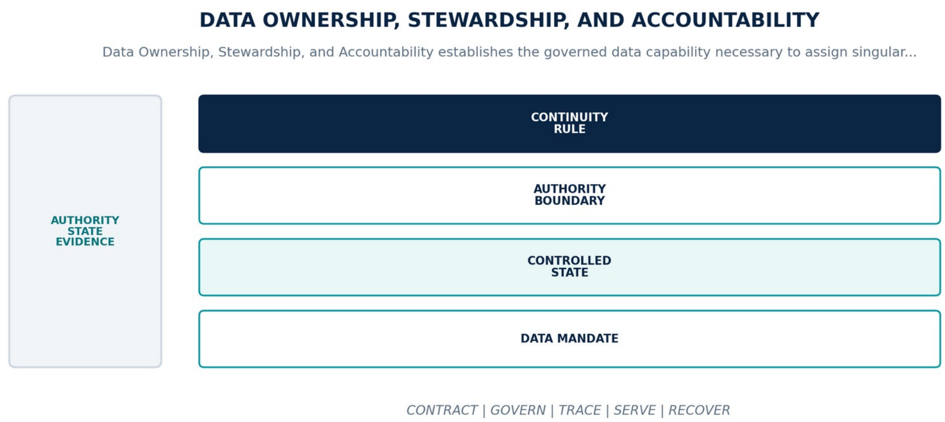

Figure 5. Data Ownership, Stewardship, and Accountability. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Assign singular accountability for data authority, stewardship, technical custody, quality, access, lifecycle, and customer obligations.
Controlled state	Data owner, steward, custodian, producer, consumer, policy authority, quality authority, escalation path, and evidence obligation.
Authority boundary	Shared platforms and managed services may distribute duties but may not create unowned data, ambiguous write authority, or silent transfer of customer control.
Continuity rule	When a material data asset or obligation has no accountable owner or conflicting authorities issue incompatible instructions, the platform shall stop authoritative mutation where safe, preserve the conflict, invoke the recorded escalation path, and keep the gap visible until accepted.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-04-001	The data-platform realization of Data Ownership, Stewardship, and Accountability SHALL assign singular accountability for data authority, stewardship, technical custody, quality, access, lifecycle, and customer obligations.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-04-002	Every production instance SHALL declare and continuously reconcile data owner, steward, custodian, producer, consumer, policy authority, quality authority, escalation path, and evidence obligation through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-04-003	The implementation SHALL enforce this authority boundary: shared platforms and managed services may distribute duties but may not create unowned data, ambiguous write authority, or silent transfer of customer control.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-04-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Ownership, Stewardship, and Accountability.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-04-005	When a material data asset or obligation has no accountable owner or conflicting authorities issue incompatible instructions, the platform SHALL stop authoritative mutation where safe, preserve the conflict, invoke the recorded escalation path, and keep the gap visible until accepted; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-04-006	Production admission and continued operation SHALL require signed responsibility matrices, ownership recertification, stewardship queue evidence, privileged-change records, and joint customer operating exercises, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A material data asset or obligation has no accountable owner or conflicting authorities issue incompatible instructions	Stop authoritative mutation where safe, preserve the conflict, invoke the recorded escalation path, and keep the gap visible until accepted.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared data owner, steward, custodian, producer, consumer, policy authority, quality authority, escalation path, and evidence obligation.
- Positive and negative boundary tests prove that shared platforms and managed services may distribute duties but may not create unowned data, ambiguous write authority, or silent transfer of customer control.
- Fault exercises inject a material data asset or obligation has no accountable owner or conflicting authorities issue incompatible instructions and verify that the platform can stop authoritative mutation where safe, preserve the conflict, invoke the recorded escalation path, and keep the gap visible until accepted.
- Independent review accepts signed responsibility matrices, ownership recertification, stewardship queue evidence, privileged-change records, and joint customer operating exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Ownership, Stewardship, and Accountability is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 11, Ch. 12, Ch. 29, Ch. 55. Primary Volume 5 AI anchors: Ch. 3, Ch. 25, Ch. 37, Ch. 47, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 4, Ch. 13, Ch. 49, Ch. 57, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-004 · C-006 · C-016 · C-036 · C-037 · C-049

5 Data Domain and Product Operating Model

Data Domain and Product Operating Model establishes the governed data capability necessary to organize reusable governed data products around stable business and intelligence domains without fragmenting the universal core. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

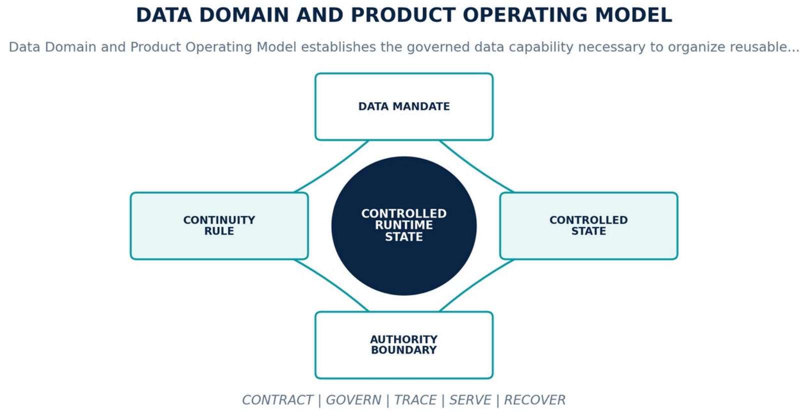

Figure 6. Data Domain and Product Operating Model. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Organize reusable governed data products around stable business and intelligence domains without fragmenting the universal core.
Controlled state	Domain identity, product identity, owner, purpose, consumers, authoritative inputs, outputs, contract, SLO, policy, cost, and lifecycle.
Authority boundary	Domain specialization may extend schemas, ontologies, policies, and products but may not fork canonical intelligence object, graph, evidence, or decision semantics.
Continuity rule	When a data product lacks a contract, duplicates canonical authority, or creates incompatible domain meaning, the platform shall deny publication, identify the canonical owner, reconcile the semantic conflict, and migrate consumers through a versioned compatibility plan.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-05-001	The data-platform realization of Data Domain and Product Operating Model SHALL organize reusable governed data products around stable business and intelligence domains without fragmenting the universal core.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-05-002	Every production instance SHALL declare and continuously reconcile domain identity, product identity, owner, purpose, consumers, authoritative inputs, outputs, contract, SLO, policy, cost, and lifecycle through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-05-003	The implementation SHALL enforce this authority boundary: domain specialization may extend schemas, ontologies, policies, and products but may not fork canonical intelligence object, graph, evidence, or decision semantics.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-05-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Domain and Product Operating Model.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-05-005	When a data product lacks a contract, duplicates canonical authority, or creates incompatible domain meaning, the platform SHALL deny publication, identify the canonical owner, reconcile the semantic conflict, and migrate consumers through a versioned compatibility plan; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-05-006	Production admission and continued operation SHALL require domain reviews, product scorecards, contract adoption, duplicate-authority detection, consumer acceptance, and retirement evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A data product lacks a contract, duplicates canonical authority, or creates incompatible domain meaning	Deny publication, identify the canonical owner, reconcile the semantic conflict, and migrate consumers through a versioned compatibility plan.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared domain identity, product identity, owner, purpose, consumers, authoritative inputs, outputs, contract, SLO, policy, cost, and lifecycle.
- Positive and negative boundary tests prove that domain specialization may extend schemas, ontologies, policies, and products but may not fork canonical intelligence object, graph, evidence, or decision semantics.
- Fault exercises inject a data product lacks a contract, duplicates canonical authority, or creates incompatible domain meaning and verify that the platform can deny publication, identify the canonical owner, reconcile the semantic conflict, and migrate consumers through a versioned compatibility plan.
- Independent review accepts domain reviews, product scorecards, contract adoption, duplicate-authority detection, consumer acceptance, and retirement evidence for each supported deployment profile.

> **CHAPTER ACCEPTANCE AND INHERITANCE**
> Data Domain and Product Operating Model is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 4, Ch. 11, Ch. 15, Ch. 23, Ch. 31, Ch. 40. Primary Volume 5 AI anchors: Ch. 8, Ch. 9, Ch. 13, Ch. 25, Ch. 27. Primary Volume 6 infrastructure anchors: Ch. 3, Ch. 31, Ch. 34, Ch. 35, Ch. 48.

CONSTITUTIONAL INHERITANCE / C-006 · C-007 · C-008 · C-009 · C-038 · C-049

6 Data Classification and Consequence Model

Data Classification and Consequence Model establishes the governed data capability necessary to bind sensitivity, rights, truth state, operational criticality, decision consequence, and handling obligations to every governed data object. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

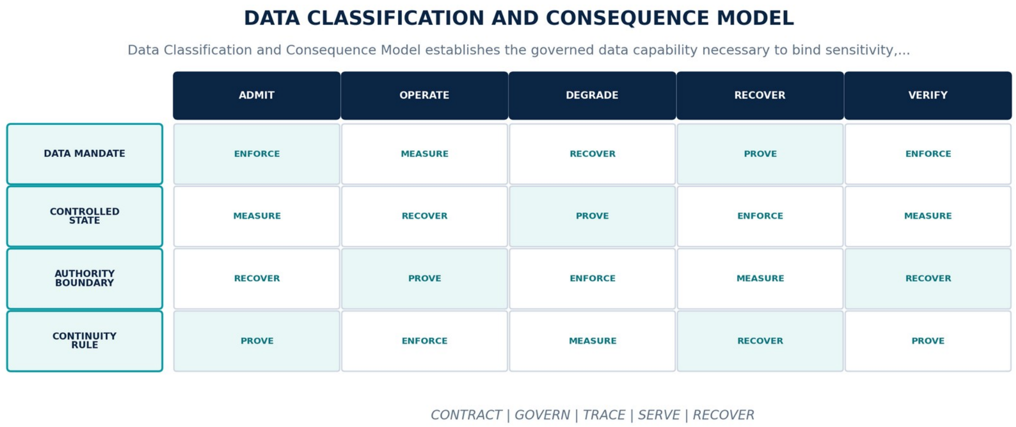

Figure 7. Data Classification and Consequence Model. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Bind sensitivity, rights, truth state, operational criticality, decision consequence, and handling obligations to every governed data object.
Controlled state	Classification label, handling policy, purpose, rights, consequence class, dissemination scope, review, expiry, and derivation rule.
Authority boundary	Classification is policy-bearing metadata enforced at collection, processing, storage, retrieval, model use, export, support, backup, and deletion boundaries.
Continuity rule	When classification is absent, conflicting, stale, or cannot be propagated through a transformation, the platform shall apply the most restrictive applicable handling, quarantine uncertain derivations, block unsafe serving, and route adjudication to the policy authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-06-001	The data-platform realization of Data Classification and Consequence Model SHALL bind sensitivity, rights, truth state, operational criticality, decision consequence, and handling obligations to every governed data object.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-06-002	Every production instance SHALL declare and continuously reconcile classification label, handling policy, purpose, rights, consequence class, dissemination scope, review, expiry, and derivation rule through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-06-003	The implementation SHALL enforce this authority boundary: classification is policy-bearing metadata enforced at collection, processing, storage, retrieval, model use, export, support, backup, and deletion boundaries.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-06-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Classification and Consequence Model.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-06-005	When classification is absent, conflicting, stale, or cannot be propagated through a transformation, the platform SHALL apply the most restrictive applicable handling, quarantine uncertain derivations, block unsafe serving, and route adjudication to the policy authority; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-06-006	Production admission and continued operation SHALL require classification fixtures, derivation tests, declassification review, cross-boundary negative tests, and end-to-end propagation evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Classification is absent, conflicting, stale, or cannot be propagated through a transformation	Apply the most restrictive applicable handling, quarantine uncertain derivations, block unsafe serving, and route adjudication to the policy authority.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared classification label, handling policy, purpose, rights, consequence class, dissemination scope, review, expiry, and derivation rule.
- Positive and negative boundary tests prove that classification is policy-bearing metadata enforced at collection, processing, storage, retrieval, model use, export, support, backup, and deletion boundaries.
- Fault exercises inject classification is absent, conflicting, stale, or cannot be propagated through a transformation and verify that the platform can apply the most restrictive applicable handling, quarantine uncertain derivations, block unsafe serving, and route adjudication to the policy authority.
- Independent review accepts classification fixtures, derivation tests, declassification review, cross-boundary negative tests, and end-to-end propagation evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Classification and Consequence Model is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 24, Ch. 27, Ch. 29, Ch. 50, Ch. 54. Primary Volume 5 AI anchors: Ch. 19, Ch. 20, Ch. 21, Ch. 29, Ch. 61. Primary Volume 6 infrastructure anchors: Ch. 13, Ch. 50, Ch. 53, Ch. 54, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-010 · C-012 · C-029 · C-035 · C-039 · C-040

7 Data Traceability and Documentation Inheritance

Data Traceability and Documentation Inheritance establishes the governed data capability necessary to connect every data requirement to constitutional invariants, canonical layers, engineering contracts, AI consumers, infrastructure services, manifests, tests, releases, and operations. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

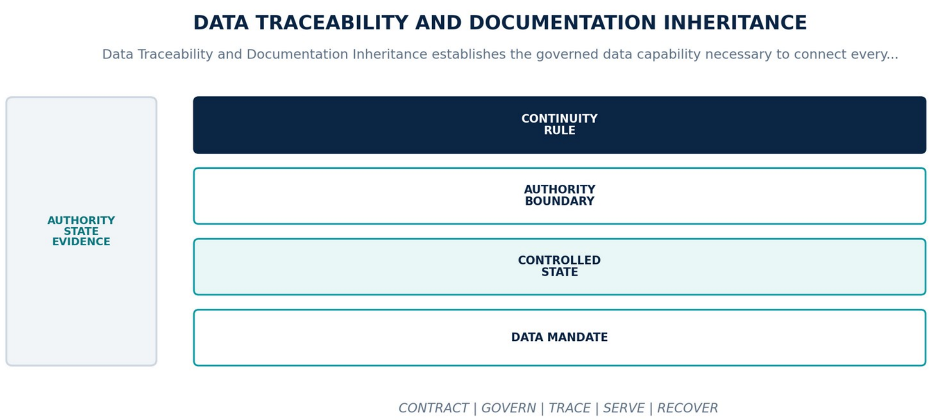

Figure 8. Data Traceability and Documentation Inheritance. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Connect every data requirement to constitutional invariants, canonical layers, engineering contracts, AI consumers, infrastructure services, manifests, tests, releases, and operations.
Controlled state	Requirement identity, source authority, owner, data component, asset, pipeline, product, evidence, release, exception, and retirement linkage.
Authority boundary	Traceability must cross documents, registries, repositories, schemas, jobs, runtime observations, incidents, exports, deletions, and historical replay without relying on prose similarity.
Continuity rule	When a production data artifact has incomplete, circular, stale, or unverifiable inheritance, the platform shall deny promotion or mark the running artifact non-conformant, expose affected products and decisions, and restore a closed evidence chain.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-07-001	The data-platform realization of Data Traceability and Documentation Inheritance SHALL connect every data requirement to constitutional invariants, canonical layers, engineering contracts, AI consumers, infrastructure services, manifests, tests, releases, and operations.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-07-002	Every production instance SHALL declare and continuously reconcile requirement identity, source authority, owner, data component, asset, pipeline, product, evidence, release, exception, and retirement linkage through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-07-003	The implementation SHALL enforce this authority boundary: traceability must cross documents, registries, repositories, schemas, jobs, runtime observations, incidents, exports, deletions, and historical replay without relying on prose similarity.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-07-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Traceability and Documentation Inheritance.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-07-005	When a production data artifact has incomplete, circular, stale, or unverifiable inheritance, the platform SHALL deny promotion or mark the running artifact non-conformant, expose affected products and decisions, and restore a closed evidence chain; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-07-006	Production admission and continued operation SHALL require bidirectional trace queries, orphan detection, release closure, runtime attestation, correction propagation, and retirement proofs, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A production data artifact has incomplete, circular, stale, or unverifiable inheritance	Deny promotion or mark the running artifact non-conformant, expose affected products and decisions, and restore a closed evidence chain.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared requirement identity, source authority, owner, data component, asset, pipeline, product, evidence, release, exception, and retirement linkage.
- Positive and negative boundary tests prove that traceability must cross documents, registries, repositories, schemas, jobs, runtime observations, incidents, exports, deletions, and historical replay without relying on prose similarity.
- Fault exercises inject a production data artifact has incomplete, circular, stale, or unverifiable inheritance and verify that the platform can deny promotion or mark the running artifact non-conformant, expose affected products and decisions, and restore a closed evidence chain.
- Independent review accepts bidirectional trace queries, orphan detection, release closure, runtime attestation, correction propagation, and retirement proofs for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Traceability and Documentation Inheritance is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 23, Ch. 28, Ch. 67, Ch. 72. Primary Volume 5 AI anchors: Ch. 6, Ch. 14, Ch. 29, Ch. 65, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 6, Ch. 45, Ch. 48, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-003 · C-012 · C-036 · C-043 · C-050 · C-052

8 Data Platform System Context

Data Platform System Context establishes the governed data capability necessary to define the end-to-end context connecting sources, enterprise systems, ingestion, processing, canonical state, graph, memory, twins, foresight, decisions, products, users, and external exchange. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

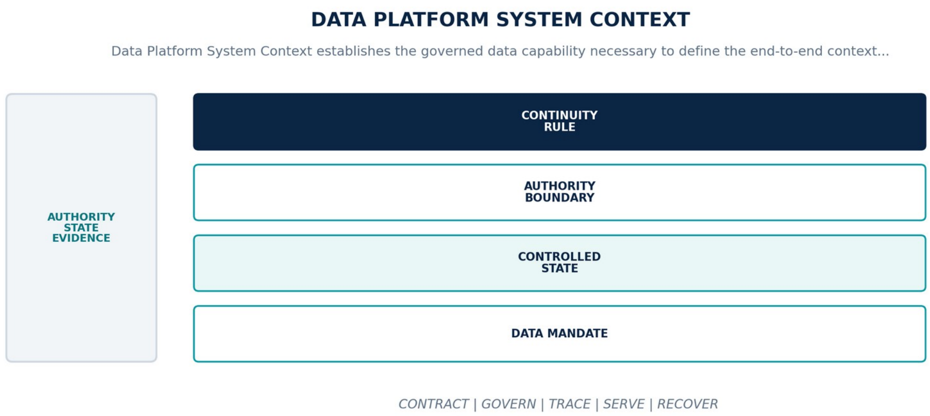

Figure 9. Data Platform System Context. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Define the end-to-end context connecting sources, enterprise systems, ingestion, processing, canonical state, graph, memory, twins, foresight, decisions, products, users, and external exchange.
Controlled state	Context node, trust boundary, data flow, authority, protocol, locality, classification, dependency, failure domain, and accountable organization.
Authority boundary	Every movement of governed data crosses an identified contract and control point; implicit trust and undocumented side channels are prohibited.
Continuity rule	When a material data flow, dependency, authority, or trust boundary is absent from the deployed context, the platform shall treat the path as denied or indeterminate, prevent new data movement, preserve in-flight evidence, and reconcile the context before admission.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-08-001	The data-platform realization of Data Platform System Context SHALL define the end-to-end context connecting sources, enterprise systems, ingestion, processing, canonical state, graph, memory, twins, foresight, decisions, products, users, and external exchange.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-08-002	Every production instance SHALL declare and continuously reconcile context node, trust boundary, data flow, authority, protocol, locality, classification, dependency, failure domain, and accountable organization through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-08-003	The implementation SHALL enforce this authority boundary: every movement of governed data crosses an identified contract and control point; implicit trust and undocumented side channels are prohibited.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-08-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Platform System Context.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-08-005	When a material data flow, dependency, authority, or trust boundary is absent from the deployed context, the platform SHALL treat the path as denied or indeterminate, prevent new data movement, preserve in-flight evidence, and reconcile the context before admission; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-08-006	Production admission and continued operation SHALL require deployed flow inventory, boundary probes, contract tests, policy traces, lineage closure, and customer acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A material data flow, dependency, authority, or trust boundary is absent from the deployed context	Treat the path as denied or indeterminate, prevent new data movement, preserve in-flight evidence, and reconcile the context before admission.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared context node, trust boundary, data flow, authority, protocol, locality, classification, dependency, failure domain, and accountable organization.
- Positive and negative boundary tests prove that every movement of governed data crosses an identified contract and control point; implicit trust and undocumented side channels are prohibited.
- Fault exercises inject a material data flow, dependency, authority, or trust boundary is absent from the deployed context and verify that the platform can treat the path as denied or indeterminate, prevent new data movement, preserve in-flight evidence, and reconcile the context before admission.
- Independent review accepts deployed flow inventory, boundary probes, contract tests, policy traces, lineage closure, and customer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Platform System Context is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 9, Ch. 13, Ch. 15, Ch. 19, Ch. 31. Primary Volume 5 AI anchors: Ch. 7, Ch. 8, Ch. 9, Ch. 13, Ch. 29. Primary Volume 6 infrastructure anchors: Ch. 7, Ch. 12, Ch. 14, Ch. 23, Ch. 31, Ch. 48.

CONSTITUTIONAL INHERITANCE / C-002 · C-008 · C-009 · C-035 · C-038 · C-042

PART II

Acquisition and Ingestion

Source contracts, external and enterprise acquisition, documents, streams, files, synchronization, validation, and raw preservation.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

9 Source Registry and Collection Contracts

Source Registry and Collection Contracts establishes the governed data capability necessary to govern every external, internal, physical, scientific, commercial, and custom source through a versioned collection contract before acquisition. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

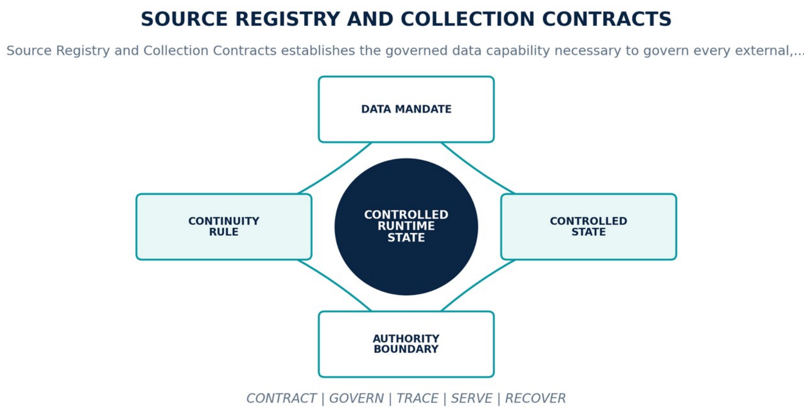

Figure 10. Source Registry and Collection Contracts. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern every external, internal, physical, scientific, commercial, and custom source through a versioned collection contract before acquisition.
Controlled state	Source identity, owner, authority, rights, purpose, endpoint, credential reference, schedule, coverage, freshness, schema, locality, and withdrawal path.
Authority boundary	Connectivity or public availability does not establish collection authority, permitted purpose, redistribution right, or fitness for strategic use.
Continuity rule	When a source contract is missing, expired, revoked, incompatible, or cannot prove authority and rights, the platform shall stop collection, preserve the last valid checkpoint and evidence, notify affected consumers, and require contract remediation before resumption.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-09-001	The data-platform realization of Source Registry and Collection Contracts SHALL govern every external, internal, physical, scientific, commercial, and custom source through a versioned collection contract before acquisition.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-09-002	Every production instance SHALL declare and continuously reconcile source identity, owner, authority, rights, purpose, endpoint, credential reference, schedule, coverage, freshness, schema, locality, and withdrawal path through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-09-003	The implementation SHALL enforce this authority boundary: connectivity or public availability does not establish collection authority, permitted purpose, redistribution right, or fitness for strategic use.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-09-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Source Registry and Collection Contracts.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-09-005	When a source contract is missing, expired, revoked, incompatible, or cannot prove authority and rights, the platform SHALL stop collection, preserve the last valid checkpoint and evidence, notify affected consumers, and require contract remediation before resumption; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-09-006	Production admission and continued operation SHALL require source registration review, rights and purpose evidence, contract tests, revocation drills, coverage validation, and customer approval, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A source contract is missing, expired, revoked, incompatible, or cannot prove authority and rights	Stop collection, preserve the last valid checkpoint and evidence, notify affected consumers, and require contract remediation before resumption.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared source identity, owner, authority, rights, purpose, endpoint, credential reference, schedule, coverage, freshness, schema, locality, and withdrawal path.
- Positive and negative boundary tests prove that connectivity or public availability does not establish collection authority, permitted purpose, redistribution right, or fitness for strategic use.
- Fault exercises inject a source contract is missing, expired, revoked, incompatible, or cannot prove authority and rights and verify that the platform can stop collection, preserve the last valid checkpoint and evidence, notify affected consumers, and require contract remediation before resumption.
- Independent review accepts source registration review, rights and purpose evidence, contract tests, revocation drills, coverage validation, and customer approval for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Source Registry and Collection Contracts is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 23, Ch. 29, Ch. 31, Ch. 54. Primary Volume 5 AI anchors: Ch. 29, Ch. 39, Ch. 40, Ch. 61, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 21, Ch. 28, Ch. 47, Ch. 50, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-012 · C-013 · C-014 · C-035 · C-037 · C-040

10 External Source Ingestion

External Source Ingestion establishes the governed data capability necessary to acquire governments, regulations, news, markets, research, papers, patents, filings, supply-chain, climate, geopolitical, social, satellite, and other authorized external data. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

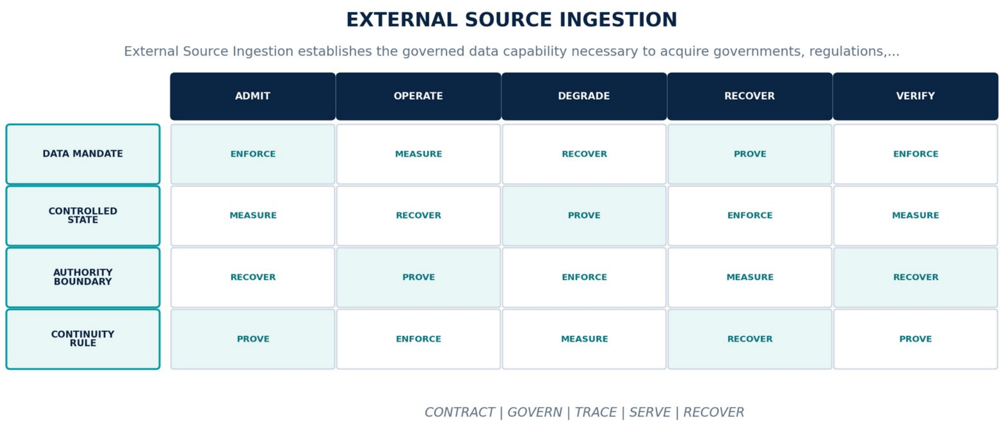

Figure 11. External Source Ingestion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Acquire governments, regulations, news, markets, research, papers, patents, filings, supply-chain, climate, geopolitical, social, satellite, and other authorized external data.
Controlled state	Acquisition identity, source contract, request or subscription, retrieved payload, evidence digest, event and observation times, status, checkpoint, and coverage result.
Authority boundary	External content remains untrusted evidence until authenticity, rights, inspection, parsing, provenance, classification, and admission controls complete.
Continuity rule	When a source is unavailable, rate-limited, manipulated, duplicated, stale, or returns content outside the approved contract, the platform shall apply contract-specific retry and backoff, quarantine unsafe content, expose coverage loss, preserve custody, and prohibit silent source substitution.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-10-001	The data-platform realization of External Source Ingestion SHALL acquire governments, regulations, news, markets, research, papers, patents, filings, supply-chain, climate, geopolitical, social, satellite, and other authorized external data.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-10-002	Every production instance SHALL declare and continuously reconcile acquisition identity, source contract, request or subscription, retrieved payload, evidence digest, event and observation times, status, checkpoint, and coverage result through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-10-003	The implementation SHALL enforce this authority boundary: external content remains untrusted evidence until authenticity, rights, inspection, parsing, provenance, classification, and admission controls complete.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-10-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for External Source Ingestion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-10-005	When a source is unavailable, rate-limited, manipulated, duplicated, stale, or returns content outside the approved contract, the platform SHALL apply contract-specific retry and backoff, quarantine unsafe content, expose coverage loss, preserve custody, and prohibit silent source substitution; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-10-006	Production admission and continued operation SHALL require representative source fixtures, authenticity and tamper tests, rate and failure exercises, correction intake, freshness evidence, and source-owner review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A source is unavailable, rate-limited, manipulated, duplicated, stale, or returns content outside the approved contract	Apply contract-specific retry and backoff, quarantine unsafe content, expose coverage loss, preserve custody, and prohibit silent source substitution.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared acquisition identity, source contract, request or subscription, retrieved payload, evidence digest, event and observation times, status, checkpoint, and coverage result.
- Positive and negative boundary tests prove that external content remains untrusted evidence until authenticity, rights, inspection, parsing, provenance, classification, and admission controls complete.
- Fault exercises inject a source is unavailable, rate-limited, manipulated, duplicated, stale, or returns content outside the approved contract and verify that the platform can apply contract-specific retry and backoff, quarantine unsafe content, expose coverage loss, preserve custody, and prohibit silent source substitution.
- Independent review accepts representative source fixtures, authenticity and tamper tests, rate and failure exercises, correction intake, freshness evidence, and source-owner review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
External Source Ingestion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 28, Ch. 31, Ch. 57. Primary Volume 5 AI anchors: Ch. 39, Ch. 40, Ch. 41, Ch. 57, Ch. 60. Primary Volume 6 infrastructure anchors: Ch. 21, Ch. 28, Ch. 30, Ch. 33, Ch. 47.

CONSTITUTIONAL INHERITANCE / C-012 · C-013 · C-014 · C-032 · C-035 · C-043

11 Enterprise Application and Database Ingestion

Enterprise Application and Database Ingestion establishes the governed data capability necessary to acquire governed operational data from ERP, CRM, finance, healthcare, manufacturing, logistics, telecom, retail, and customer databases without taking ownership from source systems. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

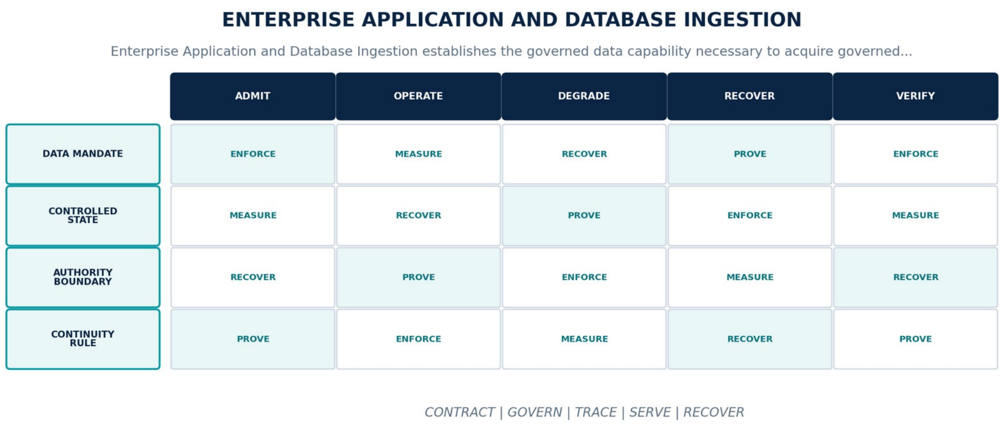

Figure 12. Enterprise Application and Database Ingestion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Acquire governed operational data from ERP, CRM, finance, healthcare, manufacturing, logistics, telecom, retail, and customer databases without taking ownership from source systems.
Controlled state	System identity, object or table scope, extraction mode, cursor, transaction boundary, source key, schema, permissions, reconciliation, and customer owner.
Authority boundary	The Eye preserves source-system authority and does not infer a consistent transaction, deletion, or business meaning beyond the declared integration contract.
Continuity rule	When permissions, transaction boundaries, source keys, schemas, or reconciliation totals become invalid, the platform shall stop affected ingestion, retain the last consistent watermark, expose incomplete scope, and reconcile with the source owner before advancing.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-11-001	The data-platform realization of Enterprise Application and Database Ingestion SHALL acquire governed operational data from ERP, CRM, finance, healthcare, manufacturing, logistics, telecom, retail, and customer databases without taking ownership from source systems.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-11-002	Every production instance SHALL declare and continuously reconcile system identity, object or table scope, extraction mode, cursor, transaction boundary, source key, schema, permissions, reconciliation, and customer owner through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-11-003	The implementation SHALL enforce this authority boundary: The Eye preserves source-system authority and does not infer a consistent transaction, deletion, or business meaning beyond the declared integration contract.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-11-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Enterprise Application and Database Ingestion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-11-005	When permissions, transaction boundaries, source keys, schemas, or reconciliation totals become invalid, the platform SHALL stop affected ingestion, retain the last consistent watermark, expose incomplete scope, and reconcile with the source owner before advancing; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-11-006	Production admission and continued operation SHALL require snapshot and incremental fixtures, transaction consistency, permission loss, schema change, reconciliation totals, and customer sign-off, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Permissions, transaction boundaries, source keys, schemas, or reconciliation totals become invalid	Stop affected ingestion, retain the last consistent watermark, expose incomplete scope, and reconcile with the source owner before advancing.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared system identity, object or table scope, extraction mode, cursor, transaction boundary, source key, schema, permissions, reconciliation, and customer owner.
- Positive and negative boundary tests prove that The Eye preserves source-system authority and does not infer a consistent transaction, deletion, or business meaning beyond the declared integration contract.
- Fault exercises inject permissions, transaction boundaries, source keys, schemas, or reconciliation totals become invalid and verify that the platform can stop affected ingestion, retain the last consistent watermark, expose incomplete scope, and reconcile with the source owner before advancing.
- Independent review accepts snapshot and incremental fixtures, transaction consistency, permission loss, schema change, reconciliation totals, and customer sign-off for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Enterprise Application and Database Ingestion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 18, Ch. 22, Ch. 25, Ch. 31. Primary Volume 5 AI anchors: Ch. 25, Ch. 26, Ch. 39, Ch. 41, Ch. 61. Primary Volume 6 infrastructure anchors: Ch. 28, Ch. 29, Ch. 32, Ch. 37, Ch. 47.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-014 · C-037 · C-038 · C-043

12 Document, Media, and Unstructured Content Ingestion

Document, Media, and Unstructured Content Ingestion establishes the governed data capability necessary to ingest documents, email, reports, images, audio, video, geospatial products, scientific artifacts, and internal knowledge with immutable originals and reviewable derivatives. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

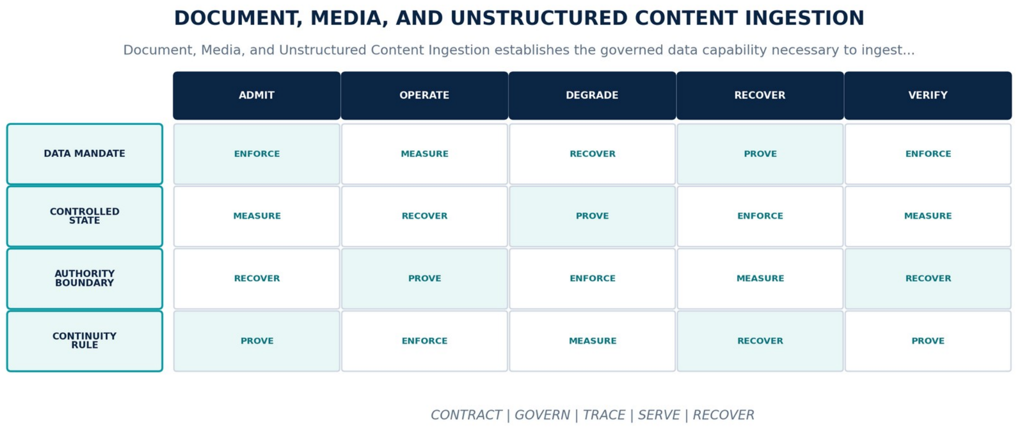

Figure 13. Document, Media, and Unstructured Content Ingestion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Ingest documents, email, reports, images, audio, video, geospatial products, scientific artifacts, and internal knowledge with immutable originals and reviewable derivatives.
Controlled state	Content identity, original bytes, media type, source, rights, custody, page or segment map, extraction plan, derivative set, and quality state.
Authority boundary	OCR, transcription, parsing, translation, chunking, and enrichment produce derived objects and may never overwrite or masquerade as the original evidence.
Continuity rule	When content is malicious, encrypted, corrupt, unsupported, partially extracted, or loses page, segment, or timestamp alignment, the platform shall quarantine the original, preserve custody, expose partial extraction, prevent unsafe downstream admission, and route controlled reprocessing or review.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-12-001	The data-platform realization of Document, Media, and Unstructured Content Ingestion SHALL ingest documents, email, reports, images, audio, video, geospatial products, scientific artifacts, and internal knowledge with immutable originals and reviewable derivatives.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-12-002	Every production instance SHALL declare and continuously reconcile content identity, original bytes, media type, source, rights, custody, page or segment map, extraction plan, derivative set, and quality state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-12-003	The implementation SHALL enforce this authority boundary: OCR, transcription, parsing, translation, chunking, and enrichment produce derived objects and may never overwrite or masquerade as the original evidence.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-12-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Document, Media, and Unstructured Content Ingestion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-12-005	When content is malicious, encrypted, corrupt, unsupported, partially extracted, or loses page, segment, or timestamp alignment, the platform SHALL quarantine the original, preserve custody, expose partial extraction, prevent unsafe downstream admission, and route controlled reprocessing or review; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-12-006	Production admission and continued operation SHALL require malware and decompression tests, format corpus, OCR and transcription evaluation, alignment checks, derivative lineage, and human sampling, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Content is malicious, encrypted, corrupt, unsupported, partially extracted, or loses page, segment, or timestamp alignment	Quarantine the original, preserve custody, expose partial extraction, prevent unsafe downstream admission, and route controlled reprocessing or review.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared content identity, original bytes, media type, source, rights, custody, page or segment map, extraction plan, derivative set, and quality state.
- Positive and negative boundary tests prove that OCR, transcription, parsing, translation, chunking, and enrichment produce derived objects and may never overwrite or masquerade as the original evidence.
- Fault exercises inject content is malicious, encrypted, corrupt, unsupported, partially extracted, or loses page, segment, or timestamp alignment and verify that the platform can quarantine the original, preserve custody, expose partial extraction, prevent unsafe downstream admission, and route controlled reprocessing or review.
- Independent review accepts malware and decompression tests, format corpus, OCR and transcription evaluation, alignment checks, derivative lineage, and human sampling for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Document, Media, and Unstructured Content Ingestion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 24, Ch. 28, Ch. 31, Ch. 32. Primary Volume 5 AI anchors: Ch. 19, Ch. 29, Ch. 41, Ch. 42, Ch. 60. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 44, Ch. 45, Ch. 47, Ch. 53.

CONSTITUTIONAL INHERITANCE / C-010 · C-012 · C-014 · C-016 · C-035 · C-045

13 Event, Stream, IoT, and Telemetry Ingestion

Event, Stream, IoT, and Telemetry Ingestion establishes the governed data capability necessary to admit high-volume event, IoT, sensor, operational, cyber, market, and telemetry streams under explicit ordering, time, quality, and backpressure contracts. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

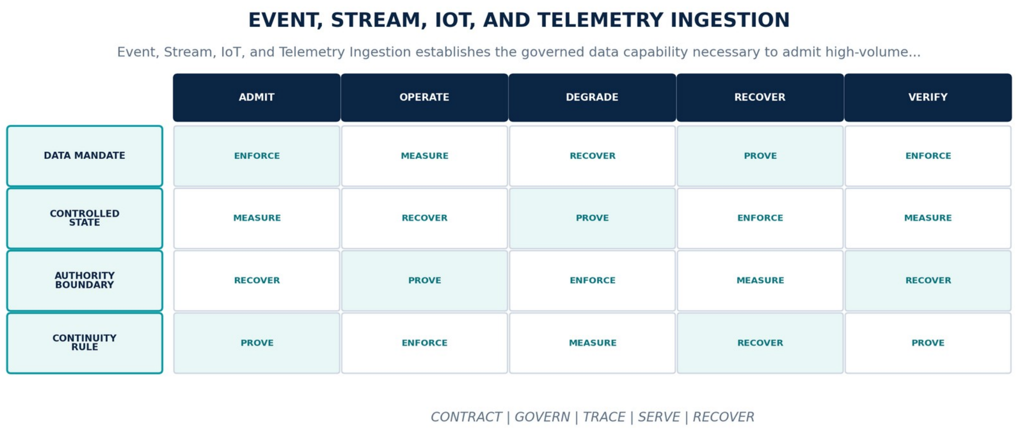

Figure 14. Event, Stream, IoT, and Telemetry Ingestion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Admit high-volume event, IoT, sensor, operational, cyber, market, and telemetry streams under explicit ordering, time, quality, and backpressure contracts.
Controlled state	Stream identity, producer, partition key, sequence, event time, ingestion time, schema, checkpoint, lateness, watermark, and loss state.
Authority boundary	Broker delivery is not canonical admission; duplicates, reordering, gaps, late data, clock error, and device trust remain explicit until resolved.
Continuity rule	When a stream overruns capacity, loses ordering or clock quality, contains gaps, or receives data from an untrusted producer, the platform shall apply bounded backpressure or shedding by consequence, quarantine suspect producers, retain loss evidence, and reconcile from durable checkpoints.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-13-001	The data-platform realization of Event, Stream, IoT, and Telemetry Ingestion SHALL admit high-volume event, IoT, sensor, operational, cyber, market, and telemetry streams under explicit ordering, time, quality, and backpressure contracts.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-13-002	Every production instance SHALL declare and continuously reconcile stream identity, producer, partition key, sequence, event time, ingestion time, schema, checkpoint, lateness, watermark, and loss state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-13-003	The implementation SHALL enforce this authority boundary: broker delivery is not canonical admission; duplicates, reordering, gaps, late data, clock error, and device trust remain explicit until resolved.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-13-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Event, Stream, IoT, and Telemetry Ingestion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-13-005	When a stream overruns capacity, loses ordering or clock quality, contains gaps, or receives data from an untrusted producer, the platform SHALL apply bounded backpressure or shedding by consequence, quarantine suspect producers, retain loss evidence, and reconcile from durable checkpoints; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-13-006	Production admission and continued operation SHALL require load and burst tests, duplicate and reordering fixtures, late-data exercises, device revocation, clock skew, replay, and loss accounting, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A stream overruns capacity, loses ordering or clock quality, contains gaps, or receives data from an untrusted producer	Apply bounded backpressure or shedding by consequence, quarantine suspect producers, retain loss evidence, and reconcile from durable checkpoints.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared stream identity, producer, partition key, sequence, event time, ingestion time, schema, checkpoint, lateness, watermark, and loss state.
- Positive and negative boundary tests prove that broker delivery is not canonical admission; duplicates, reordering, gaps, late data, clock error, and device trust remain explicit until resolved.
- Fault exercises inject a stream overruns capacity, loses ordering or clock quality, contains gaps, or receives data from an untrusted producer and verify that the platform can apply bounded backpressure or shedding by consequence, quarantine suspect producers, retain loss evidence, and reconcile from durable checkpoints.
- Independent review accepts load and burst tests, duplicate and reordering fixtures, late-data exercises, device revocation, clock skew, replay, and loss accounting for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Event, Stream, IoT, and Telemetry Ingestion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 26, Ch. 31, Ch. 57, Ch. 58. Primary Volume 5 AI anchors: Ch. 39, Ch. 44, Ch. 57, Ch. 60, Ch. 68. Primary Volume 6 infrastructure anchors: Ch. 20, Ch. 21, Ch. 27, Ch. 30, Ch. 37.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-014 · C-035 · C-039 · C-043

14 Batch, File, API, RSS, and Custom Source Ingestion

Batch, File, API, RSS, and Custom Source Ingestion establishes the governed data capability necessary to provide governed reusable ingestion patterns for scheduled files, APIs, RSS, secure transfer, bulk imports, and customer-defined adapters. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

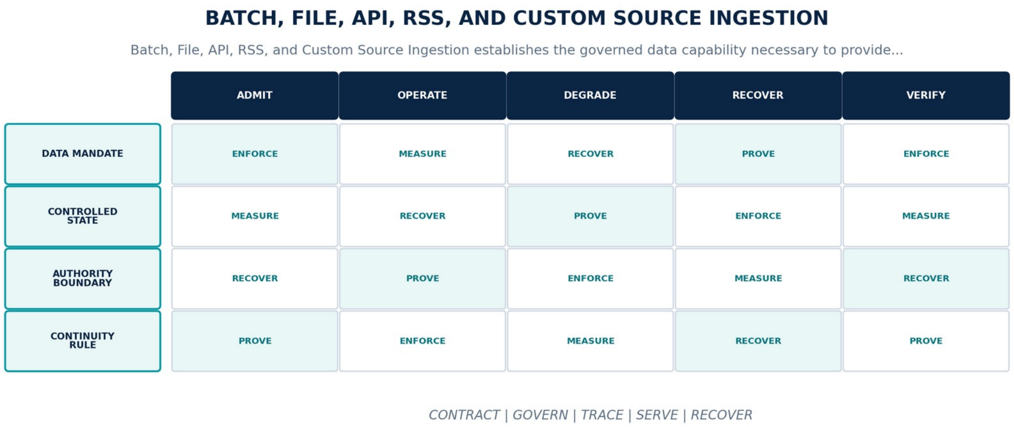

Figure 15. Batch, File, API, RSS, and Custom Source Ingestion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Provide governed reusable ingestion patterns for scheduled files, APIs, RSS, secure transfer, bulk imports, and customer-defined adapters.
Controlled state	Transfer identity, contract, manifest, checksum, schema, partition set, cursor, idempotency key, receipt, rejection set, and completion state.
Authority boundary	Custom adapters execute inside the same identity, network, credential, policy, evidence, quality, and lifecycle boundaries as first-party connectors.
Continuity rule	When a transfer is partial, duplicated, tampered, incompatible, misrouted, or cannot prove completion and reconciliation, the platform shall hold publication, isolate the transfer, preserve receipts and rejected records, resume from the committed boundary, and reconcile totals.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-14-001	The data-platform realization of Batch, File, API, RSS, and Custom Source Ingestion SHALL provide governed reusable ingestion patterns for scheduled files, APIs, RSS, secure transfer, bulk imports, and customer-defined adapters.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-14-002	Every production instance SHALL declare and continuously reconcile transfer identity, contract, manifest, checksum, schema, partition set, cursor, idempotency key, receipt, rejection set, and completion state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-14-003	The implementation SHALL enforce this authority boundary: custom adapters execute inside the same identity, network, credential, policy, evidence, quality, and lifecycle boundaries as first-party connectors.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-14-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Batch, File, API, RSS, and Custom Source Ingestion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-14-005	When a transfer is partial, duplicated, tampered, incompatible, misrouted, or cannot prove completion and reconciliation, the platform SHALL hold publication, isolate the transfer, preserve receipts and rejected records, resume from the committed boundary, and reconcile totals; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-14-006	Production admission and continued operation SHALL require golden adapter suite, checksum and manifest validation, partial-transfer recovery, idempotency, schema incompatibility, and connector certification, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A transfer is partial, duplicated, tampered, incompatible, misrouted, or cannot prove completion and reconciliation	Hold publication, isolate the transfer, preserve receipts and rejected records, resume from the committed boundary, and reconcile totals.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared transfer identity, contract, manifest, checksum, schema, partition set, cursor, idempotency key, receipt, rejection set, and completion state.
- Positive and negative boundary tests prove that custom adapters execute inside the same identity, network, credential, policy, evidence, quality, and lifecycle boundaries as first-party connectors.
- Fault exercises inject a transfer is partial, duplicated, tampered, incompatible, misrouted, or cannot prove completion and reconciliation and verify that the platform can hold publication, isolate the transfer, preserve receipts and rejected records, resume from the committed boundary, and reconcile totals.
- Independent review accepts golden adapter suite, checksum and manifest validation, partial-transfer recovery, idempotency, schema incompatibility, and connector certification for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Batch, File, API, RSS, and Custom Source Ingestion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 16, Ch. 17, Ch. 20, Ch. 22, Ch. 23, Ch. 31. Primary Volume 5 AI anchors: Ch. 17, Ch. 36, Ch. 39, Ch. 40, Ch. 60. Primary Volume 6 infrastructure anchors: Ch. 28, Ch. 37, Ch. 47, Ch. 48, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-012 · C-013 · C-014 · C-038 · C-039 · C-043

15 Change Data Capture and Synchronization

Change Data Capture and Synchronization establishes the governed data capability necessary to capture and reconcile source changes, inserts, updates, deletes, corrections, and tombstones while preserving transaction order and source authority. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

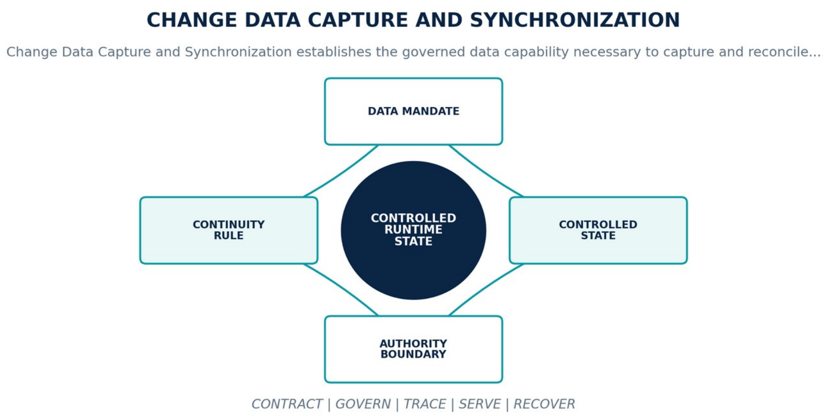

Figure 16. Change Data Capture and Synchronization. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Capture and reconcile source changes, inserts, updates, deletes, corrections, and tombstones while preserving transaction order and source authority.
Controlled state	Capture stream, source log position, transaction identity, before and after image, source key, operation, commit time, checkpoint, and reconciliation state.
Authority boundary	Change capture may replicate source state but may not invent missing transactions, reinterpret deletion, or bypass the canonical correction lifecycle.
Continuity rule	When log continuity breaks, checkpoints diverge, duplicates or gaps emerge, or source retention expires before capture, the platform shall pause dependent publication, retain the last verified position, resnapshot the bounded scope, reconcile transactions, and expose the discontinuity.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-15-001	The data-platform realization of Change Data Capture and Synchronization SHALL capture and reconcile source changes, inserts, updates, deletes, corrections, and tombstones while preserving transaction order and source authority.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-15-002	Every production instance SHALL declare and continuously reconcile capture stream, source log position, transaction identity, before and after image, source key, operation, commit time, checkpoint, and reconciliation state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-15-003	The implementation SHALL enforce this authority boundary: change capture may replicate source state but may not invent missing transactions, reinterpret deletion, or bypass the canonical correction lifecycle.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-15-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Change Data Capture and Synchronization.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-15-005	When log continuity breaks, checkpoints diverge, duplicates or gaps emerge, or source retention expires before capture, the platform SHALL pause dependent publication, retain the last verified position, resnapshot the bounded scope, reconcile transactions, and expose the discontinuity; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-15-006	Production admission and continued operation SHALL require transactional fixtures, log-gap simulation, resnapshot and catch-up, delete propagation, ordering, reconciliation, and source-owner acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Log continuity breaks, checkpoints diverge, duplicates or gaps emerge, or source retention expires before capture	Pause dependent publication, retain the last verified position, resnapshot the bounded scope, reconcile transactions, and expose the discontinuity.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared capture stream, source log position, transaction identity, before and after image, source key, operation, commit time, checkpoint, and reconciliation state.
- Positive and negative boundary tests prove that change capture may replicate source state but may not invent missing transactions, reinterpret deletion, or bypass the canonical correction lifecycle.
- Fault exercises inject log continuity breaks, checkpoints diverge, duplicates or gaps emerge, or source retention expires before capture and verify that the platform can pause dependent publication, retain the last verified position, resnapshot the bounded scope, reconcile transactions, and expose the discontinuity.
- Independent review accepts transactional fixtures, log-gap simulation, resnapshot and catch-up, delete propagation, ordering, reconciliation, and source-owner acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Change Data Capture and Synchronization is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 26, Ch. 30, Ch. 31. Primary Volume 5 AI anchors: Ch. 26, Ch. 27, Ch. 38, Ch. 39, Ch. 68. Primary Volume 6 infrastructure anchors: Ch. 20, Ch. 27, Ch. 32, Ch. 37, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-014 · C-030 · C-038 · C-043

16 Intake Validation, Quarantine, and Raw Preservation

Intake Validation, Quarantine, and Raw Preservation establishes the governed data capability necessary to validate identity, rights, integrity, schema, safety, classification, locality, and quality before preserving immutable raw evidence and admitting downstream work. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

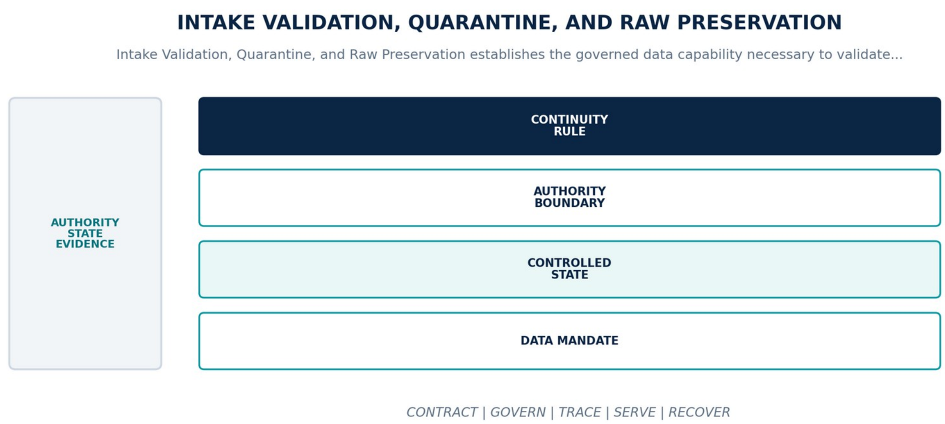

Figure 17. Intake Validation, Quarantine, and Raw Preservation. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Validate identity, rights, integrity, schema, safety, classification, locality, and quality before preserving immutable raw evidence and admitting downstream work.
Controlled state	Intake case, validation results, quarantine status, raw object identity, digest, custody events, admission decision, rejection reason, and release authority.
Authority boundary	Quarantined content, failed validations, and rejected records remain isolated and cannot enter transformation, retrieval, training, or product-serving paths.
Continuity rule	When validation services are unavailable, a digest mismatches, malware is detected, or quarantine isolation becomes suspect, the platform shall fail closed, preserve the item in the restricted evidence boundary, revoke affected derivatives, and resume only after independent validation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-16-001	The data-platform realization of Intake Validation, Quarantine, and Raw Preservation SHALL validate identity, rights, integrity, schema, safety, classification, locality, and quality before preserving immutable raw evidence and admitting downstream work.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-16-002	Every production instance SHALL declare and continuously reconcile intake case, validation results, quarantine status, raw object identity, digest, custody events, admission decision, rejection reason, and release authority through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-16-003	The implementation SHALL enforce this authority boundary: quarantined content, failed validations, and rejected records remain isolated and cannot enter transformation, retrieval, training, or product-serving paths.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-16-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Intake Validation, Quarantine, and Raw Preservation.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-16-005	When validation services are unavailable, a digest mismatches, malware is detected, or quarantine isolation becomes suspect, the platform SHALL fail closed, preserve the item in the restricted evidence boundary, revoke affected derivatives, and resume only after independent validation; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-16-006	Production admission and continued operation SHALL require negative admission tests, malware corpus, digest and custody verification, quarantine escape exercises, policy review, and evidence restore, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Validation services are unavailable, a digest mismatches, malware is detected, or quarantine isolation becomes suspect	Fail closed, preserve the item in the restricted evidence boundary, revoke affected derivatives, and resume only after independent validation.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared intake case, validation results, quarantine status, raw object identity, digest, custody events, admission decision, rejection reason, and release authority.
- Positive and negative boundary tests prove that quarantined content, failed validations, and rejected records remain isolated and cannot enter transformation, retrieval, training, or product-serving paths.
- Fault exercises inject validation services are unavailable, a digest mismatches, malware is detected, or quarantine isolation becomes suspect and verify that the platform can fail closed, preserve the item in the restricted evidence boundary, revoke affected derivatives, and resume only after independent validation.
- Independent review accepts negative admission tests, malware corpus, digest and custody verification, quarantine escape exercises, policy review, and evidence restore for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Intake Validation, Quarantine, and Raw Preservation is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 21, Ch. 24, Ch. 28, Ch. 31, Ch. 56. Primary Volume 5 AI anchors: Ch. 20, Ch. 21, Ch. 29, Ch. 39, Ch. 60. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 47, Ch. 53, Ch. 54, Ch. 56.

CONSTITUTIONAL INHERITANCE / C-010 · C-012 · C-035 · C-039 · C-040 · C-043

PART III

Contracts, Semantics, and Temporal Truth

Canonical contracts, schema evolution, intelligence objects, identity, time, truth state, provenance, and correction.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

17 Canonical Data Contract

Canonical Data Contract establishes the governed data capability necessary to define every produced or consumed dataset, stream, object, index, feature, and export through a stable machine-readable contract. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

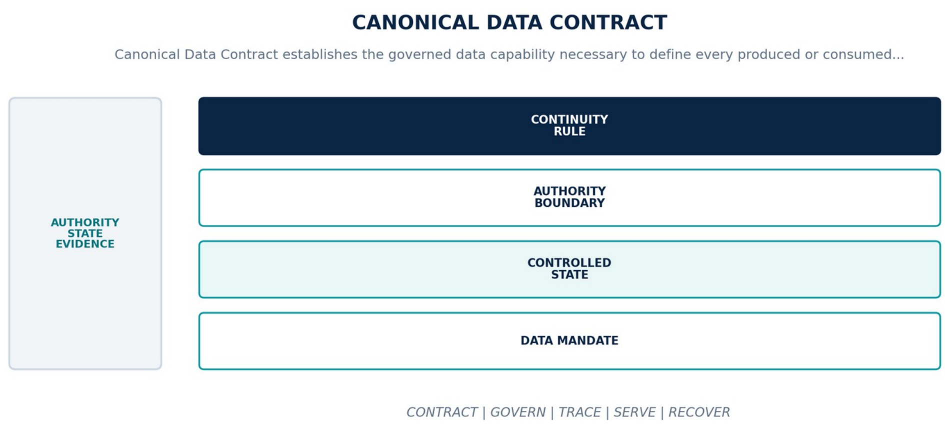

Figure 18. Canonical Data Contract. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Define every produced or consumed dataset, stream, object, index, feature, and export through a stable machine-readable contract.
Controlled state	Contract identity, owner, purpose, schema, keys, time semantics, truth state, provenance, quality, policy, compatibility, SLO, and lifecycle.
Authority boundary	Physical tables, files, topics, and APIs are implementations of contracts and cannot become the semantic authority by convention.
Continuity rule	When a producer publishes data that violates its contract or a consumer depends on undeclared behavior, the platform shall reject or isolate non-conformant records, stop incompatible publication, notify owners, and restore service through an approved contract revision.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-17-001	The data-platform realization of Canonical Data Contract SHALL define every produced or consumed dataset, stream, object, index, feature, and export through a stable machine-readable contract.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-17-002	Every production instance SHALL declare and continuously reconcile contract identity, owner, purpose, schema, keys, time semantics, truth state, provenance, quality, policy, compatibility, SLO, and lifecycle through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-17-003	The implementation SHALL enforce this authority boundary: physical tables, files, topics, and APIs are implementations of contracts and cannot become the semantic authority by convention.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-17-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Canonical Data Contract.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-17-005	When a producer publishes data that violates its contract or a consumer depends on undeclared behavior, the platform SHALL reject or isolate non-conformant records, stop incompatible publication, notify owners, and restore service through an approved contract revision; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-17-006	Production admission and continued operation SHALL require contract linting, producer and consumer tests, golden fixtures, compatibility analysis, runtime validation, and adoption evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A producer publishes data that violates its contract or a consumer depends on undeclared behavior	Reject or isolate non-conformant records, stop incompatible publication, notify owners, and restore service through an approved contract revision.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared contract identity, owner, purpose, schema, keys, time semantics, truth state, provenance, quality, policy, compatibility, SLO, and lifecycle.
- Positive and negative boundary tests prove that physical tables, files, topics, and APIs are implementations of contracts and cannot become the semantic authority by convention.
- Fault exercises inject a producer publishes data that violates its contract or a consumer depends on undeclared behavior and verify that the platform can reject or isolate non-conformant records, stop incompatible publication, notify owners, and restore service through an approved contract revision.
- Independent review accepts contract linting, producer and consumer tests, golden fixtures, compatibility analysis, runtime validation, and adoption evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Canonical Data Contract is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 23, Ch. 24, Ch. 28. Primary Volume 5 AI anchors: Ch. 19, Ch. 20, Ch. 26, Ch. 29, Ch. 63. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 32, Ch. 37, Ch. 48, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-012 · C-014 · C-035 · C-038

18 Schema Registry and Compatibility

Schema Registry and Compatibility establishes the governed data capability necessary to govern schema identity, versioning, evolution, compatibility, migration, deprecation, and consumer impact across all data interfaces. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

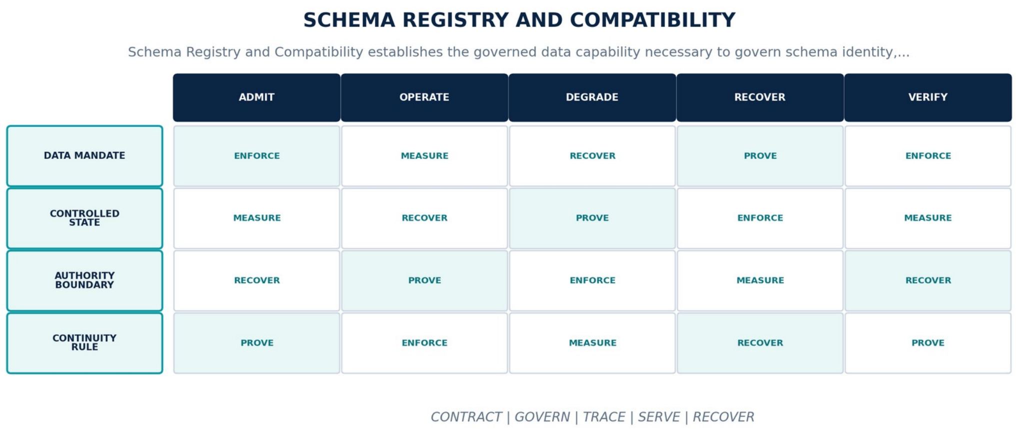

Figure 19. Schema Registry and Compatibility. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern schema identity, versioning, evolution, compatibility, migration, deprecation, and consumer impact across all data interfaces.
Controlled state	Schema subject, version, fields, constraints, compatibility policy, owners, consumers, migration, deprecation, and effective window.
Authority boundary	Schema change cannot remove policy-bearing meaning, temporal semantics, provenance, identifiers, or required consumer behavior without controlled migration.
Continuity rule	When a schema change is incompatible, a consumer is unknown, or registry state diverges from deployed producers and consumers, the platform shall block promotion, freeze the last compatible contract, enumerate affected consumers, and execute an approved migration or parallel version.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-18-001	The data-platform realization of Schema Registry and Compatibility SHALL govern schema identity, versioning, evolution, compatibility, migration, deprecation, and consumer impact across all data interfaces.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-18-002	Every production instance SHALL declare and continuously reconcile schema subject, version, fields, constraints, compatibility policy, owners, consumers, migration, deprecation, and effective window through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-18-003	The implementation SHALL enforce this authority boundary: schema change cannot remove policy-bearing meaning, temporal semantics, provenance, identifiers, or required consumer behavior without controlled migration.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-18-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Schema Registry and Compatibility.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-18-005	When a schema change is incompatible, a consumer is unknown, or registry state diverges from deployed producers and consumers, the platform SHALL block promotion, freeze the last compatible contract, enumerate affected consumers, and execute an approved migration or parallel version; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-18-006	Production admission and continued operation SHALL require compatibility fixtures, consumer inventory, forward and backward tests, dual-read or dual-write exercises, and deprecation closure, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A schema change is incompatible, a consumer is unknown, or registry state diverges from deployed producers and consumers	Block promotion, freeze the last compatible contract, enumerate affected consumers, and execute an approved migration or parallel version.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared schema subject, version, fields, constraints, compatibility policy, owners, consumers, migration, deprecation, and effective window.
- Positive and negative boundary tests prove that schema change cannot remove policy-bearing meaning, temporal semantics, provenance, identifiers, or required consumer behavior without controlled migration.
- Fault exercises inject a schema change is incompatible, a consumer is unknown, or registry state diverges from deployed producers and consumers and verify that the platform can block promotion, freeze the last compatible contract, enumerate affected consumers, and execute an approved migration or parallel version.
- Independent review accepts compatibility fixtures, consumer inventory, forward and backward tests, dual-read or dual-write exercises, and deprecation closure for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Schema Registry and Compatibility is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 23, Ch. 24, Ch. 30, Ch. 67. Primary Volume 5 AI anchors: Ch. 18, Ch. 20, Ch. 26, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 32, Ch. 37, Ch. 45, Ch. 48, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-035 · C-038 · C-043 · C-052

19 Canonical Intelligence Object Model

Canonical Intelligence Object Model establishes the governed data capability necessary to implement the common governed object envelope for evidence, observations, claims, events, entities, relationships, assessments, forecasts, scenarios, simulations, decisions, actions, outcomes, and lessons. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

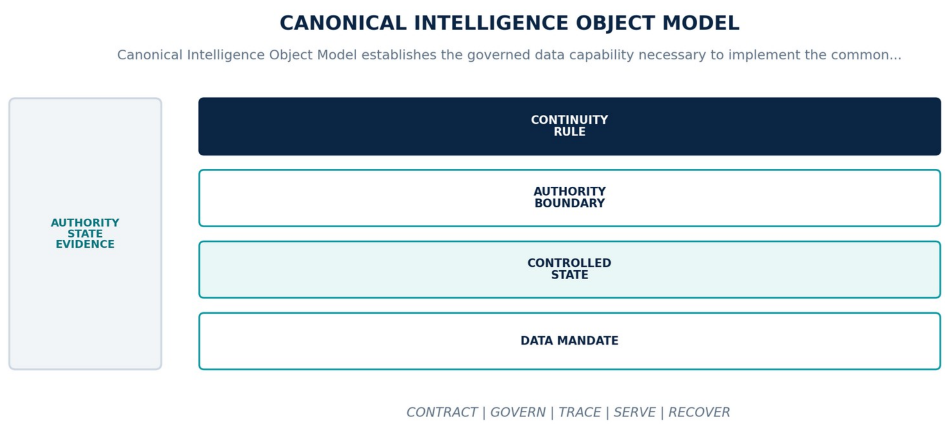

Figure 20. Canonical Intelligence Object Model. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Implement the common governed object envelope for evidence, observations, claims, events, entities, relationships, assessments, forecasts, scenarios, simulations, decisions, actions, outcomes, and lessons.
Controlled state	Object identity, type, tenant, domain, version, lifecycle, temporal coordinates, truth state, provenance, policy, quality, ownership, and content reference.
Authority boundary	Object types may specialize payloads but may not omit or reinterpret the canonical header, evidence linkage, correction, policy, or lifecycle contract.
Continuity rule	When an object lacks required header semantics, carries contradictory authority, or cannot resolve its referenced versions, the platform shall deny canonical admission, retain the proposal and errors, prevent downstream promotion, and route repair to the owning component.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-19-001	The data-platform realization of Canonical Intelligence Object Model SHALL implement the common governed object envelope for evidence, observations, claims, events, entities, relationships, assessments, forecasts, scenarios, simulations, decisions, actions, outcomes, and lessons.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-19-002	Every production instance SHALL declare and continuously reconcile object identity, type, tenant, domain, version, lifecycle, temporal coordinates, truth state, provenance, policy, quality, ownership, and content reference through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-19-003	The implementation SHALL enforce this authority boundary: object types may specialize payloads but may not omit or reinterpret the canonical header, evidence linkage, correction, policy, or lifecycle contract.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-19-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Canonical Intelligence Object Model.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-19-005	When an object lacks required header semantics, carries contradictory authority, or cannot resolve its referenced versions, the platform SHALL deny canonical admission, retain the proposal and errors, prevent downstream promotion, and route repair to the owning component; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-19-006	Production admission and continued operation SHALL require schema fixtures for every object type, lifecycle tests, cross-type references, policy propagation, correction replay, and conformance review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An object lacks required header semantics, carries contradictory authority, or cannot resolve its referenced versions	Deny canonical admission, retain the proposal and errors, prevent downstream promotion, and route repair to the owning component.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared object identity, type, tenant, domain, version, lifecycle, temporal coordinates, truth state, provenance, policy, quality, ownership, and content reference.
- Positive and negative boundary tests prove that object types may specialize payloads but may not omit or reinterpret the canonical header, evidence linkage, correction, policy, or lifecycle contract.
- Fault exercises inject an object lacks required header semantics, carries contradictory authority, or cannot resolve its referenced versions and verify that the platform can deny canonical admission, retain the proposal and errors, prevent downstream promotion, and route repair to the owning component.
- Independent review accepts schema fixtures for every object type, lifecycle tests, cross-type references, policy propagation, correction replay, and conformance review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Canonical Intelligence Object Model is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28. Primary Volume 5 AI anchors: Ch. 13, Ch. 19, Ch. 20, Ch. 26, Ch. 29. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 34, Ch. 35, Ch. 36.

CONSTITUTIONAL INHERITANCE / C-008 · C-009 · C-010 · C-011 · C-012 · C-035

20 Identity, Naming, and Tenant Scoping

Identity, Naming, and Tenant Scoping establishes the governed data capability necessary to provide stable globally unique and tenant-safe identities for sources, objects, entities, versions, assets, products, pipelines, and exchanges. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

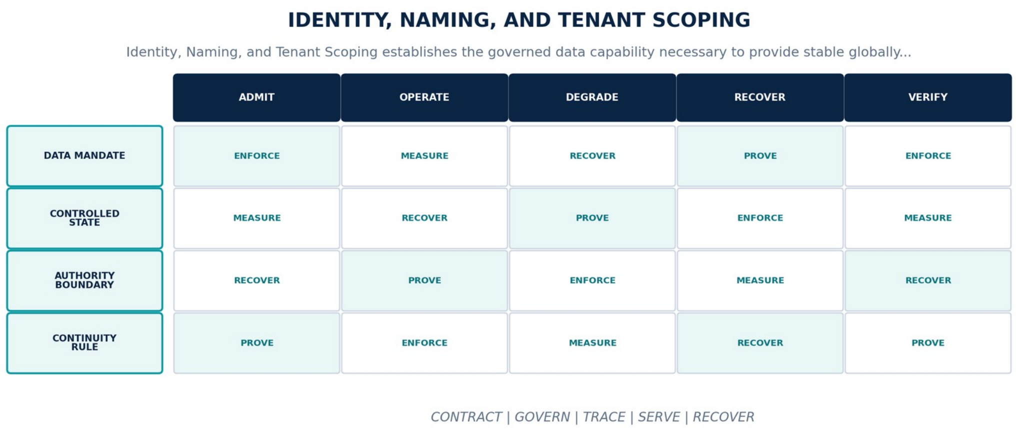

Figure 21. Identity, Naming, and Tenant Scoping. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Provide stable globally unique and tenant-safe identities for sources, objects, entities, versions, assets, products, pipelines, and exchanges.
Controlled state	Identifier namespace, issuer, tenant and domain scope, natural keys, aliases, version identity, merge or split state, and resolution history.
Authority boundary	Storage location, display name, source key, model guess, or network scope may not substitute for canonical identity and tenant context.
Continuity rule	When identity collides, tenant scope is absent, aliases conflict, or a merge or split cannot be safely resolved, the platform shall fail closed, isolate affected records, preserve candidate identities and lineage, and require governed stewardship before authoritative use.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-20-001	The data-platform realization of Identity, Naming, and Tenant Scoping SHALL provide stable globally unique and tenant-safe identities for sources, objects, entities, versions, assets, products, pipelines, and exchanges.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-20-002	Every production instance SHALL declare and continuously reconcile identifier namespace, issuer, tenant and domain scope, natural keys, aliases, version identity, merge or split state, and resolution history through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-20-003	The implementation SHALL enforce this authority boundary: storage location, display name, source key, model guess, or network scope may not substitute for canonical identity and tenant context.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-20-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Identity, Naming, and Tenant Scoping.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-20-005	When identity collides, tenant scope is absent, aliases conflict, or a merge or split cannot be safely resolved, the platform SHALL fail closed, isolate affected records, preserve candidate identities and lineage, and require governed stewardship before authoritative use; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-20-006	Production admission and continued operation SHALL require collision and cross-tenant negative tests, alias history, merge and split replay, import and export round trips, and stewardship acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity collides, tenant scope is absent, aliases conflict, or a merge or split cannot be safely resolved	Fail closed, isolate affected records, preserve candidate identities and lineage, and require governed stewardship before authoritative use.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared identifier namespace, issuer, tenant and domain scope, natural keys, aliases, version identity, merge or split state, and resolution history.
- Positive and negative boundary tests prove that storage location, display name, source key, model guess, or network scope may not substitute for canonical identity and tenant context.
- Fault exercises inject identity collides, tenant scope is absent, aliases conflict, or a merge or split cannot be safely resolved and verify that the platform can fail closed, isolate affected records, preserve candidate identities and lineage, and require governed stewardship before authoritative use.
- Independent review accepts collision and cross-tenant negative tests, alias history, merge and split replay, import and export round trips, and stewardship acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Identity, Naming, and Tenant Scoping is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 25, Ch. 30, Ch. 34, Ch. 51, Ch. 55. Primary Volume 5 AI anchors: Ch. 27, Ch. 29, Ch. 43, Ch. 45, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 13, Ch. 34, Ch. 49, Ch. 54, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-007 · C-009 · C-011 · C-016 · C-035 · C-039

21 Temporal and Bitemporal Data

Temporal and Bitemporal Data establishes the governed data capability necessary to preserve event time, observation time, validity time, and system-record time so historical world and institutional states can be reconstructed without hindsight contamination. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

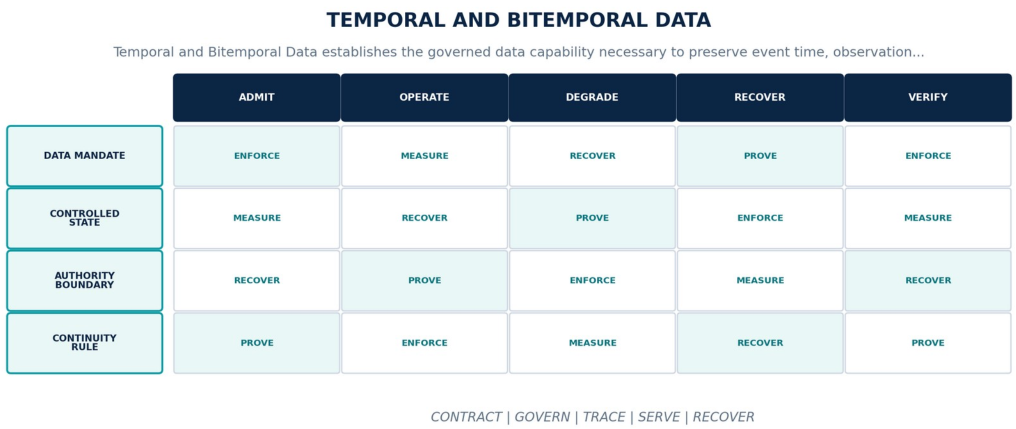

Figure 22. Temporal and Bitemporal Data. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Preserve event time, observation time, validity time, and system-record time so historical world and institutional states can be reconstructed without hindsight contamination.
Controlled state	Temporal identity, interval, precision, timezone, calendar, source clock, clock quality, ingestion watermark, correction time, and temporal lineage.
Authority boundary	Latest value storage, overwrite updates, or processing time alone cannot represent canonical temporal truth for material data.
Continuity rule	When timestamps are missing, clocks disagree, intervals overlap improperly, or late data changes an already consumed historical state, the platform shall mark temporal quality, isolate invalid intervals, version the correction, identify affected products and decisions, and rebuild from the appropriate time boundary.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-21-001	The data-platform realization of Temporal and Bitemporal Data SHALL preserve event time, observation time, validity time, and system-record time so historical world and institutional states can be reconstructed without hindsight contamination.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-21-002	Every production instance SHALL declare and continuously reconcile temporal identity, interval, precision, timezone, calendar, source clock, clock quality, ingestion watermark, correction time, and temporal lineage through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-21-003	The implementation SHALL enforce this authority boundary: latest value storage, overwrite updates, or processing time alone cannot represent canonical temporal truth for material data.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-21-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Temporal and Bitemporal Data.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-21-005	When timestamps are missing, clocks disagree, intervals overlap improperly, or late data changes an already consumed historical state, the platform SHALL mark temporal quality, isolate invalid intervals, version the correction, identify affected products and decisions, and rebuild from the appropriate time boundary; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-21-006	Production admission and continued operation SHALL require bitemporal fixtures, clock-skew tests, late and out-of-order events, historical snapshots, correction propagation, and Decision Replay validation, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Timestamps are missing, clocks disagree, intervals overlap improperly, or late data changes an already consumed historical state	Mark temporal quality, isolate invalid intervals, version the correction, identify affected products and decisions, and rebuild from the appropriate time boundary.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared temporal identity, interval, precision, timezone, calendar, source clock, clock quality, ingestion watermark, correction time, and temporal lineage.
- Positive and negative boundary tests prove that latest value storage, overwrite updates, or processing time alone cannot represent canonical temporal truth for material data.
- Fault exercises inject timestamps are missing, clocks disagree, intervals overlap improperly, or late data changes an already consumed historical state and verify that the platform can mark temporal quality, isolate invalid intervals, version the correction, identify affected products and decisions, and rebuild from the appropriate time boundary.
- Independent review accepts bitemporal fixtures, clock-skew tests, late and out-of-order events, historical snapshots, correction propagation, and Decision Replay validation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Temporal and Bitemporal Data is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 26, Ch. 30, Ch. 48, Ch. 64. Primary Volume 5 AI anchors: Ch. 26, Ch. 29, Ch. 38, Ch. 48, Ch. 63. Primary Volume 6 infrastructure anchors: Ch. 27, Ch. 31, Ch. 34, Ch. 36, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-012 · C-025 · C-035 · C-043

22 Truth State, Confidence, and Uncertainty

Truth State, Confidence, and Uncertainty establishes the governed data capability necessary to encode whether data is observed, asserted, extracted, inferred, assessed, simulated, decided, corrected, disputed, or withdrawn together with calibrated confidence and uncertainty. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

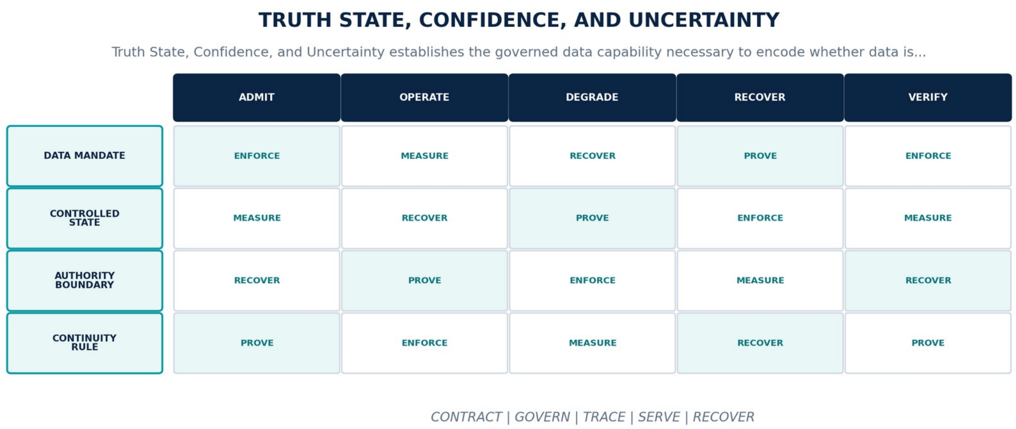

Figure 23. Truth State, Confidence, and Uncertainty. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Encode whether data is observed, asserted, extracted, inferred, assessed, simulated, decided, corrected, disputed, or withdrawn together with calibrated confidence and uncertainty.
Controlled state	Truth-state label, asserting authority, evidence set, method, confidence measure, uncertainty form, contradiction links, review state, and expiry.
Authority boundary	Confidence is not truth, frequency is not corroboration, and synthetic or model-derived data may never be silently promoted to observed fact.
Continuity rule	When truth state is missing, confidence is uncalibrated, evidence conflicts, or a downstream product collapses distinct epistemic states, the platform shall block or downgrade admission, preserve alternatives and dissent, route material ambiguity to review, and propagate the reduced fitness.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-22-001	The data-platform realization of Truth State, Confidence, and Uncertainty SHALL encode whether data is observed, asserted, extracted, inferred, assessed, simulated, decided, corrected, disputed, or withdrawn together with calibrated confidence and uncertainty.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-22-002	Every production instance SHALL declare and continuously reconcile truth-state label, asserting authority, evidence set, method, confidence measure, uncertainty form, contradiction links, review state, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-22-003	The implementation SHALL enforce this authority boundary: confidence is not truth, frequency is not corroboration, and synthetic or model-derived data may never be silently promoted to observed fact.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-22-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Truth State, Confidence, and Uncertainty.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-22-005	When truth state is missing, confidence is uncalibrated, evidence conflicts, or a downstream product collapses distinct epistemic states, the platform SHALL block or downgrade admission, preserve alternatives and dissent, route material ambiguity to review, and propagate the reduced fitness; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-22-006	Production admission and continued operation SHALL require truth-state fixtures, contradiction cases, calibration evidence, synthetic-isolation tests, review sampling, and downstream propagation checks, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Truth state is missing, confidence is uncalibrated, evidence conflicts, or a downstream product collapses distinct epistemic states	Block or downgrade admission, preserve alternatives and dissent, route material ambiguity to review, and propagate the reduced fitness.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared truth-state label, asserting authority, evidence set, method, confidence measure, uncertainty form, contradiction links, review state, and expiry.
- Positive and negative boundary tests prove that confidence is not truth, frequency is not corroboration, and synthetic or model-derived data may never be silently promoted to observed fact.
- Fault exercises inject truth state is missing, confidence is uncalibrated, evidence conflicts, or a downstream product collapses distinct epistemic states and verify that the platform can block or downgrade admission, preserve alternatives and dissent, route material ambiguity to review, and propagate the reduced fitness.
- Independent review accepts truth-state fixtures, contradiction cases, calibration evidence, synthetic-isolation tests, review sampling, and downstream propagation checks for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Truth State, Confidence, and Uncertainty is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 27, Ch. 28, Ch. 32, Ch. 46. Primary Volume 5 AI anchors: Ch. 4, Ch. 20, Ch. 29, Ch. 46, Ch. 63, Ch. 64. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 35, Ch. 44, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-012 · C-021 · C-035 · C-045

23 Provenance, Lineage, and Chain of Custody

Provenance, Lineage, and Chain of Custody establishes the governed data capability necessary to maintain end-to-end evidence from source acquisition through every transformation, human judgment, model use, merge, product, export, and correction. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

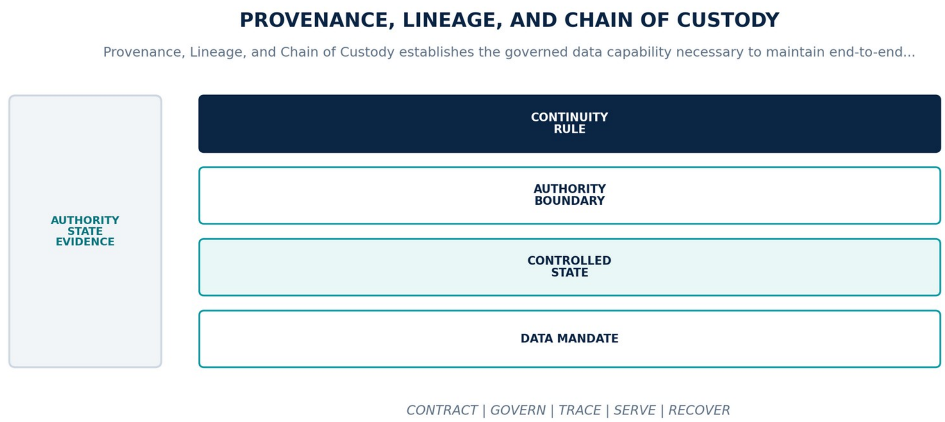

Figure 24. Provenance, Lineage, and Chain of Custody. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Maintain end-to-end evidence from source acquisition through every transformation, human judgment, model use, merge, product, export, and correction.
Controlled state	Lineage event, input and output versions, actor, method, code, model, prompt, environment, time, policy, custody, quality, and evidence digest.
Authority boundary	Lineage is canonical evidence and cannot be reconstructed only from mutable logs, filenames, table names, or post hoc explanation.
Continuity rule	When a material object, product, or decision input has incomplete lineage, broken custody, or unverifiable artifact identity, the platform shall mark the output unfit, block consequential use and export, preserve the gap, and recompute or adjudicate from verified inputs.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-23-001	The data-platform realization of Provenance, Lineage, and Chain of Custody SHALL maintain end-to-end evidence from source acquisition through every transformation, human judgment, model use, merge, product, export, and correction.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-23-002	Every production instance SHALL declare and continuously reconcile lineage event, input and output versions, actor, method, code, model, prompt, environment, time, policy, custody, quality, and evidence digest through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-23-003	The implementation SHALL enforce this authority boundary: lineage is canonical evidence and cannot be reconstructed only from mutable logs, filenames, table names, or post hoc explanation.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-23-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Provenance, Lineage, and Chain of Custody.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-23-005	When a material object, product, or decision input has incomplete lineage, broken custody, or unverifiable artifact identity, the platform SHALL mark the output unfit, block consequential use and export, preserve the gap, and recompute or adjudicate from verified inputs; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-23-006	Production admission and continued operation SHALL require lineage closure queries, digest verification, custody transfer exercises, model and human attribution, export lineage, and historical reconstruction, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A material object, product, or decision input has incomplete lineage, broken custody, or unverifiable artifact identity	Mark the output unfit, block consequential use and export, preserve the gap, and recompute or adjudicate from verified inputs.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared lineage event, input and output versions, actor, method, code, model, prompt, environment, time, policy, custody, quality, and evidence digest.
- Positive and negative boundary tests prove that lineage is canonical evidence and cannot be reconstructed only from mutable logs, filenames, table names, or post hoc explanation.
- Fault exercises inject a material object, product, or decision input has incomplete lineage, broken custody, or unverifiable artifact identity and verify that the platform can mark the output unfit, block consequential use and export, preserve the gap, and recompute or adjudicate from verified inputs.
- Independent review accepts lineage closure queries, digest verification, custody transfer exercises, model and human attribution, export lineage, and historical reconstruction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Provenance, Lineage, and Chain of Custody is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 24, Ch. 28, Ch. 30, Ch. 55. Primary Volume 5 AI anchors: Ch. 4, Ch. 18, Ch. 29, Ch. 46, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 45, Ch. 48, Ch. 55, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-025 · C-035 · C-036 · C-043

24 Correction, Withdrawal, Supersession, and Reconciliation

Correction, Withdrawal, Supersession, and Reconciliation establishes the governed data capability necessary to apply non-destructive corrections and withdrawals across canonical objects, projections, products, models, warnings, decisions, exports, and historical reconstruction. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

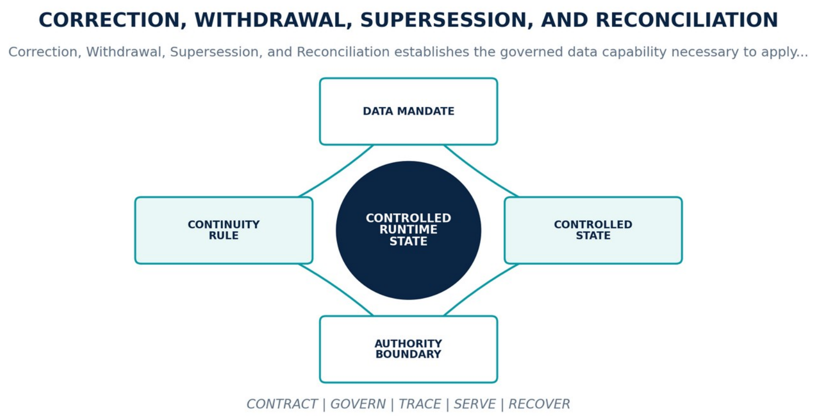

Figure 25. Correction, Withdrawal, Supersession, and Reconciliation. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Apply non-destructive corrections and withdrawals across canonical objects, projections, products, models, warnings, decisions, exports, and historical reconstruction.
Controlled state	Correction identity, authority, target versions, reason, effective time, replacement, impacted graph, propagation state, consumer acknowledgement, and closure.
Authority boundary	Correction does not erase what was previously observed or decided; historical views retain the exact state of knowledge and later amendment.
Continuity rule	When impact scope is unknown, propagation is partial, a consumer cannot reconcile, or a withdrawn object remains decision-active, the platform shall freeze affected publication, expose impact, quarantine stale derivatives, replay from the corrected boundary, and obtain closure from material consumers.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-24-001	The data-platform realization of Correction, Withdrawal, Supersession, and Reconciliation SHALL apply non-destructive corrections and withdrawals across canonical objects, projections, products, models, warnings, decisions, exports, and historical reconstruction.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-24-002	Every production instance SHALL declare and continuously reconcile correction identity, authority, target versions, reason, effective time, replacement, impacted graph, propagation state, consumer acknowledgement, and closure through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-24-003	The implementation SHALL enforce this authority boundary: correction does not erase what was previously observed or decided; historical views retain the exact state of knowledge and later amendment.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-24-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Correction, Withdrawal, Supersession, and Reconciliation.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-24-005	When impact scope is unknown, propagation is partial, a consumer cannot reconcile, or a withdrawn object remains decision-active, the platform SHALL freeze affected publication, expose impact, quarantine stale derivatives, replay from the corrected boundary, and obtain closure from material consumers; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-24-006	Production admission and continued operation SHALL require correction fan-out tests, projection rebuild, consumer acknowledgement, export amendment, decision-impact analysis, and reconciliation audit, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Impact scope is unknown, propagation is partial, a consumer cannot reconcile, or a withdrawn object remains decision-active	Freeze affected publication, expose impact, quarantine stale derivatives, replay from the corrected boundary, and obtain closure from material consumers.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared correction identity, authority, target versions, reason, effective time, replacement, impacted graph, propagation state, consumer acknowledgement, and closure.
- Positive and negative boundary tests prove that correction does not erase what was previously observed or decided; historical views retain the exact state of knowledge and later amendment.
- Fault exercises inject impact scope is unknown, propagation is partial, a consumer cannot reconcile, or a withdrawn object remains decision-active and verify that the platform can freeze affected publication, expose impact, quarantine stale derivatives, replay from the corrected boundary, and obtain closure from material consumers.
- Independent review accepts correction fan-out tests, projection rebuild, consumer acknowledgement, export amendment, decision-impact analysis, and reconciliation audit for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Correction, Withdrawal, Supersession, and Reconciliation is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 26, Ch. 28, Ch. 30, Ch. 48. Primary Volume 5 AI anchors: Ch. 26, Ch. 29, Ch. 38, Ch. 68, Ch. 69. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 32, Ch. 34, Ch. 35, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-015 · C-025 · C-030 · C-043

PART IV

Storage and Processing Architecture

State tiers, lakehouse, operational stores, evidence, messaging, batch, stream, query, cache, and projections.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

25 Data Storage Architecture and State Tiers

Data Storage Architecture and State Tiers establishes the governed data capability necessary to map each governed data asset to an authority, consistency, durability, locality, protection, access, lifecycle, and recovery class before selecting technology. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

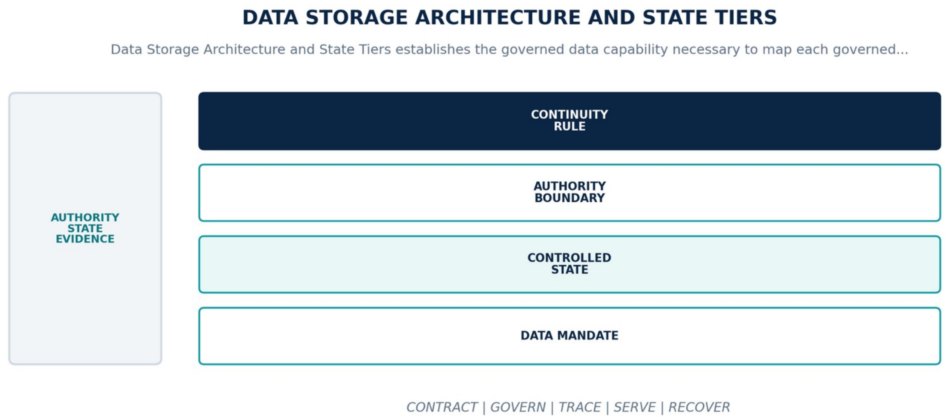

Figure 26. Data Storage Architecture and State Tiers. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Map each governed data asset to an authority, consistency, durability, locality, protection, access, lifecycle, and recovery class before selecting technology.
Controlled state	State class, authority owner, storage tier, consistency, partitioning, replication, residency, encryption, retention, RPO, RTO, and rebuild basis.
Authority boundary	Storage products implement state contracts; no engine, bucket, table, topic, or index owns strategic meaning by deployment convention.
Continuity rule	When a state class is unresolved, incompatible with its store, or cannot meet locality, durability, consistency, or recovery obligations, the platform shall deny placement or publication, preserve authoritative copies, expose degraded capability, and migrate only through a verified state plan.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-25-001	The data-platform realization of Data Storage Architecture and State Tiers SHALL map each governed data asset to an authority, consistency, durability, locality, protection, access, lifecycle, and recovery class before selecting technology.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-25-002	Every production instance SHALL declare and continuously reconcile state class, authority owner, storage tier, consistency, partitioning, replication, residency, encryption, retention, RPO, RTO, and rebuild basis through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-25-003	The implementation SHALL enforce this authority boundary: storage products implement state contracts; no engine, bucket, table, topic, or index owns strategic meaning by deployment convention.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-25-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Storage Architecture and State Tiers.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-25-005	When a state class is unresolved, incompatible with its store, or cannot meet locality, durability, consistency, or recovery obligations, the platform SHALL deny placement or publication, preserve authoritative copies, expose degraded capability, and migrate only through a verified state plan; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-25-006	Production admission and continued operation SHALL require state-class registry, technology qualification, consistency and failure tests, locality proof, restore exercises, and customer acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A state class is unresolved, incompatible with its store, or cannot meet locality, durability, consistency, or recovery obligations	Deny placement or publication, preserve authoritative copies, expose degraded capability, and migrate only through a verified state plan.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared state class, authority owner, storage tier, consistency, partitioning, replication, residency, encryption, retention, RPO, RTO, and rebuild basis.
- Positive and negative boundary tests prove that storage products implement state contracts; no engine, bucket, table, topic, or index owns strategic meaning by deployment convention.
- Fault exercises inject a state class is unresolved, incompatible with its store, or cannot meet locality, durability, consistency, or recovery obligations and verify that the platform can deny placement or publication, preserve authoritative copies, expose degraded capability, and migrate only through a verified state plan.
- Independent review accepts state-class registry, technology qualification, consistency and failure tests, locality proof, restore exercises, and customer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Storage Architecture and State Tiers is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 24, Ch. 30, Ch. 57, Ch. 61. Primary Volume 5 AI anchors: Ch. 13, Ch. 25, Ch. 26, Ch. 29, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 32, Ch. 33, Ch. 38, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-035 · C-037 · C-043

26 Lakehouse and Analytical Storage

Lakehouse and Analytical Storage establishes the governed data capability necessary to provide governed open analytical storage for raw, standardized, curated, feature, evaluation, and product-ready datasets with transactional metadata and reproducible snapshots. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

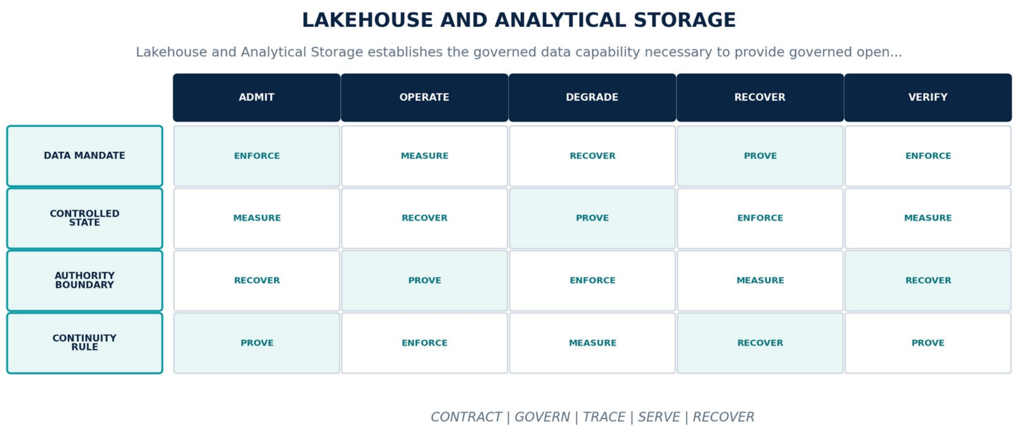

Figure 27. Lakehouse and Analytical Storage. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Provide governed open analytical storage for raw, standardized, curated, feature, evaluation, and product-ready datasets with transactional metadata and reproducible snapshots.
Controlled state	Catalog namespace, table or dataset identity, schema, partition, snapshot, manifest, owner, quality, policy, lineage, compaction, and retention state.
Authority boundary	Analytical convenience may not weaken tenant isolation, open export, temporal truth, record-level policy, correction, or canonical state ownership.
Continuity rule	When metadata and files diverge, a snapshot is corrupt, compaction loses history, or concurrent writes violate the table contract, the platform shall fence writers, preserve the last consistent snapshot, isolate suspect files, repair from transaction history, and reconcile downstream products.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-26-001	The data-platform realization of Lakehouse and Analytical Storage SHALL provide governed open analytical storage for raw, standardized, curated, feature, evaluation, and product-ready datasets with transactional metadata and reproducible snapshots.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-26-002	Every production instance SHALL declare and continuously reconcile catalog namespace, table or dataset identity, schema, partition, snapshot, manifest, owner, quality, policy, lineage, compaction, and retention state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-26-003	The implementation SHALL enforce this authority boundary: analytical convenience may not weaken tenant isolation, open export, temporal truth, record-level policy, correction, or canonical state ownership.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-26-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Lakehouse and Analytical Storage.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-26-005	When metadata and files diverge, a snapshot is corrupt, compaction loses history, or concurrent writes violate the table contract, the platform SHALL fence writers, preserve the last consistent snapshot, isolate suspect files, repair from transaction history, and reconcile downstream products; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-26-006	Production admission and continued operation SHALL require transaction and concurrency tests, snapshot time travel, schema evolution, compaction recovery, open-format portability, and scale benchmarks, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Metadata and files diverge, a snapshot is corrupt, compaction loses history, or concurrent writes violate the table contract	Fence writers, preserve the last consistent snapshot, isolate suspect files, repair from transaction history, and reconcile downstream products.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared catalog namespace, table or dataset identity, schema, partition, snapshot, manifest, owner, quality, policy, lineage, compaction, and retention state.
- Positive and negative boundary tests prove that analytical convenience may not weaken tenant isolation, open export, temporal truth, record-level policy, correction, or canonical state ownership.
- Fault exercises inject metadata and files diverge, a snapshot is corrupt, compaction loses history, or concurrent writes violate the table contract and verify that the platform can fence writers, preserve the last consistent snapshot, isolate suspect files, repair from transaction history, and reconcile downstream products.
- Independent review accepts transaction and concurrency tests, snapshot time travel, schema evolution, compaction recovery, open-format portability, and scale benchmarks for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Lakehouse and Analytical Storage is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 18, Ch. 23, Ch. 30, Ch. 58. Primary Volume 5 AI anchors: Ch. 19, Ch. 23, Ch. 25, Ch. 26, Ch. 66. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 35, Ch. 45, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-035 · C-037 · C-038 · C-043

27 Transactional and Operational Data Stores

Transactional and Operational Data Stores establishes the governed data capability necessary to host consistency-sensitive registries, workflows, policies, identities, cases, and lifecycle state behind singular service ownership. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

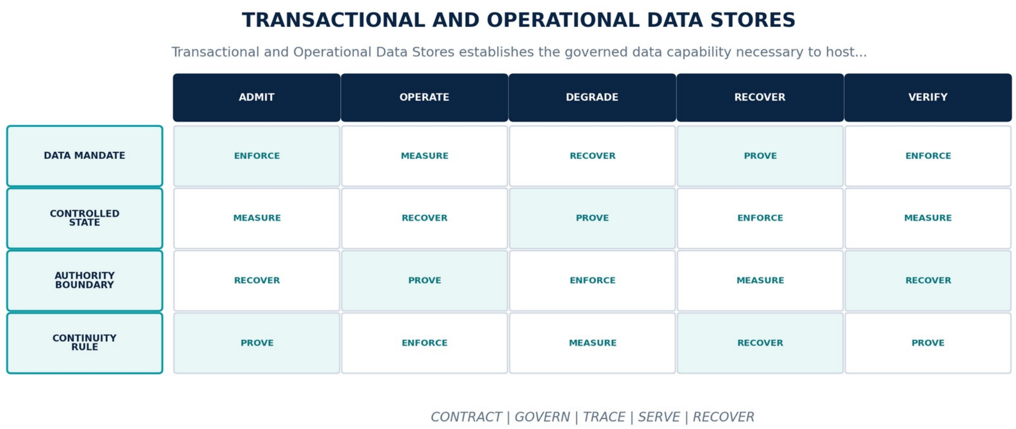

Figure 28. Transactional and Operational Data Stores. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Host consistency-sensitive registries, workflows, policies, identities, cases, and lifecycle state behind singular service ownership.
Controlled state	Database identity, owning service, schema version, transaction boundary, consistency, partition, replica, migration, backup, and health state.
Authority boundary	Cross-service direct writes, shared-table ownership, unmanaged triggers, and application-side authority bypass are prohibited.
Continuity rule	When quorum, ownership, migration, replication, or transaction integrity becomes uncertain, the platform shall fence unsafe writers, enter the declared restricted mode, restore a consistent owner, and reconcile replicas and consumers before reopening.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-27-001	The data-platform realization of Transactional and Operational Data Stores SHALL host consistency-sensitive registries, workflows, policies, identities, cases, and lifecycle state behind singular service ownership.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-27-002	Every production instance SHALL declare and continuously reconcile database identity, owning service, schema version, transaction boundary, consistency, partition, replica, migration, backup, and health state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-27-003	The implementation SHALL enforce this authority boundary: cross-service direct writes, shared-table ownership, unmanaged triggers, and application-side authority bypass are prohibited.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-27-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Transactional and Operational Data Stores.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-27-005	When quorum, ownership, migration, replication, or transaction integrity becomes uncertain, the platform SHALL fence unsafe writers, enter the declared restricted mode, restore a consistent owner, and reconcile replicas and consumers before reopening; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-27-006	Production admission and continued operation SHALL require transaction isolation, failover, fencing, migration, point-in-time restore, replica lag, and ownership negative tests, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Quorum, ownership, migration, replication, or transaction integrity becomes uncertain	Fence unsafe writers, enter the declared restricted mode, restore a consistent owner, and reconcile replicas and consumers before reopening.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared database identity, owning service, schema version, transaction boundary, consistency, partition, replica, migration, backup, and health state.
- Positive and negative boundary tests prove that cross-service direct writes, shared-table ownership, unmanaged triggers, and application-side authority bypass are prohibited.
- Fault exercises inject quorum, ownership, migration, replication, or transaction integrity becomes uncertain and verify that the platform can fence unsafe writers, enter the declared restricted mode, restore a consistent owner, and reconcile replicas and consumers before reopening.
- Independent review accepts transaction isolation, failover, fencing, migration, point-in-time restore, replica lag, and ownership negative tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Transactional and Operational Data Stores is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 12, Ch. 22, Ch. 30, Ch. 59. Primary Volume 5 AI anchors: Ch. 10, Ch. 15, Ch. 25, Ch. 38, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 32, Ch. 61, Ch. 62, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-006 · C-011 · C-016 · C-035 · C-039 · C-043

28 Object, Evidence, and Archive Storage

Object, Evidence, and Archive Storage establishes the governed data capability necessary to preserve immutable source evidence, documents, media, large artifacts, exports, archives, and custody packages with integrity and lifecycle controls. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

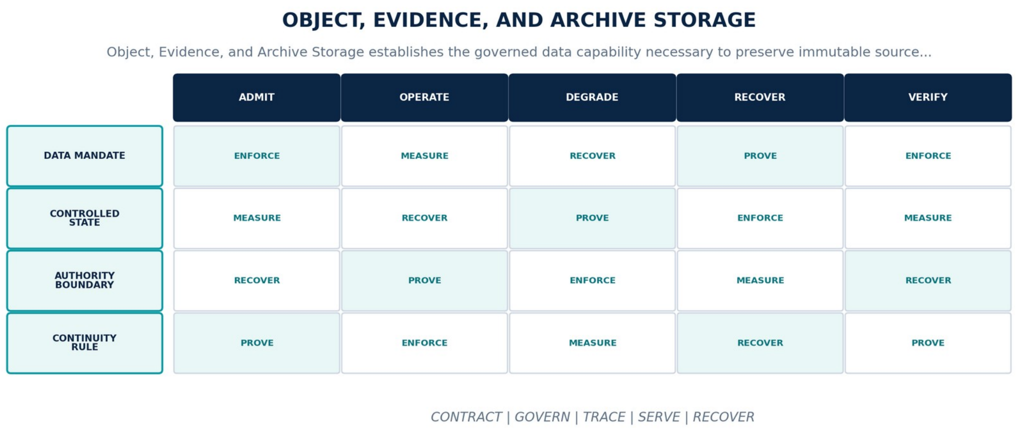

Figure 29. Object, Evidence, and Archive Storage. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Preserve immutable source evidence, documents, media, large artifacts, exports, archives, and custody packages with integrity and lifecycle controls.
Controlled state	Object identity, version, digest, source, custody, classification, encryption key, locality, retention, legal hold, replication, and restore state.
Authority boundary	Object immutability protects evidence history but does not prevent governed correction, tombstone, retention, deletion, or supersession metadata.
Continuity rule	When an object is missing, corrupt, misclassified, outside residency, or referenced by metadata that cannot prove byte integrity, the platform shall quarantine the reference, protect remaining replicas, stop dependent publication, recover from verified copies, and preserve the integrity incident.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-28-001	The data-platform realization of Object, Evidence, and Archive Storage SHALL preserve immutable source evidence, documents, media, large artifacts, exports, archives, and custody packages with integrity and lifecycle controls.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-28-002	Every production instance SHALL declare and continuously reconcile object identity, version, digest, source, custody, classification, encryption key, locality, retention, legal hold, replication, and restore state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-28-003	The implementation SHALL enforce this authority boundary: object immutability protects evidence history but does not prevent governed correction, tombstone, retention, deletion, or supersession metadata.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-28-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Object, Evidence, and Archive Storage.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-28-005	When an object is missing, corrupt, misclassified, outside residency, or referenced by metadata that cannot prove byte integrity, the platform SHALL quarantine the reference, protect remaining replicas, stop dependent publication, recover from verified copies, and preserve the integrity incident; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-28-006	Production admission and continued operation SHALL require digest and custody verification, immutability controls, large-object restore, legal hold, locality, deletion, and archive retrieval exercises, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An object is missing, corrupt, misclassified, outside residency, or referenced by metadata that cannot prove byte integrity	Quarantine the reference, protect remaining replicas, stop dependent publication, recover from verified copies, and preserve the integrity incident.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared object identity, version, digest, source, custody, classification, encryption key, locality, retention, legal hold, replication, and restore state.
- Positive and negative boundary tests prove that object immutability protects evidence history but does not prevent governed correction, tombstone, retention, deletion, or supersession metadata.
- Fault exercises inject an object is missing, corrupt, misclassified, outside residency, or referenced by metadata that cannot prove byte integrity and verify that the platform can quarantine the reference, protect remaining replicas, stop dependent publication, recover from verified copies, and preserve the integrity incident.
- Independent review accepts digest and custody verification, immutability controls, large-object restore, legal hold, locality, deletion, and archive retrieval exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Object, Evidence, and Archive Storage is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 28, Ch. 29, Ch. 30, Ch. 61, Ch. 64. Primary Volume 5 AI anchors: Ch. 25, Ch. 29, Ch. 61, Ch. 65, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 51, Ch. 53, Ch. 57, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-012 · C-015 · C-016 · C-037 · C-040 · C-043

29 Event Streaming and Durable Messaging

Event Streaming and Durable Messaging establishes the governed data capability necessary to carry data changes, domain events, work, corrections, quality signals, and lifecycle notifications through explicit delivery, ordering, replay, and retention semantics. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

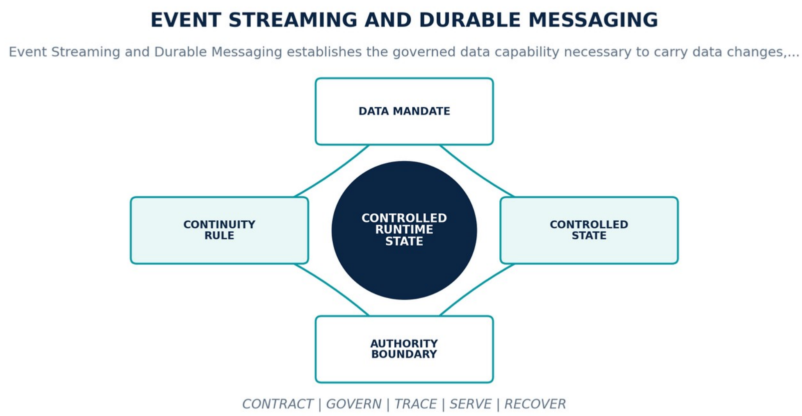

Figure 30. Event Streaming and Durable Messaging. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Carry data changes, domain events, work, corrections, quality signals, and lifecycle notifications through explicit delivery, ordering, replay, and retention semantics.
Controlled state	Topic or queue identity, producer, consumer, schema, partition key, delivery mode, acknowledgement, offset, retention, dead letter, and replay state.
Authority boundary	Message delivery does not prove business commitment, canonical admission, exactly-once institutional effect, or human approval.
Continuity rule	When brokers lose quorum, consumers fall behind, poison messages recur, ordering breaks, or replay would duplicate external effects, the platform shall apply backpressure, isolate poison data, preserve offsets, stop unsafe consumers, replay idempotently, and reconcile canonical effects.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-29-001	The data-platform realization of Event Streaming and Durable Messaging SHALL carry data changes, domain events, work, corrections, quality signals, and lifecycle notifications through explicit delivery, ordering, replay, and retention semantics.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-29-002	Every production instance SHALL declare and continuously reconcile topic or queue identity, producer, consumer, schema, partition key, delivery mode, acknowledgement, offset, retention, dead letter, and replay state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-29-003	The implementation SHALL enforce this authority boundary: message delivery does not prove business commitment, canonical admission, exactly-once institutional effect, or human approval.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-29-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Event Streaming and Durable Messaging.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-29-005	When brokers lose quorum, consumers fall behind, poison messages recur, ordering breaks, or replay would duplicate external effects, the platform SHALL apply backpressure, isolate poison data, preserve offsets, stop unsafe consumers, replay idempotently, and reconcile canonical effects; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-29-006	Production admission and continued operation SHALL require delivery and ordering fixtures, broker failover, consumer recovery, dead-letter governance, replay, idempotency, and effect reconciliation, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Brokers lose quorum, consumers fall behind, poison messages recur, ordering breaks, or replay would duplicate external effects	Apply backpressure, isolate poison data, preserve offsets, stop unsafe consumers, replay idempotently, and reconcile canonical effects.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared topic or queue identity, producer, consumer, schema, partition key, delivery mode, acknowledgement, offset, retention, dead letter, and replay state.
- Positive and negative boundary tests prove that message delivery does not prove business commitment, canonical admission, exactly-once institutional effect, or human approval.
- Fault exercises inject brokers lose quorum, consumers fall behind, poison messages recur, ordering breaks, or replay would duplicate external effects and verify that the platform can apply backpressure, isolate poison data, preserve offsets, stop unsafe consumers, replay idempotently, and reconcile canonical effects.
- Independent review accepts delivery and ordering fixtures, broker failover, consumer recovery, dead-letter governance, replay, idempotency, and effect reconciliation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Event Streaming and Durable Messaging is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 30, Ch. 41, Ch. 59. Primary Volume 5 AI anchors: Ch. 34, Ch. 35, Ch. 38, Ch. 39, Ch. 68. Primary Volume 6 infrastructure anchors: Ch. 37, Ch. 38, Ch. 41, Ch. 61, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-002 · C-011 · C-012 · C-035 · C-038 · C-043

30 Batch Processing and Orchestration

Batch Processing and Orchestration establishes the governed data capability necessary to execute reproducible bounded transformations, backfills, graph loads, feature builds, evaluations, indexing, exports, and lifecycle operations. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

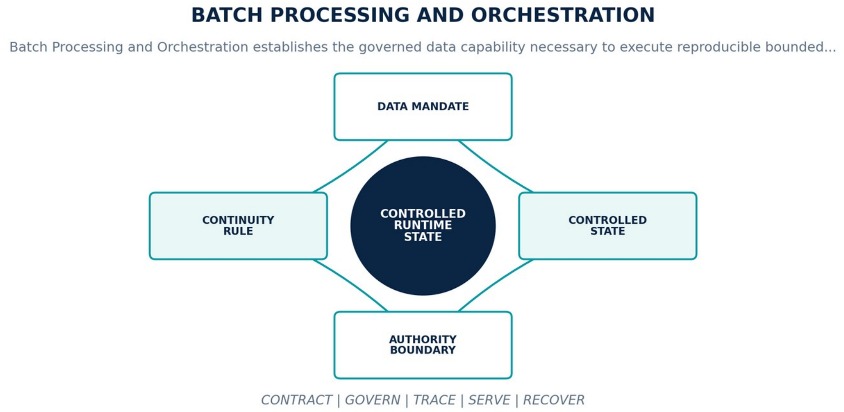

Figure 31. Batch Processing and Orchestration. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Execute reproducible bounded transformations, backfills, graph loads, feature builds, evaluations, indexing, exports, and lifecycle operations.
Controlled state	Job identity, pipeline version, input snapshot, partitions, code and configuration, policy, resources, checkpoints, outputs, metrics, and terminal state.
Authority boundary	Retries and parallel execution may not create duplicate authority, unbounded side effects, inconsistent partitions, or results without exact input lineage.
Continuity rule	When a job partially commits, inputs change during execution, partitions diverge, or recovery cannot identify the last safe boundary, the platform shall stop publication, preserve checkpoints and partial outputs, roll back or resume idempotently, and reconcile complete scope before promotion.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-30-001	The data-platform realization of Batch Processing and Orchestration SHALL execute reproducible bounded transformations, backfills, graph loads, feature builds, evaluations, indexing, exports, and lifecycle operations.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-30-002	Every production instance SHALL declare and continuously reconcile job identity, pipeline version, input snapshot, partitions, code and configuration, policy, resources, checkpoints, outputs, metrics, and terminal state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-30-003	The implementation SHALL enforce this authority boundary: retries and parallel execution may not create duplicate authority, unbounded side effects, inconsistent partitions, or results without exact input lineage.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-30-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Batch Processing and Orchestration.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-30-005	When a job partially commits, inputs change during execution, partitions diverge, or recovery cannot identify the last safe boundary, the platform SHALL stop publication, preserve checkpoints and partial outputs, roll back or resume idempotently, and reconcile complete scope before promotion; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-30-006	Production admission and continued operation SHALL require reproducibility, partition and skew tests, partial-commit recovery, backfill comparison, idempotency, cancellation, and lineage closure, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A job partially commits, inputs change during execution, partitions diverge, or recovery cannot identify the last safe boundary	Stop publication, preserve checkpoints and partial outputs, roll back or resume idempotently, and reconcile complete scope before promotion.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared job identity, pipeline version, input snapshot, partitions, code and configuration, policy, resources, checkpoints, outputs, metrics, and terminal state.
- Positive and negative boundary tests prove that retries and parallel execution may not create duplicate authority, unbounded side effects, inconsistent partitions, or results without exact input lineage.
- Fault exercises inject a job partially commits, inputs change during execution, partitions diverge, or recovery cannot identify the last safe boundary and verify that the platform can stop publication, preserve checkpoints and partial outputs, roll back or resume idempotently, and reconcile complete scope before promotion.
- Independent review accepts reproducibility, partition and skew tests, partial-commit recovery, backfill comparison, idempotency, cancellation, and lineage closure for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Batch Processing and Orchestration is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 28, Ch. 30, Ch. 41, Ch. 58. Primary Volume 5 AI anchors: Ch. 19, Ch. 38, Ch. 66, Ch. 68, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 20, Ch. 22, Ch. 37, Ch. 41, Ch. 45.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-023 · C-035 · C-043 · C-049

31 Stream Processing and Complex Event Handling

Stream Processing and Complex Event Handling establishes the governed data capability necessary to derive governed near-real-time signals, windows, patterns, weak signals, warnings, and material changes from ordered event streams. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

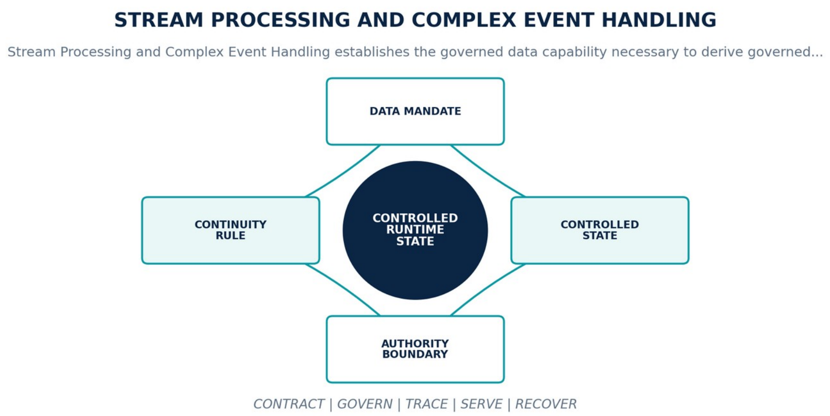

Figure 32. Stream Processing and Complex Event Handling. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Derive governed near-real-time signals, windows, patterns, weak signals, warnings, and material changes from ordered event streams.
Controlled state	Processor identity, topology version, source offsets, event-time windows, watermark, state, rules or models, outputs, checkpoints, and fitness.
Authority boundary	Low latency may not conceal late data, partial windows, state loss, duplicate emissions, model drift, or uncertainty in material signals.
Continuity rule	When watermarks stall, state corrupts, offsets diverge, output duplicates, or a rule or model is no longer fit, the platform shall suspend affected outputs, recover from a compatible checkpoint, replay within policy, retract invalid signals, and expose detection gaps.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-31-001	The data-platform realization of Stream Processing and Complex Event Handling SHALL derive governed near-real-time signals, windows, patterns, weak signals, warnings, and material changes from ordered event streams.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-31-002	Every production instance SHALL declare and continuously reconcile processor identity, topology version, source offsets, event-time windows, watermark, state, rules or models, outputs, checkpoints, and fitness through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-31-003	The implementation SHALL enforce this authority boundary: low latency may not conceal late data, partial windows, state loss, duplicate emissions, model drift, or uncertainty in material signals.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-31-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Stream Processing and Complex Event Handling.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-31-005	When watermarks stall, state corrupts, offsets diverge, output duplicates, or a rule or model is no longer fit, the platform SHALL suspend affected outputs, recover from a compatible checkpoint, replay within policy, retract invalid signals, and expose detection gaps; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-31-006	Production admission and continued operation SHALL require late-event fixtures, state and checkpoint recovery, duplicate suppression, rule versioning, load tests, weak-signal evaluation, and retraction, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Watermarks stall, state corrupts, offsets diverge, output duplicates, or a rule or model is no longer fit	Suspend affected outputs, recover from a compatible checkpoint, replay within policy, retract invalid signals, and expose detection gaps.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared processor identity, topology version, source offsets, event-time windows, watermark, state, rules or models, outputs, checkpoints, and fitness.
- Positive and negative boundary tests prove that low latency may not conceal late data, partial windows, state loss, duplicate emissions, model drift, or uncertainty in material signals.
- Fault exercises inject watermarks stall, state corrupts, offsets diverge, output duplicates, or a rule or model is no longer fit and verify that the platform can suspend affected outputs, recover from a compatible checkpoint, replay within policy, retract invalid signals, and expose detection gaps.
- Independent review accepts late-event fixtures, state and checkpoint recovery, duplicate suppression, rule versioning, load tests, weak-signal evaluation, and retraction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Stream Processing and Complex Event Handling is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 27, Ch. 41, Ch. 47, Ch. 58. Primary Volume 5 AI anchors: Ch. 48, Ch. 52, Ch. 57, Ch. 63, Ch. 68. Primary Volume 6 infrastructure anchors: Ch. 20, Ch. 37, Ch. 38, Ch. 45, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-011 · C-014 · C-032 · C-033 · C-035 · C-043

32 Query, Caching, Materialization, and Projection

Query, Caching, Materialization, and Projection establishes the governed data capability necessary to serve policy-filtered operational, analytical, graph, search, and executive queries through rebuildable projections with explicit consistency and freshness. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

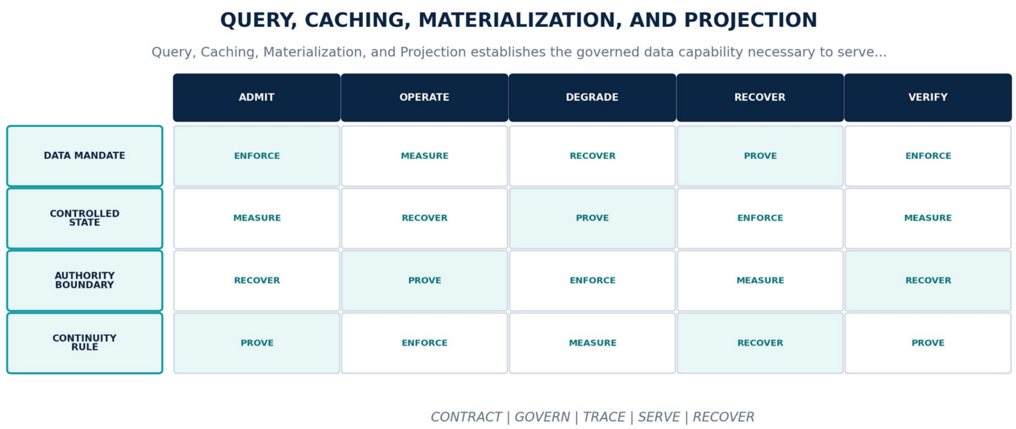

Figure 33. Query, Caching, Materialization, and Projection. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Serve policy-filtered operational, analytical, graph, search, and executive queries through rebuildable projections with explicit consistency and freshness.
Controlled state	Query contract, authoritative source, projection definition, source revision, cache key, invalidation, consistency, freshness, policy, and rebuild state.
Authority boundary	Indexes, caches, marts, views, embeddings, and materializations remain derived state and may not become silent alternate truth.
Continuity rule	When a projection is stale, partially rebuilt, policy-inconsistent, or cannot resolve its authoritative source revision, the platform shall bypass or quarantine the projection, expose staleness, serve a policy-equivalent authoritative path where available, and rebuild deterministically.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-32-001	The data-platform realization of Query, Caching, Materialization, and Projection SHALL serve policy-filtered operational, analytical, graph, search, and executive queries through rebuildable projections with explicit consistency and freshness.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-32-002	Every production instance SHALL declare and continuously reconcile query contract, authoritative source, projection definition, source revision, cache key, invalidation, consistency, freshness, policy, and rebuild state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-32-003	The implementation SHALL enforce this authority boundary: indexes, caches, marts, views, embeddings, and materializations remain derived state and may not become silent alternate truth.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-32-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Query, Caching, Materialization, and Projection.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-32-005	When a projection is stale, partially rebuilt, policy-inconsistent, or cannot resolve its authoritative source revision, the platform SHALL bypass or quarantine the projection, expose staleness, serve a policy-equivalent authoritative path where available, and rebuild deterministically; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-32-006	Production admission and continued operation SHALL require source-to-projection reconciliation, invalidation, policy filtering, stale-read fixtures, deterministic rebuild, and performance evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A projection is stale, partially rebuilt, policy-inconsistent, or cannot resolve its authoritative source revision	Bypass or quarantine the projection, expose staleness, serve a policy-equivalent authoritative path where available, and rebuild deterministically.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared query contract, authoritative source, projection definition, source revision, cache key, invalidation, consistency, freshness, policy, and rebuild state.
- Positive and negative boundary tests prove that indexes, caches, marts, views, embeddings, and materializations remain derived state and may not become silent alternate truth.
- Fault exercises inject a projection is stale, partially rebuilt, policy-inconsistent, or cannot resolve its authoritative source revision and verify that the platform can bypass or quarantine the projection, expose staleness, serve a policy-equivalent authoritative path where available, and rebuild deterministically.
- Independent review accepts source-to-projection reconciliation, invalidation, policy filtering, stale-read fixtures, deterministic rebuild, and performance evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Query, Caching, Materialization, and Projection is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 18, Ch. 21, Ch. 23, Ch. 30. Primary Volume 5 AI anchors: Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 29. Primary Volume 6 infrastructure anchors: Ch. 34, Ch. 35, Ch. 38, Ch. 44, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-009 · C-011 · C-014 · C-016 · C-035 · C-043

PART V

Knowledge, Memory, and Strategic State

Knowledge Graph, semantic governance, identity resolution, claims, enterprise memory, retrieval, twins, and strategic state packages.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

33 Knowledge Graph Data Architecture

Knowledge Graph Data Architecture establishes the governed data capability necessary to realize the Knowledge Graph as the connected semantic heart linking canonical identity, evidence, time, truth, relationships, strategy, and downstream intelligence. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

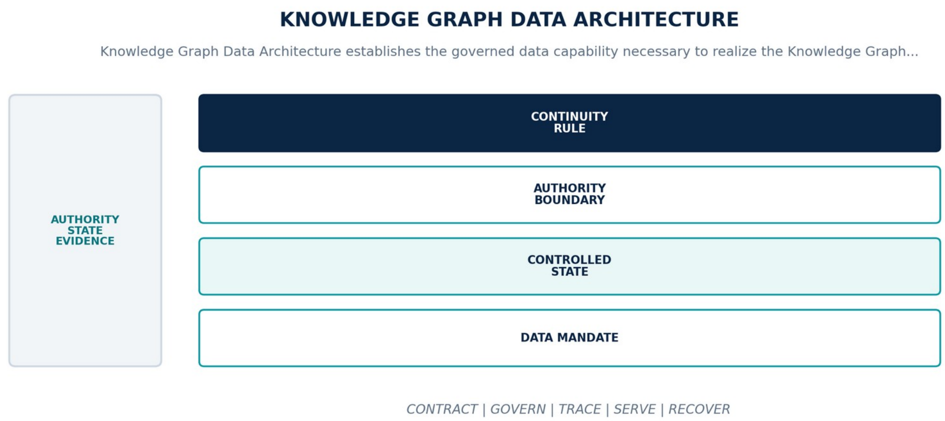

Figure 34. Knowledge Graph Data Architecture. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Realize the Knowledge Graph as the connected semantic heart linking canonical identity, evidence, time, truth, relationships, strategy, and downstream intelligence.
Controlled state	Graph namespace, ontology version, node and edge identity, temporal interval, truth state, provenance, policy, revision, partition, and stewardship.
Authority boundary	The graph owns semantic identity and governed relationships but does not absorb immutable evidence, service-owned workflow state, or synthetic branches into undifferentiated facts.
Continuity rule	When a graph revision is partial, provenance is missing, ontology and data versions conflict, or graph consensus and validation become unavailable, the platform shall reject atomic mutation, queue controlled revisions, serve the last valid graph with staleness, and reconcile through the revision log.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-33-001	The data-platform realization of Knowledge Graph Data Architecture SHALL realize the Knowledge Graph as the connected semantic heart linking canonical identity, evidence, time, truth, relationships, strategy, and downstream intelligence.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-33-002	Every production instance SHALL declare and continuously reconcile graph namespace, ontology version, node and edge identity, temporal interval, truth state, provenance, policy, revision, partition, and stewardship through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-33-003	The implementation SHALL enforce this authority boundary: the graph owns semantic identity and governed relationships but does not absorb immutable evidence, service-owned workflow state, or synthetic branches into undifferentiated facts.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-33-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Knowledge Graph Data Architecture.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-33-005	When a graph revision is partial, provenance is missing, ontology and data versions conflict, or graph consensus and validation become unavailable, the platform SHALL reject atomic mutation, queue controlled revisions, serve the last valid graph with staleness, and reconcile through the revision log; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-33-006	Production admission and continued operation SHALL require atomic revision fixtures, temporal traversal, provenance closure, policy filtering, partition recovery, merge and split replay, and graph quality evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A graph revision is partial, provenance is missing, ontology and data versions conflict, or graph consensus and validation become unavailable	Reject atomic mutation, queue controlled revisions, serve the last valid graph with staleness, and reconcile through the revision log.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared graph namespace, ontology version, node and edge identity, temporal interval, truth state, provenance, policy, revision, partition, and stewardship.
- Positive and negative boundary tests prove that the graph owns semantic identity and governed relationships but does not absorb immutable evidence, service-owned workflow state, or synthetic branches into undifferentiated facts.
- Fault exercises inject a graph revision is partial, provenance is missing, ontology and data versions conflict, or graph consensus and validation become unavailable and verify that the platform can reject atomic mutation, queue controlled revisions, serve the last valid graph with staleness, and reconcile through the revision log.
- Independent review accepts atomic revision fixtures, temporal traversal, provenance closure, policy filtering, partition recovery, merge and split replay, and graph quality evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Knowledge Graph Data Architecture is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 28, Ch. 30, Ch. 34. Primary Volume 5 AI anchors: Ch. 13, Ch. 24, Ch. 27, Ch. 29, Ch. 45. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 34, Ch. 35, Ch. 44, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-008 · C-009 · C-010 · C-011 · C-012 · C-017

34 Ontology, Taxonomy, and Semantic Governance

Ontology, Taxonomy, and Semantic Governance establishes the governed data capability necessary to govern shared concepts, types, relationships, constraints, classifications, mappings, and domain extensions through versioned semantic authority. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

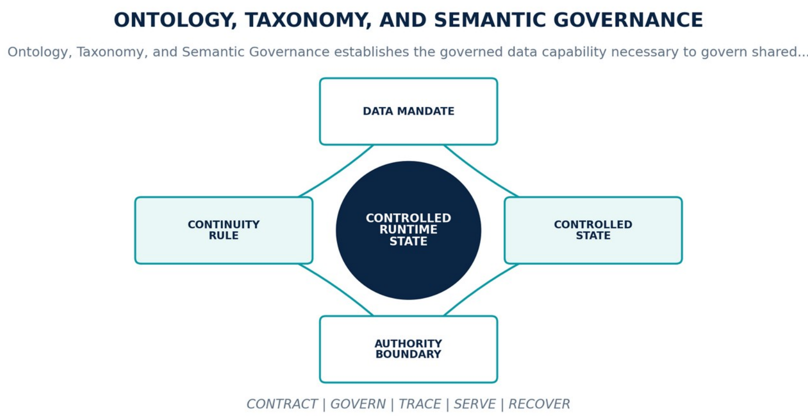

Figure 35. Ontology, Taxonomy, and Semantic Governance. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern shared concepts, types, relationships, constraints, classifications, mappings, and domain extensions through versioned semantic authority.
Controlled state	Ontology identity, version, owner, term and relation definitions, constraints, mappings, compatibility, migration, review, and effective time.
Authority boundary	Domain extensions may add specificity but may not redefine constitutional concepts, canonical layer responsibilities, or shared object semantics.
Continuity rule	When semantic changes are incompatible, mappings become ambiguous, or deployed data and consumers cannot agree on an effective ontology, the platform shall block promotion, preserve the prior version, expose impacted objects and products, and execute governed migration with stewardship approval.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-34-001	The data-platform realization of Ontology, Taxonomy, and Semantic Governance SHALL govern shared concepts, types, relationships, constraints, classifications, mappings, and domain extensions through versioned semantic authority.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-34-002	Every production instance SHALL declare and continuously reconcile ontology identity, version, owner, term and relation definitions, constraints, mappings, compatibility, migration, review, and effective time through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-34-003	The implementation SHALL enforce this authority boundary: domain extensions may add specificity but may not redefine constitutional concepts, canonical layer responsibilities, or shared object semantics.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-34-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Ontology, Taxonomy, and Semantic Governance.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-34-005	When semantic changes are incompatible, mappings become ambiguous, or deployed data and consumers cannot agree on an effective ontology, the platform SHALL block promotion, preserve the prior version, expose impacted objects and products, and execute governed migration with stewardship approval; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-34-006	Production admission and continued operation SHALL require reasoner and constraint tests, competency queries, mapping validation, migration rehearsal, consumer impact, and ontology board acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Semantic changes are incompatible, mappings become ambiguous, or deployed data and consumers cannot agree on an effective ontology	Block promotion, preserve the prior version, expose impacted objects and products, and execute governed migration with stewardship approval.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared ontology identity, version, owner, term and relation definitions, constraints, mappings, compatibility, migration, review, and effective time.
- Positive and negative boundary tests prove that domain extensions may add specificity but may not redefine constitutional concepts, canonical layer responsibilities, or shared object semantics.
- Fault exercises inject semantic changes are incompatible, mappings become ambiguous, or deployed data and consumers cannot agree on an effective ontology and verify that the platform can block promotion, preserve the prior version, expose impacted objects and products, and execute governed migration with stewardship approval.
- Independent review accepts reasoner and constraint tests, competency queries, mapping validation, migration rehearsal, consumer impact, and ontology board acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Ontology, Taxonomy, and Semantic Governance is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 6, Ch. 23, Ch. 30, Ch. 34, Ch. 72. Primary Volume 5 AI anchors: Ch. 13, Ch. 27, Ch. 45, Ch. 46, Ch. 47. Primary Volume 6 infrastructure anchors: Ch. 34, Ch. 45, Ch. 48, Ch. 65, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-007 · C-008 · C-009 · C-010 · C-035 · C-051

35 Entity Resolution and Master Identity

Entity Resolution and Master Identity establishes the governed data capability necessary to resolve companies, countries, governments, products, technologies, people, supply chains, markets, regulations, events, risks, objectives, capabilities, and other entities without forced certainty. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

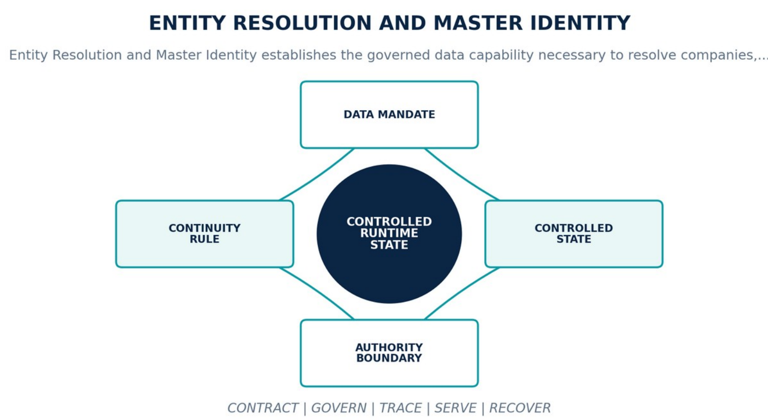

Figure 36. Entity Resolution and Master Identity. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Resolve companies, countries, governments, products, technologies, people, supply chains, markets, regulations, events, risks, objectives, capabilities, and other entities without forced certainty.
Controlled state	Entity identity, candidate set, source identifiers, features, rules and models, confidence, evidence, steward, merge or split decision, and history.
Authority boundary	Identity resolution proposes and supports decisions; high-impact merges, splits, and cross-tenant links require governed authority and remain reversible.
Continuity rule	When candidates are ambiguous, evidence conflicts, a merge causes contamination, or identity quality falls outside the accepted envelope, the platform shall retain unresolved candidates, suspend unsafe consolidation, reverse affected merges, propagate corrections, and route review to accountable stewards.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-35-001	The data-platform realization of Entity Resolution and Master Identity SHALL resolve companies, countries, governments, products, technologies, people, supply chains, markets, regulations, events, risks, objectives, capabilities, and other entities without forced certainty.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-35-002	Every production instance SHALL declare and continuously reconcile entity identity, candidate set, source identifiers, features, rules and models, confidence, evidence, steward, merge or split decision, and history through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-35-003	The implementation SHALL enforce this authority boundary: identity resolution proposes and supports decisions; high-impact merges, splits, and cross-tenant links require governed authority and remain reversible.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-35-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Entity Resolution and Master Identity.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-35-005	When candidates are ambiguous, evidence conflicts, a merge causes contamination, or identity quality falls outside the accepted envelope, the platform SHALL retain unresolved candidates, suspend unsafe consolidation, reverse affected merges, propagate corrections, and route review to accountable stewards; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-35-006	Production admission and continued operation SHALL require precision and recall evaluation, ambiguous and adversarial fixtures, merge and split replay, cross-tenant isolation, correction impact, and steward sampling, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Candidates are ambiguous, evidence conflicts, a merge causes contamination, or identity quality falls outside the accepted envelope	Retain unresolved candidates, suspend unsafe consolidation, reverse affected merges, propagate corrections, and route review to accountable stewards.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared entity identity, candidate set, source identifiers, features, rules and models, confidence, evidence, steward, merge or split decision, and history.
- Positive and negative boundary tests prove that identity resolution proposes and supports decisions; high-impact merges, splits, and cross-tenant links require governed authority and remain reversible.
- Fault exercises inject candidates are ambiguous, evidence conflicts, a merge causes contamination, or identity quality falls outside the accepted envelope and verify that the platform can retain unresolved candidates, suspend unsafe consolidation, reverse affected merges, propagate corrections, and route review to accountable stewards.
- Independent review accepts precision and recall evaluation, ambiguous and adversarial fixtures, merge and split replay, cross-tenant isolation, correction impact, and steward sampling for each supported deployment profile.

> **CHAPTER ACCEPTANCE AND INHERITANCE**
> Entity Resolution and Master Identity is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 25, Ch. 27, Ch. 28, Ch. 30, Ch. 34, Ch. 46. Primary Volume 5 AI anchors: Ch. 27, Ch. 43, Ch. 45, Ch. 46, Ch. 66. Primary Volume 6 infrastructure anchors: Ch. 34, Ch. 35, Ch. 44, Ch. 54, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-009 · C-010 · C-011 · C-012 · C-016 · C-045

36 Relationship, Event, Claim, and Assessment Data

Relationship, Event, Claim, and Assessment Data establishes the governed data capability necessary to represent material relationships, events, claims, contradictions, corroboration, assessments, and judgments as typed temporal objects with evidence and accountable origin. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

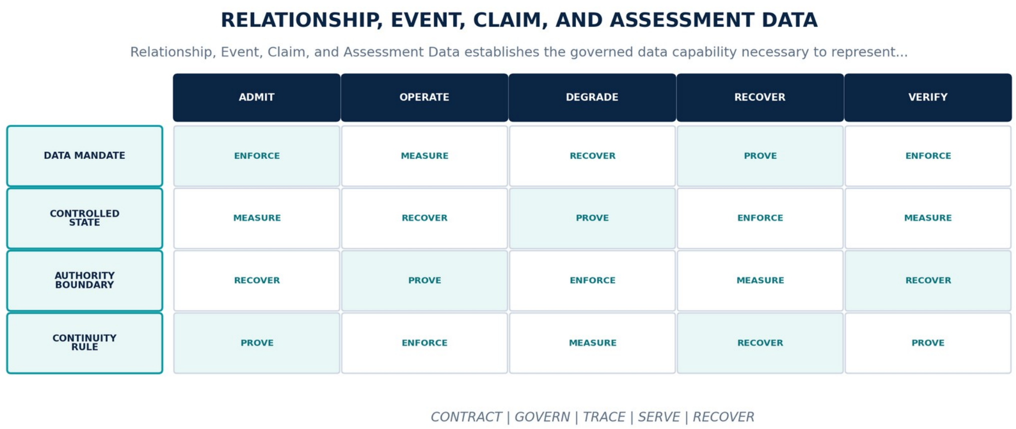

Figure 37. Relationship, Event, Claim, and Assessment Data. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Represent material relationships, events, claims, contradictions, corroboration, assessments, and judgments as typed temporal objects with evidence and accountable origin.
Controlled state	Object identity, subject and object, type, temporal scope, assertion authority, evidence, method, truth state, confidence, contradiction set, and review.
Authority boundary	Extracted, inferred, assessed, and human-judged relationships remain distinct and cannot be collapsed into one unqualified edge or fact.
Continuity rule	When required evidence, temporal scope, type constraints, or truth state is missing or contradictory, the platform shall reject canonical promotion or mark the object disputed, preserve alternatives, route material review, and prevent unsupported downstream certainty.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-36-001	The data-platform realization of Relationship, Event, Claim, and Assessment Data SHALL represent material relationships, events, claims, contradictions, corroboration, assessments, and judgments as typed temporal objects with evidence and accountable origin.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-36-002	Every production instance SHALL declare and continuously reconcile object identity, subject and object, type, temporal scope, assertion authority, evidence, method, truth state, confidence, contradiction set, and review through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-36-003	The implementation SHALL enforce this authority boundary: extracted, inferred, assessed, and human-judged relationships remain distinct and cannot be collapsed into one unqualified edge or fact.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-36-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Relationship, Event, Claim, and Assessment Data.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-36-005	When required evidence, temporal scope, type constraints, or truth state is missing or contradictory, the platform SHALL reject canonical promotion or mark the object disputed, preserve alternatives, route material review, and prevent unsupported downstream certainty; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-36-006	Production admission and continued operation SHALL require type and constraint fixtures, contradiction sets, temporal validity, provenance closure, adjudication workflow, and downstream interpretation tests, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Required evidence, temporal scope, type constraints, or truth state is missing or contradictory	Reject canonical promotion or mark the object disputed, preserve alternatives, route material review, and prevent unsupported downstream certainty.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared object identity, subject and object, type, temporal scope, assertion authority, evidence, method, truth state, confidence, contradiction set, and review.
- Positive and negative boundary tests prove that extracted, inferred, assessed, and human-judged relationships remain distinct and cannot be collapsed into one unqualified edge or fact.
- Fault exercises inject required evidence, temporal scope, type constraints, or truth state is missing or contradictory and verify that the platform can reject canonical promotion or mark the object disputed, preserve alternatives, route material review, and prevent unsupported downstream certainty.
- Independent review accepts type and constraint fixtures, contradiction sets, temporal validity, provenance closure, adjudication workflow, and downstream interpretation tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Relationship, Event, Claim, and Assessment Data is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 32, Ch. 34. Primary Volume 5 AI anchors: Ch. 20, Ch. 27, Ch. 44, Ch. 46, Ch. 63. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 34, Ch. 35, Ch. 44, Ch. 48.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-010 · C-011 · C-012 · C-035

37 Enterprise Memory and Strategic Record

Enterprise Memory and Strategic Record establishes the governed data capability necessary to preserve permission-aware institutional memory across evidence, analysis, communications, decisions, actions, outcomes, assumptions, and lessons. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

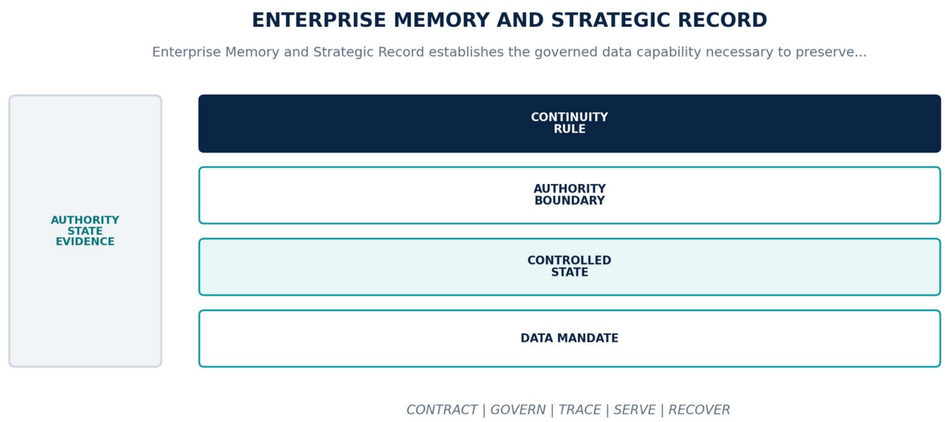

Figure 38. Enterprise Memory and Strategic Record. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Preserve permission-aware institutional memory across evidence, analysis, communications, decisions, actions, outcomes, assumptions, and lessons.
Controlled state	Memory object, version, owner, participants, content and evidence, temporal scope, policy, retention, correction, index state, and strategic record class.
Authority boundary	Retrieval frequency, embedding similarity, model output, or user annotation does not convert content into accepted organizational truth.
Continuity rule	When memory content, permissions, retention, version history, or indexes diverge or historical reconstruction becomes incomplete, the platform shall restrict access, serve metadata-only or last valid state with warnings, rebuild projections, and pause destructive lifecycle actions until scope is proven.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-37-001	The data-platform realization of Enterprise Memory and Strategic Record SHALL preserve permission-aware institutional memory across evidence, analysis, communications, decisions, actions, outcomes, assumptions, and lessons.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-37-002	Every production instance SHALL declare and continuously reconcile memory object, version, owner, participants, content and evidence, temporal scope, policy, retention, correction, index state, and strategic record class through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-37-003	The implementation SHALL enforce this authority boundary: retrieval frequency, embedding similarity, model output, or user annotation does not convert content into accepted organizational truth.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-37-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Enterprise Memory and Strategic Record.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-37-005	When memory content, permissions, retention, version history, or indexes diverge or historical reconstruction becomes incomplete, the platform SHALL restrict access, serve metadata-only or last valid state with warnings, rebuild projections, and pause destructive lifecycle actions until scope is proven; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-37-006	Production admission and continued operation SHALL require permission and purpose tests, temporal reconstruction, correction replay, legal hold, index rebuild, export, deletion verification, and customer review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Memory content, permissions, retention, version history, or indexes diverge or historical reconstruction becomes incomplete	Restrict access, serve metadata-only or last valid state with warnings, rebuild projections, and pause destructive lifecycle actions until scope is proven.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared memory object, version, owner, participants, content and evidence, temporal scope, policy, retention, correction, index state, and strategic record class.
- Positive and negative boundary tests prove that retrieval frequency, embedding similarity, model output, or user annotation does not convert content into accepted organizational truth.
- Fault exercises inject memory content, permissions, retention, version history, or indexes diverge or historical reconstruction becomes incomplete and verify that the platform can restrict access, serve metadata-only or last valid state with warnings, rebuild projections, and pause destructive lifecycle actions until scope is proven.
- Independent review accepts permission and purpose tests, temporal reconstruction, correction replay, legal hold, index rebuild, export, deletion verification, and customer review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Enterprise Memory and Strategic Record is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 30, Ch. 33, Ch. 48, Ch. 54, Ch. 64. Primary Volume 5 AI anchors: Ch. 12, Ch. 25, Ch. 26, Ch. 29, Ch. 38. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 35, Ch. 44, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-025 · C-037 · C-040

38 Search, Vector, and Retrieval Data

Search, Vector, and Retrieval Data establishes the governed data capability necessary to build policy-aware lexical, semantic, vector, multimodal, temporal, geospatial, and graph retrieval projections from governed canonical sources. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

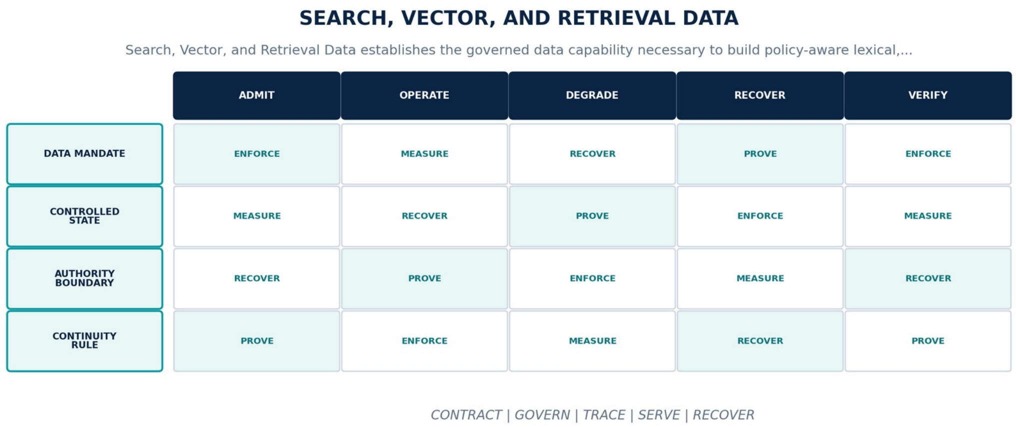

Figure 39. Search, Vector, and Retrieval Data. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Build policy-aware lexical, semantic, vector, multimodal, temporal, geospatial, and graph retrieval projections from governed canonical sources.
Controlled state	Index identity, source snapshot, chunk or unit identity, embedding model, representation version, policy labels, freshness, quality, and rebuild state.
Authority boundary	Retrieval scores are relevance signals, not truth or authority; result assembly must preserve source identity, policy, provenance, temporal state, and uncertainty.
Continuity rule	When an index is stale, incomplete, policy-inconsistent, poisoned, or built with an unapproved representation version, the platform shall withdraw the index or affected partitions, fall back to a policy-equivalent path, rebuild from verified sources, and expose retrieval degradation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-38-001	The data-platform realization of Search, Vector, and Retrieval Data SHALL build policy-aware lexical, semantic, vector, multimodal, temporal, geospatial, and graph retrieval projections from governed canonical sources.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-38-002	Every production instance SHALL declare and continuously reconcile index identity, source snapshot, chunk or unit identity, embedding model, representation version, policy labels, freshness, quality, and rebuild state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-38-003	The implementation SHALL enforce this authority boundary: retrieval scores are relevance signals, not truth or authority; result assembly must preserve source identity, policy, provenance, temporal state, and uncertainty.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-38-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Search, Vector, and Retrieval Data.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-38-005	When an index is stale, incomplete, policy-inconsistent, poisoned, or built with an unapproved representation version, the platform SHALL withdraw the index or affected partitions, fall back to a policy-equivalent path, rebuild from verified sources, and expose retrieval degradation; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-38-006	Production admission and continued operation SHALL require relevance and permission evaluation, stale and poisoned index tests, embedding migration, source deletion, deterministic rebuild, and citation closure, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An index is stale, incomplete, policy-inconsistent, poisoned, or built with an unapproved representation version	Withdraw the index or affected partitions, fall back to a policy-equivalent path, rebuild from verified sources, and expose retrieval degradation.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared index identity, source snapshot, chunk or unit identity, embedding model, representation version, policy labels, freshness, quality, and rebuild state.
- Positive and negative boundary tests prove that retrieval scores are relevance signals, not truth or authority; result assembly must preserve source identity, policy, provenance, temporal state, and uncertainty.
- Fault exercises inject an index is stale, incomplete, policy-inconsistent, poisoned, or built with an unapproved representation version and verify that the platform can withdraw the index or affected partitions, fall back to a policy-equivalent path, rebuild from verified sources, and expose retrieval degradation.
- Independent review accepts relevance and permission evaluation, stale and poisoned index tests, embedding migration, source deletion, deterministic rebuild, and citation closure for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Search, Vector, and Retrieval Data is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 23, Ch. 28, Ch. 30, Ch. 33, Ch. 46. Primary Volume 5 AI anchors: Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 29, Ch. 66. Primary Volume 6 infrastructure anchors: Ch. 35, Ch. 38, Ch. 44, Ch. 45, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-014 · C-016 · C-035 · C-045

39 Digital Twin State and Time Travel

Digital Twin State and Time Travel establishes the governed data capability necessary to persist living observed twin state, evidence-grounded estimates, model bindings, snapshots, branches, dependencies, validation, and reconciliation history. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

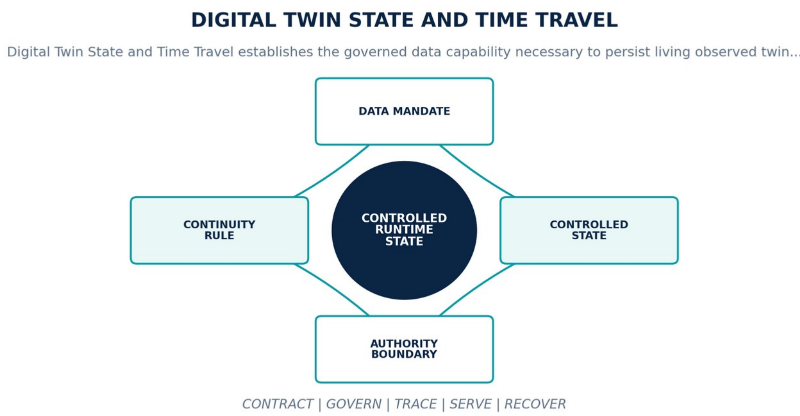

Figure 40. Digital Twin State and Time Travel. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Persist living observed twin state, evidence-grounded estimates, model bindings, snapshots, branches, dependencies, validation, and reconciliation history.
Controlled state	Twin identity, boundary, observed snapshot, synthetic branch, graph revision, evidence set, model version, assumptions, confidence, validity, and checkpoint.
Authority boundary	Observed and synthetic twin state remain physically and semantically separated; branches never write back without governed reconciliation.
Continuity rule	When a twin cannot resolve its evidence, graph, model, assumptions, operating envelope, or branch isolation, the platform shall freeze the last validated observed snapshot, mark staleness and missing variables, contain synthetic branches, and require reconciliation before use.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-39-001	The data-platform realization of Digital Twin State and Time Travel SHALL persist living observed twin state, evidence-grounded estimates, model bindings, snapshots, branches, dependencies, validation, and reconciliation history.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-39-002	Every production instance SHALL declare and continuously reconcile twin identity, boundary, observed snapshot, synthetic branch, graph revision, evidence set, model version, assumptions, confidence, validity, and checkpoint through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-39-003	The implementation SHALL enforce this authority boundary: observed and synthetic twin state remain physically and semantically separated; branches never write back without governed reconciliation.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-39-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Digital Twin State and Time Travel.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-39-005	When a twin cannot resolve its evidence, graph, model, assumptions, operating envelope, or branch isolation, the platform SHALL freeze the last validated observed snapshot, mark staleness and missing variables, contain synthetic branches, and require reconciliation before use; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-39-006	Production admission and continued operation SHALL require snapshot and branch fixtures, time travel, isolation, model binding, dependency impact, validation, reconciliation, and simulation handoff, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A twin cannot resolve its evidence, graph, model, assumptions, operating envelope, or branch isolation	Freeze the last validated observed snapshot, mark staleness and missing variables, contain synthetic branches, and require reconciliation before use.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared twin identity, boundary, observed snapshot, synthetic branch, graph revision, evidence set, model version, assumptions, confidence, validity, and checkpoint.
- Positive and negative boundary tests prove that observed and synthetic twin state remain physically and semantically separated; branches never write back without governed reconciliation.
- Fault exercises inject a twin cannot resolve its evidence, graph, model, assumptions, operating envelope, or branch isolation and verify that the platform can freeze the last validated observed snapshot, mark staleness and missing variables, contain synthetic branches, and require reconciliation before use.
- Independent review accepts snapshot and branch fixtures, time travel, isolation, model binding, dependency impact, validation, reconciliation, and simulation handoff for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Digital Twin State and Time Travel is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 26, Ch. 28, Ch. 30, Ch. 35, Ch. 38. Primary Volume 5 AI anchors: Ch. 13, Ch. 26, Ch. 28, Ch. 49, Ch. 50. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 36, Ch. 38, Ch. 45, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-018 · C-019 · C-023 · C-035

40 Forecast, Scenario, Simulation, Decision, and Outcome Data

Forecast, Scenario, Simulation, Decision, and Outcome Data establishes the governed data capability necessary to preserve versioned strategic state packages linking forecasts, scenario branches, simulation runs, recommendations, human decisions, commitments, outcomes, and learning. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

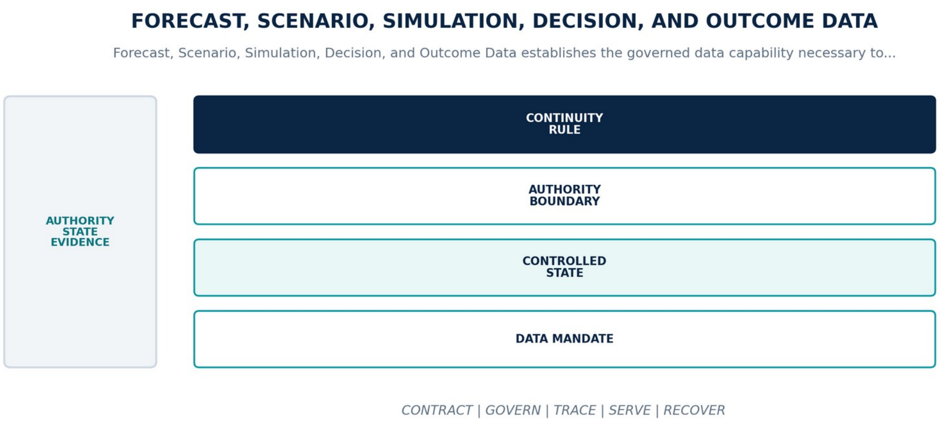

Figure 41. Forecast, Scenario, Simulation, Decision, and Outcome Data. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Preserve versioned strategic state packages linking forecasts, scenario branches, simulation runs, recommendations, human decisions, commitments, outcomes, and learning.
Controlled state	Package identity, objective, graph and twin revisions, evidence, assumptions, models, options, approvals, decision-time snapshot, commitments, outcomes, and review.
Authority boundary	Foresight and decision objects remain distinct lifecycle states; simulation results and recommendations may not become decisions, and later outcomes may not rewrite prior knowledge.
Continuity rule	When a strategic package has unresolved versions, missing human authority, withdrawn inputs, or broken outcome and replay linkage, the platform shall block commitment or mark the package unfit, preserve the historical snapshot, reopen through the governed lifecycle, and propagate impact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-40-001	The data-platform realization of Forecast, Scenario, Simulation, Decision, and Outcome Data SHALL preserve versioned strategic state packages linking forecasts, scenario branches, simulation runs, recommendations, human decisions, commitments, outcomes, and learning.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-40-002	Every production instance SHALL declare and continuously reconcile package identity, objective, graph and twin revisions, evidence, assumptions, models, options, approvals, decision-time snapshot, commitments, outcomes, and review through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-40-003	The implementation SHALL enforce this authority boundary: foresight and decision objects remain distinct lifecycle states; simulation results and recommendations may not become decisions, and later outcomes may not rewrite prior knowledge.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-40-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Forecast, Scenario, Simulation, Decision, and Outcome Data.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-40-005	When a strategic package has unresolved versions, missing human authority, withdrawn inputs, or broken outcome and replay linkage, the platform SHALL block commitment or mark the package unfit, preserve the historical snapshot, reopen through the governed lifecycle, and propagate impact; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-40-006	Production admission and continued operation SHALL require end-to-end strategic-loop fixtures, exact version resolution, human approval, decision replay, outcome attribution, correction impact, and learning admission, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A strategic package has unresolved versions, missing human authority, withdrawn inputs, or broken outcome and replay linkage	Block commitment or mark the package unfit, preserve the historical snapshot, reopen through the governed lifecycle, and propagate impact.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared package identity, objective, graph and twin revisions, evidence, assumptions, models, options, approvals, decision-time snapshot, commitments, outcomes, and review.
- Positive and negative boundary tests prove that foresight and decision objects remain distinct lifecycle states; simulation results and recommendations may not become decisions, and later outcomes may not rewrite prior knowledge.
- Fault exercises inject a strategic package has unresolved versions, missing human authority, withdrawn inputs, or broken outcome and replay linkage and verify that the platform can block commitment or mark the package unfit, preserve the historical snapshot, reopen through the governed lifecycle, and propagate impact.
- Independent review accepts end-to-end strategic-loop fixtures, exact version resolution, human approval, decision replay, outcome attribution, correction impact, and learning admission for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Forecast, Scenario, Simulation, Decision, and Outcome Data is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 28, Ch. 35, Ch. 36, Ch. 37, Ch. 38, Ch. 39, Ch. 48. Primary Volume 5 AI anchors: Ch. 48, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 69. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 36, Ch. 38, Ch. 43, Ch. 46, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-002 · C-004 · C-005 · C-020 · C-023 · C-025

PART VI

Data Products, Serving, and Interoperability

Governed data products, APIs, events, analytics, AI datasets, context products, exchange, and marketplaces.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

41 Data Product Architecture

Data Product Architecture establishes the governed data capability necessary to publish reusable governed data products with explicit purpose, ownership, contracts, quality, policy, lineage, service levels, consumers, cost, and lifecycle. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

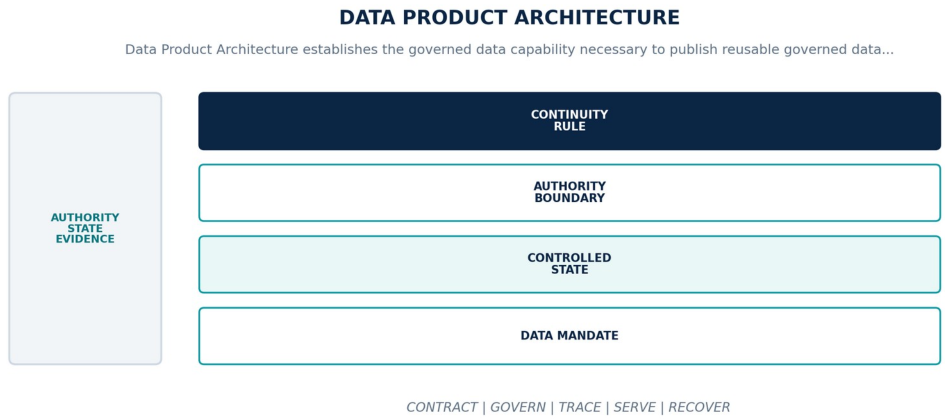

Figure 42. Data Product Architecture. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Publish reusable governed data products with explicit purpose, ownership, contracts, quality, policy, lineage, service levels, consumers, cost, and lifecycle.
Controlled state	Product identity, domain, owner, inputs, outputs, contract versions, serving modes, quality, SLOs, policy, consumers, release, and retirement.
Authority boundary	A data product is an accountable operational contract, not a renamed table, report, dashboard, ad hoc extract, or unmanaged semantic layer.
Continuity rule	When a product fails quality or SLO, loses ownership, violates policy, or diverges from its published contract, the platform shall declare degradation or withdrawal, notify consumers, preserve last valid versions, and restore through controlled release or migration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-41-001	The data-platform realization of Data Product Architecture SHALL publish reusable governed data products with explicit purpose, ownership, contracts, quality, policy, lineage, service levels, consumers, cost, and lifecycle.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-41-002	Every production instance SHALL declare and continuously reconcile product identity, domain, owner, inputs, outputs, contract versions, serving modes, quality, SLOs, policy, consumers, release, and retirement through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-41-003	The implementation SHALL enforce this authority boundary: a data product is an accountable operational contract, not a renamed table, report, dashboard, ad hoc extract, or unmanaged semantic layer.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-41-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Product Architecture.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-41-005	When a product fails quality or SLO, loses ownership, violates policy, or diverges from its published contract, the platform SHALL declare degradation or withdrawal, notify consumers, preserve last valid versions, and restore through controlled release or migration; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-41-006	Production admission and continued operation SHALL require product scorecard, consumer contract tests, lineage and policy closure, SLO evidence, cost attribution, adoption, and retirement review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A product fails quality or SLO, loses ownership, violates policy, or diverges from its published contract	Declare degradation or withdrawal, notify consumers, preserve last valid versions, and restore through controlled release or migration.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared product identity, domain, owner, inputs, outputs, contract versions, serving modes, quality, SLOs, policy, consumers, release, and retirement.
- Positive and negative boundary tests prove that a data product is an accountable operational contract, not a renamed table, report, dashboard, ad hoc extract, or unmanaged semantic layer.
- Fault exercises inject a product fails quality or SLO, loses ownership, violates policy, or diverges from its published contract and verify that the platform can declare degradation or withdrawal, notify consumers, preserve last valid versions, and restore through controlled release or migration.
- Independent review accepts product scorecard, consumer contract tests, lineage and policy closure, SLO evidence, cost attribution, adoption, and retirement review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Product Architecture is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 15, Ch. 23, Ch. 57, Ch. 64, Ch. 71. Primary Volume 5 AI anchors: Ch. 9, Ch. 13, Ch. 20, Ch. 25, Ch. 66. Primary Volume 6 infrastructure anchors: Ch. 40, Ch. 45, Ch. 48, Ch. 59, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-006 · C-014 · C-038 · C-049

42 Data APIs and Query Services

Data APIs and Query Services establishes the governed data capability necessary to serve operational and analytical data through versioned commands, queries, APIs, and governed bulk interfaces with purpose-aware authorization. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

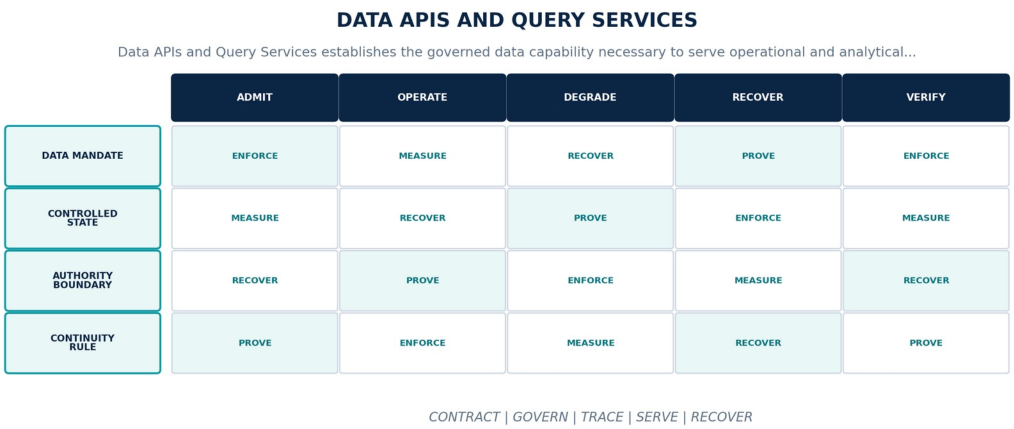

Figure 43. Data APIs and Query Services. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Serve operational and analytical data through versioned commands, queries, APIs, and governed bulk interfaces with purpose-aware authorization.
Controlled state	Service identity, contract version, query semantics, consistency, pagination, filters, policy obligations, quotas, receipts, and error or partial-result state.
Authority boundary	API success does not imply complete, fresh, policy-authorized, or decision-fit data; partial, stale, denied, abstained, and unavailable states remain explicit.
Continuity rule	When authorization, consistency, freshness, schema, pagination, or dependency state cannot satisfy the contract, the platform shall fail closed for protected access, return typed partial or stale state where allowed, preserve receipts, and prevent silent truncation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-42-001	The data-platform realization of Data APIs and Query Services SHALL serve operational and analytical data through versioned commands, queries, APIs, and governed bulk interfaces with purpose-aware authorization.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-42-002	Every production instance SHALL declare and continuously reconcile service identity, contract version, query semantics, consistency, pagination, filters, policy obligations, quotas, receipts, and error or partial-result state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-42-003	The implementation SHALL enforce this authority boundary: API success does not imply complete, fresh, policy-authorized, or decision-fit data; partial, stale, denied, abstained, and unavailable states remain explicit.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-42-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data APIs and Query Services.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-42-005	When authorization, consistency, freshness, schema, pagination, or dependency state cannot satisfy the contract, the platform SHALL fail closed for protected access, return typed partial or stale state where allowed, preserve receipts, and prevent silent truncation; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-42-006	Production admission and continued operation SHALL require contract and negative authorization tests, consistency and pagination fixtures, load and quota tests, partial-result semantics, and audit evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Authorization, consistency, freshness, schema, pagination, or dependency state cannot satisfy the contract	Fail closed for protected access, return typed partial or stale state where allowed, preserve receipts, and prevent silent truncation.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared service identity, contract version, query semantics, consistency, pagination, filters, policy obligations, quotas, receipts, and error or partial-result state.
- Positive and negative boundary tests prove that API success does not imply complete, fresh, policy-authorized, or decision-fit data; partial, stale, denied, abstained, and unavailable states remain explicit.
- Fault exercises inject authorization, consistency, freshness, schema, pagination, or dependency state cannot satisfy the contract and verify that the platform can fail closed for protected access, return typed partial or stale state where allowed, preserve receipts, and prevent silent truncation.
- Independent review accepts contract and negative authorization tests, consistency and pagination fixtures, load and quota tests, partial-result semantics, and audit evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data APIs and Query Services is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 16, Ch. 18, Ch. 20, Ch. 21, Ch. 50. Primary Volume 5 AI anchors: Ch. 19, Ch. 20, Ch. 21, Ch. 23, Ch. 29. Primary Volume 6 infrastructure anchors: Ch. 24, Ch. 25, Ch. 30, Ch. 48, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-003 · C-014 · C-035 · C-038 · C-039 · C-043

43 Event Products and Subscriptions

Event Products and Subscriptions establishes the governed data capability necessary to publish versioned change, signal, correction, lifecycle, quality, and strategic events to governed consumers through registered subscriptions. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

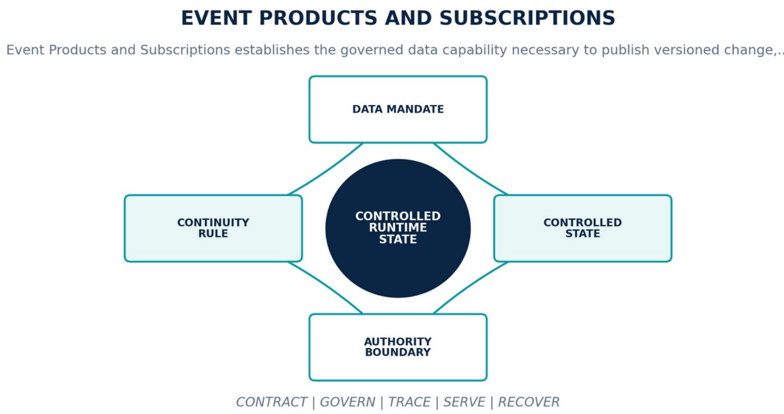

Figure 44. Event Products and Subscriptions. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Publish versioned change, signal, correction, lifecycle, quality, and strategic events to governed consumers through registered subscriptions.
Controlled state	Event product, schema, producer, subject, ordering key, delivery, retention, subscription, filter, consumer checkpoint, policy, and revocation.
Authority boundary	Subscriptions grant only declared data, purpose, time, tenant, and consequence scope; consumers may not infer broader access from topic reachability.
Continuity rule	When a subscription is unauthorized, schema-incompatible, lagging beyond policy, or cannot process correction and replay semantics, the platform shall revoke or pause delivery, preserve offsets, constrain retention, notify owner, and require conformance before resumption.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-43-001	The data-platform realization of Event Products and Subscriptions SHALL publish versioned change, signal, correction, lifecycle, quality, and strategic events to governed consumers through registered subscriptions.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-43-002	Every production instance SHALL declare and continuously reconcile event product, schema, producer, subject, ordering key, delivery, retention, subscription, filter, consumer checkpoint, policy, and revocation through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-43-003	The implementation SHALL enforce this authority boundary: subscriptions grant only declared data, purpose, time, tenant, and consequence scope; consumers may not infer broader access from topic reachability.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-43-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Event Products and Subscriptions.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-43-005	When a subscription is unauthorized, schema-incompatible, lagging beyond policy, or cannot process correction and replay semantics, the platform SHALL revoke or pause delivery, preserve offsets, constrain retention, notify owner, and require conformance before resumption; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-43-006	Production admission and continued operation SHALL require subscription authorization, schema compatibility, lag and replay, correction propagation, revocation, tenant isolation, and consumer acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A subscription is unauthorized, schema-incompatible, lagging beyond policy, or cannot process correction and replay semantics	Revoke or pause delivery, preserve offsets, constrain retention, notify owner, and require conformance before resumption.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared event product, schema, producer, subject, ordering key, delivery, retention, subscription, filter, consumer checkpoint, policy, and revocation.
- Positive and negative boundary tests prove that subscriptions grant only declared data, purpose, time, tenant, and consequence scope; consumers may not infer broader access from topic reachability.
- Fault exercises inject a subscription is unauthorized, schema-incompatible, lagging beyond policy, or cannot process correction and replay semantics and verify that the platform can revoke or pause delivery, preserve offsets, constrain retention, notify owner, and require conformance before resumption.
- Independent review accepts subscription authorization, schema compatibility, lag and replay, correction propagation, revocation, tenant isolation, and consumer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Event Products and Subscriptions is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 20, Ch. 22, Ch. 23, Ch. 50. Primary Volume 5 AI anchors: Ch. 34, Ch. 35, Ch. 38, Ch. 57, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 25, Ch. 37, Ch. 48, Ch. 50, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-012 · C-014 · C-035 · C-038 · C-039 · C-043

44 Analytics and Semantic Serving

Analytics and Semantic Serving establishes the governed data capability necessary to provide governed metrics, dimensions, measures, cohorts, relationships, temporal views, and analytical models without creating a parallel truth system. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

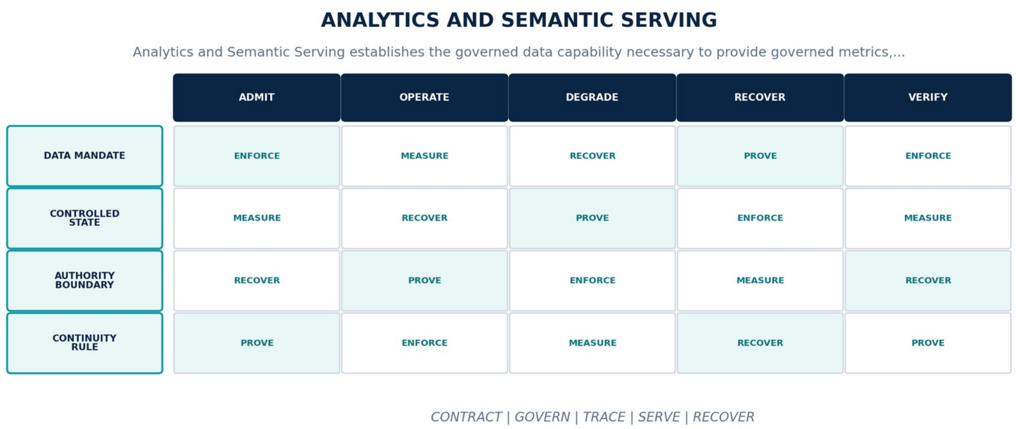

Figure 45. Analytics and Semantic Serving. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Provide governed metrics, dimensions, measures, cohorts, relationships, temporal views, and analytical models without creating a parallel truth system.
Controlled state	Semantic model identity, owner, metric definition, grain, dimensions, filters, source revisions, policy, quality, certification, and effective time.
Authority boundary	Dashboards and BI tools are access modes; calculated metrics and semantic models remain versioned products linked to canonical data and evidence.
Continuity rule	When metric definitions conflict, source grain is incompatible, certification expires, or analytical outputs cannot reproduce from declared inputs, the platform shall withdraw certification, expose affected reports and decisions, freeze the last valid model, and reconcile definitions under semantic governance.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-44-001	The data-platform realization of Analytics and Semantic Serving SHALL provide governed metrics, dimensions, measures, cohorts, relationships, temporal views, and analytical models without creating a parallel truth system.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-44-002	Every production instance SHALL declare and continuously reconcile semantic model identity, owner, metric definition, grain, dimensions, filters, source revisions, policy, quality, certification, and effective time through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-44-003	The implementation SHALL enforce this authority boundary: dashboards and BI tools are access modes; calculated metrics and semantic models remain versioned products linked to canonical data and evidence.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-44-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Analytics and Semantic Serving.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-44-005	When metric definitions conflict, source grain is incompatible, certification expires, or analytical outputs cannot reproduce from declared inputs, the platform SHALL withdraw certification, expose affected reports and decisions, freeze the last valid model, and reconcile definitions under semantic governance; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-44-006	Production admission and continued operation SHALL require metric fixtures, grain and aggregation tests, reproducibility, policy filtering, definition diff, historical recalculation, and owner acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Metric definitions conflict, source grain is incompatible, certification expires, or analytical outputs cannot reproduce from declared inputs	Withdraw certification, expose affected reports and decisions, freeze the last valid model, and reconcile definitions under semantic governance.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared semantic model identity, owner, metric definition, grain, dimensions, filters, source revisions, policy, quality, certification, and effective time.
- Positive and negative boundary tests prove that dashboards and BI tools are access modes; calculated metrics and semantic models remain versioned products linked to canonical data and evidence.
- Fault exercises inject metric definitions conflict, source grain is incompatible, certification expires, or analytical outputs cannot reproduce from declared inputs and verify that the platform can withdraw certification, expose affected reports and decisions, freeze the last valid model, and reconcile definitions under semantic governance.
- Independent review accepts metric fixtures, grain and aggregation tests, reproducibility, policy filtering, definition diff, historical recalculation, and owner acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Analytics and Semantic Serving is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 23, Ch. 26, Ch. 27, Ch. 40. Primary Volume 5 AI anchors: Ch. 9, Ch. 13, Ch. 19, Ch. 56, Ch. 64. Primary Volume 6 infrastructure anchors: Ch. 35, Ch. 44, Ch. 46, Ch. 48, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-011 · C-014 · C-034 · C-047

45 Feature, Training, Evaluation, and Feedback Data

Feature, Training, Evaluation, and Feedback Data establishes the governed data capability necessary to govern datasets and features used to train, adapt, evaluate, calibrate, red-team, monitor, and learn from models and agents. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

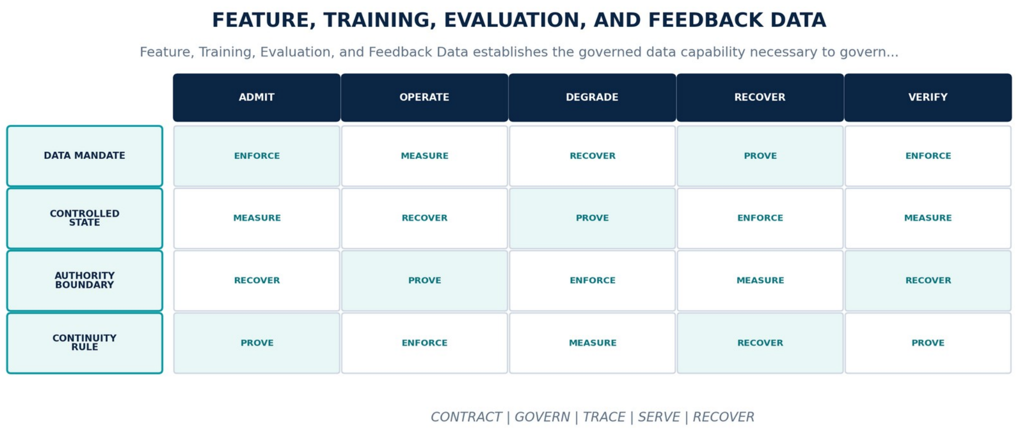

Figure 46. Feature, Training, Evaluation, and Feedback Data. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern datasets and features used to train, adapt, evaluate, calibrate, red-team, monitor, and learn from models and agents.
Controlled state	Dataset identity, purpose, population, snapshot, labels, features, rights, provenance, splits, leakage controls, quality, bias, approval, and expiry.
Authority boundary	Production, training, evaluation, red-team, synthetic, feedback, and outcome data remain isolated by purpose and may not be silently reused or contaminated.
Continuity rule	When rights, provenance, labels, split integrity, representativeness, leakage, or evaluation independence becomes invalid, the platform shall revoke the dataset or feature revision, stop affected model promotion, identify dependent artifacts, and rebuild or re-evaluate from approved data.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-45-001	The data-platform realization of Feature, Training, Evaluation, and Feedback Data SHALL govern datasets and features used to train, adapt, evaluate, calibrate, red-team, monitor, and learn from models and agents.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-45-002	Every production instance SHALL declare and continuously reconcile dataset identity, purpose, population, snapshot, labels, features, rights, provenance, splits, leakage controls, quality, bias, approval, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-45-003	The implementation SHALL enforce this authority boundary: production, training, evaluation, red-team, synthetic, feedback, and outcome data remain isolated by purpose and may not be silently reused or contaminated.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-45-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Feature, Training, Evaluation, and Feedback Data.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-45-005	When rights, provenance, labels, split integrity, representativeness, leakage, or evaluation independence becomes invalid, the platform SHALL revoke the dataset or feature revision, stop affected model promotion, identify dependent artifacts, and rebuild or re-evaluate from approved data; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-45-006	Production admission and continued operation SHALL require dataset cards, rights review, lineage closure, leakage and contamination tests, slice metrics, split reproducibility, and model-impact analysis, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Rights, provenance, labels, split integrity, representativeness, leakage, or evaluation independence becomes invalid	Revoke the dataset or feature revision, stop affected model promotion, identify dependent artifacts, and rebuild or re-evaluate from approved data.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared dataset identity, purpose, population, snapshot, labels, features, rights, provenance, splits, leakage controls, quality, bias, approval, and expiry.
- Positive and negative boundary tests prove that production, training, evaluation, red-team, synthetic, feedback, and outcome data remain isolated by purpose and may not be silently reused or contaminated.
- Fault exercises inject rights, provenance, labels, split integrity, representativeness, leakage, or evaluation independence becomes invalid and verify that the platform can revoke the dataset or feature revision, stop affected model promotion, identify dependent artifacts, and rebuild or re-evaluate from approved data.
- Independent review accepts dataset cards, rights review, lineage closure, leakage and contamination tests, slice metrics, split reproducibility, and model-impact analysis for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Feature, Training, Evaluation, and Feedback Data is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 28, Ch. 29, Ch. 46, Ch. 48, Ch. 54, Ch. 67. Primary Volume 5 AI anchors: Ch. 14, Ch. 15, Ch. 19, Ch. 61, Ch. 66, Ch. 69, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 43, Ch. 44, Ch. 45, Ch. 53, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-012 · C-031 · C-037 · C-040 · C-044 · C-045

46 Context Assembly and Retrieval Products

Context Assembly and Retrieval Products establishes the governed data capability necessary to assemble purpose-bound evidence, graph, memory, twin, policy, and user context for agents, models, analysts, and executive workflows. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

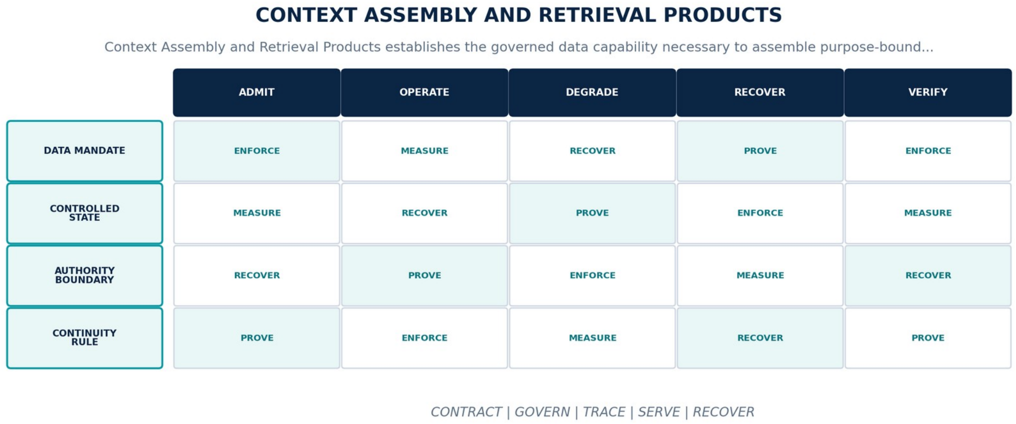

Figure 47. Context Assembly and Retrieval Products. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Assemble purpose-bound evidence, graph, memory, twin, policy, and user context for agents, models, analysts, and executive workflows.
Controlled state	Context manifest, request purpose, principal, object versions, retrieval methods, ranking, filters, token or size budget, omissions, citations, and expiry.
Authority boundary	Context assembly does not grant new access, erase contradictory evidence, hide truncation, or convert retrieved content into instructions or accepted truth.
Continuity rule	When policy, freshness, relevance, token budget, injection defense, or citation closure cannot satisfy the task contract, the platform shall abstain or return a typed partial context, isolate malicious content, expose omissions, and prevent downstream admission as complete evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-46-001	The data-platform realization of Context Assembly and Retrieval Products SHALL assemble purpose-bound evidence, graph, memory, twin, policy, and user context for agents, models, analysts, and executive workflows.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-46-002	Every production instance SHALL declare and continuously reconcile context manifest, request purpose, principal, object versions, retrieval methods, ranking, filters, token or size budget, omissions, citations, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-46-003	The implementation SHALL enforce this authority boundary: context assembly does not grant new access, erase contradictory evidence, hide truncation, or convert retrieved content into instructions or accepted truth.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-46-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Context Assembly and Retrieval Products.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-46-005	When policy, freshness, relevance, token budget, injection defense, or citation closure cannot satisfy the task contract, the platform SHALL abstain or return a typed partial context, isolate malicious content, expose omissions, and prevent downstream admission as complete evidence; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-46-006	Production admission and continued operation SHALL require permission and purpose tests, retrieval evaluation, injection corpus, truncation and omission behavior, citation closure, freshness, and replay, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy, freshness, relevance, token budget, injection defense, or citation closure cannot satisfy the task contract	Abstain or return a typed partial context, isolate malicious content, expose omissions, and prevent downstream admission as complete evidence.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared context manifest, request purpose, principal, object versions, retrieval methods, ranking, filters, token or size budget, omissions, citations, and expiry.
- Positive and negative boundary tests prove that context assembly does not grant new access, erase contradictory evidence, hide truncation, or convert retrieved content into instructions or accepted truth.
- Fault exercises inject policy, freshness, relevance, token budget, injection defense, or citation closure cannot satisfy the task contract and verify that the platform can abstain or return a typed partial context, isolate malicious content, expose omissions, and prevent downstream admission as complete evidence.
- Independent review accepts permission and purpose tests, retrieval evaluation, injection corpus, truncation and omission behavior, citation closure, freshness, and replay for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Context Assembly and Retrieval Products is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 28, Ch. 32, Ch. 33, Ch. 44. Primary Volume 5 AI anchors: Ch. 12, Ch. 19, Ch. 23, Ch. 24, Ch. 29, Ch. 38. Primary Volume 6 infrastructure anchors: Ch. 42, Ch. 44, Ch. 47, Ch. 50, Ch. 53.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-016 · C-028 · C-035 · C-040

47 Data Exchange, Export, and Portability

Data Exchange, Export, and Portability establishes the governed data capability necessary to provide complete governed export, import, exchange, replication, and portability paths using open formats and verifiable manifests. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

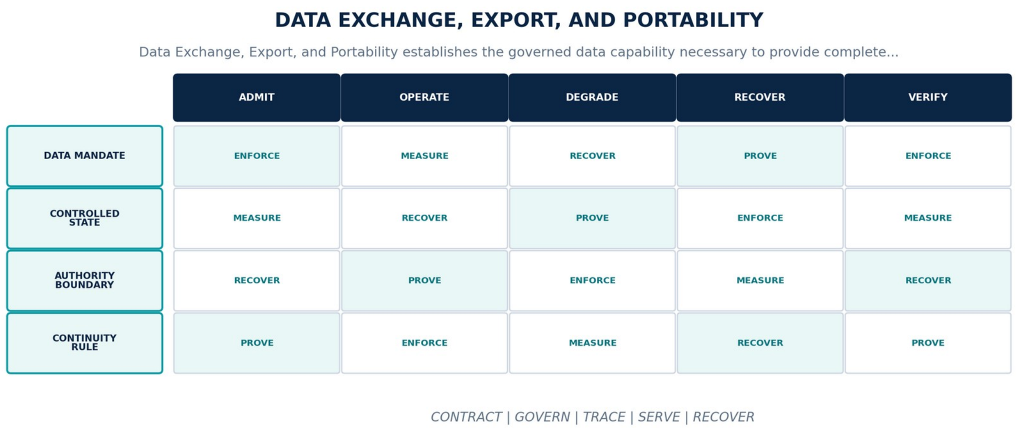

Figure 48. Data Exchange, Export, and Portability. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Provide complete governed export, import, exchange, replication, and portability paths using open formats and verifiable manifests.
Controlled state	Exchange identity, requester, purpose, scope, contract versions, object and relationship closure, policy, encryption, destination, receipt, and expiry.
Authority boundary	Exports preserve identity, temporal truth, provenance, corrections, policy labels, graph links, and manifest integrity; proprietary stores may not trap customer knowledge.
Continuity rule	When scope, rights, destination, completeness, integrity, re-import compatibility, or revocation cannot be proven, the platform shall deny or quarantine the exchange, preserve the request and evidence, repair the manifest, and require recipient acknowledgement before closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-47-001	The data-platform realization of Data Exchange, Export, and Portability SHALL provide complete governed export, import, exchange, replication, and portability paths using open formats and verifiable manifests.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-47-002	Every production instance SHALL declare and continuously reconcile exchange identity, requester, purpose, scope, contract versions, object and relationship closure, policy, encryption, destination, receipt, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-47-003	The implementation SHALL enforce this authority boundary: exports preserve identity, temporal truth, provenance, corrections, policy labels, graph links, and manifest integrity; proprietary stores may not trap customer knowledge.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-47-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Exchange, Export, and Portability.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-47-005	When scope, rights, destination, completeness, integrity, re-import compatibility, or revocation cannot be proven, the platform SHALL deny or quarantine the exchange, preserve the request and evidence, repair the manifest, and require recipient acknowledgement before closure; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-47-006	Production admission and continued operation SHALL require round-trip fixtures, large-scale export, relationship closure, checksum and signature verification, policy filtering, revocation, and customer acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Scope, rights, destination, completeness, integrity, re-import compatibility, or revocation cannot be proven	Deny or quarantine the exchange, preserve the request and evidence, repair the manifest, and require recipient acknowledgement before closure.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared exchange identity, requester, purpose, scope, contract versions, object and relationship closure, policy, encryption, destination, receipt, and expiry.
- Positive and negative boundary tests prove that exports preserve identity, temporal truth, provenance, corrections, policy labels, graph links, and manifest integrity; proprietary stores may not trap customer knowledge.
- Fault exercises inject scope, rights, destination, completeness, integrity, re-import compatibility, or revocation cannot be proven and verify that the platform can deny or quarantine the exchange, preserve the request and evidence, repair the manifest, and require recipient acknowledgement before closure.
- Independent review accepts round-trip fixtures, large-scale export, relationship closure, checksum and signature verification, policy filtering, revocation, and customer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Exchange, Export, and Portability is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 23, Ch. 25, Ch. 28, Ch. 29, Ch. 30. Primary Volume 5 AI anchors: Ch. 17, Ch. 21, Ch. 25, Ch. 61, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 11, Ch. 28, Ch. 29, Ch. 47, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-012 · C-037 · C-038 · C-039 · C-041 · C-043

48 Marketplace Data Packages and Extension Contracts

Marketplace Data Packages and Extension Contracts establishes the governed data capability necessary to govern reusable data products, schemas, ontologies, connectors, mappings, features, and scenario datasets distributed through controlled marketplaces. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

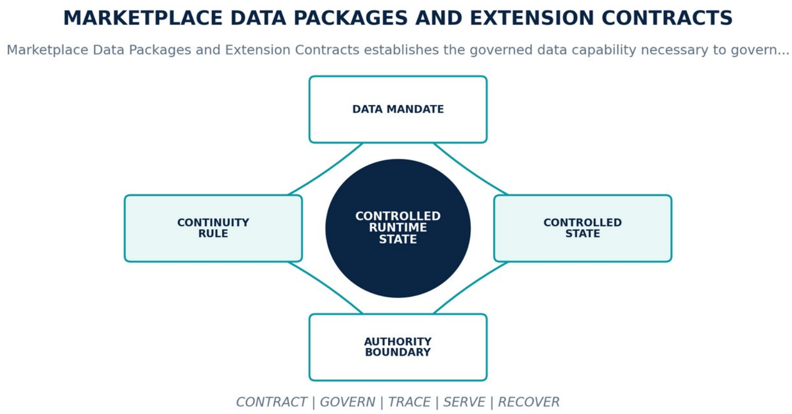

Figure 49. Marketplace Data Packages and Extension Contracts. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern reusable data products, schemas, ontologies, connectors, mappings, features, and scenario datasets distributed through controlled marketplaces.
Controlled state	Package identity, publisher, license, contents, contracts, permissions, dependencies, signatures, evaluation, sandbox profile, monitoring, revocation, and support.
Authority boundary	Marketplace installation grants no implicit access to customer data, memory, graph, keys, models, tools, or external destinations.
Continuity rule	When a package is unsigned, malicious, incompatible, over-privileged, unlicensed, unfit, or revoked, the platform shall block installation or execution, quarantine the package, identify affected data and products, revoke access, and restore a trusted version.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-48-001	The data-platform realization of Marketplace Data Packages and Extension Contracts SHALL govern reusable data products, schemas, ontologies, connectors, mappings, features, and scenario datasets distributed through controlled marketplaces.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-48-002	Every production instance SHALL declare and continuously reconcile package identity, publisher, license, contents, contracts, permissions, dependencies, signatures, evaluation, sandbox profile, monitoring, revocation, and support through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-48-003	The implementation SHALL enforce this authority boundary: marketplace installation grants no implicit access to customer data, memory, graph, keys, models, tools, or external destinations.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-48-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Marketplace Data Packages and Extension Contracts.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-48-005	When a package is unsigned, malicious, incompatible, over-privileged, unlicensed, unfit, or revoked, the platform SHALL block installation or execution, quarantine the package, identify affected data and products, revoke access, and restore a trusted version; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-48-006	Production admission and continued operation SHALL require signature and provenance, sandbox, permission denial, compatibility, data-access isolation, quality evaluation, revocation, and uninstall evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A package is unsigned, malicious, incompatible, over-privileged, unlicensed, unfit, or revoked	Block installation or execution, quarantine the package, identify affected data and products, revoke access, and restore a trusted version.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared package identity, publisher, license, contents, contracts, permissions, dependencies, signatures, evaluation, sandbox profile, monitoring, revocation, and support.
- Positive and negative boundary tests prove that marketplace installation grants no implicit access to customer data, memory, graph, keys, models, tools, or external destinations.
- Fault exercises inject a package is unsigned, malicious, incompatible, over-privileged, unlicensed, unfit, or revoked and verify that the platform can block installation or execution, quarantine the package, identify affected data and products, revoke access, and restore a trusted version.
- Independent review accepts signature and provenance, sandbox, permission denial, compatibility, data-access isolation, quality evaluation, revocation, and uninstall evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Marketplace Data Packages and Extension Contracts is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 29, Ch. 44, Ch. 50, Ch. 56, Ch. 67. Primary Volume 5 AI anchors: Ch. 17, Ch. 30, Ch. 36, Ch. 46, Ch. 47, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 42, Ch. 47, Ch. 48, Ch. 50, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-007 · C-028 · C-035 · C-039 · C-046 · C-049

PART VII

Governance, Trust, and Lifecycle

Catalog, access, privacy, sovereignty, confidentiality, retention, stewardship, and accountability.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

49 Metadata Catalog and Data Discovery

Metadata Catalog and Data Discovery establishes the governed data capability necessary to maintain a complete authoritative catalog of data assets, contracts, schemas, owners, lineage, policy, quality, service levels, consumers, cost, and lifecycle. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

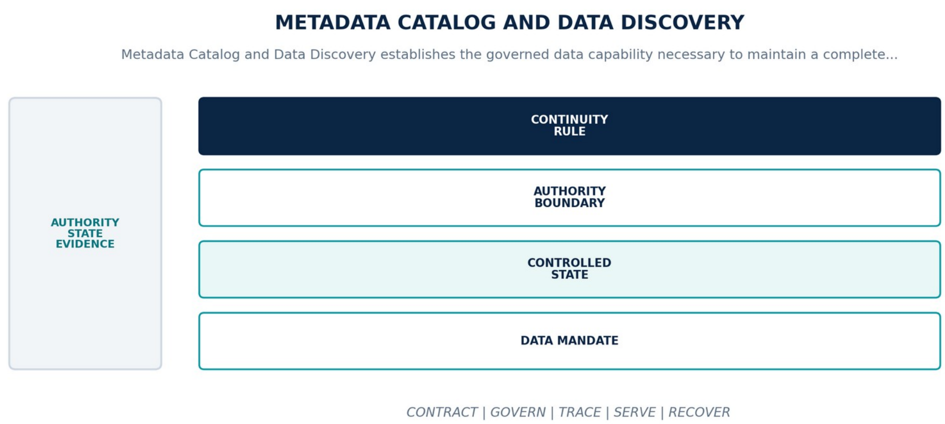

Figure 50. Metadata Catalog and Data Discovery. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Maintain a complete authoritative catalog of data assets, contracts, schemas, owners, lineage, policy, quality, service levels, consumers, cost, and lifecycle.
Controlled state	Catalog entry, asset identity, owner, classification, contracts, locations, lineage, glossary, quality, SLO, consumers, release, and lifecycle state.
Authority boundary	Catalog metadata may describe and index authority but cannot replace the owning source, contract registry, policy service, or canonical object state.
Continuity rule	When catalog coverage, ownership, lineage, policy, or runtime observations are missing, stale, duplicated, or inconsistent, the platform shall mark affected assets undiscoverable or untrusted, stop new consumption where required, reconcile authoritative registries, and expose coverage debt.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-49-001	The data-platform realization of Metadata Catalog and Data Discovery SHALL maintain a complete authoritative catalog of data assets, contracts, schemas, owners, lineage, policy, quality, service levels, consumers, cost, and lifecycle.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-49-002	Every production instance SHALL declare and continuously reconcile catalog entry, asset identity, owner, classification, contracts, locations, lineage, glossary, quality, SLO, consumers, release, and lifecycle state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-49-003	The implementation SHALL enforce this authority boundary: catalog metadata may describe and index authority but cannot replace the owning source, contract registry, policy service, or canonical object state.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-49-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Metadata Catalog and Data Discovery.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-49-005	When catalog coverage, ownership, lineage, policy, or runtime observations are missing, stale, duplicated, or inconsistent, the platform SHALL mark affected assets undiscoverable or untrusted, stop new consumption where required, reconcile authoritative registries, and expose coverage debt; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-49-006	Production admission and continued operation SHALL require inventory reconciliation, orphan detection, ownership recertification, lineage coverage, search permission, runtime sync, and retirement evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Catalog coverage, ownership, lineage, policy, or runtime observations are missing, stale, duplicated, or inconsistent	Mark affected assets undiscoverable or untrusted, stop new consumption where required, reconcile authoritative registries, and expose coverage debt.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared catalog entry, asset identity, owner, classification, contracts, locations, lineage, glossary, quality, SLO, consumers, release, and lifecycle state.
- Positive and negative boundary tests prove that catalog metadata may describe and index authority but cannot replace the owning source, contract registry, policy service, or canonical object state.
- Fault exercises inject catalog coverage, ownership, lineage, policy, or runtime observations are missing, stale, duplicated, or inconsistent and verify that the platform can mark affected assets undiscoverable or untrusted, stop new consumption where required, reconcile authoritative registries, and expose coverage debt.
- Independent review accepts inventory reconciliation, orphan detection, ownership recertification, lineage coverage, search permission, runtime sync, and retirement evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Metadata Catalog and Data Discovery is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 11, Ch. 23, Ch. 28, Ch. 29, Ch. 64. Primary Volume 5 AI anchors: Ch. 6, Ch. 13, Ch. 25, Ch. 29, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 14, Ch. 40, Ch. 48, Ch. 58, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 · C-012 · C-016 · C-035 · C-038 · C-049

50 Policy, Access, and Purpose Enforcement

Policy, Access, and Purpose Enforcement establishes the governed data capability necessary to enforce identity, tenant, role, attribute, purpose, classification, rights, locality, time, consequence, and obligation policy at every data boundary. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

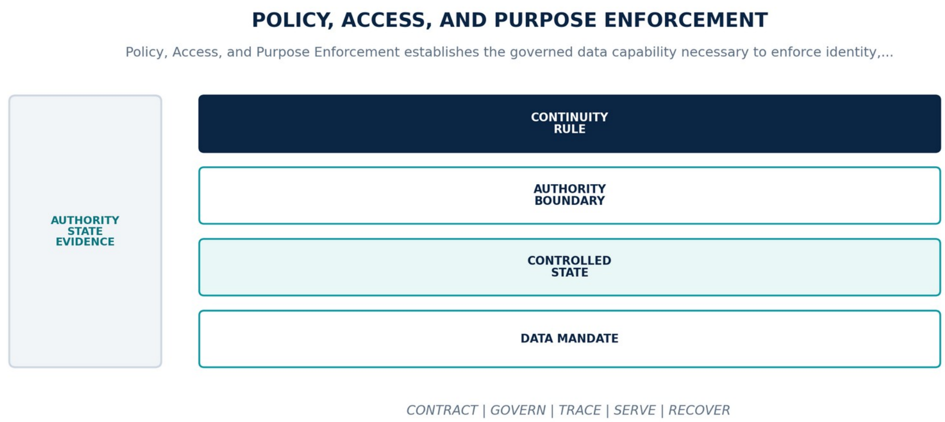

Figure 51. Policy, Access, and Purpose Enforcement. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Enforce identity, tenant, role, attribute, purpose, classification, rights, locality, time, consequence, and obligation policy at every data boundary.
Controlled state	Policy decision, principal, resource, action, purpose, context, policy version, result, obligations, enforcement receipt, and review state.
Authority boundary	Applications, pipelines, agents, models, administrators, and storage engines cannot self-authorize or silently bypass independent policy enforcement.
Continuity rule	When identity, purpose, policy, classification, or enforcement evidence is missing, conflicting, stale, or unavailable, the platform shall fail closed for protected operations, use only explicit bounded cached decisions where authorized, revoke suspect grants, and preserve denial evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-50-001	The data-platform realization of Policy, Access, and Purpose Enforcement SHALL enforce identity, tenant, role, attribute, purpose, classification, rights, locality, time, consequence, and obligation policy at every data boundary.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-50-002	Every production instance SHALL declare and continuously reconcile policy decision, principal, resource, action, purpose, context, policy version, result, obligations, enforcement receipt, and review state through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-50-003	The implementation SHALL enforce this authority boundary: applications, pipelines, agents, models, administrators, and storage engines cannot self-authorize or silently bypass independent policy enforcement.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-50-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Policy, Access, and Purpose Enforcement.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-50-005	When identity, purpose, policy, classification, or enforcement evidence is missing, conflicting, stale, or unavailable, the platform SHALL fail closed for protected operations, use only explicit bounded cached decisions where authorized, revoke suspect grants, and preserve denial evidence; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-50-006	Production admission and continued operation SHALL require positive and negative authorization, purpose change, policy outage, revocation propagation, row and field filtering, and privileged-access review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity, purpose, policy, classification, or enforcement evidence is missing, conflicting, stale, or unavailable	Fail closed for protected operations, use only explicit bounded cached decisions where authorized, revoke suspect grants, and preserve denial evidence.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared policy decision, principal, resource, action, purpose, context, policy version, result, obligations, enforcement receipt, and review state.
- Positive and negative boundary tests prove that applications, pipelines, agents, models, administrators, and storage engines cannot self-authorize or silently bypass independent policy enforcement.
- Fault exercises inject identity, purpose, policy, classification, or enforcement evidence is missing, conflicting, stale, or unavailable and verify that the platform can fail closed for protected operations, use only explicit bounded cached decisions where authorized, revoke suspect grants, and preserve denial evidence.
- Independent review accepts positive and negative authorization, purpose change, policy outage, revocation propagation, row and field filtering, and privileged-access review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Policy, Access, and Purpose Enforcement is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 25, Ch. 29, Ch. 49, Ch. 50, Ch. 55. Primary Volume 5 AI anchors: Ch. 21, Ch. 30, Ch. 36, Ch. 37, Ch. 61, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 49, Ch. 50, Ch. 54, Ch. 57, Ch. 66.

CONSTITUTIONAL INHERITANCE / C-028 · C-035 · C-036 · C-037 · C-039 · C-040

51 Privacy, Minimization, Consent, and Lawful Basis

Privacy, Minimization, Consent, and Lawful Basis establishes the governed data capability necessary to make collection, use, inference, sharing, retention, correction, access, and deletion purpose-bound, proportionate, reviewable, and enforceable. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

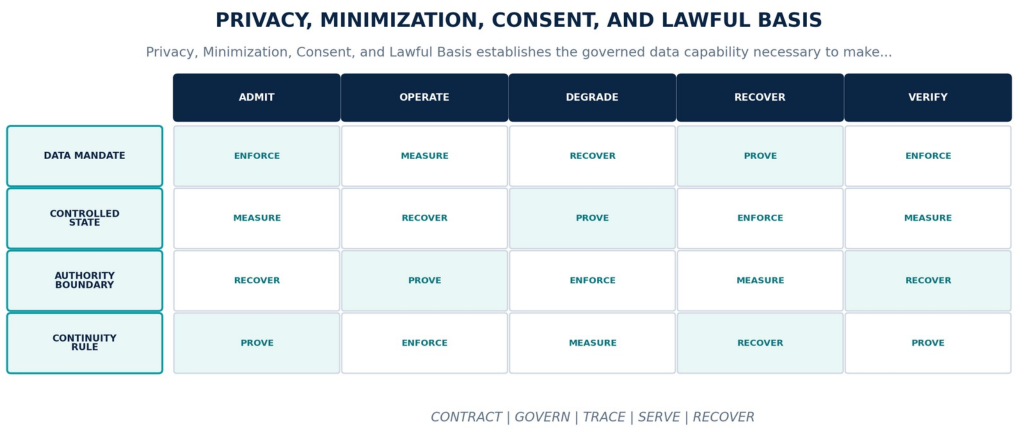

Figure 52. Privacy, Minimization, Consent, and Lawful Basis. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Make collection, use, inference, sharing, retention, correction, access, and deletion purpose-bound, proportionate, reviewable, and enforceable.
Controlled state	Data subject or protected class context, purpose, lawful basis, consent where applicable, minimization rule, use restriction, access request, and expiry.
Authority boundary	Availability, model utility, analytics value, or operational convenience does not authorize collection or secondary use beyond the approved purpose.
Continuity rule	When purpose, lawful basis, consent, minimization, or data-subject obligations cannot be established or fulfilled, the platform shall stop the affected processing, restrict access, preserve the compliance case, execute required correction or deletion, and notify accountable authorities.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-51-001	The data-platform realization of Privacy, Minimization, Consent, and Lawful Basis SHALL make collection, use, inference, sharing, retention, correction, access, and deletion purpose-bound, proportionate, reviewable, and enforceable.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-51-002	Every production instance SHALL declare and continuously reconcile data subject or protected class context, purpose, lawful basis, consent where applicable, minimization rule, use restriction, access request, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-51-003	The implementation SHALL enforce this authority boundary: availability, model utility, analytics value, or operational convenience does not authorize collection or secondary use beyond the approved purpose.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-51-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Privacy, Minimization, Consent, and Lawful Basis.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-51-005	When purpose, lawful basis, consent, minimization, or data-subject obligations cannot be established or fulfilled, the platform SHALL stop the affected processing, restrict access, preserve the compliance case, execute required correction or deletion, and notify accountable authorities; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-51-006	Production admission and continued operation SHALL require purpose and field minimization tests, consent changes, subject request exercises, inference review, secondary-use denial, and deletion verification, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Purpose, lawful basis, consent, minimization, or data-subject obligations cannot be established or fulfilled	Stop the affected processing, restrict access, preserve the compliance case, execute required correction or deletion, and notify accountable authorities.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared data subject or protected class context, purpose, lawful basis, consent where applicable, minimization rule, use restriction, access request, and expiry.
- Positive and negative boundary tests prove that availability, model utility, analytics value, or operational convenience does not authorize collection or secondary use beyond the approved purpose.
- Fault exercises inject purpose, lawful basis, consent, minimization, or data-subject obligations cannot be established or fulfilled and verify that the platform can stop the affected processing, restrict access, preserve the compliance case, execute required correction or deletion, and notify accountable authorities.
- Independent review accepts purpose and field minimization tests, consent changes, subject request exercises, inference review, secondary-use denial, and deletion verification for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Privacy, Minimization, Consent, and Lawful Basis is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 50, Ch. 54, Ch. 55, Ch. 64. Primary Volume 5 AI anchors: Ch. 19, Ch. 21, Ch. 25, Ch. 61, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 50, Ch. 53, Ch. 54, Ch. 57, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-016 · C-029 · C-037 · C-039 · C-040 · C-050

52 Residency, Sovereignty, and Cross-Border Control

Residency, Sovereignty, and Cross-Border Control establishes the governed data capability necessary to bind storage, processing, replication, indexes, models, telemetry, backups, support, and exchange to approved jurisdictional and administrative boundaries. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

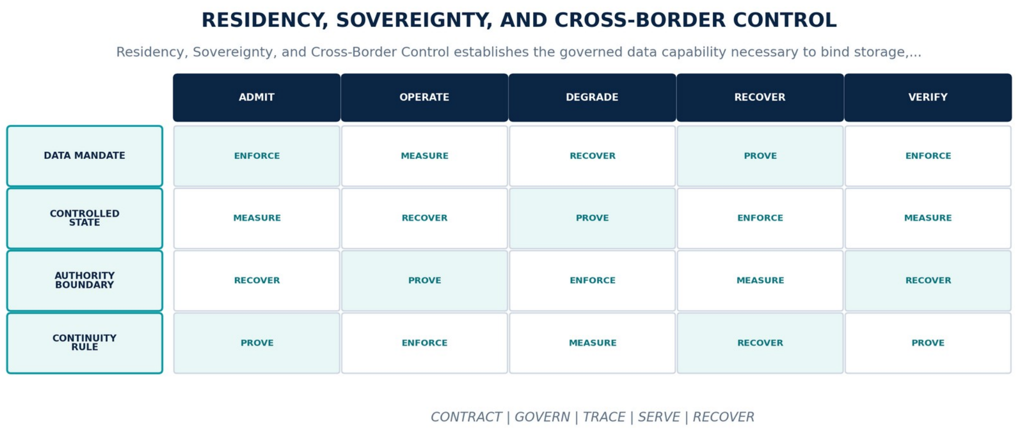

Figure 53. Residency, Sovereignty, and Cross-Border Control. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Bind storage, processing, replication, indexes, models, telemetry, backups, support, and exchange to approved jurisdictional and administrative boundaries.
Controlled state	Sovereignty profile, residency class, processing locality, key authority, administrator jurisdiction, transfer mechanism, exception, and evidence.
Authority boundary	Residency applies end to end across primary, replica, cache, archive, backup, recovery, support, provider, and derived data paths.
Continuity rule	When placement, processing, support, model use, recovery, or exchange would cross an unapproved boundary, the platform shall deny the operation, choose an approved local path or explicit reduced capability, and preserve the blocked reason and affected scope.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-52-001	The data-platform realization of Residency, Sovereignty, and Cross-Border Control SHALL bind storage, processing, replication, indexes, models, telemetry, backups, support, and exchange to approved jurisdictional and administrative boundaries.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-52-002	Every production instance SHALL declare and continuously reconcile sovereignty profile, residency class, processing locality, key authority, administrator jurisdiction, transfer mechanism, exception, and evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-52-003	The implementation SHALL enforce this authority boundary: residency applies end to end across primary, replica, cache, archive, backup, recovery, support, provider, and derived data paths.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-52-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Residency, Sovereignty, and Cross-Border Control.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-52-005	When placement, processing, support, model use, recovery, or exchange would cross an unapproved boundary, the platform SHALL deny the operation, choose an approved local path or explicit reduced capability, and preserve the blocked reason and affected scope; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-52-006	Production admission and continued operation SHALL require placement and egress proofs, key and administrator tests, backup locality, provider denial, disconnected operation, and recovery exercises, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Placement, processing, support, model use, recovery, or exchange would cross an unapproved boundary	Deny the operation, choose an approved local path or explicit reduced capability, and preserve the blocked reason and affected scope.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared sovereignty profile, residency class, processing locality, key authority, administrator jurisdiction, transfer mechanism, exception, and evidence.
- Positive and negative boundary tests prove that residency applies end to end across primary, replica, cache, archive, backup, recovery, support, provider, and derived data paths.
- Fault exercises inject placement, processing, support, model use, recovery, or exchange would cross an unapproved boundary and verify that the platform can deny the operation, choose an approved local path or explicit reduced capability, and preserve the blocked reason and affected scope.
- Independent review accepts placement and egress proofs, key and administrator tests, backup locality, provider denial, disconnected operation, and recovery exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Residency, Sovereignty, and Cross-Border Control is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 29, Ch. 52, Ch. 53, Ch. 54, Ch. 61. Primary Volume 5 AI anchors: Ch. 21, Ch. 25, Ch. 61, Ch. 71, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 11, Ch. 29, Ch. 51, Ch. 53, Ch. 57, Ch. 63.

CONSTITUTIONAL INHERITANCE / C-037 · C-039 · C-040 · C-041 · C-042 · C-043

53 Encryption, Tokenization, Masking, and Confidentiality

Encryption, Tokenization, Masking, and Confidentiality establishes the governed data capability necessary to protect data at rest, in transit, in use, in lower environments, and in shared outputs through policy-bound cryptographic and transformation controls. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

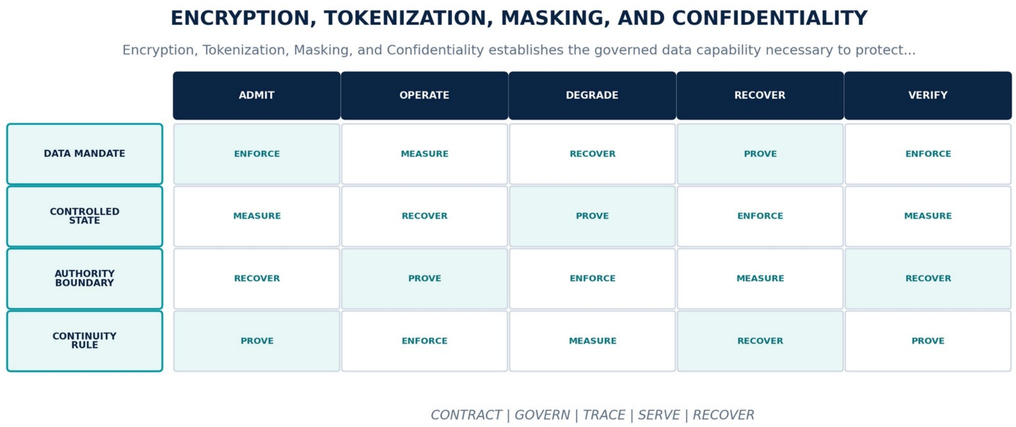

Figure 54. Encryption, Tokenization, Masking, and Confidentiality. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Protect data at rest, in transit, in use, in lower environments, and in shared outputs through policy-bound cryptographic and transformation controls.
Controlled state	Protection profile, data class, key identity, token vault, masking rule, confidential execution requirement, rotation, revocation, and evidence.
Authority boundary	Masking and tokenization preserve controlled usability but do not change classification, purpose, rights, retention, or re-identification risk.
Continuity rule	When keys, certificates, masks, tokens, attestation, or re-identification controls become invalid or unavailable, the platform shall stop protected processing, revoke exposed authority, quarantine affected derivatives, rotate or recover controls, and assess data impact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-53-001	The data-platform realization of Encryption, Tokenization, Masking, and Confidentiality SHALL protect data at rest, in transit, in use, in lower environments, and in shared outputs through policy-bound cryptographic and transformation controls.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-53-002	Every production instance SHALL declare and continuously reconcile protection profile, data class, key identity, token vault, masking rule, confidential execution requirement, rotation, revocation, and evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-53-003	The implementation SHALL enforce this authority boundary: masking and tokenization preserve controlled usability but do not change classification, purpose, rights, retention, or re-identification risk.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-53-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Encryption, Tokenization, Masking, and Confidentiality.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-53-005	When keys, certificates, masks, tokens, attestation, or re-identification controls become invalid or unavailable, the platform SHALL stop protected processing, revoke exposed authority, quarantine affected derivatives, rotate or recover controls, and assess data impact; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-53-006	Production admission and continued operation SHALL require encryption posture, key rotation and recovery, tokenization determinism, mask quality, re-identification testing, and confidential execution attestation, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Keys, certificates, masks, tokens, attestation, or re-identification controls become invalid or unavailable	Stop protected processing, revoke exposed authority, quarantine affected derivatives, rotate or recover controls, and assess data impact.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared protection profile, data class, key identity, token vault, masking rule, confidential execution requirement, rotation, revocation, and evidence.
- Positive and negative boundary tests prove that masking and tokenization preserve controlled usability but do not change classification, purpose, rights, retention, or re-identification risk.
- Fault exercises inject keys, certificates, masks, tokens, attestation, or re-identification controls become invalid or unavailable and verify that the platform can stop protected processing, revoke exposed authority, quarantine affected derivatives, rotate or recover controls, and assess data impact.
- Independent review accepts encryption posture, key rotation and recovery, tokenization determinism, mask quality, re-identification testing, and confidential execution attestation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Encryption, Tokenization, Masking, and Confidentiality is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 49, Ch. 50, Ch. 52, Ch. 54, Ch. 55. Primary Volume 5 AI anchors: Ch. 21, Ch. 36, Ch. 60, Ch. 61, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 49, Ch. 51, Ch. 52, Ch. 53, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-037 · C-039 · C-040 · C-041

54 Retention, Legal Hold, Archive, and Deletion

Retention, Legal Hold, Archive, and Deletion establishes the governed data capability necessary to execute lawful retention, archive, legal hold, customer export, deletion, and residual-reference closure across canonical and derived data. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

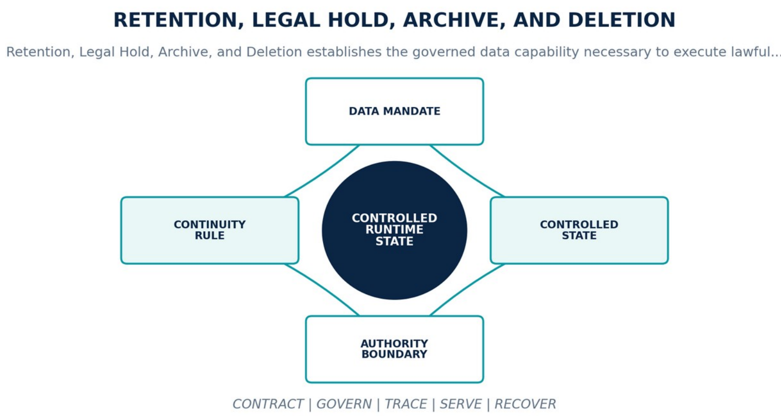

Figure 55. Retention, Legal Hold, Archive, and Deletion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Execute lawful retention, archive, legal hold, customer export, deletion, and residual-reference closure across canonical and derived data.
Controlled state	Lifecycle policy, trigger, retention period, hold, archive tier, deletion scope, authority, execution, residual references, verification, and evidence.
Authority boundary	Deletion is not complete when derived copies, indexes, features, caches, backups, exports, or external processors remain outside the declared disposition.
Continuity rule	When scope is uncertain, legal holds conflict, a copy is unavailable, or deletion cannot be verified without corrupting historical accountability, the platform shall pause unsafe deletion, protect held data, resolve authority and scope, execute in dependency order, and record residual risk and closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-54-001	The data-platform realization of Retention, Legal Hold, Archive, and Deletion SHALL execute lawful retention, archive, legal hold, customer export, deletion, and residual-reference closure across canonical and derived data.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-54-002	Every production instance SHALL declare and continuously reconcile lifecycle policy, trigger, retention period, hold, archive tier, deletion scope, authority, execution, residual references, verification, and evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-54-003	The implementation SHALL enforce this authority boundary: deletion is not complete when derived copies, indexes, features, caches, backups, exports, or external processors remain outside the declared disposition.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-54-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Retention, Legal Hold, Archive, and Deletion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-54-005	When scope is uncertain, legal holds conflict, a copy is unavailable, or deletion cannot be verified without corrupting historical accountability, the platform SHALL pause unsafe deletion, protect held data, resolve authority and scope, execute in dependency order, and record residual risk and closure; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-54-006	Production admission and continued operation SHALL require retention timers, hold precedence, archive retrieval, delete propagation, backup expiry, external processor receipts, and verification evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Scope is uncertain, legal holds conflict, a copy is unavailable, or deletion cannot be verified without corrupting historical accountability	Pause unsafe deletion, protect held data, resolve authority and scope, execute in dependency order, and record residual risk and closure.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared lifecycle policy, trigger, retention period, hold, archive tier, deletion scope, authority, execution, residual references, verification, and evidence.
- Positive and negative boundary tests prove that deletion is not complete when derived copies, indexes, features, caches, backups, exports, or external processors remain outside the declared disposition.
- Fault exercises inject scope is uncertain, legal holds conflict, a copy is unavailable, or deletion cannot be verified without corrupting historical accountability and verify that the platform can pause unsafe deletion, protect held data, resolve authority and scope, execute in dependency order, and record residual risk and closure.
- Independent review accepts retention timers, hold precedence, archive retrieval, delete propagation, backup expiry, external processor receipts, and verification evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Retention, Legal Hold, Archive, and Deletion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 30, Ch. 54, Ch. 55, Ch. 61, Ch. 64. Primary Volume 5 AI anchors: Ch. 25, Ch. 29, Ch. 61, Ch. 65, Ch. 69. Primary Volume 6 infrastructure anchors: Ch. 33, Ch. 51, Ch. 53, Ch. 57, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-025 · C-037 · C-040

55 Data Stewardship, Approval, and Exception Management

Data Stewardship, Approval, and Exception Management establishes the governed data capability necessary to operate durable queues and decision rights for quality, identity, semantics, classification, access, correction, lifecycle, and data exceptions. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

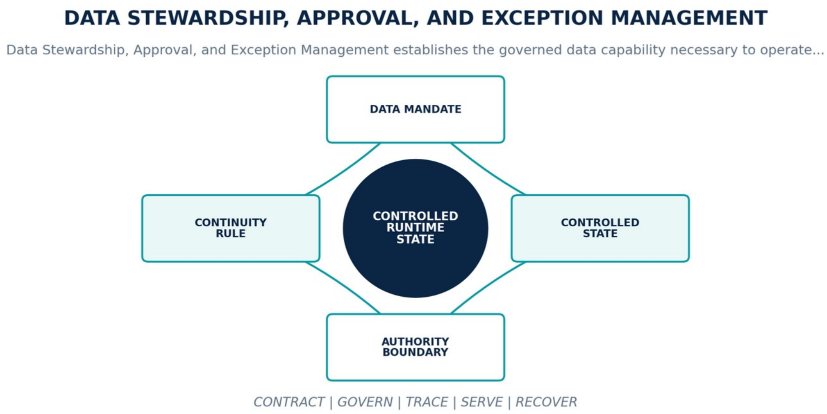

Figure 56. Data Stewardship, Approval, and Exception Management. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Operate durable queues and decision rights for quality, identity, semantics, classification, access, correction, lifecycle, and data exceptions.
Controlled state	Stewardship case, subject, owner, evidence, proposed action, reviewers, segregation of duties, decision, conditions, expiry, and closure.
Authority boundary	Automation may recommend and execute pre-approved reversible steps; named humans retain authority for material adjudication and exception acceptance.
Continuity rule	When a case lacks owner, evidence, timely review, independent approval, or an exception exceeds its scope or expiry, the platform shall escalate by consequence, constrain affected use, block unapproved changes, and keep the unresolved condition visible until closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-55-001	The data-platform realization of Data Stewardship, Approval, and Exception Management SHALL operate durable queues and decision rights for quality, identity, semantics, classification, access, correction, lifecycle, and data exceptions.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-55-002	Every production instance SHALL declare and continuously reconcile stewardship case, subject, owner, evidence, proposed action, reviewers, segregation of duties, decision, conditions, expiry, and closure through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-55-003	The implementation SHALL enforce this authority boundary: automation may recommend and execute pre-approved reversible steps; named humans retain authority for material adjudication and exception acceptance.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-55-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Stewardship, Approval, and Exception Management.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-55-005	When a case lacks owner, evidence, timely review, independent approval, or an exception exceeds its scope or expiry, the platform SHALL escalate by consequence, constrain affected use, block unapproved changes, and keep the unresolved condition visible until closure; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-55-006	Production admission and continued operation SHALL require workflow and SLA evidence, segregation tests, authority verification, exception ageing, override review, and sampled decision quality, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A case lacks owner, evidence, timely review, independent approval, or an exception exceeds its scope or expiry	Escalate by consequence, constrain affected use, block unapproved changes, and keep the unresolved condition visible until closure.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared stewardship case, subject, owner, evidence, proposed action, reviewers, segregation of duties, decision, conditions, expiry, and closure.
- Positive and negative boundary tests prove that automation may recommend and execute pre-approved reversible steps; named humans retain authority for material adjudication and exception acceptance.
- Fault exercises inject a case lacks owner, evidence, timely review, independent approval, or an exception exceeds its scope or expiry and verify that the platform can escalate by consequence, constrain affected use, block unapproved changes, and keep the unresolved condition visible until closure.
- Independent review accepts workflow and SLA evidence, segregation tests, authority verification, exception ageing, override review, and sampled decision quality for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Stewardship, Approval, and Exception Management is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 3, Ch. 29, Ch. 42, Ch. 50, Ch. 55, Ch. 72. Primary Volume 5 AI anchors: Ch. 3, Ch. 37, Ch. 47, Ch. 64, Ch. 65. Primary Volume 6 infrastructure anchors: Ch. 4, Ch. 49, Ch. 56, Ch. 64, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-016 · C-028 · C-036 · C-043 · C-050

56 Data Audit, Non-Repudiation, and Accountability

Data Audit, Non-Repudiation, and Accountability establishes the governed data capability necessary to produce tamper-evident attributable records for material collection, access, transformation, correction, approval, export, deletion, release, and administrative action. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

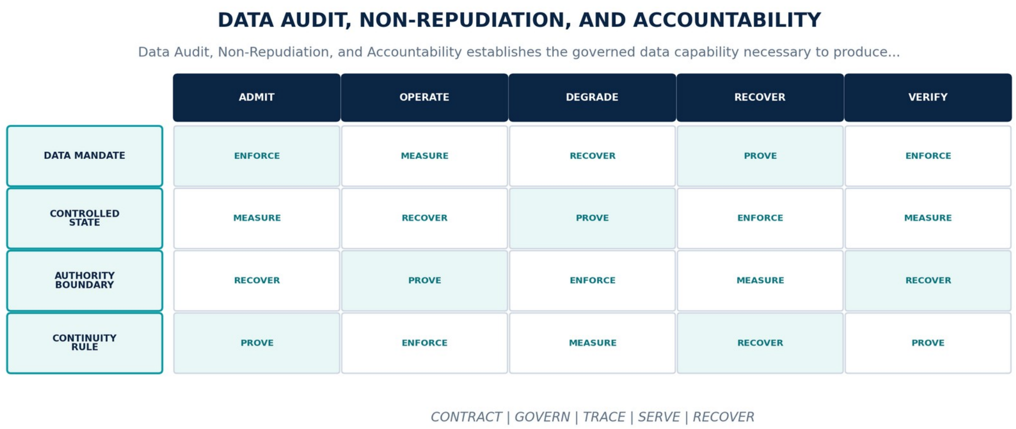

Figure 57. Data Audit, Non-Repudiation, and Accountability. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Produce tamper-evident attributable records for material collection, access, transformation, correction, approval, export, deletion, release, and administrative action.
Controlled state	Audit event, actor and workload identity, action, target versions, purpose, policy, time, result, obligations, correlation, integrity anchor, and custody.
Authority boundary	Operational logs support diagnosis but do not replace protected audit evidence, human approval records, data lineage, or customer-visible receipts.
Continuity rule	When audit coverage, identity attribution, time, integrity, correlation, retention, or custody becomes incomplete or suspect, the platform shall declare the accountability gap, restrict high-consequence operations, preserve alternate evidence, repair coverage, and investigate affected scope.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-56-001	The data-platform realization of Data Audit, Non-Repudiation, and Accountability SHALL produce tamper-evident attributable records for material collection, access, transformation, correction, approval, export, deletion, release, and administrative action.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-56-002	Every production instance SHALL declare and continuously reconcile audit event, actor and workload identity, action, target versions, purpose, policy, time, result, obligations, correlation, integrity anchor, and custody through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-56-003	The implementation SHALL enforce this authority boundary: operational logs support diagnosis but do not replace protected audit evidence, human approval records, data lineage, or customer-visible receipts.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-56-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Audit, Non-Repudiation, and Accountability.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-56-005	When audit coverage, identity attribution, time, integrity, correlation, retention, or custody becomes incomplete or suspect, the platform SHALL declare the accountability gap, restrict high-consequence operations, preserve alternate evidence, repair coverage, and investigate affected scope; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-56-006	Production admission and continued operation SHALL require coverage mapping, integrity verification, privileged and bulk-access exercises, clock correlation, export and deletion receipts, and audit reconstruction, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Audit coverage, identity attribution, time, integrity, correlation, retention, or custody becomes incomplete or suspect	Declare the accountability gap, restrict high-consequence operations, preserve alternate evidence, repair coverage, and investigate affected scope.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared audit event, actor and workload identity, action, target versions, purpose, policy, time, result, obligations, correlation, integrity anchor, and custody.
- Positive and negative boundary tests prove that operational logs support diagnosis but do not replace protected audit evidence, human approval records, data lineage, or customer-visible receipts.
- Fault exercises inject audit coverage, identity attribution, time, integrity, correlation, retention, or custody becomes incomplete or suspect and verify that the platform can declare the accountability gap, restrict high-consequence operations, preserve alternate evidence, repair coverage, and investigate affected scope.
- Independent review accepts coverage mapping, integrity verification, privileged and bulk-access exercises, clock correlation, export and deletion receipts, and audit reconstruction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Audit, Non-Repudiation, and Accountability is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 28, Ch. 49, Ch. 50, Ch. 55, Ch. 62, Ch. 63. Primary Volume 5 AI anchors: Ch. 29, Ch. 36, Ch. 65, Ch. 68, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 49, Ch. 55, Ch. 56, Ch. 58, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-012 · C-025 · C-035 · C-036 · C-039 · C-043

PART VIII

Quality, Reliability, and Operations

Quality, freshness, observability, service levels, degradation, recovery, incidents, and lifecycle operations.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

57 Data Quality Architecture

Data Quality Architecture establishes the governed data capability necessary to treat quality as governed multidimensional product state across accuracy, completeness, validity, consistency, uniqueness, timeliness, provenance, representativeness, and fitness. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

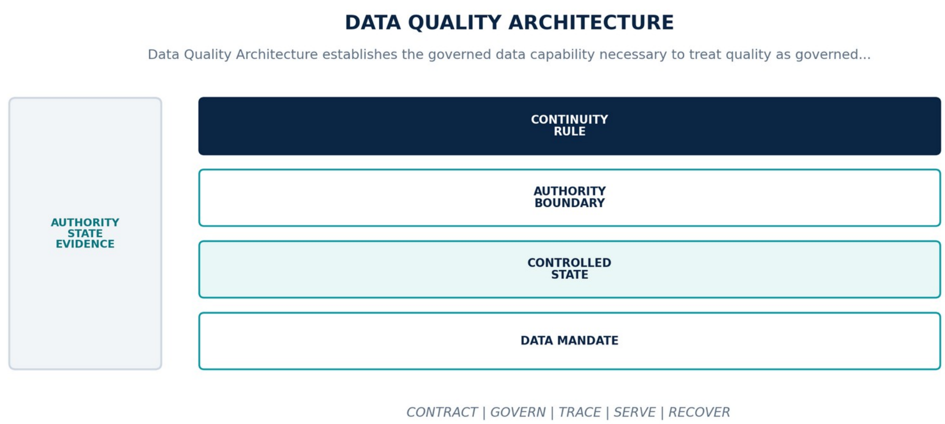

Figure 58. Data Quality Architecture. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Treat quality as governed multidimensional product state across accuracy, completeness, validity, consistency, uniqueness, timeliness, provenance, representativeness, and fitness.
Controlled state	Quality rule, dimension, scope, population, threshold, method, owner, observation, exception, impact, remediation, and evidence.
Authority boundary	Quality scores inform fitness but may not hide failed dimensions, affected records, uncertainty, policy violations, or decision consequence behind one aggregate.
Continuity rule	When quality falls outside contract, measurement is invalid, or impact cannot be bounded to products and decisions, the platform shall declare degradation, quarantine affected partitions or objects, notify consumers, remediate from authoritative sources, and reverify before release.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-57-001	The data-platform realization of Data Quality Architecture SHALL treat quality as governed multidimensional product state across accuracy, completeness, validity, consistency, uniqueness, timeliness, provenance, representativeness, and fitness.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-57-002	Every production instance SHALL declare and continuously reconcile quality rule, dimension, scope, population, threshold, method, owner, observation, exception, impact, remediation, and evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-57-003	The implementation SHALL enforce this authority boundary: quality scores inform fitness but may not hide failed dimensions, affected records, uncertainty, policy violations, or decision consequence behind one aggregate.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-57-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Quality Architecture.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-57-005	When quality falls outside contract, measurement is invalid, or impact cannot be bounded to products and decisions, the platform SHALL declare degradation, quarantine affected partitions or objects, notify consumers, remediate from authoritative sources, and reverify before release; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-57-006	Production admission and continued operation SHALL require rule fixtures, distribution and anomaly tests, slice analysis, impact lineage, remediation replay, consumer acknowledgement, and governance review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Quality falls outside contract, measurement is invalid, or impact cannot be bounded to products and decisions	Declare degradation, quarantine affected partitions or objects, notify consumers, remediate from authoritative sources, and reverify before release.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared quality rule, dimension, scope, population, threshold, method, owner, observation, exception, impact, remediation, and evidence.
- Positive and negative boundary tests prove that quality scores inform fitness but may not hide failed dimensions, affected records, uncertainty, policy violations, or decision consequence behind one aggregate.
- Fault exercises inject quality falls outside contract, measurement is invalid, or impact cannot be bounded to products and decisions and verify that the platform can declare degradation, quarantine affected partitions or objects, notify consumers, remediate from authoritative sources, and reverify before release.
- Independent review accepts rule fixtures, distribution and anomaly tests, slice analysis, impact lineage, remediation replay, consumer acknowledgement, and governance review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Quality Architecture is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 27, Ch. 28, Ch. 46, Ch. 57, Ch. 62, Ch. 71. Primary Volume 5 AI anchors: Ch. 14, Ch. 63, Ch. 66, Ch. 68, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 58, Ch. 59, Ch. 60, Ch. 64, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-005 · C-014 · C-034 · C-035 · C-045 · C-049

58 Freshness, Completeness, and Coverage

Freshness, Completeness, and Coverage establishes the governed data capability necessary to measure whether sources, data products, graph, memory, twins, indexes, features, and strategic packages are current and sufficiently complete for declared purposes. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

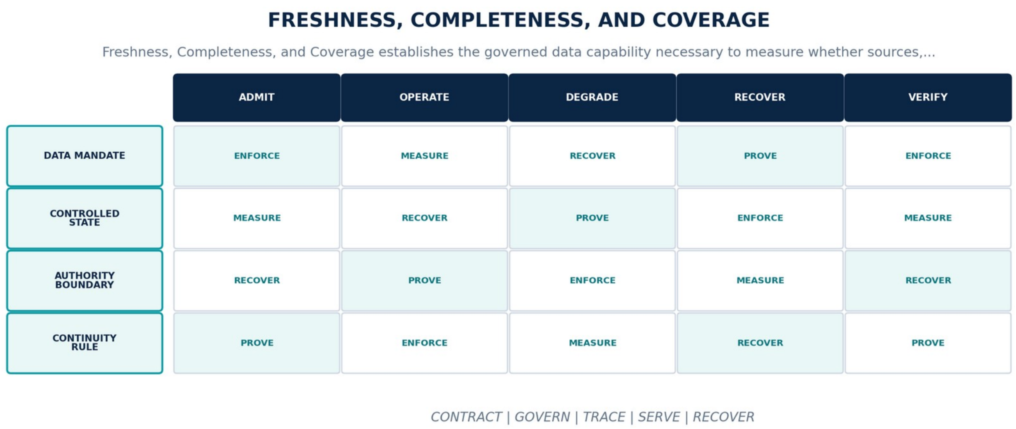

Figure 59. Freshness, Completeness, and Coverage. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Measure whether sources, data products, graph, memory, twins, indexes, features, and strategic packages are current and sufficiently complete for declared purposes.
Controlled state	Freshness contract, observation time, expected arrival, watermark, expiry, completeness denominator, coverage universe, blind spot, and consumer impact.
Authority boundary	Successful pipeline execution does not prove source freshness, world coverage, semantic completeness, or decision fitness.
Continuity rule	When arrivals are late, expected scope is absent, coverage is unknown, or expiry invalidates dependent products, the platform shall surface staleness and blind spots, suppress unsafe outputs, trigger alternate authorized collection or human review, and propagate fitness impact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-58-001	The data-platform realization of Freshness, Completeness, and Coverage SHALL measure whether sources, data products, graph, memory, twins, indexes, features, and strategic packages are current and sufficiently complete for declared purposes.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-58-002	Every production instance SHALL declare and continuously reconcile freshness contract, observation time, expected arrival, watermark, expiry, completeness denominator, coverage universe, blind spot, and consumer impact through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-58-003	The implementation SHALL enforce this authority boundary: successful pipeline execution does not prove source freshness, world coverage, semantic completeness, or decision fitness.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-58-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Freshness, Completeness, and Coverage.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-58-005	When arrivals are late, expected scope is absent, coverage is unknown, or expiry invalidates dependent products, the platform SHALL surface staleness and blind spots, suppress unsafe outputs, trigger alternate authorized collection or human review, and propagate fitness impact; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-58-006	Production admission and continued operation SHALL require freshness and coverage fixtures, missing-source simulations, denominator validation, blind-spot review, expiry propagation, and user comprehension, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Arrivals are late, expected scope is absent, coverage is unknown, or expiry invalidates dependent products	Surface staleness and blind spots, suppress unsafe outputs, trigger alternate authorized collection or human review, and propagate fitness impact.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared freshness contract, observation time, expected arrival, watermark, expiry, completeness denominator, coverage universe, blind spot, and consumer impact.
- Positive and negative boundary tests prove that successful pipeline execution does not prove source freshness, world coverage, semantic completeness, or decision fitness.
- Fault exercises inject arrivals are late, expected scope is absent, coverage is unknown, or expiry invalidates dependent products and verify that the platform can surface staleness and blind spots, suppress unsafe outputs, trigger alternate authorized collection or human review, and propagate fitness impact.
- Independent review accepts freshness and coverage fixtures, missing-source simulations, denominator validation, blind-spot review, expiry propagation, and user comprehension for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Freshness, Completeness, and Coverage is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 31, Ch. 47, Ch. 57, Ch. 62. Primary Volume 5 AI anchors: Ch. 39, Ch. 40, Ch. 57, Ch. 63, Ch. 68. Primary Volume 6 infrastructure anchors: Ch. 28, Ch. 30, Ch. 45, Ch. 58, Ch. 59.

CONSTITUTIONAL INHERITANCE / C-013 · C-014 · C-032 · C-033 · C-035 · C-049

59 Data Observability and Lineage Monitoring

Data Observability and Lineage Monitoring establishes the governed data capability necessary to observe data flows, schemas, volumes, distributions, quality, freshness, lineage, policies, costs, and consumer impact as one causal operating model. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

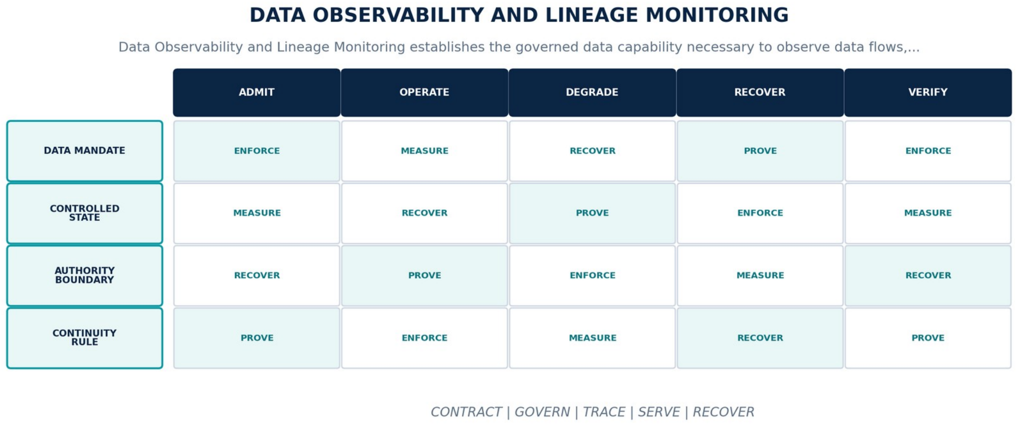

Figure 60. Data Observability and Lineage Monitoring. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Observe data flows, schemas, volumes, distributions, quality, freshness, lineage, policies, costs, and consumer impact as one causal operating model.
Controlled state	Observation identity, asset, pipeline run, contract version, lineage span, metric, anomaly, affected graph, alert, owner, and resolution.
Authority boundary	Telemetry is operational evidence and may not replace canonical data, protected audit, quality authority, or source-of-truth reconciliation.
Continuity rule	When observability coverage is lost, lineage breaks, anomaly detectors drift, or telemetry overload obscures material data state, the platform shall declare observability degradation, preserve priority signals, restrict risky change, reconstruct from alternate evidence, and restore causal coverage.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-59-001	The data-platform realization of Data Observability and Lineage Monitoring SHALL observe data flows, schemas, volumes, distributions, quality, freshness, lineage, policies, costs, and consumer impact as one causal operating model.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-59-002	Every production instance SHALL declare and continuously reconcile observation identity, asset, pipeline run, contract version, lineage span, metric, anomaly, affected graph, alert, owner, and resolution through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-59-003	The implementation SHALL enforce this authority boundary: telemetry is operational evidence and may not replace canonical data, protected audit, quality authority, or source-of-truth reconciliation.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-59-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Observability and Lineage Monitoring.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-59-005	When observability coverage is lost, lineage breaks, anomaly detectors drift, or telemetry overload obscures material data state, the platform SHALL declare observability degradation, preserve priority signals, restrict risky change, reconstruct from alternate evidence, and restore causal coverage; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-59-006	Production admission and continued operation SHALL require end-to-end trace fixtures, telemetry loss and overload, lineage break detection, anomaly validation, privacy review, and incident reconstruction, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Observability coverage is lost, lineage breaks, anomaly detectors drift, or telemetry overload obscures material data state	Declare observability degradation, preserve priority signals, restrict risky change, reconstruct from alternate evidence, and restore causal coverage.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared observation identity, asset, pipeline run, contract version, lineage span, metric, anomaly, affected graph, alert, owner, and resolution.
- Positive and negative boundary tests prove that telemetry is operational evidence and may not replace canonical data, protected audit, quality authority, or source-of-truth reconciliation.
- Fault exercises inject observability coverage is lost, lineage breaks, anomaly detectors drift, or telemetry overload obscures material data state and verify that the platform can declare observability degradation, preserve priority signals, restrict risky change, reconstruct from alternate evidence, and restore causal coverage.
- Independent review accepts end-to-end trace fixtures, telemetry loss and overload, lineage break detection, anomaly validation, privacy review, and incident reconstruction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Observability and Lineage Monitoring is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 28, Ch. 57, Ch. 62, Ch. 63, Ch. 64. Primary Volume 5 AI anchors: Ch. 14, Ch. 65, Ch. 66, Ch. 68, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 45, Ch. 56, Ch. 58, Ch. 59, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-012 · C-014 · C-035 · C-039 · C-043 · C-049

60 Data SLOs, Capacity, and Performance

Data SLOs, Capacity, and Performance establishes the governed data capability necessary to govern latency, throughput, freshness, quality, durability, recovery, query, ingestion, processing, export, and cost objectives by consequence and deployment scope. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

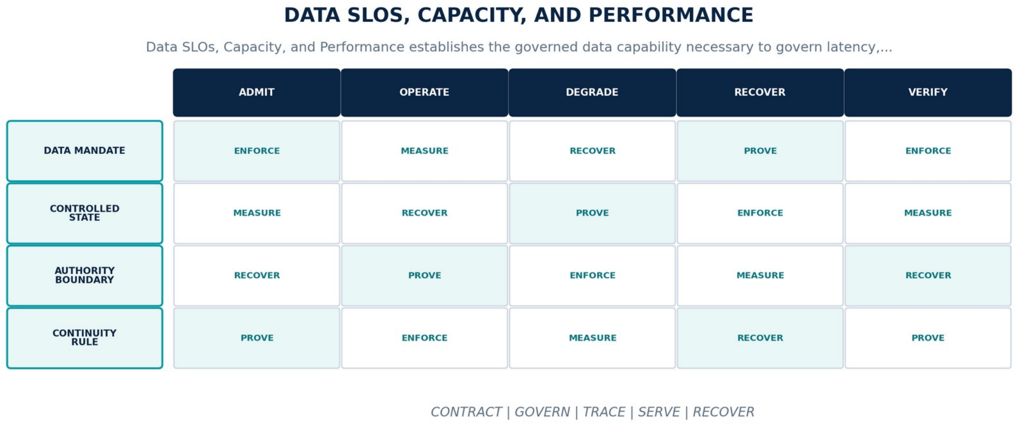

Figure 61. Data SLOs, Capacity, and Performance. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Govern latency, throughput, freshness, quality, durability, recovery, query, ingestion, processing, export, and cost objectives by consequence and deployment scope.
Controlled state	Indicator, objective, population, window, exclusions, budget, demand model, headroom, bottleneck, burn state, owner, and action policy.
Authority boundary	SLOs cover user-visible and control-critical data journeys including stale, partial, denied, late, missing, recovering, and incorrect states.
Continuity rule	When an error budget burns beyond policy, demand exceeds safe headroom, or measurement cannot support a fitness claim, the platform shall halt or restrict change, apply consequence-based admission and shedding, expose affected journeys, and prioritize recovery and capacity.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-60-001	The data-platform realization of Data SLOs, Capacity, and Performance SHALL govern latency, throughput, freshness, quality, durability, recovery, query, ingestion, processing, export, and cost objectives by consequence and deployment scope.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-60-002	Every production instance SHALL declare and continuously reconcile indicator, objective, population, window, exclusions, budget, demand model, headroom, bottleneck, burn state, owner, and action policy through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-60-003	The implementation SHALL enforce this authority boundary: SLOs cover user-visible and control-critical data journeys including stale, partial, denied, late, missing, recovering, and incorrect states.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-60-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data SLOs, Capacity, and Performance.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-60-005	When an error budget burns beyond policy, demand exceeds safe headroom, or measurement cannot support a fitness claim, the platform SHALL halt or restrict change, apply consequence-based admission and shedding, expose affected journeys, and prioritize recovery and capacity; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-60-006	Production admission and continued operation SHALL require indicator validation, representative load, burst and recovery tests, queue and skew analysis, error-budget simulation, and capacity forecast accuracy, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An error budget burns beyond policy, demand exceeds safe headroom, or measurement cannot support a fitness claim	Halt or restrict change, apply consequence-based admission and shedding, expose affected journeys, and prioritize recovery and capacity.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared indicator, objective, population, window, exclusions, budget, demand model, headroom, bottleneck, burn state, owner, and action policy.
- Positive and negative boundary tests prove that SLOs cover user-visible and control-critical data journeys including stale, partial, denied, late, missing, recovering, and incorrect states.
- Fault exercises inject an error budget burns beyond policy, demand exceeds safe headroom, or measurement cannot support a fitness claim and verify that the platform can halt or restrict change, apply consequence-based admission and shedding, expose affected journeys, and prioritize recovery and capacity.
- Independent review accepts indicator validation, representative load, burst and recovery tests, queue and skew analysis, error-budget simulation, and capacity forecast accuracy for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data SLOs, Capacity, and Performance is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 57, Ch. 58, Ch. 59, Ch. 62, Ch. 63. Primary Volume 5 AI anchors: Ch. 22, Ch. 63, Ch. 66, Ch. 68, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 22, Ch. 30, Ch. 58, Ch. 59, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-014 · C-035 · C-038 · C-043 · C-045 · C-049

61 Availability, Durability, and Degraded Data Modes

Availability, Durability, and Degraded Data Modes establishes the governed data capability necessary to maintain safe useful data services through component, cell, zone, site, source, schema, pipeline, store, and dependency failures without fabricating completeness. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

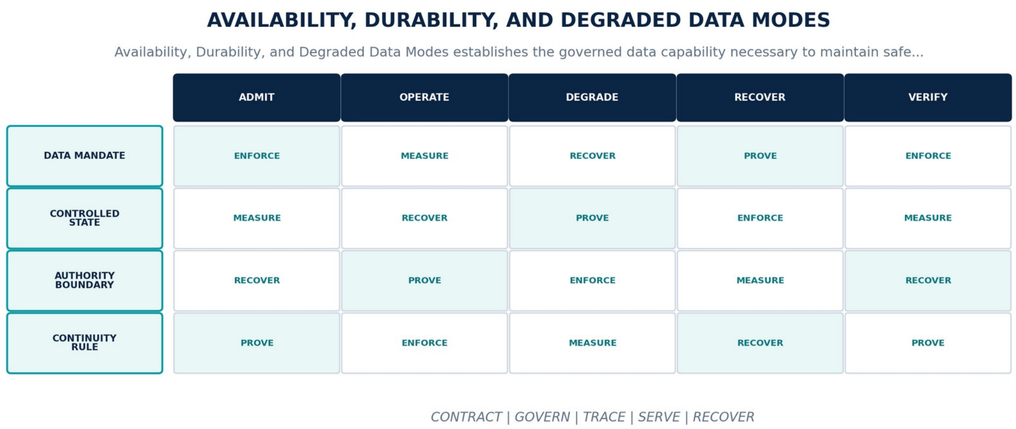

Figure 62. Availability, Durability, and Degraded Data Modes. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Maintain safe useful data services through component, cell, zone, site, source, schema, pipeline, store, and dependency failures without fabricating completeness.
Controlled state	Availability profile, recovery unit, redundancy, quorum, source and projection state, degraded mode, trust limits, owner, entry, and exit evidence.
Authority boundary	Redundancy is valid only when copies do not share undeclared failure, release, policy, key, locality, source, or corruption domains.
Continuity rule	When failures exceed redundancy, authoritative state is uncertain, or normal service would violate truth, policy, quality, or sovereignty, the platform shall enter the explicit safer data mode, fence unsafe writes, mark stale or partial results, suspend consequential products, and reconcile before exit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-61-001	The data-platform realization of Availability, Durability, and Degraded Data Modes SHALL maintain safe useful data services through component, cell, zone, site, source, schema, pipeline, store, and dependency failures without fabricating completeness.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-61-002	Every production instance SHALL declare and continuously reconcile availability profile, recovery unit, redundancy, quorum, source and projection state, degraded mode, trust limits, owner, entry, and exit evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-61-003	The implementation SHALL enforce this authority boundary: redundancy is valid only when copies do not share undeclared failure, release, policy, key, locality, source, or corruption domains.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-61-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Availability, Durability, and Degraded Data Modes.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-61-005	When failures exceed redundancy, authoritative state is uncertain, or normal service would violate truth, policy, quality, or sovereignty, the platform SHALL enter the explicit safer data mode, fence unsafe writes, mark stale or partial results, suspend consequential products, and reconcile before exit; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-61-006	Production admission and continued operation SHALL require fault injection from component through site, source loss, corruption, quorum and fencing, degraded journeys, failback, and consumer impact, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Failures exceed redundancy, authoritative state is uncertain, or normal service would violate truth, policy, quality, or sovereignty	Enter the explicit safer data mode, fence unsafe writes, mark stale or partial results, suspend consequential products, and reconcile before exit.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared availability profile, recovery unit, redundancy, quorum, source and projection state, degraded mode, trust limits, owner, entry, and exit evidence.
- Positive and negative boundary tests prove that redundancy is valid only when copies do not share undeclared failure, release, policy, key, locality, source, or corruption domains.
- Fault exercises inject failures exceed redundancy, authoritative state is uncertain, or normal service would violate truth, policy, quality, or sovereignty and verify that the platform can enter the explicit safer data mode, fence unsafe writes, mark stale or partial results, suspend consequential products, and reconcile before exit.
- Independent review accepts fault injection from component through site, source loss, corruption, quorum and fencing, degraded journeys, failback, and consumer impact for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Availability, Durability, and Degraded Data Modes is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 30, Ch. 57, Ch. 59, Ch. 60, Ch. 61. Primary Volume 5 AI anchors: Ch. 20, Ch. 26, Ch. 63, Ch. 68, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 61, Ch. 62, Ch. 66.

CONSTITUTIONAL INHERITANCE / C-010 · C-014 · C-035 · C-041 · C-043 · C-049

62 Backup, Restore, and Disaster Recovery

Backup, Restore, and Disaster Recovery establishes the governed data capability necessary to protect every durable data class with immutable policy-matched recovery points, tested restore, dependency ordering, integrity validation, and consumer reconciliation. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

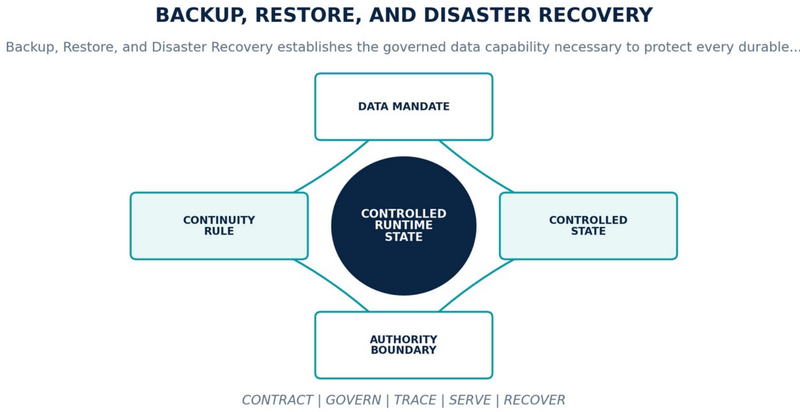

Figure 63. Backup, Restore, and Disaster Recovery. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Protect every durable data class with immutable policy-matched recovery points, tested restore, dependency ordering, integrity validation, and consumer reconciliation.
Controlled state	Recovery unit, source revision, consistency point, backup set, encryption, locality, retention, legal hold, RPO, RTO, restore, and reconciliation.
Authority boundary	Copied bytes are not recovered data service until identity, schemas, policies, keys, lineage, versions, projections, consumers, sovereignty, and quality reconcile.
Continuity rule	When a backup is missing, corrupt, unverifiable, outside locality, incompatible, or cannot restore within declared objectives, the platform shall declare the degraded recovery posture, protect remaining copies, constrain destructive change, select an alternate path, and prove usable restoration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-62-001	The data-platform realization of Backup, Restore, and Disaster Recovery SHALL protect every durable data class with immutable policy-matched recovery points, tested restore, dependency ordering, integrity validation, and consumer reconciliation.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-62-002	Every production instance SHALL declare and continuously reconcile recovery unit, source revision, consistency point, backup set, encryption, locality, retention, legal hold, RPO, RTO, restore, and reconciliation through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-62-003	The implementation SHALL enforce this authority boundary: copied bytes are not recovered data service until identity, schemas, policies, keys, lineage, versions, projections, consumers, sovereignty, and quality reconcile.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-62-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Backup, Restore, and Disaster Recovery.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-62-005	When a backup is missing, corrupt, unverifiable, outside locality, incompatible, or cannot restore within declared objectives, the platform SHALL declare the degraded recovery posture, protect remaining copies, constrain destructive change, select an alternate path, and prove usable restoration; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-62-006	Production admission and continued operation SHALL require automated restore validation, corruption and deletion drills, key recovery, point-in-time restore, graph and projection rebuild, regional recovery, and failback, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A backup is missing, corrupt, unverifiable, outside locality, incompatible, or cannot restore within declared objectives	Declare the degraded recovery posture, protect remaining copies, constrain destructive change, select an alternate path, and prove usable restoration.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared recovery unit, source revision, consistency point, backup set, encryption, locality, retention, legal hold, RPO, RTO, restore, and reconciliation.
- Positive and negative boundary tests prove that copied bytes are not recovered data service until identity, schemas, policies, keys, lineage, versions, projections, consumers, sovereignty, and quality reconcile.
- Fault exercises inject a backup is missing, corrupt, unverifiable, outside locality, incompatible, or cannot restore within declared objectives and verify that the platform can declare the degraded recovery posture, protect remaining copies, constrain destructive change, select an alternate path, and prove usable restoration.
- Independent review accepts automated restore validation, corruption and deletion drills, key recovery, point-in-time restore, graph and projection rebuild, regional recovery, and failback for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Backup, Restore, and Disaster Recovery is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 30, Ch. 61, Ch. 62, Ch. 64. Primary Volume 5 AI anchors: Ch. 25, Ch. 26, Ch. 38, Ch. 65, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 33, Ch. 36, Ch. 62, Ch. 63.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-015 · C-041 · C-043 · C-049

63 Data Incident, Breach, and Correction Response

Data Incident, Breach, and Correction Response establishes the governed data capability necessary to coordinate detection, scope, authority, containment, evidence, correction, communication, recovery, decision impact, and lessons for data incidents. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

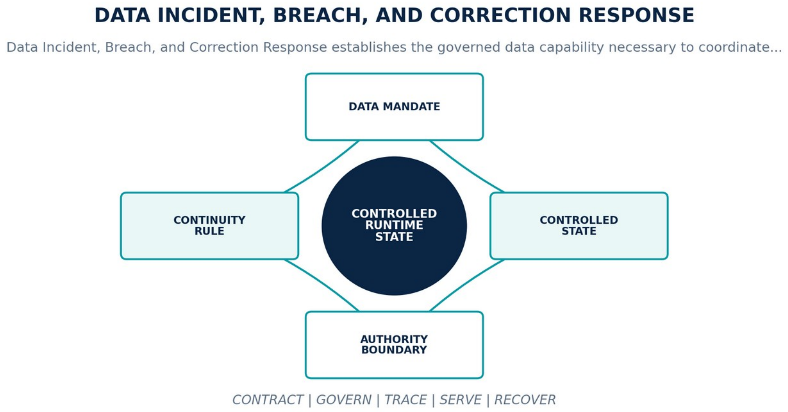

Figure 64. Data Incident, Breach, and Correction Response. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Coordinate detection, scope, authority, containment, evidence, correction, communication, recovery, decision impact, and lessons for data incidents.
Controlled state	Incident identity, severity, command roles, affected assets and subjects, timeline, access and lineage evidence, decisions, actions, notifications, and closure.
Authority boundary	Automation may detect and execute pre-authorized reversible controls; named humans retain incident command, notification, risk acceptance, and return-to-service authority.
Continuity rule	When scope, ownership, breach status, correction impact, or affected decisions remain uncertain during an active event, the platform shall assume the safer consequence class, establish command, constrain access and propagation, preserve evidence, and update scope as facts change.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-63-001	The data-platform realization of Data Incident, Breach, and Correction Response SHALL coordinate detection, scope, authority, containment, evidence, correction, communication, recovery, decision impact, and lessons for data incidents.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-63-002	Every production instance SHALL declare and continuously reconcile incident identity, severity, command roles, affected assets and subjects, timeline, access and lineage evidence, decisions, actions, notifications, and closure through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-63-003	The implementation SHALL enforce this authority boundary: automation may detect and execute pre-authorized reversible controls; named humans retain incident command, notification, risk acceptance, and return-to-service authority.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-63-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Incident, Breach, and Correction Response.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-63-005	When scope, ownership, breach status, correction impact, or affected decisions remain uncertain during an active event, the platform SHALL assume the safer consequence class, establish command, constrain access and propagation, preserve evidence, and update scope as facts change; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-63-006	Production admission and continued operation SHALL require tabletop and live exercises, breach and corruption scenarios, correction fan-out, decision-impact analysis, custody, communication, and closure review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Scope, ownership, breach status, correction impact, or affected decisions remain uncertain during an active event	Assume the safer consequence class, establish command, constrain access and propagation, preserve evidence, and update scope as facts change.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared incident identity, severity, command roles, affected assets and subjects, timeline, access and lineage evidence, decisions, actions, notifications, and closure.
- Positive and negative boundary tests prove that automation may detect and execute pre-authorized reversible controls; named humans retain incident command, notification, risk acceptance, and return-to-service authority.
- Fault exercises inject scope, ownership, breach status, correction impact, or affected decisions remain uncertain during an active event and verify that the platform can assume the safer consequence class, establish command, constrain access and propagation, preserve evidence, and update scope as facts change.
- Independent review accepts tabletop and live exercises, breach and corruption scenarios, correction fan-out, decision-impact analysis, custody, communication, and closure review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Incident, Breach, and Correction Response is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 30, Ch. 48, Ch. 55, Ch. 62, Ch. 63. Primary Volume 5 AI anchors: Ch. 37, Ch. 47, Ch. 65, Ch. 68, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 66.

CONSTITUTIONAL INHERITANCE / C-004 · C-012 · C-025 · C-036 · C-039 · C-043

64 Operational Data Lifecycle and Housekeeping

Operational Data Lifecycle and Housekeeping establishes the governed data capability necessary to operate compaction, partitioning, reindexing, archiving, vacuuming, expiry, deduplication, reconciliation, and cleanup as controlled evidence-bearing workflows. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

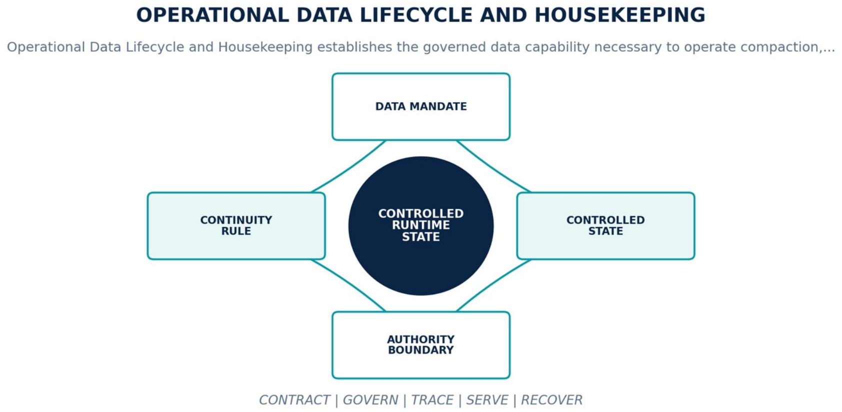

Figure 65. Operational Data Lifecycle and Housekeeping. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Operate compaction, partitioning, reindexing, archiving, vacuuming, expiry, deduplication, reconciliation, and cleanup as controlled evidence-bearing workflows.
Controlled state	Operation identity, target scope, authority, plan, dependencies, resource budget, approvals, checkpoints, observations, rollback, verification, and closure.
Authority boundary	Maintenance may change physical representation but may not alter canonical meaning, temporal history, lineage, policy, legal hold, or replayability.
Continuity rule	When an operation exceeds scope, corrupts state, blocks recovery, or cannot safely roll back or complete, the platform shall stop mutation, fence affected state, preserve before and after evidence, restore or roll forward through an approved plan, and reconcile consumers.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-64-001	The data-platform realization of Operational Data Lifecycle and Housekeeping SHALL operate compaction, partitioning, reindexing, archiving, vacuuming, expiry, deduplication, reconciliation, and cleanup as controlled evidence-bearing workflows.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-64-002	Every production instance SHALL declare and continuously reconcile operation identity, target scope, authority, plan, dependencies, resource budget, approvals, checkpoints, observations, rollback, verification, and closure through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-64-003	The implementation SHALL enforce this authority boundary: maintenance may change physical representation but may not alter canonical meaning, temporal history, lineage, policy, legal hold, or replayability.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-64-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Operational Data Lifecycle and Housekeeping.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-64-005	When an operation exceeds scope, corrupts state, blocks recovery, or cannot safely roll back or complete, the platform SHALL stop mutation, fence affected state, preserve before and after evidence, restore or roll forward through an approved plan, and reconcile consumers; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-64-006	Production admission and continued operation SHALL require dry run, production-like scale, interruption and resume, corruption detection, rollback or roll-forward, resource limits, and lifecycle audit, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An operation exceeds scope, corrupts state, blocks recovery, or cannot safely roll back or complete	Stop mutation, fence affected state, preserve before and after evidence, restore or roll forward through an approved plan, and reconcile consumers.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared operation identity, target scope, authority, plan, dependencies, resource budget, approvals, checkpoints, observations, rollback, verification, and closure.
- Positive and negative boundary tests prove that maintenance may change physical representation but may not alter canonical meaning, temporal history, lineage, policy, legal hold, or replayability.
- Fault exercises inject an operation exceeds scope, corrupts state, blocks recovery, or cannot safely roll back or complete and verify that the platform can stop mutation, fence affected state, preserve before and after evidence, restore or roll forward through an approved plan, and reconcile consumers.
- Independent review accepts dry run, production-like scale, interruption and resume, corruption detection, rollback or roll-forward, resource limits, and lifecycle audit for each supported deployment profile.

> **CHAPTER ACCEPTANCE AND INHERITANCE**
> Operational Data Lifecycle and Housekeeping is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 30, Ch. 61, Ch. 63, Ch. 64. Primary Volume 5 AI anchors: Ch. 25, Ch. 38, Ch. 68, Ch. 69, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 31, Ch. 38, Ch. 62, Ch. 65, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-035 · C-043 · C-049

PART IX

Delivery, Economics, and Governance

Pipeline engineering, environments, verification, migration, deployment parity, economics, readiness, and baseline governance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

65 Data Pipeline Engineering and Repository Standards

Data Pipeline Engineering and Repository Standards establishes the governed data capability necessary to organize data platform source, contracts, schemas, transformations, quality rules, migrations, fixtures, policies, and ownership as reviewable modular repositories. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

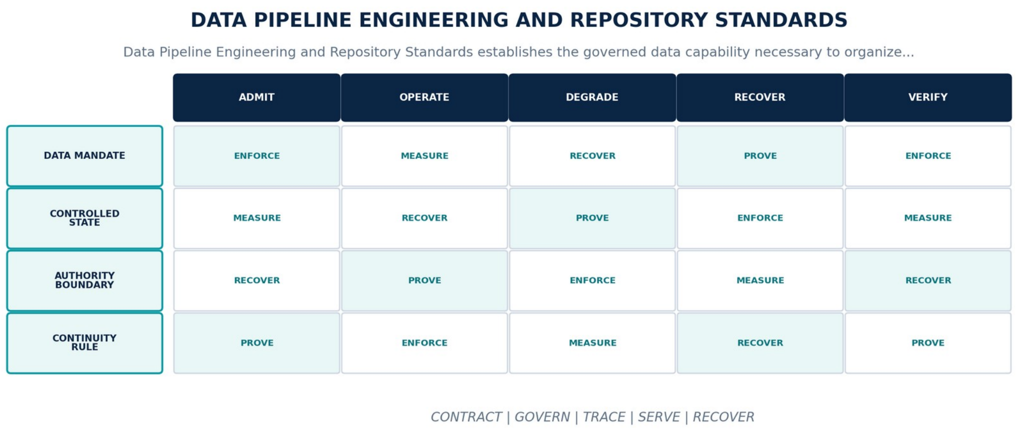

Figure 66. Data Pipeline Engineering and Repository Standards. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Organize data platform source, contracts, schemas, transformations, quality rules, migrations, fixtures, policies, and ownership as reviewable modular repositories.
Controlled state	Module identity, owning team, source revision, dependencies, contract and schema set, tests, build, release, deployment targets, and lifecycle.
Authority boundary	Repository boundaries follow authority and ownership; shared libraries may support but may not create cross-owner writes or hidden data contracts.
Continuity rule	When source provenance, ownership, dependencies, contract references, or reproducible build state is incomplete, the platform shall block build or promotion, expose the unresolved dependency, restore module closure, and require owner and reviewer acceptance.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-65-001	The data-platform realization of Data Pipeline Engineering and Repository Standards SHALL organize data platform source, contracts, schemas, transformations, quality rules, migrations, fixtures, policies, and ownership as reviewable modular repositories.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.
DP-65-002	Every production instance SHALL declare and continuously reconcile module identity, owning team, source revision, dependencies, contract and schema set, tests, build, release, deployment targets, and lifecycle through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-65-003	The implementation SHALL enforce this authority boundary: repository boundaries follow authority and ownership; shared libraries may support but may not create cross-owner writes or hidden data contracts.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-65-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Pipeline Engineering and Repository Standards.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-65-005	When source provenance, ownership, dependencies, contract references, or reproducible build state is incomplete, the platform SHALL block build or promotion, expose the unresolved dependency, restore module closure, and require owner and reviewer acceptance; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-65-006	Production admission and continued operation SHALL require repository linting, dependency and license closure, reproducible build, ownership review, contract mapping, and retirement evidence, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Source provenance, ownership, dependencies, contract references, or reproducible build state is incomplete	Block build or promotion, expose the unresolved dependency, restore module closure, and require owner and reviewer acceptance.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared module identity, owning team, source revision, dependencies, contract and schema set, tests, build, release, deployment targets, and lifecycle.
- Positive and negative boundary tests prove that repository boundaries follow authority and ownership; shared libraries may support but may not create cross-owner writes or hidden data contracts.
- Fault exercises inject source provenance, ownership, dependencies, contract references, or reproducible build state is incomplete and verify that the platform can block build or promotion, expose the unresolved dependency, restore module closure, and require owner and reviewer acceptance.
- Independent review accepts repository linting, dependency and license closure, reproducible build, ownership review, contract mapping, and retirement evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Pipeline Engineering and Repository Standards is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 11, Ch. 23, Ch. 65, Ch. 67, Ch. 68. Primary Volume 5 AI anchors: Ch. 6, Ch. 15, Ch. 18, Ch. 66, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 40, Ch. 45, Ch. 55, Ch. 67, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-006 · C-008 · C-012 · C-035 · C-038 · C-052

66 Configuration, Infrastructure, and Environment Promotion

Configuration, Infrastructure, and Environment Promotion establishes the governed data capability necessary to promote data capabilities through controlled configuration, infrastructure, contract, schema, policy, and environment states with semantic parity. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

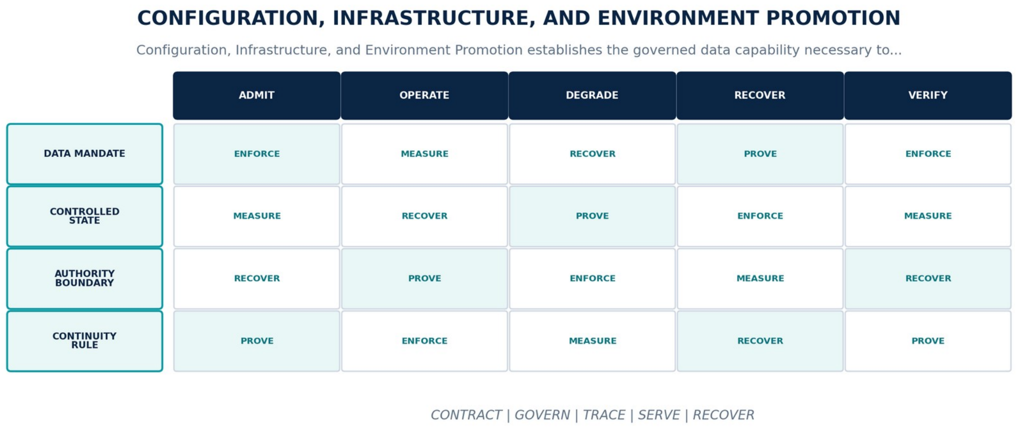

Figure 67. Configuration, Infrastructure, and Environment Promotion. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Promote data capabilities through controlled configuration, infrastructure, contract, schema, policy, and environment states with semantic parity.
Controlled state	Environment, deployment profile, desired configuration, contract and schema versions, infrastructure bindings, secrets, policy, data fixtures, drift, and promotion.
Authority boundary	Configuration may vary by locality and scale but cannot create environment-specific truth, policy, quality, correction, or export semantics.
Continuity rule	When desired and observed state drift, an environment lacks certified dependencies, or promotion would cross an unverified semantic difference, the platform shall hold promotion, preserve the last approved state, expose the diff, reconcile dependencies, and rerun shared conformance fixtures.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-66-001	The data-platform realization of Configuration, Infrastructure, and Environment Promotion SHALL promote data capabilities through controlled configuration, infrastructure, contract, schema, policy, and environment states with semantic parity.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-66-002	Every production instance SHALL declare and continuously reconcile environment, deployment profile, desired configuration, contract and schema versions, infrastructure bindings, secrets, policy, data fixtures, drift, and promotion through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-66-003	The implementation SHALL enforce this authority boundary: configuration may vary by locality and scale but cannot create environment-specific truth, policy, quality, correction, or export semantics.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.
DP-66-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Configuration, Infrastructure, and Environment Promotion.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-66-005	When desired and observed state drift, an environment lacks certified dependencies, or promotion would cross an unverified semantic difference, the platform SHALL hold promotion, preserve the last approved state, expose the diff, reconcile dependencies, and rerun shared conformance fixtures; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-66-006	Production admission and continued operation SHALL require configuration schema tests, drift detection, cross-environment data fixtures, secret and policy validation, deployment parity, and rollback, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Desired and observed state drift, an environment lacks certified dependencies, or promotion would cross an unverified semantic difference	Hold promotion, preserve the last approved state, expose the diff, reconcile dependencies, and rerun shared conformance fixtures.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared environment, deployment profile, desired configuration, contract and schema versions, infrastructure bindings, secrets, policy, data fixtures, drift, and promotion.
- Positive and negative boundary tests prove that configuration may vary by locality and scale but cannot create environment-specific truth, policy, quality, correction, or export semantics.
- Fault exercises inject desired and observed state drift, an environment lacks certified dependencies, or promotion would cross an unverified semantic difference and verify that the platform can hold promotion, preserve the last approved state, expose the diff, reconcile dependencies, and rerun shared conformance fixtures.
- Independent review accepts configuration schema tests, drift detection, cross-environment data fixtures, secret and policy validation, deployment parity, and rollback for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Configuration, Infrastructure, and Environment Promotion is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 23, Ch. 66, Ch. 68, Ch. 69, Ch. 70. Primary Volume 5 AI anchors: Ch. 18, Ch. 21, Ch. 66, Ch. 70, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 3, Ch. 9, Ch. 10, Ch. 11, Ch. 67, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-035 · C-037 · C-041 · C-042 · C-043 · C-049

67 Test Data, Fixtures, and Verification Architecture

Test Data, Fixtures, and Verification Architecture establishes the governed data capability necessary to verify data behavior through versioned synthetic, masked, sampled, adversarial, historical, and golden datasets isolated from production authority. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

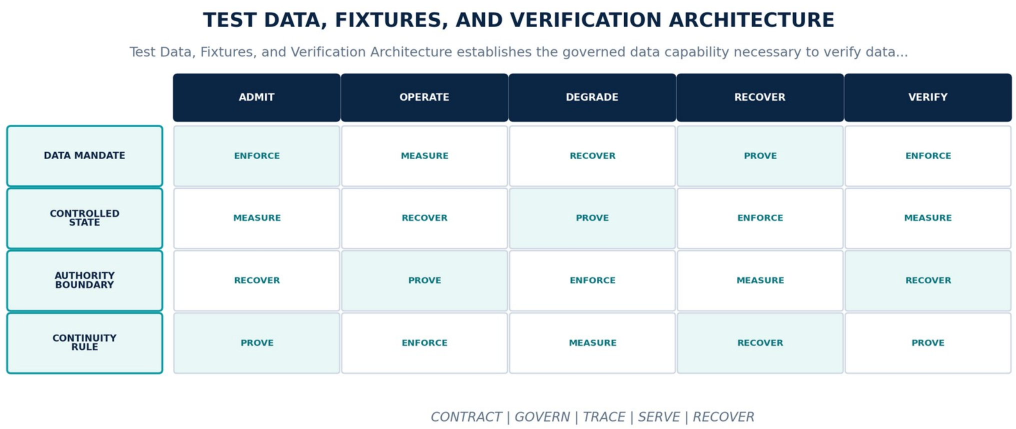

Figure 68. Test Data, Fixtures, and Verification Architecture. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Verify data behavior through versioned synthetic, masked, sampled, adversarial, historical, and golden datasets isolated from production authority.
Controlled state	Fixture identity, purpose, source or generator, schema, seed, classification, expected results, environment scope, expiry, and evidence.
Authority boundary	Test data may model production conditions but may not expose protected production content, leak across tenants, or become model training or canonical truth by reuse.
Continuity rule	When fixtures are stale, non-representative, contaminated, privacy-unsafe, irreproducible, or cannot distinguish expected failure states, the platform shall invalidate affected evidence, quarantine the fixture, rebuild from approved sources or generators, and rerun dependent verification.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-67-001	The data-platform realization of Test Data, Fixtures, and Verification Architecture SHALL verify data behavior through versioned synthetic, masked, sampled, adversarial, historical, and golden datasets isolated from production authority.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-67-002	Every production instance SHALL declare and continuously reconcile fixture identity, purpose, source or generator, schema, seed, classification, expected results, environment scope, expiry, and evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-67-003	The implementation SHALL enforce this authority boundary: test data may model production conditions but may not expose protected production content, leak across tenants, or become model training or canonical truth by reuse.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-67-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Test Data, Fixtures, and Verification Architecture.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-67-005	When fixtures are stale, non-representative, contaminated, privacy-unsafe, irreproducible, or cannot distinguish expected failure states, the platform SHALL invalidate affected evidence, quarantine the fixture, rebuild from approved sources or generators, and rerun dependent verification; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-67-006	Production admission and continued operation SHALL require contract, quality, temporal, lineage, policy, scale, migration, recovery, adversarial, privacy, and deployment-parity test suites, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Fixtures are stale, non-representative, contaminated, privacy-unsafe, irreproducible, or cannot distinguish expected failure states	Invalidate affected evidence, quarantine the fixture, rebuild from approved sources or generators, and rerun dependent verification.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared fixture identity, purpose, source or generator, schema, seed, classification, expected results, environment scope, expiry, and evidence.
- Positive and negative boundary tests prove that test data may model production conditions but may not expose protected production content, leak across tenants, or become model training or canonical truth by reuse.
- Fault exercises inject fixtures are stale, non-representative, contaminated, privacy-unsafe, irreproducible, or cannot distinguish expected failure states and verify that the platform can invalidate affected evidence, quarantine the fixture, rebuild from approved sources or generators, and rerun dependent verification.
- Independent review accepts contract, quality, temporal, lineage, policy, scale, migration, recovery, adversarial, privacy, and deployment-parity test suites for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Test Data, Fixtures, and Verification Architecture is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 27, Ch. 46, Ch. 54, Ch. 67, Ch. 71. Primary Volume 5 AI anchors: Ch. 60, Ch. 61, Ch. 66, Ch. 67, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 45, Ch. 53, Ch. 55, Ch. 68, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-010 · C-023 · C-035 · C-040 · C-045 · C-049

68 Data Release, Migration, Backfill, and Rollback

Data Release, Migration, Backfill, and Rollback establishes the governed data capability necessary to release dependency-closed data changes through evidence-gated migrations, backfills, dual operation, cutover, rollback or roll-forward, and consumer reconciliation. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

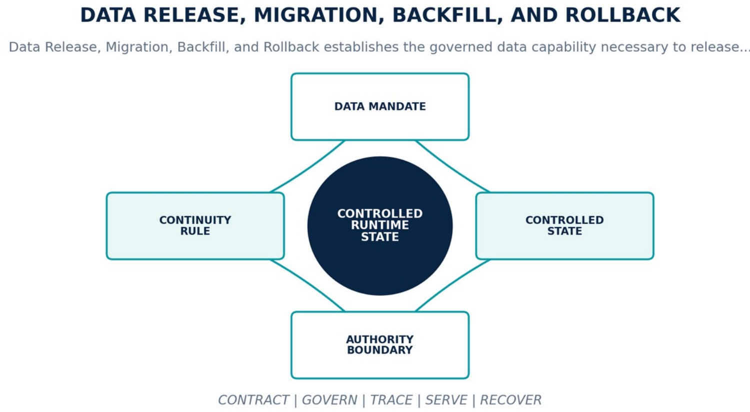

Figure 69. Data Release, Migration, Backfill, and Rollback. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Release dependency-closed data changes through evidence-gated migrations, backfills, dual operation, cutover, rollback or roll-forward, and consumer reconciliation.
Controlled state	Release identity, contracts, schemas, code, policies, datasets, migration plan, backfill scope, checkpoints, consumers, approvals, rollout, and rollback target.
Authority boundary	A data release is the exact resolved system configuration and state transition; mutable tags, floating schemas, and out-of-band production changes are non-conformant.
Continuity rule	When migration or backfill diverges, validation fails, consumers disagree, or rollback would lose authoritative changes, the platform shall halt rollout, contain the cohort, preserve old and new state, reconcile deltas, and execute the approved rollback or roll-forward strategy.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-68-001	The data-platform realization of Data Release, Migration, Backfill, and Rollback SHALL release dependency-closed data changes through evidence-gated migrations, backfills, dual operation, cutover, rollback or roll-forward, and consumer reconciliation.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-68-002	Every production instance SHALL declare and continuously reconcile release identity, contracts, schemas, code, policies, datasets, migration plan, backfill scope, checkpoints, consumers, approvals, rollout, and rollback target through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-68-003	The implementation SHALL enforce this authority boundary: a data release is the exact resolved system configuration and state transition; mutable tags, floating schemas, and out-of-band production changes are non-conformant.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-68-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Release, Migration, Backfill, and Rollback.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-68-005	When migration or backfill diverges, validation fails, consumers disagree, or rollback would lose authoritative changes, the platform SHALL halt rollout, contain the cohort, preserve old and new state, reconcile deltas, and execute the approved rollback or roll-forward strategy; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-68-006	Production admission and continued operation SHALL require dependency closure, representative rehearsal, dual-read and dual-write validation, backfill totals, canary, rollback, correction, and consumer acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Migration or backfill diverges, validation fails, consumers disagree, or rollback would lose authoritative changes	Halt rollout, contain the cohort, preserve old and new state, reconcile deltas, and execute the approved rollback or roll-forward strategy.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared release identity, contracts, schemas, code, policies, datasets, migration plan, backfill scope, checkpoints, consumers, approvals, rollout, and rollback target.
- Positive and negative boundary tests prove that a data release is the exact resolved system configuration and state transition; mutable tags, floating schemas, and out-of-band production changes are non-conformant.
- Fault exercises inject migration or backfill diverges, validation fails, consumers disagree, or rollback would lose authoritative changes and verify that the platform can halt rollout, contain the cohort, preserve old and new state, reconcile deltas, and execute the approved rollback or roll-forward strategy.
- Independent review accepts dependency closure, representative rehearsal, dual-read and dual-write validation, backfill totals, canary, rollback, correction, and consumer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Release, Migration, Backfill, and Rollback is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 23, Ch. 30, Ch. 68, Ch. 69, Ch. 71. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 66, Ch. 68, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 45, Ch. 48, Ch. 65, Ch. 67, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-035 · C-043 · C-049 · C-052

69 Deployment Parity, Disconnected Sync, and Air Gaps

Deployment Parity, Disconnected Sync, and Air Gaps establishes the governed data capability necessary to operate and synchronize the common data platform across SaaS, Private Cloud, On-Premise, disconnected, and air-gapped deployments without hidden cloud dependencies. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

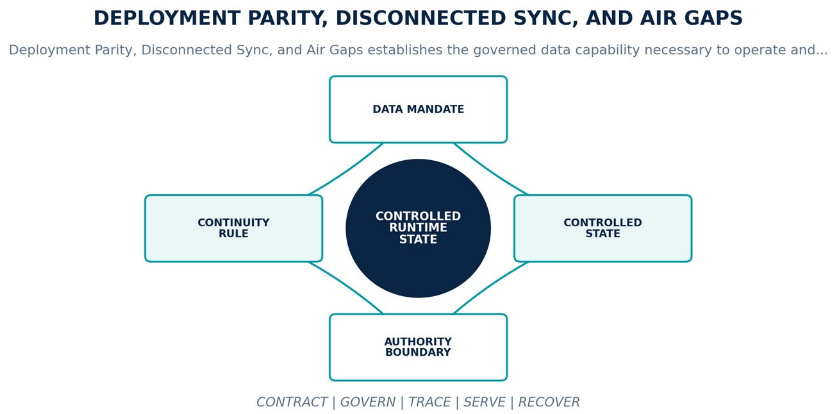

Figure 70. Deployment Parity, Disconnected Sync, and Air Gaps. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Operate and synchronize the common data platform across SaaS, Private Cloud, On-Premise, disconnected, and air-gapped deployments without hidden cloud dependencies.
Controlled state	Deployment profile, local authority, source set, package manifest, trusted time, transfer path, synchronization ledger, conflict policy, and reconciliation.
Authority boundary	Offline operation preserves object, time, truth, provenance, policy, correction, export, and human-authority semantics even when freshness and capability are reduced.
Continuity rule	When trusted time, signed packages, transfer inspection, local dependencies, synchronization evidence, or conflict resolution is unavailable, the platform shall remain in the declared local continuity mode, reject unverifiable exchange, preserve pending changes, and reconcile only through approved ceremony.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-69-001	The data-platform realization of Deployment Parity, Disconnected Sync, and Air Gaps SHALL operate and synchronize the common data platform across SaaS, Private Cloud, On-Premise, disconnected, and air-gapped deployments without hidden cloud dependencies.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-69-002	Every production instance SHALL declare and continuously reconcile deployment profile, local authority, source set, package manifest, trusted time, transfer path, synchronization ledger, conflict policy, and reconciliation through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-69-003	The implementation SHALL enforce this authority boundary: offline operation preserves object, time, truth, provenance, policy, correction, export, and human-authority semantics even when freshness and capability are reduced.	Policy / sovereignty test	Signed control result bound to identity, purpose, tenant, data class, locality, policy, exact data versions, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-69-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Deployment Parity, Disconnected Sync, and Air Gaps.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-69-005	When trusted time, signed packages, transfer inspection, local dependencies, synchronization evidence, or conflict resolution is unavailable, the platform SHALL remain in the declared local continuity mode, reject unverifiable exchange, preserve pending changes, and reconcile only through approved ceremony; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-69-006	Production admission and continued operation SHALL require zero-egress operation, offline install and update, removable-media ceremony, clock discontinuity, delayed synchronization, conflicts, export, and recovery, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Trusted time, signed packages, transfer inspection, local dependencies, synchronization evidence, or conflict resolution is unavailable	Remain in the declared local continuity mode, reject unverifiable exchange, preserve pending changes, and reconcile only through approved ceremony.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared deployment profile, local authority, source set, package manifest, trusted time, transfer path, synchronization ledger, conflict policy, and reconciliation.
- Positive and negative boundary tests prove that offline operation preserves object, time, truth, provenance, policy, correction, export, and human-authority semantics even when freshness and capability are reduced.
- Fault exercises inject trusted time, signed packages, transfer inspection, local dependencies, synchronization evidence, or conflict resolution is unavailable and verify that the platform can remain in the declared local continuity mode, reject unverifiable exchange, preserve pending changes, and reconcile only through approved ceremony.
- Independent review accepts zero-egress operation, offline install and update, removable-media ceremony, clock discontinuity, delayed synchronization, conflicts, export, and recovery for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Deployment Parity, Disconnected Sync, and Air Gaps is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 23, Ch. 30, Ch. 54, Ch. 70, Ch. 71. Primary Volume 5 AI anchors: Ch. 21, Ch. 26, Ch. 60, Ch. 70, Ch. 71. Primary Volume 6 infrastructure anchors: Ch. 9, Ch. 10, Ch. 11, Ch. 27, Ch. 57, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-011 · C-037 · C-038 · C-041 · C-042 · C-043

70 Data Cost, Capacity, Sustainability, and Unit Economics

Data Cost, Capacity, Sustainability, and Unit Economics establishes the governed data capability necessary to make storage, processing, transfer, retrieval, graph, indexing, quality, recovery, licensing, energy, and operational cost visible by product, domain, customer, and workload. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

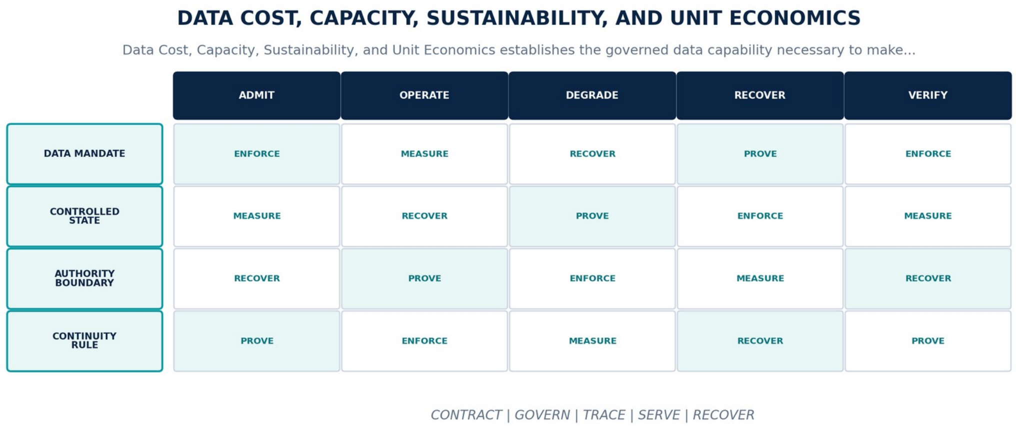

Figure 71. Data Cost, Capacity, Sustainability, and Unit Economics. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Make storage, processing, transfer, retrieval, graph, indexing, quality, recovery, licensing, energy, and operational cost visible by product, domain, customer, and workload.
Controlled state	Cost ledger, allocation key, resource measure, rate, budget, forecast, anomaly, retention tier, energy, carbon, optimization, and decision evidence.
Authority boundary	Economic optimization may not weaken isolation, provenance, quality, freshness, durability, recovery, sovereignty, accessibility, or customer control.
Continuity rule	When cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a contract, SLO, policy, or recovery posture, the platform shall stop unsafe optimization, expose trade-offs, apply approved demand and lifecycle controls, and route material variance to accountable owners.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-70-001	The data-platform realization of Data Cost, Capacity, Sustainability, and Unit Economics SHALL make storage, processing, transfer, retrieval, graph, indexing, quality, recovery, licensing, energy, and operational cost visible by product, domain, customer, and workload.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-70-002	Every production instance SHALL declare and continuously reconcile cost ledger, allocation key, resource measure, rate, budget, forecast, anomaly, retention tier, energy, carbon, optimization, and decision evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-70-003	The implementation SHALL enforce this authority boundary: economic optimization may not weaken isolation, provenance, quality, freshness, durability, recovery, sovereignty, accessibility, or customer control.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-70-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Cost, Capacity, Sustainability, and Unit Economics.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-70-005	When cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a contract, SLO, policy, or recovery posture, the platform SHALL stop unsafe optimization, expose trade-offs, apply approved demand and lifecycle controls, and route material variance to accountable owners; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-70-006	Production admission and continued operation SHALL require allocation reconciliation, unit-cost benchmarks, workload and retention sensitivity, anomaly tests, capacity forecasts, and control-preserving optimization review, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a contract, SLO, policy, or recovery posture	Stop unsafe optimization, expose trade-offs, apply approved demand and lifecycle controls, and route material variance to accountable owners.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared cost ledger, allocation key, resource measure, rate, budget, forecast, anomaly, retention tier, energy, carbon, optimization, and decision evidence.
- Positive and negative boundary tests prove that economic optimization may not weaken isolation, provenance, quality, freshness, durability, recovery, sovereignty, accessibility, or customer control.
- Fault exercises inject cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a contract, SLO, policy, or recovery posture and verify that the platform can stop unsafe optimization, expose trade-offs, apply approved demand and lifecycle controls, and route material variance to accountable owners.
- Independent review accepts allocation reconciliation, unit-cost benchmarks, workload and retention sensitivity, anomaly tests, capacity forecasts, and control-preserving optimization review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Cost, Capacity, Sustainability, and Unit Economics is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 57, Ch. 58, Ch. 64, Ch. 71, Ch. 72. Primary Volume 5 AI anchors: Ch. 22, Ch. 25, Ch. 66, Ch. 68, Ch. 70. Primary Volume 6 infrastructure anchors: Ch. 22, Ch. 31, Ch. 59, Ch. 60, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-006 · C-014 · C-035 · C-037 · C-043 · C-049

71 Data Platform Acceptance and Operational Readiness

Data Platform Acceptance and Operational Readiness establishes the governed data capability necessary to prove that a specific customer deployment can collect, govern, process, serve, explain, secure, fail, restore, migrate, export, and retire its contracted data scope. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

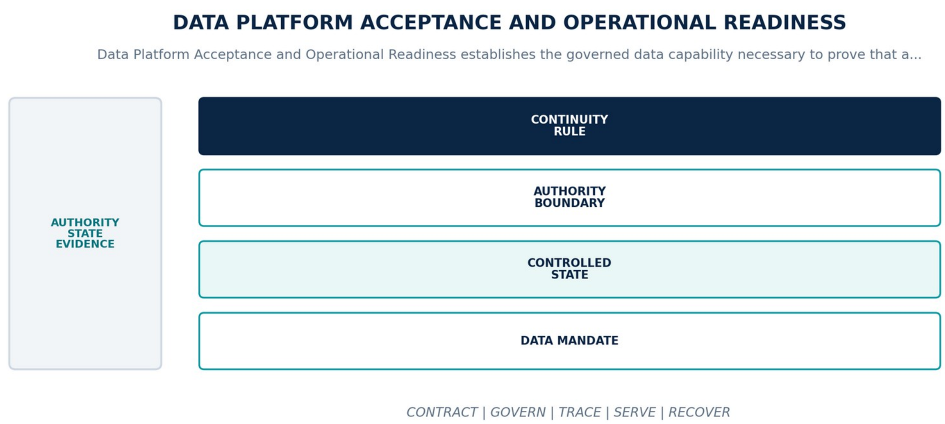

Figure 72. Data Platform Acceptance and Operational Readiness. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Prove that a specific customer deployment can collect, govern, process, serve, explain, secure, fail, restore, migrate, export, and retire its contracted data scope.
Controlled state	Acceptance scope, deployment, sources, contracts, schemas, assets, pipelines, products, policies, tests, operators, open conditions, evidence, and expiry.
Authority boundary	Acceptance is bound to exact data classes, products, workloads, integrations, infrastructure, releases, consequence levels, responsibilities, and operating period.
Continuity rule	When a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required data procedures, the platform shall withhold production authority or restrict accepted scope, record conditions, remediate, and retest through joint customer and platform authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-71-001	The data-platform realization of Data Platform Acceptance and Operational Readiness SHALL prove that a specific customer deployment can collect, govern, process, serve, explain, secure, fail, restore, migrate, export, and retire its contracted data scope.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-71-002	Every production instance SHALL declare and continuously reconcile acceptance scope, deployment, sources, contracts, schemas, assets, pipelines, products, policies, tests, operators, open conditions, evidence, and expiry through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-71-003	The implementation SHALL enforce this authority boundary: acceptance is bound to exact data classes, products, workloads, integrations, infrastructure, releases, consequence levels, responsibilities, and operating period.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
DP-71-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Platform Acceptance and Operational Readiness.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-71-005	When a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required data procedures, the platform SHALL withhold production authority or restrict accepted scope, record conditions, remediate, and retest through joint customer and platform authority; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-71-006	Production admission and continued operation SHALL require source through decision-path fixtures, quality and policy, scale, resilience, restore, migration, export, deletion, operations drills, and signed acceptance, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required data procedures	Withhold production authority or restrict accepted scope, record conditions, remediate, and retest through joint customer and platform authority.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared acceptance scope, deployment, sources, contracts, schemas, assets, pipelines, products, policies, tests, operators, open conditions, evidence, and expiry.
- Positive and negative boundary tests prove that acceptance is bound to exact data classes, products, workloads, integrations, infrastructure, releases, consequence levels, responsibilities, and operating period.
- Fault exercises inject a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required data procedures and verify that the platform can withhold production authority or restrict accepted scope, record conditions, remediate, and retest through joint customer and platform authority.
- Independent review accepts source through decision-path fixtures, quality and policy, scale, resilience, restore, migration, export, deletion, operations drills, and signed acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Platform Acceptance and Operational Readiness is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 57, Ch. 61, Ch. 63, Ch. 67, Ch. 71. Primary Volume 5 AI anchors: Ch. 66, Ch. 68, Ch. 70, Ch. 71, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 59, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-004 · C-006 · C-035 · C-037 · C-042 · C-049

72 Data Platform Governance and Baseline Closure

Data Platform Governance and Baseline Closure establishes the governed data capability necessary to keep the data architecture coherent through ownership, standards, architecture decisions, metrics, exceptions, conformance, lifecycle evidence, review, and controlled amendment. It fixes authority, ownership, temporal meaning, truth state, provenance, policy, quality, failure behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

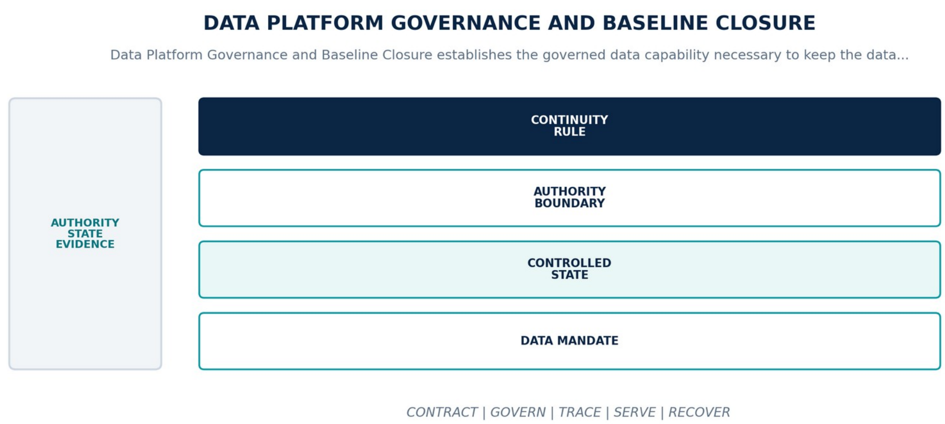

Figure 73. Data Platform Governance and Baseline Closure. The diagram connects data mandate, controlled state, authority boundary, continuity rule as one controlled data contract across admission, operation, degradation, recovery, and assurance.

Data Platform contract

CONTRACT ELEMENT	BINDING DEFINITION
Data mandate	Keep the data architecture coherent through ownership, standards, architecture decisions, metrics, exceptions, conformance, lifecycle evidence, review, and controlled amendment.
Controlled state	Architecture decision, standard, owner, exception, risk acceptance, review cadence, conformance state, amendment, debt, and closure evidence.
Authority boundary	Technologies and physical designs may evolve behind frozen data contracts; local implementation cannot redefine the Strategic Intelligence Operating System or canonical truth.
Continuity rule	When material divergence, unmanaged data debt, repeated exception, obsolete control, or unresolved ownership threatens the baseline, the platform shall constrain affected change and use, assign accountable remediation, elevate semantic conflicts, and amend only through the governing authority chain.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-72-001	The data-platform realization of Data Platform Governance and Baseline Closure SHALL keep the data architecture coherent through ownership, standards, architecture decisions, metrics, exceptions, conformance, lifecycle evidence, review, and controlled amendment.	Automated conformance test	Versioned fixture and signed result for exact contracts, schemas, code, data versions, policies, infrastructure, and deployment profile.
DP-72-002	Every production instance SHALL declare and continuously reconcile architecture decision, standard, owner, exception, risk acceptance, review cadence, conformance state, amendment, debt, and closure evidence through versioned contracts, observed state, ownership, lineage, and audit evidence.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-72-003	The implementation SHALL enforce this authority boundary: technologies and physical designs may evolve behind frozen data contracts; local implementation cannot redefine the Strategic Intelligence Operating System or canonical truth.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
DP-72-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same object, temporal, truth-state, provenance, policy, correction, export, failure, replay, and human-authority semantics for Data Platform Governance and Baseline Closure.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-72-005	When material divergence, unmanaged data debt, repeated exception, obsolete control, or unresolved ownership threatens the baseline, the platform SHALL constrain affected change and use, assign accountable remediation, elevate semantic conflicts, and amend only through the governing authority chain; it SHALL NOT infer complete, current, authorized, correct, reconciled, or recovered data state.	Fault / recovery exercise	Fault trace, affected data graph, authority boundary, degraded mode, recovery, reconciliation, consumer impact, and accountable closure.
DP-72-006	Production admission and continued operation SHALL require architecture review, exception ageing, quality and conformance trends, customer control, Decision Replay, retirement evidence, and executive attestation, bound to the exact contracts, schemas, code, data versions, policies, infrastructure, customer scope, deployment profile, and period under review.	Quality / service-level test	Representative data population, distributions, thresholds, blind spots, bottlenecks, error budget, and admitted fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Material divergence, unmanaged data debt, repeated exception, obsolete control, or unresolved ownership threatens the baseline	Constrain affected change and use, assign accountable remediation, elevate semantic conflicts, and amend only through the governing authority chain.
Contract, policy, lineage, or quality evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer data fitness from successful process execution.
Deployment-profile behavior diverges from the common data contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned data manifest and runtime inventory prove the declared architecture decision, standard, owner, exception, risk acceptance, review cadence, conformance state, amendment, debt, and closure evidence.
- Positive and negative boundary tests prove that technologies and physical designs may evolve behind frozen data contracts; local implementation cannot redefine the Strategic Intelligence Operating System or canonical truth.
- Fault exercises inject material divergence, unmanaged data debt, repeated exception, obsolete control, or unresolved ownership threatens the baseline and verify that the platform can constrain affected change and use, assign accountable remediation, elevate semantic conflicts, and amend only through the governing authority chain.
- Independent review accepts architecture review, exception ageing, quality and conformance trends, customer control, Decision Replay, retirement evidence, and executive attestation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE
Data Platform Governance and Baseline Closure is conformant only when every DP requirement is implemented for the declared scope, data failure states are observable and recoverable, and evidence resolves to the exact released data configuration. Primary Volume 4 engineering anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 63, Ch. 72. Primary Volume 5 AI anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 69, Ch. 70, Ch. 72. Primary Volume 6 infrastructure anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 64, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-009 · C-051 · C-052

PART APP

Data Platform Reference Appendices

Normative component, zone, source, contract, object, temporal, product, manifest, quality, failure, control, lifecycle, traceability, decision, and terminology catalogs.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

A Data Platform Component Catalog

The 104 stable data component identities realize authority, intake, contracts, semantics, storage, processing, knowledge, products, trust, quality, operations, delivery, and economics. Technologies may implement these identities but do not replace their contracts.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
DAT-GV-01	Authority and Governance	Data authority registry	Registry / control	Governance domain	Own authoritative data responsibilities, write rights, and delegated duties.
DAT-GV-02	Authority and Governance	Data domain registry	Registry / control	Governance domain	Own domain identities, boundaries, accountable leaders, and extension contracts.
DAT-GV-03	Authority and Governance	Data product registry	Registry / control	Governance domain	Own product purpose, contracts, consumers, SLOs, cost, and lifecycle.
DAT-GV-04	Authority and Governance	Data ownership service	Registry / control	Governance domain	Resolve owner, steward, custodian, producer, consumer, and escalation state.
DAT-GV-05	Authority and Governance	Classification registry	Registry / control	Governance domain	Own sensitivity, consequence, handling, dissemination, and derivation rules.
DAT-GV-06	Authority and Governance	Data architecture registry	Registry / control	Governance domain	Own standards, architecture decisions, exceptions, and conformance state.
DAT-GV-07	Authority and Governance	Stewardship case service	Registry / control	Governance domain	Execute identity, semantic, quality, access, correction, and lifecycle adjudication.
DAT-GV-08	Authority and Governance	Data exception service	Registry / control	Governance domain	Own narrow exceptions, risk acceptance, controls, expiry, and closure.
DAT-GV-09	Authority and Governance	Conformance service	Registry / control	Governance domain	Evaluate data requirements, manifests, contracts, releases, and runtime evidence.
DAT-GV-10	Authority and Governance	Data responsibility portal	Registry / control	Governance domain	Expose customer and platform duties, approvals, evidence, and escalation.
DAT-IN-01	Acquisition and Intake	Source registry and contract service	Runtime / evidence	Connector or intake cell	Own source authority, rights, purpose, endpoints, freshness, and withdrawal.
DAT-IN-02	Acquisition and Intake	Connector runtime	Runtime / evidence	Connector or intake cell	Execute isolated first-party and custom collection adapters.
DAT-IN-03	Acquisition and Intake	Collection scheduler	Runtime / evidence	Connector or intake cell	Coordinate schedules, subscriptions, leases, retries, and priority.
DAT-IN-04	Acquisition and Intake	External source receiver	Runtime / evidence	Connector or intake cell	Acquire authorized government, news, market, research, patent, and filing data.
DAT-IN-05	Acquisition and Intake	Enterprise system receiver	Runtime / evidence	Connector or intake cell	Acquire ERP, CRM, database, document, and operational-system data.
DAT-IN-06	Acquisition and Intake	File and bulk-transfer receiver	Runtime / evidence	Connector or intake cell	Admit manifested files, secure transfers, and bulk exchange packages.
DAT-IN-07	Acquisition and Intake	API and RSS receiver	Runtime / evidence	Connector or intake cell	Acquire versioned API and feed data under rate and cursor contracts.
DAT-IN-08	Acquisition and Intake	Event and IoT receiver	Runtime / evidence	Connector or intake cell	Admit ordered streams, devices, telemetry, and sensor evidence.
DAT-IN-09	Acquisition and Intake	Change data capture service	Runtime / evidence	Connector or intake cell	Capture transactional changes, deletes, tombstones, and source positions.
DAT-IN-10	Acquisition and Intake	Document and media intake	Runtime / evidence	Connector or intake cell	Preserve originals and create aligned parsing, OCR, transcription, and media derivatives.
DAT-IN-11	Acquisition and Intake	Content inspection service	Runtime / evidence	Connector or intake cell	Detect malicious, corrupt, unsupported, or policy-prohibited content.
DAT-IN-12	Acquisition and Intake	Intake validation service	Runtime / evidence	Connector or intake cell	Validate integrity, schema, rights, classification, locality, and quality.
DAT-IN-13	Acquisition and Intake	Quarantine service	Runtime / evidence	Connector or intake cell	Isolate untrusted or non-conformant content and controlled reinspection.
DAT-IN-14	Acquisition and Intake	Raw preservation service	Runtime / evidence	Connector or intake cell	Commit immutable original evidence, digests, custody, and intake receipts.
DAT-SM-01	Contracts and Semantics	Data contract registry	Registry / canonical state	Semantic domain	Own dataset, stream, object, feature, index, and export contracts.
DAT-SM-02	Contracts and Semantics	Schema registry	Registry / canonical state	Semantic domain	Own schema identity, compatibility, migration, deprecation, and consumer impact.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
DAT-SM-03	Contracts and Semantics	Canonical object service	Registry / canonical state	Semantic domain	Validate and admit intelligence objects with the common governed header.
DAT-SM-04	Contracts and Semantics	Identity and namespace service	Registry / canonical state	Semantic domain	Issue stable tenant-safe identifiers for assets, objects, and versions.
DAT-SM-05	Contracts and Semantics	Temporal semantics service	Registry / canonical state	Semantic domain	Validate event, observation, validity, and system-record time.
DAT-SM-06	Contracts and Semantics	Truth-state registry	Registry / canonical state	Semantic domain	Own epistemic labels, confidence forms, uncertainty, and promotion rules.
DAT-SM-07	Contracts and Semantics	Provenance and lineage service	Registry / canonical state	Semantic domain	Record inputs, outputs, actors, methods, versions, policies, and custody.
DAT-SM-08	Contracts and Semantics	Correction and withdrawal service	Registry / canonical state	Semantic domain	Coordinate non-destructive correction, withdrawal, supersession, and impact.
DAT-SM-09	Contracts and Semantics	Compatibility analyzer	Registry / canonical state	Semantic domain	Evaluate producer, consumer, schema, ontology, and contract changes.
DAT-SM-10	Contracts and Semantics	Semantic mapping service	Registry / canonical state	Semantic domain	Own governed mappings between domain and canonical concepts.
DAT-SM-11	Contracts and Semantics	Data policy metadata service	Registry / canonical state	Semantic domain	Bind classification, purpose, rights, retention, and obligations to data.
DAT-SM-12	Contracts and Semantics	Contract fixture service	Registry / canonical state	Semantic domain	Publish golden positive, negative, historical, and migration fixtures.
DAT-ST-01	Storage and Processing	State-class registry	Canonical / derived state	Data state cell	Resolve authority, consistency, durability, locality, protection, and recovery.
DAT-ST-02	Storage and Processing	Lakehouse catalog	Canonical / derived state	Data state cell	Own open analytical tables, snapshots, partitions, and transaction metadata.
DAT-ST-03	Storage and Processing	Analytical object store	Canonical / derived state	Data state cell	Persist raw, standardized, curated, feature, and evaluation data.
DAT-ST-04	Storage and Processing	Operational database platform	Canonical / derived state	Data state cell	Host service-owned consistency-sensitive data.
DAT-ST-05	Storage and Processing	Evidence object store	Canonical / derived state	Data state cell	Retain immutable source, document, media, and custody objects.
DAT-ST-06	Storage and Processing	Archive and cold-tier store	Canonical / derived state	Data state cell	Retain governed historical data under retrieval and lifecycle contracts.
DAT-ST-07	Storage and Processing	Event streaming platform	Canonical / derived state	Data state cell	Deliver ordered durable events with replay and retention.
DAT-ST-08	Storage and Processing	Durable queue platform	Canonical / derived state	Data state cell	Deliver bounded work, retries, delays, and dead-letter state.
DAT-ST-09	Storage and Processing	Batch orchestration service	Canonical / derived state	Data state cell	Execute versioned transformations, backfills, exports, and rebuilds.
DAT-ST-10	Storage and Processing	Stream processing service	Canonical / derived state	Data state cell	Execute event-time stateful processing, windows, rules, and signals.
DAT-ST-11	Storage and Processing	Distributed query service	Canonical / derived state	Data state cell	Serve federated policy-aware analytical queries.
DAT-ST-12	Storage and Processing	Materialization service	Canonical / derived state	Data state cell	Build governed marts, views, aggregates, and reusable projections.
DAT-ST-13	Storage and Processing	Cache platform	Canonical / derived state	Data state cell	Serve source-versioned temporary acceleration with explicit invalidation.
DAT-ST-14	Storage and Processing	Checkpoint store	Canonical / derived state	Data state cell	Persist pipeline, stream, exchange, and recovery progress.
DAT-KN-01	Knowledge and Strategic State	Ontology registry	Canonical / product state	Knowledge cell	Own semantic versions, constraints, mappings, migrations, and stewardship.
DAT-KN-02	Knowledge and Strategic State	Entity resolution service	Canonical / product state	Knowledge cell	Resolve canonical identities, candidates, merges, splits, and evidence.
DAT-KN-03	Knowledge and Strategic State	Knowledge Graph write service	Canonical / product state	Knowledge cell	Validate atomic temporal graph revisions and provenance edges.
DAT-KN-04	Knowledge and Strategic State	Knowledge Graph store	Canonical / product state	Knowledge cell	Persist canonical entities, relationships, claims, events, and strategy context.
DAT-KN-05	Knowledge and Strategic State	Graph projection service	Canonical / product state	Knowledge cell	Build rebuildable traversal and analytical graph projections.
DAT-KN-06	Knowledge and Strategic State	Enterprise Memory service	Canonical / product state	Knowledge cell	Persist versioned permission-aware institutional and strategic records.
DAT-KN-07	Knowledge and Strategic State	Document and artifact repository	Canonical / product state	Knowledge cell	Retain governed internal documents, communications, and artifacts.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
DAT-KN-08	Knowledge and Strategic State	Search index service	Canonical / product state	Knowledge cell	Build policy-aware lexical, temporal, structured, and geospatial indexes.
DAT-KN-09	Knowledge and Strategic State	Vector index service	Canonical / product state	Knowledge cell	Build versioned multimodal semantic retrieval projections.
DAT-KN-10	Knowledge and Strategic State	Digital Twin state service	Canonical / product state	Knowledge cell	Persist definitions, observed snapshots, branches, models, and validation.
DAT-KN-11	Knowledge and Strategic State	Strategic package service	Canonical / product state	Knowledge cell	Persist forecasts, scenarios, simulations, decisions, commitments, and outcomes.
DAT-KN-12	Knowledge and Strategic State	Decision Replay archive	Canonical / product state	Knowledge cell	Preserve exact decision-time knowledge and later outcome linkage.
DAT-SV-01	Products and Serving	Data product control service	Product / serving state	Serving cell	Register, release, monitor, degrade, and retire governed data products.
DAT-SV-02	Products and Serving	Data API gateway	Product / serving state	Serving cell	Serve typed purpose-aware queries and bulk interfaces.
DAT-SV-03	Products and Serving	Event product gateway	Product / serving state	Serving cell	Publish governed event products and subscription state.
DAT-SV-04	Products and Serving	Semantic analytics service	Product / serving state	Serving cell	Serve certified metrics, dimensions, measures, and temporal views.
DAT-SV-05	Products and Serving	Feature data service	Product / serving state	Serving cell	Serve versioned online and offline features with point-in-time correctness.
DAT-SV-06	Products and Serving	AI dataset service	Product / serving state	Serving cell	Own training, evaluation, feedback, red-team, and calibration datasets.
DAT-SV-07	Products and Serving	Context assembly service	Product / serving state	Serving cell	Build evidence-bound purpose-filtered context manifests.
DAT-SV-08	Products and Serving	Data exchange service	Product / serving state	Serving cell	Create signed exports, imports, replication, and portability packages.
DAT-SV-09	Products and Serving	Marketplace package service	Product / serving state	Serving cell	Distribute signed governed data extensions and their lifecycle.
DAT-SV-10	Products and Serving	Consumer registry	Product / serving state	Serving cell	Own product subscriptions, contract versions, usage, impact, and migration state.
DAT-TR-01	Trust and Lifecycle	Metadata catalog	Policy / protected evidence	Trust domain	Discover assets, owners, contracts, lineage, policy, quality, SLOs, and lifecycle.
DAT-TR-02	Trust and Lifecycle	Purpose and access policy service	Policy / protected evidence	Trust domain	Evaluate identity, purpose, tenant, classification, rights, and locality.
DAT-TR-03	Trust and Lifecycle	Data policy enforcement gateway	Policy / protected evidence	Trust domain	Apply decisions and obligations at data access and processing boundaries.
DAT-TR-04	Trust and Lifecycle	Privacy and minimization service	Policy / protected evidence	Trust domain	Own field, content, consent, lawful-basis, and secondary-use controls.
DAT-TR-05	Trust and Lifecycle	Residency and sovereignty controller	Policy / protected evidence	Trust domain	Enforce storage, processing, key, support, transfer, and recovery locality.
DAT-TR-06	Trust and Lifecycle	Tokenization service	Policy / protected evidence	Trust domain	Replace protected values with policy-bound reversible or irreversible tokens.
DAT-TR-07	Trust and Lifecycle	Masking and de-identification service	Policy / protected evidence	Trust domain	Produce controlled lower-environment and analytical representations.
DAT-TR-08	Trust and Lifecycle	Retention policy service	Policy / protected evidence	Trust domain	Compute lifecycle actions, expiry, archive, and review state.
DAT-TR-09	Trust and Lifecycle	Legal hold service	Policy / protected evidence	Trust domain	Preserve scoped data and prevent conflicting lifecycle actions.
DAT-TR-10	Trust and Lifecycle	Deletion orchestrator	Policy / protected evidence	Trust domain	Delete canonical and derived copies in dependency order and verify residue.
DAT-TR-11	Trust and Lifecycle	Immutable data audit store	Policy / protected evidence	Trust domain	Retain attributable tamper-evident data access and change evidence.
DAT-TR-12	Trust and Lifecycle	Customer rights and request service	Policy / protected evidence	Trust domain	Execute access, correction, export, restriction, and deletion requests.
DAT-OP-01	Quality and Operations	Data quality rules service	Telemetry / control	Operations domain	Execute versioned quality dimensions, thresholds, exceptions, and evidence.
DAT-OP-02	Quality and Operations	Freshness and coverage monitor	Telemetry / control	Operations domain	Measure expected arrivals, watermarks, completeness, expiry, and blind spots.
DAT-OP-03	Quality and Operations	Data observability service	Telemetry / control	Operations domain	Correlate contracts, runs, lineage, volumes, distributions, quality, and consumers.
DAT-OP-04	Quality and Operations	Data SLO service	Telemetry / control	Operations domain	Compute indicators, objectives, error budgets, burn, and response state.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
DAT-OP-05	Quality and Operations	Data capacity service	Telemetry / control	Operations domain	Model demand, storage, processing, transfer, query, headroom, and bottlenecks.
DAT-OP-06	Quality and Operations	Data degraded-mode controller	Telemetry / control	Operations domain	Declare, communicate, enforce, and exit controlled data modes.
DAT-OP-07	Quality and Operations	Backup controller	Telemetry / control	Operations domain	Create, verify, retain, and expire policy-matched recovery sets.
DAT-OP-08	Quality and Operations	Restore orchestrator	Telemetry / control	Operations domain	Restore data, contracts, schemas, keys, lineage, and consumers in order.
DAT-OP-09	Quality and Operations	Data incident platform	Telemetry / control	Operations domain	Manage incident command, scope, evidence, actions, communication, and closure.
DAT-OP-10	Quality and Operations	Correction impact service	Telemetry / control	Operations domain	Identify and track products, models, warnings, decisions, and exports affected.
DAT-OP-11	Quality and Operations	Operational lifecycle service	Telemetry / control	Operations domain	Execute compaction, reindex, archive, expiry, deduplication, and cleanup.
DAT-OP-12	Quality and Operations	Data operations evidence vault	Telemetry / control	Operations domain	Retain readiness, change, incident, recovery, and acceptance evidence.
DAT-DL-01	Delivery and Economics	Data source repository service	Registry / durable workflow	Delivery domain	Own pipeline, schema, contract, rule, migration, and fixture source identity.
DAT-DL-02	Delivery and Economics	Data build and provenance service	Registry / durable workflow	Delivery domain	Build reproducible artifacts and attest source and dependencies.
DAT-DL-03	Delivery and Economics	Data release registry	Registry / durable workflow	Delivery domain	Own dependency-closed data release identity and lifecycle.
DAT-DL-04	Delivery and Economics	Migration and backfill controller	Registry / durable workflow	Delivery domain	Execute rehearsed migrations, checkpoints, validation, and reconciliation.
DAT-DL-05	Delivery and Economics	Data promotion controller	Registry / durable workflow	Delivery domain	Promote cohorts through evidence gates and rollback boundaries.
DAT-DL-06	Delivery and Economics	Deployment parity service	Registry / durable workflow	Delivery domain	Run common data fixtures across SaaS, Private Cloud, and On-Premise.
DAT-DL-07	Delivery and Economics	Data cost and sustainability ledger	Registry / durable workflow	Delivery domain	Allocate storage, processing, transfer, retrieval, energy, and cost.
DAT-DL-08	Delivery and Economics	Data readiness service	Registry / durable workflow	Delivery domain	Own deployment acceptance scope, tests, operators, conditions, and expiry.

Component conformance rules

RULE	BINDING APPLICATION
Authority precedes technology	A product, database, framework, managed service, or internal implementation becomes conformant only through a stable DAT component identity and approved manifest.
Ownership remains singular	Each authoritative lifecycle has one accountable component even where stewardship, custody, hosting, processing, and support duties are shared.
Data state is explicit	Evidence, authoritative, product, derived, temporary, audit, archive, and recovery state are declared separately and cannot be inferred from physical storage.
Semantics survive substitution	A replacement must preserve object, time, truth, provenance, policy, correction, failure, export, replay, and lifecycle contracts.
Evidence follows release	Conformance, quality, privacy, security, scale, recovery, deployment-parity, and retirement evidence resolves to the exact component and released data configuration.

CATALOG AUTHORITY

The component catalog freezes accountable data responsibilities, not vendor products. Technology can change only behind the same authority, state, semantic, policy, failure, recovery, evidence, and deployment-parity boundaries.

B Data Zone and State Tier Catalog

Data zones are authority and lifecycle boundaries, not merely storage folders. Every movement across zones carries contract, identity, temporal, truth-state, provenance, policy, quality, and reconciliation evidence.

ID	ZONE OR TIER	AUTHORITY	CONTENTS	BINDING CONTRACT
DZ-01	External source boundary	External authority	Untrusted source payloads	Collection contract and intake controls
DZ-02	Enterprise source boundary	Customer system authority	Operational source data	Integration contract and reconciliation
DZ-03	Transfer quarantine	Intake authority	Unvalidated packages and content	No downstream access before admission
DZ-04	Raw evidence tier	Observation authority	Immutable originals and custody	Append-only versioned evidence
DZ-05	Standardized data tier	Data contract owner	Parsed and normalized records	Schema, lineage, truth state, quality
DZ-06	Curated domain tier	Domain product owner	Validated reusable domain products	Product contract and SLO
DZ-07	Canonical object tier	Canonical component owner	Governed intelligence objects	Common object and lifecycle contract
DZ-08	Knowledge Graph tier	Graph authority	Temporal entities and relationships	Ontology, provenance, revision control
DZ-09	Enterprise Memory tier	Memory authority	Institutional and strategic records	Permission, retention, correction
DZ-10	Digital Twin tier	Twin authority	Observed snapshots and synthetic branches	Branch isolation and validation
DZ-11	Strategic state tier	Foresight and decision owners	Forecast, scenario, simulation, decision data	Exact version and human authority
DZ-12	Feature and AI dataset tier	AI data owner	Training, evaluation, feedback, features	Purpose isolation and dataset approval
DZ-13	Search projection tier	Source owner remains authoritative	Lexical and structured indexes	Rebuildable and visibly stale
DZ-14	Vector projection tier	Source owner remains authoritative	Embeddings and multimodal representations	Model-versioned and rebuildable
DZ-15	Semantic analytics tier	Metric owner	Certified measures and dimensions	Grain, source revision, effective time
DZ-16	Serving tier	Data product owner	API, event, query, and context products	Purpose, policy, SLO, receipts
DZ-17	Exchange staging tier	Exchange authority	Signed export and import packages	Scope, rights, destination, expiry
DZ-18	Archive tier	Lifecycle authority	Historical low-access data	Retention, hold, retrieval, deletion
DZ-19	Recovery tier	Recovery authority	Immutable backup and restore sets	RPO, RTO, locality, reconciliation
DZ-20	Audit evidence tier	Audit authority	Tamper-evident actions and receipts	Independent integrity and custody

C Source and Ingestion Pattern Catalog

Source patterns standardize acquisition without flattening source-specific authority, rights, time, correction, quality, rate, and reconciliation semantics.

ID	PATTERN	ACQUISITION MODE	CRITICAL SEMANTICS	MINIMUM EVIDENCE
SP-01	Government and regulation feed	Schedule / subscription	Effective dates, jurisdiction, withdrawal	Coverage and legal-authority review
SP-02	News and media feed	API / RSS / crawl	Publication, update, correction, provenance	Freshness, duplication, manipulation tests
SP-03	Financial market feed	Streaming / batch	Exchange time, sequence, entitlement	Gap, latency, entitlement reconciliation
SP-04	Research and scientific corpus	API / file / repository	Version, authorship, retraction, license	Retraction and citation validation
SP-05	Patent and filing corpus	API / bulk file	Filing identity, amendment, jurisdiction	Completeness and amendment replay
SP-06	Supply-chain feed	API / EDI / file	Parties, shipments, routes, confidence	Identity and transaction reconciliation
SP-07	Satellite and geospatial product	Object / API	Acquisition time, footprint, sensor, processing	Geospatial alignment and custody
SP-08	Climate and weather feed	Streaming / forecast package	Observation versus model, horizon, revision	Temporal and model separation
SP-09	Geopolitical event feed	API / analyst package	Event, claim, assessment, confidence	Truth-state and provenance review
SP-10	Social signal source	API / approved collection	Purpose, minimization, sampling, deletion	Privacy and representativeness review
SP-11	ERP integration	CDC / API / batch	Transaction boundary, business keys, deletes	Source totals and watermark reconciliation
SP-12	CRM integration	CDC / API	Object lifecycle, consent, ownership	Permission and delete propagation
SP-13	Operational database	CDC / snapshot	Log position, transaction order, schema	Gap and resnapshot exercises
SP-14	Document repository	Event / API / bulk	Version, permissions, folders, deletion	Permission and version reconciliation
SP-15	Email and collaboration	API / export	Participants, thread, attachment, retention	Purpose, rights, and access tests
SP-16	IoT and sensor stream	Broker / edge	Device identity, event time, sequence, calibration	Device trust and clock tests
SP-17	Cyber telemetry	Stream / batch	Asset, event, detector, retention, privilege	Security isolation and loss accounting
SP-18	File transfer	SFTP / managed exchange	Manifest, checksum, partitions, receipt	Partial and duplicate transfer recovery
SP-19	Bulk historical load	Object package	Snapshot date, scope, schema, totals	Manifest and reconciliation
SP-20	Custom REST API	Request / pagination	Contract, cursor, rate, retries, rights	Adapter certification suite
SP-21	RSS or Atom feed	Polling	Entry identity, update, deletion, source health	Deduplication and correction
SP-22	Manual governed upload	Human workflow	Uploader, purpose, approval, manifest	Authority and malware inspection
SP-23	Disconnected exchange	Signed package	Trusted time, media, transfer authority	Air-gap ceremony and reconciliation
SP-24	Marketplace source package	Signed install	Publisher, license, permissions, dependencies	Sandbox, revocation, and isolation

D Data Contract and Schema Compatibility Matrix

Compatibility is semantic as well as structural. Stable bytes or fields do not make a change compatible when identity, time, truth state, policy, rights, quality, or consumer meaning changes.

ID	CHANGE	COMPATIBILITY	REQUIRED EVIDENCE	REQUIRED PATH
DC-01	Add optional field	Backward compatible	Consumer default and unknown-field tests	Ordinary controlled release
DC-02	Add required field	Incompatible without population plan	Historical backfill and producer readiness	Parallel version or migration
DC-03	Remove field	Incompatible	Consumer inventory and deprecation evidence	Deprecate then major version
DC-04	Rename field	Incompatible semantic change	Alias and round-trip fixtures	Parallel field or major version
DC-05	Widen numeric type	Conditionally compatible	Range, precision, and consumer tests	Evidence-gated minor version
DC-06	Narrow numeric type	Incompatible	Overflow and historical-value analysis	Major version and migration
DC-07	Change enum values	Conditional	Unknown-value consumer behavior	Additive minor or major version
DC-08	Change semantic definition	Incompatible even if shape is stable	Metric and decision impact analysis	New field or major version
DC-09	Change key or identity	High-risk incompatible	Merge, split, replay, and referential tests	Governed identity migration
DC-10	Change temporal meaning	High-risk incompatible	Historical reconstruction and late-data tests	New contract version
DC-11	Change truth-state rules	High-risk incompatible	Epistemic and downstream fitness tests	Governance approval and major version
DC-12	Change classification or rights	Policy change	Access and export negative tests	Immediate enforcement with impact review
DC-13	Partitioning change	Physically compatible only	Reconciliation, performance, and rollback	Controlled migration
DC-14	Retention change	Lifecycle change	Hold, archive, delete, and replay impact	Policy authority approval
DC-15	Producer replacement	Contract-dependent	Golden fixtures and lineage continuity	Cohort promotion
DC-16	Consumer retirement	Compatible after closure	Subscription, export, and historical replay checks	Recorded retirement

E Canonical Intelligence Object Field Dictionary

The canonical object header travels with every material intelligence object. Specialized payloads may add fields but cannot remove or redefine required identity, time, truth, provenance, policy, quality, correction, and lifecycle semantics.

FIELD	BINDING DEFINITION	PRESENCE RULE
object_id	Stable globally unique object identity.	Required by the canonical object profile
object_type	Canonical intelligence object type.	Required by the canonical object profile
tenant_id	Customer isolation and administrative scope.	Required by the canonical object profile
domain_id	Governed intelligence or business domain scope.	Required by the canonical object profile
object_version	Immutable version identity.	Required by the canonical object profile
lifecycle_state	Proposed, admitted, active, disputed, corrected, withdrawn, archived, or deleted state.	Required by the canonical object profile
owning_component	Canonical layer component with write authority.	Required by the canonical object profile
accountable_owner	Human or organizational owner.	Required by the canonical object profile
source_object_ids	Direct source and parent object versions.	Required by the canonical object profile
event_time	Time the represented event occurred.	Required by the canonical object profile
observation_time	Time evidence was observed or acquired.	Required by the canonical object profile
valid_from	Start of represented validity.	Required by the canonical object profile
valid_to	End of represented validity.	Required by the canonical object profile
recorded_at	Time the platform committed this version.	Required by the canonical object profile
time_precision	Declared temporal precision and ambiguity.	Required by the canonical object profile
source_clock_quality	Quality and trust of source time.	Required by the canonical object profile
truth_state	Observed, asserted, extracted, inferred, assessed, synthetic, decided, disputed, or withdrawn.	Required by the canonical object profile
confidence	Calibrated confidence representation where applicable.	Required by the canonical object profile
uncertainty	Distribution, interval, scenario set, qualitative limits, or unknown state.	Required by the canonical object profile
evidence_refs	Immutable evidence object references.	Required by the canonical object profile
provenance_ref	End-to-end transformation and custody graph reference.	Required by the canonical object profile
method_ref	Rule, code, model, prompt, human process, or instrument version.	Required by the canonical object profile
contradiction_refs	Known incompatible claims or assessments.	Required by the canonical object profile
corroboration_refs	Independent supporting evidence and claims.	Required by the canonical object profile
classification	Sensitivity and handling classification.	Required by the canonical object profile

FIELD	BINDING DEFINITION	PRESENCE RULE
purpose_scope	Permitted processing purpose.	Required by the canonical object profile
rights_profile	License, consent, contractual, and redistribution rights.	Required by the canonical object profile
residency_profile	Storage and processing locality obligations.	Required by the canonical object profile
retention_profile	Retention, legal hold, archive, and deletion policy.	Required by the canonical object profile
access_policy_ref	Versioned authorization policy.	Required by the canonical object profile
quality_profile	Applicable data-quality contract.	Required by the canonical object profile
quality_state	Observed quality results and exceptions.	Required by the canonical object profile
freshness_state	Age, watermark, expiry, and staleness state.	Required by the canonical object profile
schema_ref	Canonical payload schema and version.	Required by the canonical object profile
ontology_ref	Semantic model and version.	Required by the canonical object profile
correction_of	Prior object version corrected by this version.	Required by the canonical object profile
supersedes	Prior version or object replaced for current use.	Required by the canonical object profile
withdrawal_reason	Reason and authority for withdrawal.	Required by the canonical object profile
audit_correlation_id	Correlation to protected audit and workflow evidence.	Required by the canonical object profile
content_ref	Payload or large-object content reference.	Required by the canonical object profile

F Temporal, Truth-State, and Provenance Profiles

Profiles prevent observed, asserted, inferred, assessed, synthetic, recommended, decided, corrected, and withdrawn state from collapsing into one undifferentiated record.

ID	OBJECT PROFILE	TRUTH STATE	TEMPORAL CONTRACT	VERSION BEHAVIOR
TT-01	Immutable observation	Observed	Event + observation + record time	Correction creates a new version
TT-02	Source assertion	Asserted	Assertion validity + observation	Withdrawal preserves history
TT-03	Extracted claim	Extracted	Evidence time + extraction record	Re-extraction remains a new version
TT-04	Inferred relationship	Inferred	Evidence validity + inference record	Model or evidence change triggers reassessment
TT-05	Human assessment	Assessed	Assessment effective window	Dissent and supersession retained
TT-06	Canonical entity	Stewarded identity	Identity validity interval	Merge and split are reversible
TT-07	Temporal relationship	Observed / asserted / inferred	Valid-from and valid-to	Overlaps require typed rules
TT-08	Metric observation	Observed calculation	Measurement window + record time	Definition version is mandatory
TT-09	Forecast package	Model output	As-of time + horizon	Expiry and withdrawal are explicit
TT-10	Scenario branch	Synthetic possibility	Branch base time + review window	Never merges into observed state
TT-11	Simulation run	Synthetic experiment	Resolved initial state + run time	Invalidation preserves prior result
TT-12	Recommendation	Advisory output	Decision context time	Does not become a decision
TT-13	Human decision	Decided	Decision time + effective time	Later evidence does not rewrite it
TT-14	Commitment	Institutional obligation	Effective interval + due dates	Closure and breach are versioned
TT-15	Outcome observation	Observed result	Outcome event + observation time	Links to prior commitments and decisions
TT-16	Correction	Corrective authority	Correction effective + record time	Preserves affected history
TT-17	Withdrawal	No longer decision-active	Withdrawal effective time	Prior use remains replayable
TT-18	Synthetic training data	Synthetic	Generation snapshot + record time	Purpose-isolated from observed truth
TT-19	Stale projection	Derived stale state	Source revision + build time	Cannot represent current authority
TT-20	Deleted record marker	Lifecycle tombstone	Deletion effective + verified time	Retains only permitted accountability metadata

G Data Product and Serving Pattern Catalog

Data products are reusable accountable operational contracts. APIs, events, tables, graphs, search, features, context, briefings, and exports are serving modes inside those contracts.

ID	PATTERN	SERVING MODE	OWNER	ACCEPTANCE FOCUS
DPD-01	Canonical object product	Typed object API + events	Canonical component	Object lifecycle and policy fixtures
DPD-02	Observation evidence product	Object + metadata	World Observation	Custody, rights, freshness, correction
DPD-03	Curated domain dataset	Open table + contract	Domain data owner	Quality, lineage, SLO, portability
DPD-04	Knowledge Graph product	Graph query + change events	Knowledge Graph	Temporal, provenance, policy, revision
DPD-05	Enterprise Memory product	Purpose-aware retrieval	Enterprise Memory	Permission, version, retention, deletion
DPD-06	Digital Twin snapshot product	Snapshot + change events	Digital Twins	Observed/synthetic isolation and validation
DPD-07	Forecast data product	Forecast package	Prediction Engine	Horizon, calibration, lineage, expiry
DPD-08	Scenario portfolio product	Versioned branch package	Scenario Intelligence	Plurality, assumptions, signposts
DPD-09	Simulation result product	Run package	Simulation Engine	Resolved versions, seeds, sensitivity
DPD-10	Decision package product	Review and approval package	Decision Intelligence	Alternatives, dissent, human authority
DPD-11	Executive briefing product	Live briefing + events	Executive Operating System	Evidence, materiality, acknowledgement
DPD-12	Certified metric product	Semantic query	Metric owner	Definition, grain, source revision
DPD-13	Event signal product	Subscription stream	Producing domain	Ordering, replay, correction, lag
DPD-14	Search index product	Search API	Source product owner	Relevance, permission, freshness, rebuild
DPD-15	Vector retrieval product	Semantic retrieval API	Source product owner	Embedding version, citation, rebuild
DPD-16	Feature product	Online + offline feature	AI data owner	Point-in-time correctness and drift
DPD-17	Evaluation dataset product	Versioned dataset	Evaluation authority	Independence, slices, leakage, expiry
DPD-18	Context manifest product	Bounded context package	Context control plane	Purpose, policy, omissions, citations
DPD-19	Customer export product	Signed open package	Customer data authority	Completeness, integrity, re-import
DPD-20	Marketplace data package	Signed installable package	Publisher + governance	License, sandbox, evaluation, revocation

H Data Asset Manifest Field Dictionary

The data asset manifest binds each governed dataset, stream, object family, graph, index, feature, product, exchange, or evidence collection to exact authority, semantics, state, policy, quality, lifecycle, recovery, deployment, and evidence.

FIELD	BINDING DEFINITION	PRESENCE RULE
asset_id	Stable governed data asset identity.	Required unless contractually inapplicable
asset_name	Canonical human-readable name.	Required unless contractually inapplicable
asset_type	Dataset, stream, object, graph, index, feature, product, export, or evidence type.	Required unless contractually inapplicable
domain_id	Owning data domain.	Required unless contractually inapplicable
canonical_layer	Owning or consuming canonical layer.	Required unless contractually inapplicable
accountable_owner	Named organizational owner.	Required unless contractually inapplicable
data_steward	Responsible stewardship authority.	Required unless contractually inapplicable
technical_custodian	Platform team operating physical data state.	Required unless contractually inapplicable
purpose	Approved business and strategic purpose.	Required unless contractually inapplicable
consumer_registry	Approved material consumers and contracts.	Required unless contractually inapplicable
authority_class	Authoritative, evidence, product, derived, temporary, or recovery state.	Required unless contractually inapplicable
source_contracts	Governing source and collection contracts.	Required unless contractually inapplicable
data_contract	Machine-readable producer and consumer contract.	Required unless contractually inapplicable
schema_refs	Payload schema identities and versions.	Required unless contractually inapplicable
ontology_refs	Semantic definitions and mappings.	Required unless contractually inapplicable
identity_keys	Canonical and natural key definitions.	Required unless contractually inapplicable
temporal_profile	Event, observation, validity, and record-time semantics.	Required unless contractually inapplicable
truth_state_profile	Permitted epistemic states and promotion rules.	Required unless contractually inapplicable
provenance_requirement	Required lineage and custody closure.	Required unless contractually inapplicable
classification	Sensitivity and handling classification.	Required unless contractually inapplicable
purpose_policy	Permitted use and secondary-use restrictions.	Required unless contractually inapplicable
rights_profile	License, consent, contract, and redistribution rules.	Required unless contractually inapplicable
residency_profile	Storage and processing locality.	Required unless contractually inapplicable
encryption_profile	At-rest, in-transit, in-use, and key requirements.	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
access_policy	Versioned authorization policy.	Required unless contractually inapplicable
quality_contract	Dimensions, rules, thresholds, and exceptions.	Required unless contractually inapplicable
freshness_contract	Expected arrival, watermark, expiry, and staleness.	Required unless contractually inapplicable
slo_profile	Availability, latency, durability, and recovery objectives.	Required unless contractually inapplicable
storage_class	Authority, consistency, durability, and recovery tier.	Required unless contractually inapplicable
physical_locations	Resolved stores, topics, tables, indexes, and archives.	Required unless contractually inapplicable
partitioning	Partition, shard, and distribution contract.	Required unless contractually inapplicable
retention_policy	Retention, archive, hold, and deletion rules.	Required unless contractually inapplicable
backup_profile	Recovery point, time, locality, and verification.	Required unless contractually inapplicable
correction_policy	Correction, withdrawal, supersession, and replay rules.	Required unless contractually inapplicable
export_profile	Permitted formats, scope closure, and destination controls.	Required unless contractually inapplicable
cost_center	Cost and resource attribution.	Required unless contractually inapplicable
release_id	Dependency-closed released data version.	Required unless contractually inapplicable
deployment_profiles	Certified SaaS, Private Cloud, and On-Premise behavior.	Required unless contractually inapplicable
lifecycle_state	Proposed through retired state.	Required unless contractually inapplicable
evidence_refs	Conformance, operational, quality, security, and acceptance evidence.	Required unless contractually inapplicable

I Data Pipeline Manifest Field Dictionary

The data pipeline manifest binds every batch, stream, CDC, exchange, migration, and lifecycle execution to exact inputs, outputs, artifacts, policies, checkpoints, failure behavior, release, and evidence.

FIELD	BINDING DEFINITION	PRESENCE RULE
pipeline_id	Stable pipeline identity.	Required unless contractually inapplicable
pipeline_version	Immutable executable definition version.	Required unless contractually inapplicable
owner	Accountable pipeline owner.	Required unless contractually inapplicable
purpose	Declared transformation or movement purpose.	Required unless contractually inapplicable
execution_mode	Batch, stream, CDC, interactive, exchange, or lifecycle mode.	Required unless contractually inapplicable
input_assets	Exact input asset contracts and versions.	Required unless contractually inapplicable
output_assets	Exact output asset contracts and versions.	Required unless contractually inapplicable
code_artifacts	Signed code, image, package, and dependency identities.	Required unless contractually inapplicable
configuration	Versioned non-secret configuration.	Required unless contractually inapplicable
schema_bindings	Input, intermediate, and output schema versions.	Required unless contractually inapplicable
ontology_bindings	Semantic models and mappings.	Required unless contractually inapplicable
policy_bindings	Access, purpose, classification, and locality policies.	Required unless contractually inapplicable
identity_profile	Workload identity and delegation requirements.	Required unless contractually inapplicable
secret_refs	Identity-bound credential references.	Required unless contractually inapplicable
network_profile	Permitted sources, services, and destinations.	Required unless contractually inapplicable
placement_profile	Tenant, domain, locality, and infrastructure constraints.	Required unless contractually inapplicable
resource_profile	Compute, memory, accelerator, storage, and transfer envelope.	Required unless contractually inapplicable
trigger	Schedule, event, subscription, command, or lifecycle trigger.	Required unless contractually inapplicable
ordering	Partition and ordering semantics.	Required unless contractually inapplicable
idempotency	Duplicate detection and effect identity.	Required unless contractually inapplicable
checkpointing	Committed recovery boundaries and compatible state.	Required unless contractually inapplicable
retry_policy	Bounded retries, backoff, and exhaustion behavior.	Required unless contractually inapplicable
compensation	Rollback, retraction, or forward-repair behavior.	Required unless contractually inapplicable
late_data_policy	Watermarks, windows, lateness, and retractions.	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
quality_gates	Pre-publication rules and thresholds.	Required unless contractually inapplicable
lineage_profile	Required input-output and execution evidence.	Required unless contractually inapplicable
observability	Metrics, logs, traces, data observations, and correlation.	Required unless contractually inapplicable
slo_profile	Latency, throughput, quality, freshness, and recovery objectives.	Required unless contractually inapplicable
failure_modes	Declared normal, partial, failed, recovering, and terminal states.	Required unless contractually inapplicable
degraded_behavior	Explicit output suppression, partial state, and user impact.	Required unless contractually inapplicable
backfill_plan	Historical scope, partitioning, validation, and reconciliation.	Required unless contractually inapplicable
migration_plan	Dual operation, cutover, compatibility, and closure.	Required unless contractually inapplicable
rollback_target	Last compatible release and state boundary.	Required unless contractually inapplicable
deployment_profiles	Certified physical realizations and parity fixtures.	Required unless contractually inapplicable
release_id	Dependency-closed release identity.	Required unless contractually inapplicable
test_evidence	Contract, quality, security, scale, failure, and recovery evidence.	Required unless contractually inapplicable
approval_state	Required technical, data, policy, and customer approvals.	Required unless contractually inapplicable
operational_runbook	Diagnosis, pause, resume, reconcile, recover, and retire procedures.	Required unless contractually inapplicable
cost_budget	Resource, transfer, storage, and cost controls.	Required unless contractually inapplicable
lifecycle_state	Proposed, released, degraded, withdrawn, superseded, or retired.	Required unless contractually inapplicable

J Data Quality and SLO Metric Catalog

Metrics establish a common operating vocabulary for contracts, sources, temporal truth, provenance, quality, graph, retrieval, pipelines, recovery, portability, economics, and deployment parity. Every SLO profile declares population, window, exclusions, threshold, budget, action, owner, and evidence.

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
DQM-001	Contract conformance rate	Records, objects, or events satisfying the active data contract.	Asset + tenant + deployment + product + consumer + window
DQM-002	Schema rejection rate	Inputs rejected for schema or constraint violation.	Asset + tenant + deployment + product + consumer + window
DQM-003	Source freshness	Age of the latest accepted source observation against contract.	Asset + tenant + deployment + product + consumer + window
DQM-004	Arrival completeness	Expected versus received partitions, records, or events.	Asset + tenant + deployment + product + consumer + window
DQM-005	Coverage ratio	Observed governed universe versus declared coverage universe.	Asset + tenant + deployment + product + consumer + window
DQM-006	Blind-spot duration	Time a material coverage gap remains unresolved.	Asset + tenant + deployment + product + consumer + window
DQM-007	Validity rate	Values satisfying type, range, format, and business constraints.	Asset + tenant + deployment + product + consumer + window
DQM-008	Accuracy estimate	Verified correspondence to authoritative or sampled ground truth.	Asset + tenant + deployment + product + consumer + window
DQM-009	Uniqueness rate	Records unique under the declared identity and grain.	Asset + tenant + deployment + product + consumer + window
DQM-010	Consistency rate	Cross-field, cross-source, and cross-product constraint agreement.	Asset + tenant + deployment + product + consumer + window
DQM-011	Referential integrity	Resolvable governed references among material objects.	Asset + tenant + deployment + product + consumer + window
DQM-012	Temporal completeness	Objects carrying required event, observation, validity, and record time.	Asset + tenant + deployment + product + consumer + window
DQM-013	Temporal anomaly rate	Invalid intervals, ordering, clock, or future-time anomalies.	Asset + tenant + deployment + product + consumer + window
DQM-014	Truth-state completeness	Material objects carrying valid epistemic labels.	Asset + tenant + deployment + product + consumer + window
DQM-015	Provenance completeness	Material outputs with closed lineage to evidence.	Asset + tenant + deployment + product + consumer + window
DQM-016	Custody integrity	Evidence transfers with verified digest and custody continuity.	Asset + tenant + deployment + product + consumer + window
DQM-017	Classification completeness	Assets and records with enforceable classification.	Asset + tenant + deployment + product + consumer + window
DQM-018	Policy enforcement success	Authorized operations satisfying all returned obligations.	Asset + tenant + deployment + product + consumer + window
DQM-019	Unauthorized access denial	Negative access attempts denied and evidenced.	Asset + tenant + deployment + product + consumer + window
DQM-020	Correction propagation time	Time from correction admission to material consumer closure.	Asset + tenant + deployment + product + consumer + window
DQM-021	Stale projection rate	Queries served from projections beyond freshness contract.	Asset + tenant + deployment + product + consumer + window
DQM-022	Projection reconciliation gap	Difference between authoritative and derived state.	Asset + tenant + deployment + product + consumer + window
DQM-023	Entity resolution precision	Accepted identity links that are correct.	Asset + tenant + deployment + product + consumer + window
DQM-024	Entity resolution recall	Resolvable identities successfully linked.	Asset + tenant + deployment + product + consumer + window

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
DQM-025	Graph edge validity	Edges satisfying type, time, provenance, and constraint rules.	Asset + tenant + deployment + product + consumer + window
DQM-026	Graph orphan rate	Canonical graph objects lacking required connections.	Asset + tenant + deployment + product + consumer + window
DQM-027	Retrieval permission correctness	Results satisfying object- and purpose-level policy.	Asset + tenant + deployment + product + consumer + window
DQM-028	Retrieval relevance	Task-appropriate relevance by governed evaluation set.	Asset + tenant + deployment + product + consumer + window
DQM-029	Citation closure	Returned claims and context linked to accessible evidence.	Asset + tenant + deployment + product + consumer + window
DQM-030	Feature point-in-time correctness	Feature values excluding future information leakage.	Asset + tenant + deployment + product + consumer + window
DQM-031	Dataset leakage rate	Training or evaluation contamination and target leakage.	Asset + tenant + deployment + product + consumer + window
DQM-032	Pipeline success rate	Runs reaching a reconciled terminal state.	Asset + tenant + deployment + product + consumer + window
DQM-033	Pipeline recovery time	Time to resume from a committed compatible boundary.	Asset + tenant + deployment + product + consumer + window
DQM-034	Stream end-to-end latency	Event time to governed product availability.	Asset + tenant + deployment + product + consumer + window
DQM-035	Consumer lag	Committed source position minus consumer checkpoint.	Asset + tenant + deployment + product + consumer + window
DQM-036	Query latency	Policy-complete user-visible query response time.	Asset + tenant + deployment + product + consumer + window
DQM-037	Data durability	Committed authoritative data retained through failure.	Asset + tenant + deployment + product + consumer + window
DQM-038	Restore verification rate	Recovery sets automatically restored and reconciled.	Asset + tenant + deployment + product + consumer + window
DQM-039	Export round-trip integrity	Exported assets re-imported without semantic loss.	Asset + tenant + deployment + product + consumer + window
DQM-040	Unit data cost	Storage, processing, transfer, and operations cost per governed unit.	Asset + tenant + deployment + product + consumer + window

K Failure Mode and Degraded Data State Matrix

The baseline failure matrix makes containment, staleness, incompleteness, quarantine, correction, recovery, reconciliation, and consumer impact explicit. Domain analyses may add detail but cannot remove these control families.

ID	FAILURE	PRIMARY DOMAIN	REQUIRED RESPONSE	MINIMUM EXERCISE
DF-01	Source unavailable	Source	Expose coverage and freshness loss; retry by contract	Source outage exercise
DF-02	Source authority revoked	Source contract	Stop collection and downstream promotion	Revocation drill
DF-03	Malicious or corrupt content	Intake	Quarantine and preserve custody	Content corpus test
DF-04	Partial file transfer	Exchange	Hold publication and resume from manifest	Interrupted transfer
DF-05	CDC log gap	Synchronization	Pause, resnapshot bounded scope, reconcile	Log-gap exercise
DF-06	Stream overload	Streaming	Backpressure or explicit consequence-based shedding	Burst test
DF-07	Clock discontinuity	Temporal	Mark time quality and constrain affected use	Clock-skew test
DF-08	Schema incompatibility	Contract	Block producer or consumer promotion	Compatibility fixture
DF-09	Identity collision	Identity	Fail closed and route stewardship	Collision test
DF-10	Truth-state collapse	Semantics	Withdraw affected outputs and restore labels	Epistemic negative test
DF-11	Lineage break	Provenance	Mark output unfit and recompute	Lineage closure test
DF-12	Correction propagation stall	Lifecycle	Freeze affected publication and escalate consumers	Correction fan-out
DF-13	Lakehouse metadata divergence	Analytical state	Fence writers and restore consistent snapshot	Transaction recovery
DF-14	Operational database quorum loss	Transactional state	Fence writes and recover consistent owner	Quorum and fencing
DF-15	Evidence object corruption	Evidence	Quarantine reference and restore verified replica	Digest corruption drill
DF-16	Broker quorum loss	Messaging	Fail over, preserve offsets, reconcile	Broker recovery
DF-17	Poison message recurrence	Messaging	Isolate, dead-letter, and bound retries	Poison message test
DF-18	Partial batch commit	Processing	Stop publication and reconcile partitions	Partial commit recovery
DF-19	Stream checkpoint corruption	Processing	Restore compatible checkpoint and replay	State recovery
DF-20	Stale or poisoned index	Retrieval	Withdraw and rebuild from canonical sources	Index rebuild
DF-21	Graph revision conflict	Knowledge Graph	Reject atomic change and reconcile log	Revision conflict test
DF-22	Memory permission drift	Enterprise Memory	Restrict access and rebuild policy projection	Permission recovery
DF-23	Twin branch leakage	Digital Twins	Contain branch and invalidate affected outputs	Branch isolation test
DF-24	Unauthorized data access	Trust	Deny, revoke, investigate, preserve audit	Negative authorization
DF-25	Residency violation attempt	Sovereignty	Deny transfer or placement	Locality test
DF-26	Key or token service failure	Confidentiality	Stop protected processing or use approved continuity	Key continuity exercise
DF-27	Deletion scope uncertainty	Lifecycle	Pause deletion and resolve dependency closure	Delete-scope test
DF-28	Quality contract breach	Data product	Declare degradation and quarantine affected scope	Quality incident
DF-29	Observability loss	Operations	Constrain risky change and preserve priority evidence	Telemetry loss
DF-30	Backup corruption	Recovery	Protect remaining copies and select alternate set	Backup corruption
DF-31	Migration divergence	Release	Stop cohort and reconcile old and new state	Migration rollback

ID	FAILURE	PRIMARY DOMAIN	REQUIRED RESPONSE	MINIMUM EXERCISE
DF-32	Disconnected sync conflict	Deployment	Preserve both histories and apply governed reconciliation	Offline conflict drill

Degraded-data command rules

COMMAND RULE	REQUIRED OPERATING BEHAVIOR
Declare before serving	Every degraded data mode identifies its trigger, authority, available and unavailable data, truth and freshness limits, affected consumers, maximum duration, and exit evidence.
Protect strategic meaning	The system does not represent missing evidence, stale sources, partial pipelines, broken lineage, failed policy, invalid models, or unreconciled recovery as complete strategic data.
Constrain propagation	Publication, training, retrieval, warning, export, and authoritative effects stop when their data authority, purpose, version, quality, or reconciliation boundary cannot be established.
Preserve evidence	Entry, suppressed work, lost observations, consumer impact, human decisions, corrections, recovery, and exit remain correlated to affected data and releases.
Exit through proof	Normal service resumes only after authoritative state, contracts, schemas, policy, lineage, quality, projections, consumers, recovery, and user-visible semantics reconcile.

FAILURE DOCTRINE

Availability is never manufactured by hiding data degradation. When the safe interpretation is uncertain, The Eye selects the more restrictive consequence class, exposes the limitation, and restores service through verified state.

L Data Security, Privacy, and Residency Control Matrix

Data controls are enforced through independent identity, policy, storage, processing, serving, lifecycle, and audit boundaries. Applications, agents, models, providers, and administrators cannot waive their authority.

ID	CONTROL	ENFORCEMENT BOUNDARY	REQUIRED EVIDENCE
DCN-01	Source authority before collection	Source registry and connector admission	Contract, rights, purpose, and approval
DCN-02	Content quarantine	Intake boundary	Inspection, isolation, release authority
DCN-03	Immutable raw preservation	Evidence store	Digest, version, custody, retention
DCN-04	Canonical contract validation	Producer and consumer boundary	Schema and contract results
DCN-05	Tenant and domain scoping	Every asset and request	Cross-tenant negative tests
DCN-06	Temporal truth validation	Canonical admission	Bitemporal fixtures and clock quality
DCN-07	Truth-state separation	Object and graph admission	Observed, inferred, assessed, synthetic tests
DCN-08	End-to-end lineage	Every material transformation	Closed input-output provenance
DCN-09	Correction propagation	Lifecycle controller	Impact, acknowledgement, and closure
DCN-10	State-class enforcement	Storage admission	Authority, consistency, locality, recovery
DCN-11	Derived-state rebuildability	Index, cache, projection	Source revision and rebuild evidence
DCN-12	Service-owned writes	Operational stores	Cross-owner write denial
DCN-13	Messaging idempotency	Producer and consumer	Effect identity and replay reconciliation
DCN-14	Graph revision atomicity	Knowledge Graph write boundary	Constraint, provenance, and rollback
DCN-15	Memory permission enforcement	Retrieval and archive	Purpose-aware object filtering
DCN-16	Twin branch isolation	Twin and simulation boundary	Observed versus synthetic separation
DCN-17	Purpose-aware access	API, query, pipeline, agent, model	Policy decision and obligation receipt
DCN-18	Data minimization	Collection, processing, context, telemetry	Field and content minimization evidence
DCN-19	Residency enforcement	Placement, transfer, backup, support	End-to-end locality proof
DCN-20	Encryption at rest	Every protected state class	Key binding and posture
DCN-21	Encryption in transit	Every protected data path	Protocol and certificate posture
DCN-22	Tokenization and masking	Declared protected uses	Transformation and re-identification tests
DCN-23	Retention and legal hold	Lifecycle boundary	Timer, hold precedence, and evidence
DCN-24	Verifiable deletion	Canonical and derived stores	Residual reference closure
DCN-25	Tamper-evident data audit	Material access and change	Integrity anchors and custody
DCN-26	Data quality gates	Pre-publication and serving	Rule, threshold, exception, remediation
DCN-27	Freshness and coverage warnings	Products and executive surfaces	Expiry and blind-spot propagation
DCN-28	Data SLO enforcement	User-visible data journeys	Indicators, budgets, and action
DCN-29	Recovery verification	Every durable state class	Automated restore and reconciliation
DCN-30	Dependency-closed data release	Build and promotion	Contracts, schemas, code, policy, data, infrastructure
DCN-31	Deployment parity	SaaS, Private Cloud, On-Premise	Common semantic fixtures
DCN-32	Human authority for material adjudication	Stewardship, exceptions, incidents	Authenticated approval and rationale

M Lifecycle, Retention, Deletion, and Recovery Matrix

Lifecycle and recovery objectives are defined per governed data class. Copied bytes are not recovered data until contracts, schemas, identity, time, truth, provenance, policies, keys, quality, projections, consumers, sovereignty, and historical replay reconcile.

ID	LIFECYCLE OR RECOVERY UNIT	RPO / DISPOSITION	RTO	BASIS	MINIMUM EVIDENCE
LR-01	Source contracts	Policy history retained	Hours	Versioned registry	Contract and revocation recovery
LR-02	Raw evidence	No acknowledged loss where required	Hours	Immutable replicated objects	Digest restore
LR-03	Canonical objects	Last committed version	Minutes to hours	Transactional history	Point-in-time restore
LR-04	Data contracts and schemas	No released identity loss	Minutes	Signed registry copies	Registry recovery
LR-05	Lineage and custody	No material committed loss	Hours	Protected event log	Closure reconstruction
LR-06	Lakehouse metadata	Last committed snapshot	Hours	Transaction log and manifests	Snapshot restore
LR-07	Analytical data files	Contract-specific	Hours to days	Replicated objects and manifests	Partition restore
LR-08	Operational databases	Seconds to hours	Minutes to hours	Logs and snapshots	Point-in-time restore
LR-09	Event streams	Committed offset	Minutes to hours	Replicated logs	Broker replay
LR-10	Pipeline checkpoints	Last compatible boundary	Minutes	Checkpoint store	Resume and reconcile
LR-11	Knowledge Graph	Declared revision	Hours	Revision log and snapshots	Graph replay
LR-12	Enterprise Memory	Declared object revision	Hours	Canonical objects and policy state	Temporal restore
LR-13	Search indexes	Rebuildable	Hours to days	Canonical sources and definitions	Deterministic rebuild
LR-14	Vector indexes	Rebuildable	Hours to days	Source snapshots and model revision	Embedding rebuild
LR-15	Digital Twins	Last consistent snapshot	Hours	Snapshots, events, model bindings	Twin reconciliation
LR-16	Strategic packages	No committed decision loss	Hours	Canonical records and evidence	Decision-path recovery
LR-17	Feature data	Point-in-time contract	Hours	Source snapshots and feature code	Feature rebuild
LR-18	AI datasets	Released dataset snapshot	Hours to days	Immutable manifests and objects	Dataset reconstruction
LR-19	Policy and classification	Last approved revision	Minutes	Signed bundles and registry	Fail-closed restore
LR-20	Retention and legal holds	No active hold loss	Hours	Protected lifecycle registry	Hold reconciliation
LR-21	Data audit	No committed audit loss	Hours	Immutable independent copies	Integrity verification
LR-22	Data quality evidence	Bounded observations	Hours	Rules, results, and exceptions	Scorecard recovery
LR-23	Customer exports	Published package retained by policy	Hours	Signed manifest and object set	Package verification
LR-24	Air-gapped deployment	Local policy objective	Hours to days	Signed offline recovery media	No-network recovery

N Requirement and Cross-Volume Traceability

Traceability resolves every stable DP requirement range upward to constitutional invariants and across the controlled engineering, AI, and infrastructure baselines. Individual requirements inherit every anchor listed for their chapter.

CH.	DATA SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTION	V4	V5	V6
1	Data Platform Authority and Scope	DP-01-001-DP-01-006	6	C-001, C-003, C-004, C-008, C-009, C-052	V4:1, V4:4, V4:6, V4:11, V4:72	V5:1, V5:6, V5:8, V5:13, V5:72	V6:1, V6:3, V6:6, V6:31, V6:72
2	Specification Hierarchy and Data Conformance	DP-02-001-DP-02-006	6	C-003, C-008, C-035, C-051, C-052	V4:2, V4:3, V4:6, V4:23, V4:72	V5:2, V5:6, V5:14, V5:70, V5:72	V6:2, V6:6, V6:48, V6:68, V6:72
3	Data Principles and Semantic Parity	DP-03-001-DP-03-006	6	C-007, C-011, C-012, C-038, C-042, C-051	V4:5, V4:14, V4:23, V4:24, V4:30	V5:7, V5:13, V5:20, V5:26, V5:71	V6:3, V6:8, V6:9, V6:10, V6:71
4	Data Ownership, Stewardship, and Accountability	DP-04-001-DP-04-006	6	C-004, C-006, C-016, C-036, C-037, C-049	V4:8, V4:11, V4:12, V4:29, V4:55	V5:3, V5:25, V5:37, V5:47, V5:65	V6:4, V6:13, V6:49, V6:57, V6:71
5	Data Domain and Product Operating Model	DP-05-001-DP-05-006	6	C-006, C-007, C-008, C-009, C-038, C-049	V4:4, V4:11, V4:15, V4:23, V4:31, V4:40	V5:8, V5:9, V5:13, V5:25, V5:27	V6:3, V6:31, V6:34, V6:35, V6:48
6	Data Classification and Consequence Model	DP-06-001-DP-06-006	6	C-010, C-012, C-029, C-035, C-039, C-040	V4:13, V4:24, V4:27, V4:29, V4:50, V4:54	V5:19, V5:20, V5:21, V5:29, V5:61	V6:13, V6:50, V6:53, V6:54, V6:57
7	Data Traceability and Documentation Inheritance	DP-07-001-DP-07-006	6	C-003, C-012, C-036, C-043, C-050, C-052	V4:6, V4:23, V4:28, V4:67, V4:72	V5:6, V5:14, V5:29, V5:65, V5:70	V6:6, V6:45, V6:48, V6:68, V6:72
8	Data Platform System Context	DP-08-001-DP-08-006	6	C-002, C-008, C-009, C-035, C-038, C-042	V4:7, V4:9, V4:13, V4:15, V4:19, V4:31	V5:7, V5:8, V5:9, V5:13, V5:29	V6:7, V6:12, V6:14, V6:23, V6:31, V6:48
9	Source Registry and Collection Contracts	DP-09-001-DP-09-006	6	C-012, C-013, C-014, C-035, C-037, C-040	V4:15, V4:23, V4:29, V4:31, V4:54	V5:29, V5:39, V5:40, V5:61, V5:65	V6:21, V6:28, V6:47, V6:50, V6:57
10	External Source Ingestion	DP-10-001-DP-10-006	6	C-012, C-013, C-014, C-032, C-035, C-043	V4:19, V4:22, V4:28, V4:31, V4:57	V5:39, V5:40, V5:41, V5:57, V5:60	V6:21, V6:28, V6:30, V6:33, V6:47
11	Enterprise Application and Database Ingestion	DP-11-001-DP-11-006	6	C-011, C-012, C-014, C-037, C-038, C-043	V4:12, V4:18, V4:22, V4:25, V4:31	V5:25, V5:26, V5:39, V5:41, V5:61	V6:28, V6:29, V6:32, V6:37, V6:47
12	Document, Media, and Unstructured Content Ingestion	DP-12-001-DP-12-006	6	C-010, C-012, C-014, C-016, C-035, C-045	V4:21, V4:24, V4:28, V4:31, V4:32	V5:19, V5:29, V5:41, V5:42, V5:60	V6:33, V6:44, V6:45, V6:47, V6:53
13	Event, Stream, IoT, and Telemetry Ingestion	DP-13-001-DP-13-006	6	C-011, C-012, C-014, C-035, C-039, C-043	V4:19, V4:22, V4:26, V4:31, V4:57, V4:58	V5:39, V5:44, V5:57, V5:60, V5:68	V6:20, V6:21, V6:27, V6:30, V6:37
14	Batch, File, API, RSS, and Custom Source Ingestion	DP-14-001-DP-14-006	6	C-012, C-013, C-014, C-038, C-039, C-043	V4:16, V4:17, V4:20, V4:22, V4:23, V4:31	V5:17, V5:36, V5:39, V5:40, V5:60	V6:28, V6:37, V6:47, V6:48, V6:55
15	Change Data Capture and Synchronization	DP-15-001-DP-15-006	6	C-011, C-012, C-014, C-030, C-038, C-043	V4:19, V4:22, V4:26, V4:30, V4:31	V5:26, V5:27, V5:38, V5:39, V5:68	V6:20, V6:27, V6:32, V6:37, V6:62
16	Intake Validation, Quarantine, and Raw Preservation	DP-16-001-DP-16-006	6	C-010, C-012, C-035, C-039, C-040, C-043	V4:13, V4:21, V4:24, V4:28, V4:31, V4:56	V5:20, V5:21, V5:29, V5:39, V5:60	V6:33, V6:47, V6:53, V6:54, V6:56
17	Canonical Data Contract	DP-17-001-DP-17-006	6	C-010, C-011, C-012, C-014, C-035, C-038	V4:15, V4:20, V4:23, V4:24, V4:28	V5:19, V5:20, V5:26, V5:29, V5:63	V6:31, V6:32, V6:37, V6:48, V6:59
18	Schema Registry and Compatibility	DP-18-001-DP-18-006	6	C-011, C-012, C-035, C-038, C-043, C-052	V4:20, V4:23, V4:24, V4:30, V4:67	V5:18, V5:20, V5:26, V5:70	V6:32, V6:37, V6:45, V6:48, V6:68
19	Canonical Intelligence Object Model	DP-19-001-DP-19-006	6	C-008, C-009, C-010, C-011, C-012, C-035	V4:20, V4:24, V4:25, V4:26, V4:27, V4:28	V5:13, V5:19, V5:20, V5:26, V5:29	V6:31, V6:33, V6:34, V6:35, V6:36
20	Identity, Naming, and Tenant Scoping	DP-20-001-DP-20-006	6	C-007, C-009, C-011, C-016, C-035, C-039	V4:25, V4:30, V4:34, V4:51, V4:55	V5:27, V5:29, V5:43, V5:45, V5:65	V6:13, V6:34, V6:49, V6:54, V6:57
21	Temporal and Bitemporal Data	DP-21-001-DP-21-006	6	C-010, C-011, C-012, C-025, C-035, C-043	V4:19, V4:26, V4:30, V4:48, V4:64	V5:26, V5:29, V5:38, V5:48, V5:63	V6:27, V6:31, V6:34, V6:36, V6:58

CH.	DATA SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTION	V4	V5	V6
22	Truth State, Confidence, and Uncertainty	DP-22-001-DP-22-006	6	C-005, C-010, C-012, C-021, C-035, C-045	V4:21, V4:27, V4:28, V4:32, V4:46	V5:4, V5:20, V5:29, V5:46, V5:63, V5:64	V6:31, V6:33, V6:35, V6:44, V6:59
23	Provenance, Lineage, and Chain of Custody	DP-23-001-DP-23-006	6	C-005, C-012, C-025, C-035, C-036, C-043	V4:6, V4:24, V4:28, V4:30, V4:55	V5:4, V5:18, V5:29, V5:46, V5:65	V6:33, V6:45, V6:48, V6:55, V6:58
24	Correction, Withdrawal, Supersession, and Reconciliation	DP-24-001-DP-24-006	6	C-011, C-012, C-015, C-025, C-030, C-043	V4:20, V4:26, V4:28, V4:30, V4:48	V5:26, V5:29, V5:38, V5:68, V5:69	V6:31, V6:32, V6:34, V6:35, V6:62
25	Data Storage Architecture and State Tiers	DP-25-001-DP-25-006	6	C-011, C-015, C-016, C-035, C-037, C-043	V4:12, V4:24, V4:30, V4:57, V4:61	V5:13, V5:25, V5:26, V5:29, V5:71	V6:31, V6:32, V6:33, V6:38, V6:62
26	Lakehouse and Analytical Storage	DP-26-001-DP-26-006	6	C-011, C-015, C-035, C-037, C-038, C-043	V4:12, V4:18, V4:23, V4:30, V4:58	V5:19, V5:23, V5:25, V5:26, V5:66	V6:31, V6:33, V6:35, V6:45, V6:60
27	Transactional and Operational Data Stores	DP-27-001-DP-27-006	6	C-006, C-011, C-016, C-035, C-039, C-043	V4:11, V4:12, V4:22, V4:30, V4:59	V5:10, V5:15, V5:25, V5:38, V5:70	V6:31, V6:32, V6:61, V6:62, V6:65
28	Object, Evidence, and Archive Storage	DP-28-001-DP-28-006	6	C-012, C-015, C-016, C-037, C-040, C-043	V4:12, V4:28, V4:29, V4:30, V4:61, V4:64	V5:25, V5:29, V5:61, V5:65, V5:71	V6:33, V6:51, V6:53, V6:57, V6:62
29	Event Streaming and Durable Messaging	DP-29-001-DP-29-006	6	C-002, C-011, C-012, C-035, C-038, C-043	V4:19, V4:22, V4:30, V4:41, V4:59	V5:34, V5:35, V5:38, V5:39, V5:68	V6:37, V6:38, V6:41, V6:61, V6:62
30	Batch Processing and Orchestration	DP-30-001-DP-30-006	6	C-011, C-012, C-023, C-035, C-043, C-049	V4:22, V4:28, V4:30, V4:41, V4:58	V5:19, V5:38, V5:66, V5:68, V5:70	V6:20, V6:22, V6:37, V6:41, V6:45
31	Stream Processing and Complex Event Handling	DP-31-001-DP-31-006	6	C-011, C-014, C-032, C-033, C-035, C-043	V4:19, V4:22, V4:27, V4:41, V4:47, V4:58	V5:48, V5:52, V5:57, V5:63, V5:68	V6:20, V6:37, V6:38, V6:45, V6:59
32	Query, Caching, Materialization, and Projection	DP-32-001-DP-32-006	6	C-009, C-011, C-014, C-016, C-035, C-043	V4:12, V4:18, V4:21, V4:23, V4:30	V5:23, V5:24, V5:25, V5:26, V5:29	V6:34, V6:35, V6:38, V6:44, V6:58
33	Knowledge Graph Data Architecture	DP-33-001-DP-33-006	6	C-008, C-009, C-010, C-011, C-012, C-017	V4:24, V4:26, V4:28, V4:30, V4:34	V5:13, V5:24, V5:27, V5:29, V5:45	V6:31, V6:34, V6:35, V6:44, V6:62
34	Ontology, Taxonomy, and Semantic Governance	DP-34-001-DP-34-006	6	C-007, C-008, C-009, C-010, C-035, C-051	V4:2, V4:6, V4:23, V4:30, V4:34, V4:72	V5:13, V5:27, V5:45, V5:46, V5:47	V6:34, V6:45, V6:48, V6:65, V6:68
35	Entity Resolution and Master Identity	DP-35-001-DP-35-006	6	C-009, C-010, C-011, C-012, C-016, C-045	V4:25, V4:27, V4:28, V4:30, V4:34, V4:46	V5:27, V5:43, V5:45, V5:46, V5:66	V6:34, V6:35, V6:44, V6:54, V6:59
36	Relationship, Event, Claim, and Assessment Data	DP-36-001-DP-36-006	6	C-005, C-009, C-010, C-011, C-012, C-035	V4:24, V4:26, V4:27, V4:28, V4:32, V4:34	V5:20, V5:27, V5:44, V5:46, V5:63	V6:31, V6:34, V6:35, V6:44, V6:48
37	Enterprise Memory and Strategic Record	DP-37-001-DP-37-006	6	C-011, C-015, C-016, C-025, C-037, C-040	V4:29, V4:30, V4:33, V4:48, V4:54, V4:64	V5:12, V5:25, V5:26, V5:29, V5:38	V6:31, V6:33, V6:35, V6:44, V6:62
38	Search, Vector, and Retrieval Data	DP-38-001-DP-38-006	6	C-005, C-012, C-014, C-016, C-035, C-045	V4:18, V4:23, V4:28, V4:30, V4:33, V4:46	V5:23, V5:24, V5:25, V5:26, V5:29, V5:66	V6:35, V6:38, V6:44, V6:45, V6:58
39	Digital Twin State and Time Travel	DP-39-001-DP-39-006	6	C-010, C-011, C-018, C-019, C-023, C-035	V4:26, V4:28, V4:30, V4:35, V4:38	V5:13, V5:26, V5:28, V5:49, V5:50	V6:31, V6:36, V6:38, V6:45, V6:62
40	Forecast, Scenario, Simulation, Decision, and Outcome Data	DP-40-001-DP-40-006	6	C-002, C-004, C-005, C-020, C-023, C-025	V4:24, V4:28, V4:35, V4:36, V4:37, V4:38, V4:39, V4:48	V5:48, V5:49, V5:50, V5:51, V5:52, V5:69	V6:33, V6:36, V6:38, V6:43, V6:46, V6:62
41	Data Product Architecture	DP-41-001-DP-41-006	6	C-001, C-003, C-006, C-014, C-038, C-049	V4:11, V4:15, V4:23, V4:57, V4:64, V4:71	V5:9, V5:13, V5:20, V5:25, V5:66	V6:40, V6:45, V6:48, V6:59, V6:70
42	Data APIs and Query Services	DP-42-001-DP-42-006	6	C-003, C-014, C-035, C-038, C-039, C-043	V4:15, V4:16, V4:18, V4:20, V4:21, V4:50	V5:19, V5:20, V5:21, V5:23, V5:29	V6:24, V6:25, V6:30, V6:48, V6:59
43	Event Products and Subscriptions	DP-43-001-DP-43-006	6	C-012, C-014, C-035, C-038, C-039, C-043	V4:19, V4:20, V4:22, V4:23, V4:50	V5:34, V5:35, V5:38, V5:57, V5:65	V6:25, V6:37, V6:48, V6:50, V6:59
44	Analytics and Semantic Serving	DP-44-001-DP-44-006	6	C-001, C-003, C-011, C-014, C-034, C-047	V4:18, V4:23, V4:26, V4:27, V4:40	V5:9, V5:13, V5:19, V5:56, V5:64	V6:35, V6:44, V6:46, V6:48, V6:58
45	Feature, Training, Evaluation, and Feedback Data	DP-45-001-DP-45-006	6	C-012, C-031, C-037, C-040, C-044, C-045	V4:28, V4:29, V4:46, V4:48, V4:54, V4:67	V5:14, V5:15, V5:19, V5:61, V5:66, V5:69, V5:70	V6:43, V6:44, V6:45, V6:53, V6:57
46	Context Assembly and Retrieval Products	DP-46-001-DP-46-006	6	C-005, C-012, C-016, C-028, C-035, C-040	V4:21, V4:28, V4:32, V4:33, V4:44	V5:12, V5:19, V5:23, V5:24, V5:29, V5:38	V6:42, V6:44, V6:47, V6:50, V6:53

CH.	DATA SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTION	V4	V5	V6
47	Data Exchange, Export, and Portability	DP-47-001-DP-47-006	6	C-012, C-037, C-038, C-039, C-041, C-043	V4:20, V4:23, V4:25, V4:28, V4:29, V4:30	V5:17, V5:21, V5:25, V5:61, V5:71	V6:11, V6:28, V6:29, V6:47, V6:57
48	Marketplace Data Packages and Extension Contracts	DP-48-001-DP-48-006	6	C-007, C-028, C-035, C-039, C-046, C-049	V4:23, V4:29, V4:44, V4:50, V4:56, V4:67	V5:17, V5:30, V5:36, V5:46, V5:47, V5:70	V6:42, V6:47, V6:48, V6:50, V6:55
49	Metadata Catalog and Data Discovery	DP-49-001-DP-49-006	6	C-006, C-012, C-016, C-035, C-038, C-049	V4:6, V4:11, V4:23, V4:28, V4:29, V4:64	V5:6, V5:13, V5:25, V5:29, V5:65	V6:14, V6:40, V6:48, V6:58, V6:71
50	Policy, Access, and Purpose Enforcement	DP-50-001-DP-50-006	6	C-028, C-035, C-036, C-037, C-039, C-040	V4:13, V4:25, V4:29, V4:49, V4:50, V4:55	V5:21, V5:30, V5:36, V5:37, V5:61, V5:65	V6:49, V6:50, V6:54, V6:57, V6:66
51	Privacy, Minimization, Consent, and Lawful Basis	DP-51-001-DP-51-006	6	C-016, C-029, C-037, C-039, C-040, C-050	V4:29, V4:50, V4:54, V4:55, V4:64	V5:19, V5:21, V5:25, V5:61, V5:65	V6:50, V6:53, V6:54, V6:57, V6:62
52	Residency, Sovereignty, and Cross-Border Control	DP-52-001-DP-52-006	6	C-037, C-039, C-040, C-041, C-042, C-043	V4:14, V4:29, V4:52, V4:53, V4:54, V4:61	V5:21, V5:25, V5:61, V5:71, V5:72	V6:11, V6:29, V6:51, V6:53, V6:57, V6:63
53	Encryption, Tokenization, Masking, and Confidentiality	DP-53-001-DP-53-006	6	C-035, C-036, C-037, C-039, C-040, C-041	V4:29, V4:49, V4:50, V4:52, V4:54, V4:55	V5:21, V5:36, V5:60, V5:61, V5:65	V6:49, V6:51, V6:52, V6:53, V6:57
54	Retention, Legal Hold, Archive, and Deletion	DP-54-001-DP-54-006	6	C-011, C-015, C-016, C-025, C-037, C-040	V4:29, V4:30, V4:54, V4:55, V4:61, V4:64	V5:25, V5:29, V5:61, V5:65, V5:69	V6:33, V6:51, V6:53, V6:57, V6:62
55	Data Stewardship, Approval, and Exception Management	DP-55-001-DP-55-006	6	C-004, C-016, C-028, C-036, C-043, C-050	V4:3, V4:29, V4:42, V4:50, V4:55, V4:72	V5:3, V5:37, V5:47, V5:64, V5:65	V6:4, V6:49, V6:56, V6:64, V6:72
56	Data Audit, Non-Repudiation, and Accountability	DP-56-001-DP-56-006	6	C-012, C-025, C-035, C-036, C-039, C-043	V4:28, V4:49, V4:50, V4:55, V4:62, V4:63	V5:29, V5:36, V5:65, V5:68, V5:72	V6:49, V6:55, V6:56, V6:58, V6:64
57	Data Quality Architecture	DP-57-001-DP-57-006	6	C-005, C-014, C-034, C-035, C-045, C-049	V4:27, V4:28, V4:46, V4:57, V4:62, V4:71	V5:14, V5:63, V5:66, V5:68, V5:72	V6:58, V6:59, V6:60, V6:64, V6:71
58	Freshness, Completeness, and Coverage	DP-58-001-DP-58-006	6	C-013, C-014, C-032, C-033, C-035, C-049	V4:21, V4:31, V4:47, V4:57, V4:62	V5:39, V5:40, V5:57, V5:63, V5:68	V6:28, V6:30, V6:45, V6:58, V6:59
59	Data Observability and Lineage Monitoring	DP-59-001-DP-59-006	6	C-012, C-014, C-035, C-039, C-043, C-049	V4:28, V4:57, V4:62, V4:63, V4:64	V5:14, V5:65, V5:66, V5:68, V5:72	V6:45, V6:56, V6:58, V6:59, V6:64
60	Data SLOs, Capacity, and Performance	DP-60-001-DP-60-006	6	C-014, C-035, C-038, C-043, C-045, C-049	V4:57, V4:58, V4:59, V4:62, V4:63	V5:22, V5:63, V5:66, V5:68, V5:71	V6:22, V6:30, V6:58, V6:59, V6:60
61	Availability, Durability, and Degraded Data Modes	DP-61-001-DP-61-006	6	C-010, C-014, C-035, C-041, C-043, C-049	V4:21, V4:30, V4:57, V4:59, V4:60, V4:61	V5:20, V5:26, V5:63, V5:68, V5:72	V6:31, V6:33, V6:61, V6:62, V6:66
62	Backup, Restore, and Disaster Recovery	DP-62-001-DP-62-006	6	C-011, C-012, C-015, C-041, C-043, C-049	V4:12, V4:30, V4:61, V4:62, V4:64	V5:25, V5:26, V5:38, V5:65, V5:72	V6:31, V6:33, V6:36, V6:62, V6:63
63	Data Incident, Breach, and Correction Response	DP-63-001-DP-63-006	6	C-004, C-012, C-025, C-036, C-039, C-043	V4:30, V4:48, V4:55, V4:62, V4:63	V5:37, V5:47, V5:65, V5:68, V5:72	V6:56, V6:58, V6:62, V6:64, V6:66
64	Operational Data Lifecycle and Housekeeping	DP-64-001-DP-64-006	6	C-011, C-015, C-016, C-035, C-043, C-049	V4:22, V4:30, V4:61, V4:63, V4:64	V5:25, V5:38, V5:68, V5:69, V5:70	V6:31, V6:38, V6:62, V6:65, V6:68
65	Data Pipeline Engineering and Repository Standards	DP-65-001-DP-65-006	6	C-006, C-008, C-012, C-035, C-038, C-052	V4:6, V4:11, V4:23, V4:65, V4:67, V4:68	V5:6, V5:15, V5:18, V5:66, V5:70	V6:40, V6:45, V6:55, V6:67, V6:68
66	Configuration, Infrastructure, and Environment Promotion	DP-66-001-DP-66-006	6	C-035, C-037, C-041, C-042, C-043, C-049	V4:14, V4:23, V4:66, V4:68, V4:69, V4:70	V5:18, V5:21, V5:66, V5:70, V5:71	V6:3, V6:9, V6:10, V6:11, V6:67, V6:68
67	Test Data, Fixtures, and Verification Architecture	DP-67-001-DP-67-006	6	C-010, C-023, C-035, C-040, C-045, C-049	V4:27, V4:46, V4:54, V4:67, V4:71	V5:60, V5:61, V5:66, V5:67, V5:70	V6:45, V6:53, V6:55, V6:68, V6:71
68	Data Release, Migration, Backfill, and Rollback	DP-68-001-DP-68-006	6	C-011, C-012, C-035, C-043, C-049, C-052	V4:22, V4:23, V4:30, V4:68, V4:69, V4:71	V5:15, V5:18, V5:66, V5:68, V5:70	V6:45, V6:48, V6:65, V6:67, V6:68
69	Deployment Parity, Disconnected Sync, and Air Gaps	DP-69-001-DP-69-006	6	C-011, C-037, C-038, C-041, C-042, C-043	V4:14, V4:23, V4:30, V4:54, V4:70, V4:71	V5:21, V5:26, V5:60, V5:70, V5:71	V6:9, V6:10, V6:11, V6:27, V6:57, V6:71
70	Data Cost, Capacity, Sustainability, and Unit Economics	DP-70-001-DP-70-006	6	C-006, C-014, C-035, C-037, C-043, C-049	V4:29, V4:57, V4:58, V4:64, V4:71, V4:72	V5:22, V5:25, V5:66, V5:68, V5:70	V6:22, V6:31, V6:59, V6:60, V6:70
71	Data Platform Acceptance and Operational Readiness	DP-71-001-DP-71-006	6	C-004, C-006, C-035, C-037, C-042, C-049	V4:14, V4:57, V4:61, V4:63, V4:67, V4:71	V5:66, V5:68, V5:70, V5:71, V5:72	V6:59, V6:61, V6:62, V6:63, V6:66, V6:71

CH.	DATA SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTION	V4	V5	V6
72	Data Platform Governance and Baseline Closure	DP-72-001-DP-72-006	6	C-001, C-003, C-004, C-009, C-051, C-052	V4:1, V4:2, V4:6, V4:63, V4:72	V5:1, V5:2, V5:6, V5:69, V5:70, V5:72	V6:1, V6:2, V6:6, V6:64, V6:71, V6:72

Constitutional Data Platform matrix

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 7 ANCHORS
C-001	Category integrity	The Eye shall be conceived, designed, sold, and operated as a Strategic Intelligence Operating System. It shall not be reduced to a chatbot, dashboard, BI tool, copilot, or reporting product.	Ch. 1, Ch. 41, Ch. 44, Ch. 72, Ch. 3, Ch. 5, Ch. 8, Ch. 40
C-002	Closed strategic loop	The product shall implement one continuous operating loop: Observe, Understand, Predict, Simulate, Decide, and Learn. No stage may be treated as a disconnected feature island.	Ch. 8, Ch. 29, Ch. 40, Ch. 1, Ch. 3, Ch. 5, Ch. 44, Ch. 72
C-003	Product boundary	Interfaces such as chat, dashboards, reports, and APIs may exist, but they are access modes to the operating system rather than the product definition.	Ch. 1, Ch. 2, Ch. 7, Ch. 41, Ch. 42, Ch. 44, Ch. 72, Ch. 3, Ch. 5, Ch. 8, Ch. 40
C-004	Human decision sovereignty	A named human authority shall make every final strategic or high-impact decision. The system may advise, prepare, and execute approved workflows, but it shall not silently assume decision rights.	Ch. 1, Ch. 4, Ch. 40, Ch. 55, Ch. 63, Ch. 71, Ch. 72, Ch. 3, Ch. 5, Ch. 8, Ch. 44
C-005	Explain every recommendation	Every material forecast, scenario, simulation result, risk, opportunity, and recommendation shall expose evidence, assumptions, uncertainty, lineage, alternatives, and accountable system versions.	Ch. 22, Ch. 23, Ch. 36, Ch. 38, Ch. 40, Ch. 46, Ch. 57, Ch. 1, Ch. 3, Ch. 5, Ch. 8, Ch. 44, Ch. 72
C-006	Enterprise operating standard	The common platform shall satisfy enterprise requirements for governance, security, scale, auditability, reliability, and lifecycle management from its first production architecture.	Ch. 4, Ch. 5, Ch. 27, Ch. 41, Ch. 49, Ch. 65, Ch. 70, Ch. 71, Ch. 1, Ch. 3, Ch. 8, Ch. 40, Ch. 44, Ch. 72
C-007	Universal core, governed specialization	A single domain-neutral core shall serve all customer classes. Domain ontologies, models, workflows, agents, and policies may extend the core without forking its constitutional semantics.	Ch. 3, Ch. 5, Ch. 20, Ch. 34, Ch. 48, Ch. 8, Ch. 17, Ch. 19, Ch. 33, Ch. 35, Ch. 36
C-008	Ten-layer architecture	The ten named architecture layers are the canonical product structure. Cross-cutting planes may support them but shall not replace, collapse, or rename them without constitutional amendment.	Ch. 1, Ch. 2, Ch. 5, Ch. 8, Ch. 19, Ch. 33, Ch. 34, Ch. 65, Ch. 3, Ch. 17, Ch. 20, Ch. 35, Ch. 36
C-009	Knowledge Graph centrality	The Knowledge Graph shall be the connected semantic heart of The Eye and the shared context substrate for memory, twins, reasoning, prediction, scenarios, simulations, and decisions.	Ch. 1, Ch. 5, Ch. 8, Ch. 19, Ch. 20, Ch. 32, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 72, Ch. 3, Ch. 17
C-010	Truth-state separation	Observed evidence, extracted claims, inferred relationships, model outputs, human judgments, and synthetic scenario data shall remain explicitly distinguishable throughout the system.	Ch. 6, Ch. 12, Ch. 16, Ch. 17, Ch. 19, Ch. 21, Ch. 22, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 39, Ch. 61, Ch. 67, Ch. 3, Ch. 5, Ch. 8, Ch. 20
C-011	Temporal truth	Material facts and relationships shall preserve event time, observation time, validity time, and system-record time so that past world states can be reconstructed without hindsight contamination.	Ch. 3, Ch. 11, Ch. 13, Ch. 15, Ch. 17, Ch. 18, Ch. 19, Ch. 20, Ch. 21, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 29, Ch. 30, Ch. 31, Ch. 32, Ch. 33, Ch. 35, Ch. 36, Ch. 37, Ch. 39, Ch. 44, Ch. 54, Ch. 62, Ch. 64, Ch. 68, Ch. 69, Ch. 5, Ch. 8, Ch. 34
C-012	End-to-end provenance	Every material intelligence object shall carry source, chain of custody, transformation history, ownership, rights, quality, and version lineage back to its evidence.	Ch. 3, Ch. 6, Ch. 7, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 13, Ch. 14, Ch. 15, Ch. 16, Ch. 17, Ch. 18, Ch. 19, Ch. 21, Ch. 22, Ch. 23, Ch. 24, Ch. 28, Ch. 29, Ch. 30, Ch. 33, Ch. 35, Ch. 36, Ch. 38, Ch. 43, Ch. 45, Ch. 46, Ch. 47, Ch. 49, Ch. 56, Ch. 59, Ch. 62, Ch. 63, Ch. 65, Ch. 68, Ch. 5, Ch. 8, Ch. 20, Ch. 34
C-013	Observation breadth	The World Observation Layer shall support the full canonical source universe and allow governed custom sources without architectural rework.	Ch. 9, Ch. 10, Ch. 14, Ch. 58, Ch. 3, Ch. 5, Ch. 8, Ch. 17, Ch. 19, Ch. 20, Ch. 33, Ch. 34, Ch. 35, Ch. 36

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 7 ANCHORS
C-014	Freshness and quality contracts	Sources, derived data, twins, models, and intelligence products shall declare measurable freshness, completeness, quality, and expiry expectations.	Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 13, Ch. 14, Ch. 15, Ch. 17, Ch. 31, Ch. 32, Ch. 38, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 70, Ch. 21, Ch. 25, Ch. 37, Ch. 39
C-015	Durable institutional memory	The Eye shall preserve institutional context, decisions, assumptions, evidence, outcomes, and lessons as durable memory, subject to lawful retention, deletion, and customer policy.	Ch. 24, Ch. 25, Ch. 26, Ch. 28, Ch. 37, Ch. 54, Ch. 62, Ch. 64, Ch. 9, Ch. 10, Ch. 21, Ch. 38, Ch. 39
C-016	Memory governance	Memory shall be permission-aware, tenant-isolated, versioned, retrievable, correctable, and protected from unverified model output becoming accepted organizational truth.	Ch. 4, Ch. 12, Ch. 20, Ch. 25, Ch. 27, Ch. 28, Ch. 32, Ch. 35, Ch. 37, Ch. 38, Ch. 46, Ch. 49, Ch. 51, Ch. 54, Ch. 55, Ch. 64, Ch. 9, Ch. 10, Ch. 21, Ch. 39
C-017	Strategy Graph	Objectives, hypotheses, decisions, initiatives, capabilities, resources, indicators, risks, opportunities, and outcomes shall be connected in a governed Strategy Graph anchored to the Knowledge Graph.	Ch. 33, Ch. 9, Ch. 10, Ch. 21, Ch. 25, Ch. 37, Ch. 38, Ch. 39
C-018	Grounded Digital Twins	Every Digital Twin shall be grounded in evidence, graph entities, temporal state, declared assumptions, and calibrated models; it shall never present synthetic state as observed fact.	Ch. 39, Ch. 9, Ch. 10, Ch. 21, Ch. 25, Ch. 37, Ch. 38
C-019	Twin versionability	Digital Twins shall support time travel, branching, comparison, replay, reconciliation, and versioned evolution across actual and simulated states.	Ch. 39, Ch. 9, Ch. 10, Ch. 21, Ch. 25, Ch. 37, Ch. 38
C-020	Canonical prediction horizons	The Prediction Engine shall support 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons with horizon-appropriate methods and uncertainty.	Ch. 40, Ch. 22, Ch. 39, Ch. 45, Ch. 46
C-021	Probabilistic forecasting	Forecasts shall communicate distributions, confidence, calibration, drivers, sensitivity, and failure conditions rather than false precision or unsupported certainty.	Ch. 22, Ch. 39, Ch. 40, Ch. 45, Ch. 46
C-022	Scenario plurality	Scenario Intelligence shall maintain multiple coherent futures, including baseline, upside, downside, disruption, and user-defined alternatives; no single forecast shall masquerade as destiny.	Ch. 22, Ch. 39, Ch. 40, Ch. 45, Ch. 46
C-023	Reproducible simulation	Every simulation shall declare initial state, assumptions, interventions, model versions, constraints, random seeds where applicable, outputs, and sensitivity so results can be reproduced and challenged.	Ch. 30, Ch. 39, Ch. 40, Ch. 67, Ch. 22, Ch. 45, Ch. 46
C-024	Decision completeness	Decision Intelligence shall compare viable options against explicit objectives, constraints, risks, opportunities, trade-offs, second-order effects, reversibility, and information value.	Ch. 22, Ch. 39, Ch. 40, Ch. 45, Ch. 46
C-025	Decision Replay	The Eye shall preserve the state of knowledge available at decision time and support replay against later evidence and outcomes without rewriting the historical record.	Ch. 21, Ch. 23, Ch. 24, Ch. 37, Ch. 40, Ch. 54, Ch. 56, Ch. 63, Ch. 20, Ch. 46, Ch. 50, Ch. 55
C-026	Objective alignment	Recommendations shall be evaluated against declared objectives, stakeholder obligations, risk appetite, policy, and strategic constraints rather than an opaque global score.	Ch. 20, Ch. 23, Ch. 24, Ch. 37, Ch. 40, Ch. 46, Ch. 50, Ch. 55
C-027	Executive Operating System	The executive surface shall organize strategic attention, decisions, scenarios, evidence, objectives, commitments, and outcomes. It shall not degrade into a passive dashboard collection.	Ch. 20, Ch. 23, Ch. 24, Ch. 37, Ch. 40, Ch. 46, Ch. 50, Ch. 55
C-028	Bounded agent authority	Every agent shall have a declared purpose, tools, data scope, permissions, budgets, escalation rules, prohibited actions, evaluation suite, and accountable owner.	Ch. 46, Ch. 48, Ch. 50, Ch. 55, Ch. 20, Ch. 23, Ch. 24, Ch. 37, Ch. 40
C-029	Supervised orchestration	Planner, Supervisor, and Workflow agents shall enforce policy, separation of duties, checkpoints, timeouts, retries, human gates, and complete execution traces across multi-agent work.	Ch. 6, Ch. 51, Ch. 20, Ch. 23, Ch. 24, Ch. 37, Ch. 40, Ch. 46, Ch. 50, Ch. 55
C-030	Model pluralism	The platform shall support multiple models and reasoning approaches behind governed interfaces so customers retain control over cost, sovereignty, performance, and vendor dependence.	Ch. 15, Ch. 24, Ch. 20, Ch. 23, Ch. 37, Ch. 40, Ch. 46, Ch. 50, Ch. 55
C-031	No silent self-modification	Continuous learning shall never permit unreviewed production models, agents, prompts, policies, ontologies, or decision logic to modify themselves silently.	Ch. 45, Ch. 31, Ch. 44, Ch. 48, Ch. 57, Ch. 58, Ch. 59, Ch. 60
C-032	Proactive intelligence	The system shall surface material changes, weak signals, risks, and opportunities before users ask, while respecting relevance, authority, attention, and notification policy.	Ch. 10, Ch. 31, Ch. 58, Ch. 44, Ch. 48, Ch. 57, Ch. 59, Ch. 60
C-033	Early-warning discipline	Warnings shall be deduplicated, prioritized, evidence-backed, confidence-aware, time-bounded, and linked to affected objectives, assets, decisions, and response playbooks.	Ch. 31, Ch. 58, Ch. 44, Ch. 48, Ch. 57, Ch. 59, Ch. 60
C-034	Decomposable Strategic Health Score	Any Strategic Health Score shall expose component measures, weights, evidence, confidence, trend, and sensitivity. The score shall never replace the underlying analysis.	Ch. 44, Ch. 57, Ch. 31, Ch. 48, Ch. 58, Ch. 59, Ch. 60

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 7 ANCHORS
C-035	Trust as a system property	Trust, provenance, identity, authorization, policy, safety, and audit shall operate across every layer and intelligence object, not as a downstream compliance add-on.	Ch. 2, Ch. 6, Ch. 8, Ch. 9, Ch. 10, Ch. 12, Ch. 13, Ch. 16, Ch. 17, Ch. 18, Ch. 19, Ch. 20, Ch. 21, Ch. 22, Ch. 23, Ch. 25, Ch. 26, Ch. 27, Ch. 29, Ch. 30, Ch. 31, Ch. 32, Ch. 34, Ch. 36, Ch. 38, Ch. 39, Ch. 42, Ch. 43, Ch. 46, Ch. 48, Ch. 49, Ch. 50, Ch. 53, Ch. 56, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 64, Ch. 65, Ch. 66, Ch. 67, Ch. 68, Ch. 70, Ch. 71, Ch. 44
C-036	Non-repudiable accountability	Material access, transformations, agent actions, approvals, decisions, overrides, releases, and exports shall produce tamper-evident audit records attributable to identities and versions.	Ch. 4, Ch. 7, Ch. 23, Ch. 50, Ch. 53, Ch. 55, Ch. 56, Ch. 63, Ch. 47, Ch. 51, Ch. 52, Ch. 54, Ch. 62, Ch. 69
C-037	Customer data control	Customers retain governance over their data, models, twins, policies, retention, residency, encryption, exports, and deletion within the chosen deployment and contractual boundary.	Ch. 4, Ch. 9, Ch. 11, Ch. 25, Ch. 26, Ch. 28, Ch. 37, Ch. 45, Ch. 47, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 66, Ch. 69, Ch. 70, Ch. 71, Ch. 56, Ch. 62
C-038	Open interoperability	The Eye shall integrate through governed APIs, events, connectors, common data formats, and export paths; core customer knowledge shall not be trapped behind proprietary-only access.	Ch. 3, Ch. 5, Ch. 8, Ch. 11, Ch. 14, Ch. 15, Ch. 17, Ch. 18, Ch. 26, Ch. 29, Ch. 41, Ch. 42, Ch. 43, Ch. 47, Ch. 49, Ch. 60, Ch. 65, Ch. 69, Ch. 4, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 56, Ch. 62
C-039	Zero-trust security	Identity, device, workload, agent, model, source, and data access shall be continuously authenticated, authorized, scoped, logged, and re-evaluated under least privilege.	Ch. 6, Ch. 13, Ch. 14, Ch. 16, Ch. 20, Ch. 27, Ch. 42, Ch. 43, Ch. 47, Ch. 48, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 56, Ch. 59, Ch. 63, Ch. 4, Ch. 54, Ch. 62, Ch. 69
C-040	Privacy and minimization	Collection and use shall be purpose-bound, proportionate, policy-enforced, and minimized, with support for consent, legal basis, access, correction, retention, and deletion obligations.	Ch. 6, Ch. 9, Ch. 16, Ch. 28, Ch. 37, Ch. 45, Ch. 46, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 67, Ch. 4, Ch. 47, Ch. 56, Ch. 62, Ch. 69
C-041	Sovereignty by design	Data residency, key ownership, network isolation, local inference, model choice, administrative control, and disconnected operation shall be supported where customer policy requires them.	Ch. 47, Ch. 52, Ch. 53, Ch. 61, Ch. 62, Ch. 66, Ch. 69, Ch. 4, Ch. 50, Ch. 51, Ch. 54, Ch. 56
C-042	Deployment semantic parity	SaaS, Private Cloud, and On-Premise deployments shall preserve the same core semantics, governance model, APIs, artifacts, and intelligence lifecycle, with documented capability differences only where unavoidable.	Ch. 3, Ch. 8, Ch. 52, Ch. 66, Ch. 69, Ch. 71, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 57, Ch. 60, Ch. 61, Ch. 62, Ch. 63
C-043	Resilient operation	The platform shall fail safely, degrade intelligibly, recover predictably, preserve evidence and audit continuity, and expose when data, models, or dependencies are unavailable or stale.	Ch. 7, Ch. 10, Ch. 11, Ch. 13, Ch. 14, Ch. 15, Ch. 16, Ch. 18, Ch. 21, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 30, Ch. 31, Ch. 32, Ch. 42, Ch. 43, Ch. 47, Ch. 52, Ch. 55, Ch. 56, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 64, Ch. 66, Ch. 68, Ch. 69, Ch. 70, Ch. 3, Ch. 17, Ch. 41, Ch. 57, Ch. 71
C-044	Governed learning loop	Learning from outcomes, feedback, overrides, and new evidence shall pass through evaluation, approval, release, monitoring, rollback, and tenant-isolation controls.	Ch. 45, Ch. 3, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 57, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71
C-045	Continuous evaluation	Models, agents, forecasts, extractors, ontologies, simulations, and recommendations shall be evaluated for accuracy, calibration, robustness, drift, bias, safety, cost, and operational fitness.	Ch. 12, Ch. 22, Ch. 35, Ch. 38, Ch. 45, Ch. 57, Ch. 60, Ch. 67, Ch. 3, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71
C-046	Governed marketplaces	Scenario and Agent Marketplace packages shall be signed, versioned, inspectable, permission-scoped, licensed, evaluated, sandboxed, monitored, revocable, and traceable to publishers.	Ch. 48, Ch. 3, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 57, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71
C-047	Experience integrity	The product experience shall expose context, uncertainty, provenance, decision state, and consequence at the point of use; critical meaning shall not be hidden behind ornamental visualization.	Ch. 44, Ch. 3, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 57, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71
C-048	Role and accessibility clarity	Experiences shall be accessible and role-appropriate for executives, analysts, operators, domain experts, administrators, and governance teams without creating separate truths.	Ch. 3, Ch. 17, Ch. 18, Ch. 41, Ch. 42, Ch. 57, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 71

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 7 ANCHORS
C-049	Operational quality contract	Reliability, performance, scale, freshness, accuracy, calibration, explainability, reproducibility, security, accessibility, and cost shall be measured as product behavior, not assumed as implementation detail.	Ch. 4, Ch. 5, Ch. 30, Ch. 41, Ch. 48, Ch. 49, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 64, Ch. 66, Ch. 67, Ch. 68, Ch. 70, Ch. 71, Ch. 3, Ch. 17, Ch. 18, Ch. 42, Ch. 63
C-050	Responsible-use boundary	The Eye shall not autonomously execute decisions that materially determine life, liberty, legal status, access to essential services, employment, credit, or equivalent high-impact rights and interests.	Ch. 7, Ch. 51, Ch. 55, Ch. 1, Ch. 2, Ch. 24, Ch. 65, Ch. 68, Ch. 71, Ch. 72
C-051	Constitutional change control	Changes to mission, product category, human authority, principles, canonical layers, core semantics, or constitutional invariants require explicit amendment, impact analysis, ratification, and versioned publication.	Ch. 2, Ch. 3, Ch. 34, Ch. 72, Ch. 1, Ch. 7, Ch. 24, Ch. 55, Ch. 65, Ch. 68, Ch. 71
C-052	Documentation inheritance	Volumes 1 through 10 shall inherit this constitution, use its canonical terms, trace material requirements to invariant IDs, and identify rather than conceal any unresolved conflict.	Ch. 1, Ch. 2, Ch. 7, Ch. 18, Ch. 65, Ch. 68, Ch. 72, Ch. 24, Ch. 55, Ch. 71

O Data Platform Architecture Decision Baseline

These decisions freeze the governing Data Platform beneath Volumes 0, 3, 4, 5, and 6. Implementations may change storage, processing, catalog, and serving technologies behind them but cannot treat a local exception as authority to change their semantics.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
DADR-001	Contracts precede physical data structures	Tables, topics, files, indexes, and APIs implement stable semantic contracts.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-002	Canonical ownership remains singular	Every authoritative lifecycle has one accountable owning component.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-003	Raw evidence is immutable and derivatives are explicit	Parsing and AI transformation never overwrite source evidence.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-004	Four temporal coordinates are first-class	Historical reconstruction does not depend on latest-value storage.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-005	Truth state is mandatory	Observed, inferred, assessed, synthetic, and decided data remain distinguishable.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-006	Provenance is canonical data	Material lineage cannot depend on mutable operational logs.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-007	Correction is non-destructive	Historical knowledge and later amendments remain replayable.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-008	State class precedes storage engine	Authority, consistency, locality, lifecycle, and recovery select technology.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-009	Open analytical formats preserve portability	Customer knowledge is not trapped in proprietary-only representation.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-010	Operational writes follow service ownership	Shared tables and cross-owner mutation are prohibited.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-011	Messaging semantics are explicit	Delivery and acknowledgement do not imply institutional commitment.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-012	Derived projections are rebuildable	Indexes, caches, marts, and embeddings never become silent truth.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-013	Knowledge Graph is the semantic heart	Identity, relationship, time, provenance, and strategy context converge there.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-014	Observed and synthetic twin state are isolated	Scenario and simulation branches cannot contaminate observed state.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-015	Strategic packages preserve exact versions	Forecasts, simulations, recommendations, decisions, and outcomes remain distinct.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-016	Data products are operational contracts	A table, report, or dashboard is not a product without ownership and SLOs.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
DADR-017	Access is identity- and purpose-bound	Reachability and possession never authorize use.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-018	Privacy and sovereignty propagate to derivatives	Indexes, features, models, backups, and exports inherit obligations.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-019	Deletion requires residual closure	Canonical, derived, recovery, and processor copies are accounted for.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-020	Quality is multidimensional product state	One aggregate score cannot conceal failed controls or uncertainty.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-021	Degraded data modes are explicit	The platform never fabricates completeness, freshness, correctness, or policy state.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-022	Backups are untrusted until restored	Recovery evidence includes schemas, keys, lineage, policies, and consumers.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-023	Data releases are dependency-closed transitions	Schemas, code, policy, data, migration, and rollback are resolved together.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.
DADR-024	One data contract spans every deployment	Physical variation cannot create SaaS, Private Cloud, or On-Premise semantics.	Implement through owned components, contracts, manifests, independent controls, immutable releases, operating evidence, recovery, audit, and customer acceptance.

P Data Platform Glossary

These terms are canonical for data authority, ingestion, contracts, temporal truth, provenance, state, knowledge, memory, products, privacy, quality, lifecycle, recovery, delivery, economics, and governance. Domain designs may add narrower terms but may not redefine the concepts below.

TERM	DATA PLATFORM DEFINITION
Analytical storage	Storage optimized for governed scans, aggregation, historical analysis, and reproducible snapshots.
Asset	A governed dataset, stream, object family, graph, index, feature, product, exchange, or evidence collection.
Authority class	The declared role of data as authoritative, evidence, product, derived, temporary, or recovery state.
Backfill	Controlled recomputation or loading of historical scope under exact contracts and reconciliation.
Bitemporal	Preserving represented validity time separately from platform record time.
Blind spot	A declared part of the coverage universe that is not currently observed with required quality.
Canonical object	A governed intelligence object carrying common identity, time, truth, provenance, policy, quality, and lifecycle.
Catalog	Authoritative metadata describing assets, owners, contracts, lineage, policy, quality, consumers, and lifecycle.
CDC	Change data capture from source transaction logs or equivalent ordered change mechanisms.
Chain of custody	Attributable evidence of possession, transfer, integrity, and handling from source through use.
Classification	Policy-bearing sensitivity, consequence, and handling designation.
Completeness	The degree to which required records, fields, partitions, relationships, or coverage are present.
Consumer contract	The versioned behavior and obligations a consumer accepts from a data product.
Context manifest	A bounded purpose-filtered list of exact evidence and data versions assembled for a task.
Correction	A new governed version that amends prior data without erasing historical state.
Coverage universe	The declared set of sources, entities, regions, periods, or phenomena a product intends to observe.
Custodian	The technical authority operating physical data state under owner and policy direction.
Data contract	A machine-readable agreement covering semantics, schema, keys, time, truth, provenance, policy, quality, SLOs, and lifecycle.
Data domain	A stable governed business or intelligence scope with accountable ownership and extension authority.
Data mesh	A federated operating pattern for domain ownership and products that remains subordinate to the universal core.
Data product	An owned reusable operational data contract with consumers, quality, policy, service levels, cost, and lifecycle.
Data quality	Measured multidimensional fitness of data for a declared purpose and consequence.
Data steward	The accountable authority for semantic, identity, quality, access, correction, or lifecycle adjudication.
Dataset snapshot	An immutable reproducible version of a governed dataset and its manifest.
Degraded data mode	An explicit operating state with declared available data, limitations, trust, freshness, and exit evidence.
Deletion verification	Evidence that approved deletion scope and residual references have been resolved.
Derived data	Data computed from authoritative or evidence state and rebuildable under a declared method.
Entity resolution	Evidence-backed identification of records referring to the same or distinct real-world entity.
Event time	Time the represented event occurred in the source or world.
Evidence object	Immutable source content and metadata with identity, rights, integrity, and custody.
Exchange manifest	Signed inventory of data, versions, contracts, policy, integrity, destination, and receipts in a transfer.
Feature	A versioned model input derived under point-in-time, lineage, policy, and quality contracts.

TERM	DATA PLATFORM DEFINITION
Freshness	Age and expected-arrival state of data relative to its declared contract.
Golden fixture	Versioned input and expected result used to prove contract behavior.
Grain	The exact entity, event, time, and aggregation level represented by one data record.
Graph revision	An atomic validated version of nodes, edges, ontology bindings, provenance, and temporal state.
Lakehouse	Open analytical storage combining object-scale data, transactional metadata, schemas, and time travel.
Late data	Data arriving after the processing watermark or expected decision window.
Lawful basis	Approved legal or contractual authority for a data processing purpose.
Legal hold	Authority preventing modification or deletion of scoped data despite ordinary lifecycle policy.
Lineage	The directed record of inputs, transformations, actors, methods, versions, and outputs.
Materialization	A rebuildable stored projection produced for serving or analysis.
Metric grain	The population, dimensions, time window, and unit for a governed measure.
Minimization	Limiting collection and use to data proportionate and necessary for an approved purpose.
Observation time	Time the platform or source observed the evidence.
Ontology	A governed semantic model of types, relationships, constraints, and mappings.
Open format	A documented portable representation that permits independent access and reimplementation.
Operational store	Consistency-sensitive service-owned storage for live canonical lifecycle state.
Point-in-time correctness	Use of only information available at the represented historical time.
Projection	Derived query or serving state rebuildable from an authoritative source and definition.
Provenance	Evidence of source, custody, method, actor, policy, quality, and version history.
Purpose limitation	Restriction that data is used only for declared authorized purposes.
Quarantine	Isolated state preventing untrusted or non-conformant data from downstream use.
Record time	Time the platform committed a version to canonical state.
Reconciliation	Comparison and repair of source, canonical, derived, transferred, or recovered data state.
Residency	Approved physical or jurisdictional location of data, processing, keys, backups, and support.
Retention	Policy controlling how long data remains active, archived, held, or eligible for deletion.
Schema	Machine-readable structure, types, fields, constraints, and compatibility identity.
Semantic parity	Identical strategic data meaning and lifecycle despite physical deployment differences.
Source contract	Versioned authority, rights, purpose, collection, quality, and correction agreement for a source.
Source of truth	The accountable component and state contract with canonical write authority for a lifecycle.
State tier	A controlled data zone defined by authority, purpose, protection, lifecycle, and recovery.
Synthetic data	Artificial data explicitly separated from observed evidence and governed by purpose.
Temporal truth	Preservation of event, observation, validity, and record time for reconstruction.
Tombstone	A durable lifecycle marker representing deletion or withdrawal without pretending the prior record never existed.
Truth state	The epistemic status of data as observed, asserted, extracted, inferred, assessed, synthetic, decided, disputed, or withdrawn.
Validity time	The interval during which the represented fact or relationship is considered valid.
Watermark	A processing claim about event-time completeness up to a declared boundary.
Withdrawal	A governed state removing an object from current active use while preserving historical accountability.

Baseline closure

Volume 7 closes when every production data capability can identify its authority, source contracts, accountable owners, canonical objects, identities, time semantics, truth states, provenance, corrections, contracts, schemas, ontologies, state tiers, pipelines, products, consumers, policies, quality, service levels, failure states, degraded modes, retention, deletion, recovery, deployment profile, release, cost, acceptance, evidence, exception, and retirement state.

END STATE

The Eye is data-complete here as one enterprise Strategic Intelligence Operating System: evidence-grounded, temporally reconstructable, semantically connected, permission-aware, correction-preserving, deployment-portable, quality-measured, recoverable through tested state, open to governed exchange, and always subordinate to accountable human authority.

CONSTITUTIONAL INHERITANCE / C-001 · C-002 · C-004 · C-005 · C-009 · C-010 · C-011 · C-012 · C-015 · C-035 · C-037 · C-043 · C-049 · C-051 · C-052